- **Abstract**

Extreme climatic conditions have given birth, through years of evolution and studies, to different passive solutions. One of them: façade perforated screens, varying from one place to another according to cultural and climatic factors.

Mashrabiya (traditional perforated screens), situated in the middle east countries, is an element that gathers both passive environmental solutions, traditional art and cultural aspects. It is also a façade component with various geometrical forms that can combine different screens typologies. Implementation of the traditional perforated screens in vernacular architecture has been conducted by designing a façade component that balances the need for shading and allowing air to penetrate the room at certain desired velocities. Reintroducing Mashrabiya in contemporary design can be the starting point towards achieving passive buildings in hot climates.

Nowadays, there is a vast reduction in the use of this powerful traditional shading tool in contemporary buildings, endangering Mashrabiya as an art. So, this thesis aims to illustrate the environmental performance of the existing traditional perforated screens and re-evaluate them in a practical way.

Reintroducing them as a passive solution can be Through systematic studies and quantification of the environmental performance of these façade components with the help of analytical tools. This study can portray the implementation of Mashrabiya in a more appropriate way, through providing design recommendations that emphasize the most applicable use for various patterns according to the location, orientation and type of the space. this can facilitate its usage for a better approach in the different environmental parameters.

Evaluating environmental performance of Mashrabiya

Generating guidelines for contemporary implementation

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- **Acknowledgment:**

For every start, there must be an end and there must be a full stop for the end of the story. So here I am making a full stop and butting an end for my scientific research that has been finished with the help of my professors who I am being grateful for. Thanks a lot for helping me with your advice and knowledge and also for teaching me to be confident in my steps especially professor Rosa Schiano-phan. I am very grateful to all the effort and support she provided and all the knowledge she shared with me.

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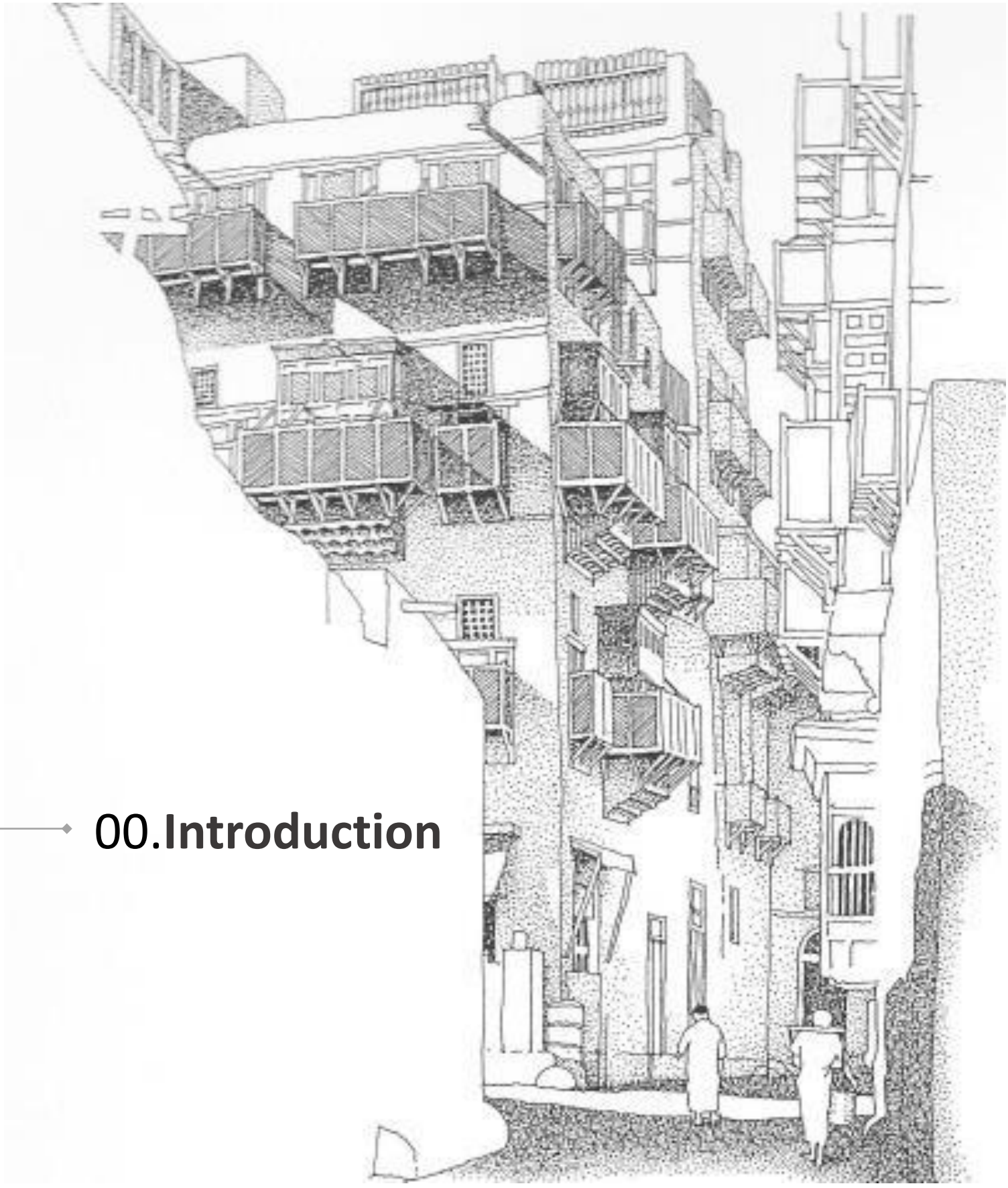


00.Introduction

Introduction:

1.Theoretical background

The word "Mashrabiya" is defined in the Oxford dictionary as a latticework of Islamic architecture. Its meaning in the Arabic language is driven from the primary function required from Mashrabiya. Where the word "Mashrabiya" in the Arabic language is driven from the word "sharab" which means drink, this is rooted in the primary function of Mashrabiya where its perforated screens allow air to flow and cool down the ceramic pots placed near the screens. From that, the users used to place a ceramic pot in front of inlet pattern, in order to achieve a multi proposal process. where the air passing gets colder and moisturized while interring the space, and the water inside the pot is cold for drinking as if it is a "natural refrigerator."

Mashrabiya as a concept is a component developed by time to balance the environmental and cultural requirements. It was based in Islamic architecture to achieve the balance between making comfort and preserving the privacy of the indoor spaces. Based on that, the typology of the Mashrabiya itself started to evolve through time, to achieve the required and desired functions it was created for.

In the past hot climatic regions used to respond to extreme climatic factors through implementing thick walls with small openings, to maintain the number of solar gains resulted from the light entering the space. Also, the urban fabric used to provide more adaptability towards these extreme climatic conditions.

In hot climates, especially the middle eastern countries, the urban fabric used to be close with the narrow streets. That brought the buildings close to each other to provide mutual shading. But this typology forced the buildings to have small and narrow openings respecting the privacy of the houses.

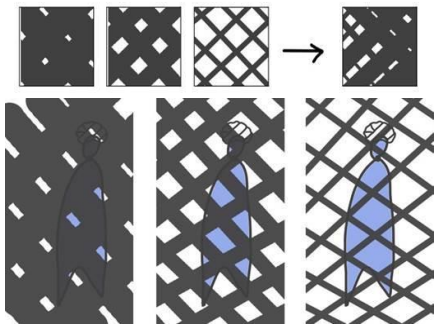


Fig 0.1.1 sketch shows how perforated screens is linked to privacy , source:(google)

According to the Islamic religion, the women privacy inside the house should be protected from the outside pedestrians, as well as other cultural aspects. This formed the shape of the vernacular architecture at this time. But through time, it was essential to allow some external factors to enter the space, like air, a proper amount of light and the prayers sounds coming from the mosque. All these factors managed to change the concept of windows in the vernacular architecture of the middle eastern countries. That's when Mashrabiya was introduced as a solution. Despite the comprehensive implementation of Mashrabiya in Islamic architecture through time, no papers had documented the Mashrabiya as a component with its wide variations.

2.Context and precedent of Mashrabiya

The idea of Mashrabiya expanded with the expansion of the Islamic culture, and the patterns implemented in it starts to vary according to the culture and the heritage of each place. And based on that many typologies of Mashrabiya were formed.

The typologies of Mashrabiya used to vary according to geometry, function and the patterns used in the component. Also, the name and the material used to vary from a region to another according to its climatic factors and cultural aspects.

As the typology of the Mashrabiya changes the environmental performance changes as well, based on that, the scope of this paper was oriented towards two main types of Mashrabiya, which are:

Andalusian Mashrabiya- based in span and morocco

Ottoman Mashrabiya based in Egypt

These two types were selected because they represent the majority of the existing Mashrabiya in the middle east, and there are huge variations between the two models. This will expand the documentation of the analysis of different patterns and forms.

Most of the countries that implemented Mashrabiya are characterized by having a hot climate. Where they used Mashrabiya in order to create balance between allowing the air to flow to the indoor spaces and preventing excessive solar gains. For example, the summer days in Egypt last more than six months, and this majorly affected the typology of the architecture to be more adapted towards extreme climatic days. Based on that, many typologies of Mashrabiya were implemented in Egyptian Islamic architecture. These typologies varied from a period to another in terms of geometry and pattern.

- Evaporation process

Where the droplets on the exterior surface of the pot help to moisturize the air

- Cooling the water inside the pot

Where the air passing by the pot cool the surface which helps in cooling the water inside, so mashrabiya acts as a passive refrigerator to cool the drinking water.

- Cooling the air coming inside the space

Where the air temperature reduces significantly by the evaporative cooling effect or by hitting the cold surface of the ceramic pot

Fig 0.1.2 sketch introducing the ceramic pot in mashrabiya , source:(author)

3.Research questions

•Based on what was found in the literature review, there was a lot of research questions, which are:
•What are the basics of geometry?
•How can the geometry of the component affect the total performance?
•How is the pattern selected to be applied for a particular façade?
•Why do some components have a variable combination of patterns?
•How do the different components contribute to the indoor comfort

4.Objectives of the study:

Due to the lake of resources that tried to document Mashrabiya with its variation and details. This study will provide a systematic recommendation to reintroduce Mashrabiya in a more practical way. During the main part and steps, the paper will be documenting different forms, shapes, and details that can help in the efficient modelling of Mashrabiya.

5.Scope of the study

Since that the main aim of the study is to quantify and evaluate the environmental performance of Mashrabiya, the paper focused on the component with its details and comparing them with each other, based on that no surrounding context was considered.
Also, the main focus was on comparing the performance in different cases, so the climatic context was fixed to test the performance in the hottest day in summer of Cairo-Egypt.

6.Observing Mashrabiya

•To understand how different typologies preform, it was important to observe these typologies and test their environmental performance in terms of shading and ventilation. For that, fieldwork was held to provide sufficient data and satisfying background of these different forms and patterns. Objectives of fieldwork were not limited on monitoring the environmental performance and understand the variations in the performance from a case to another. It was also for documenting the dimensions of each pattern and the details of the different components. Afterwards, a systematic dimensional sketch were made in order to guarantee the accuracy in the modelling of the patterns and components in different software. During the fieldwork, a survey was held among the craftsmen and carpenters to understand how they used to model the Mashrabiya and state what patterns should be applied. This step was essential to know how the majority of contemporary Mashrabiya examples were modelled.

Fig 0.1.3 photos taken inside the carpenters workshops , source:(author)

From the fieldwork, it was observed the drop implementation of Mashrabiya and the dis-connectivity between the contemporary models of Mashrabiya and the traditional examples stated in the papers. From this, there was a capability of stating some disadvantages that might lead to this drop-in implementation. Also, from understanding the traditional examples, it can be noted that the contemporary perforated screens made for low cost (typical) houses are far away from the concept of Mashrabiya.

7.Methodology

This study depends on empirical evidence and fieldwork documentation besides the literature review of previous paper works and books that contributed a lot to understanding Mashrabiya. Based on literature review, fieldwork and systematic quantification of the environmental performance of Mashrabiya.
After observing and concluding a lot of facts about Mashrabiya, the analysing process was shaped according to the outcomes of the fieldwork. It was found that Mashrabiya performance varies significantly by changing the smallest details of it, as well as the change in the integral component. This shaped the process of quantification of Mashrabiya performance to be divided into two main parts:
A. **Micro-level:** where the main focus will be on the patterns and how they differ in shape and performance. To quantify these shapes, seven patterns were selected based on its variation, where each pattern represent a representative example of a category of patterns similar to it in form, dimensions and perforation percentage. Then the seven pattern were correlated with each other in order to understand the variation in the performances in terms of shading and ventilating.

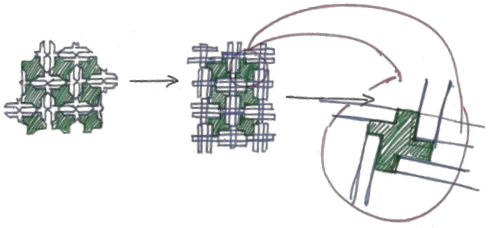


Fig 0.1.4 drawn concept of micro level methodology, source:(author)

B. Macro-level (component level): where the main focus was to understand how the Mashrabiya preforms as a component with different typologies. In consequence, different geometries were tested with various forms and patterns. In this part, the study aimed to understand the outcomes of combining different patterns in the same component and set the baseline of understanding the different combinations.

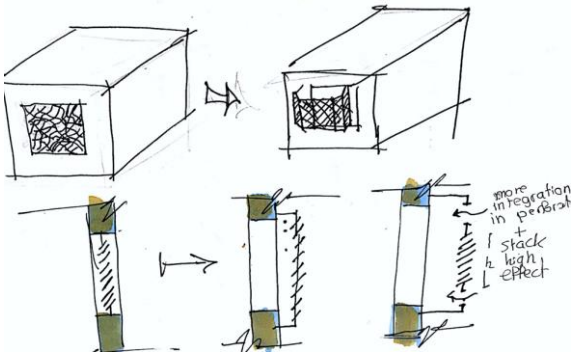


Fig 0.1.5 drawn concept of macro level methodology , source:(author)

C. Site-level: this case is to observe how the analytical tools will visualise the performance of a living example of Mashrabiya, with putting site, surrounding context, and climatic conditions into consideration. This case had its objectives and expected outcomes. As the desired approach of this step is proving some hypothesis that was made during the fieldwork.

8. summary of results

Based on the analytical quantification of Mashrabiya in terms of screens and component. It was found that Mashrabiya, as a façade component, can have an impressive impact in improving the indoor conditions of the space. But it also can have a negative effect by not preforming efficiently if the wrong pattern is applied to the façade.

9. Providing design guidelines

Based on the conclusion made from analysing and quantifying the environmental performance of Mashrabiya, in terms of geometry, combinations of patterns and adding additional process to the component. It was found that Mashrabiya, as a component, can have a negative impact on the indoor space if it is inappropriately applied. If a wrong pattern is applied to the wrong façade, its performance efficiency will be massively reduced. For this, environmental designing is a tool that can reintroduce Mashrabiya in the right way. Offering a matrix of patterns preference and recommendations based on systematic quantification of its environmental performance on different parameters.

01.Theoretical background

Literature review – precedent –research questions and hypothesis

1.1 Mashrabiya in vernacular architecture:

Mashrabiya is rooted in vernacular architecture in order to balance the need of ventilation and cooling the space and shading the indoor spaces by preventing the undesired solar radiations that eventually contributes to the uncomfortableness in the resultant temperature of the indoor space.

Mashrabiya, as an element, doesn't only reflect the cleverness in vernacular architecture. It also shows essential facts about Islamic architecture and how conservative society shaped the architecture to be more adaptable to social and cultural aspects. That created the cradle of Mashrabiya as an idea.

Islamic architecture can be defined in a more synthetic expression as art that merges both functionalism and sustainable design. And this is conducted to how the Islamic architecture is considered to be the outcome of significant architectural examples that merged culture, art, comfort and function. Also adapted to environmental aspects and local resources at the same time. (yannas, 2013)

The enhanced factors in Islamic architecture were implemented to be responsive to extreme climatic conditions while adhering with all the commonly flowed principles of Islam as a common religion in the regions that implemented this architecture.

Egyptian Islamic architecture was mainly determined by:

- Egyptian Islamic architecture was mainly determined by:
-
- Islamic aspects represented in principles of *Holy Quran* which have undeniably formed the main shape of Islamic architecture. (yannas, 2013)
- Extreme climatic conditions the differed from regions to another in the same country.
- Social culture, where Egypt was considered to be different in its cultural and social forms from the other surrounding countries in the middle east.
- Bedouin and Berbers' peaceful trading and aggressive incursions. (yannas, 2013)
- Scarce water and building resources. (yannas, 2013)

These principles and the mentioned factors shaped the traditional architecture in Egypt from the urban typologies of the smallest villages in the country until the shape of the windows in the building.

Fig 1.1.1 vernacular urban typology (source: yannas, 2013)

1.2 The principles of creating a window on a façade in vernacular architecture:

During the extended *Ottoman* and *Fatimith* period, the social aspects and religious believes stipulated that villages were to be built to one fixed height. The fixed height was a symbolic idea towards metaphorize the equality among all the society layers. And demolishing the ideology of hierarchical societies. Also, the urban typology that tried to overcome the extreme climatic conditions forced the buildings to be closer to each other to provide mutual shading. (yannas, 2013)

Fig 1.2.1 sketch of vernacular urban typology (source: Thahab, 2014)

This urban typology created a conflict between the building design and the cultural and religious aspects, which encouraged respecting the privacy of the indoor spaces. This was hard to be achieved within such an urban typology where buildings are in the same height and close to each other. They had to pay attention to light and air. And also allow the *Mahazzin's* voice (prayer voice) to reach their hearing whenever they are inside the space.

All these factors contributed to starting the idea of Mashrabiya in Egypt, where its typologies evolved more according to each modification that occurs in the social and cultural aspect.

The typical wooden bay window that appeared in Lebanon at the end of the 70s and the middle of 80s was considered to be one of the typologies in Mashrabiya that gathered a combination of different patterns in complex geometry. This was to allow seeing the streets without being seen; this combination window was evolved in Lebanon because of three factors: economic conditions, local aspect and personal preferences. (Ragette, 1980)

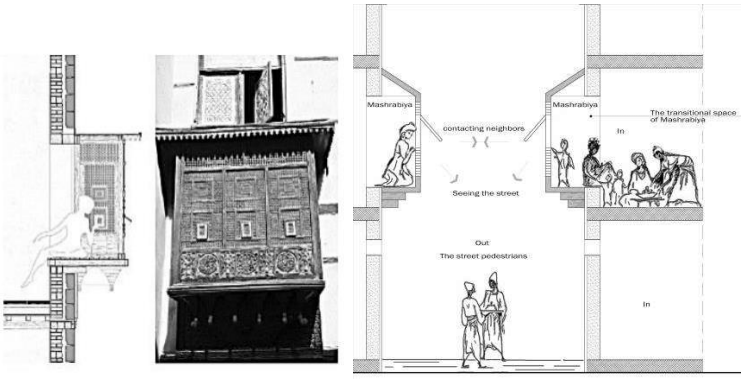
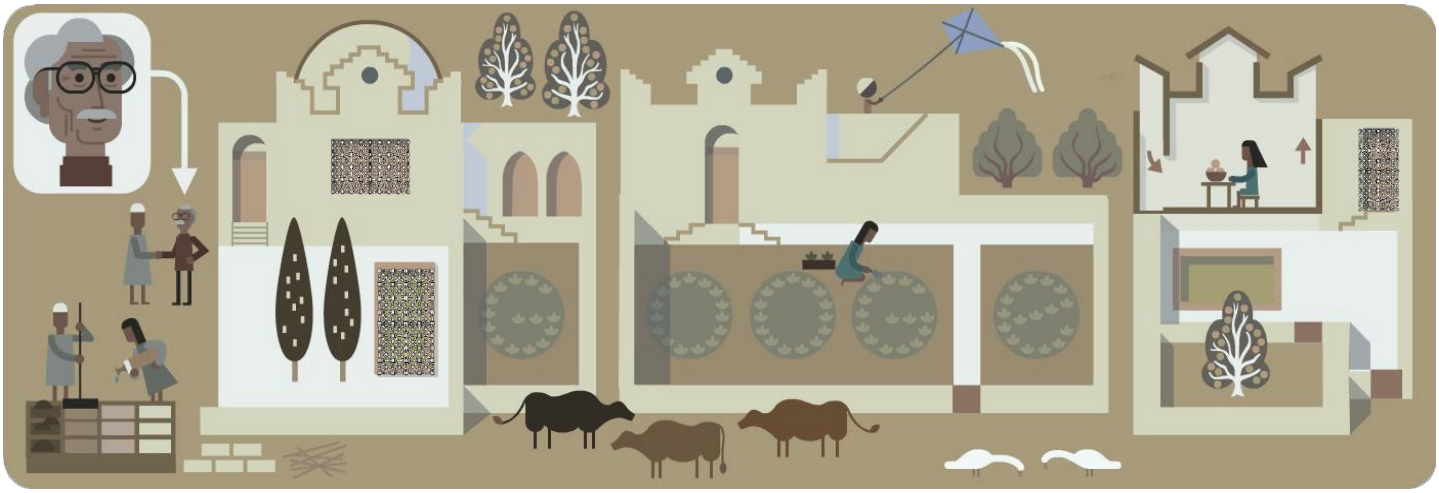


Fig 1.2.2 The Mashrabiya' section, and the transitional social space (Source: Ficarelli, 2008)

1.3. Hassan Fathi reviving Mashrabiya

Past and present within environmental context



1.3.1.Hassan Fathi architecture and Mashrabiya.

Hassan Fathi was considered to be one of the most famous Egyptian architects who were concerned to preserve the heritage of Egyptian vernacular architecture. He also wanted to introduce a naturally passive building using only the available resources.

"In his interpretations of the local vernacular Prof. Hassan Fathy confirmed to the authentic forms, materials and symbols for the most part. He used and refined many traditional building features such as formal (public) and informal (private) courtyards, fountains, arcades, vaults, pendentives, squinches, lightly-coloured thick mud-brick walls, domes as well as wind-catchers and wood lattice screens to the fullest extent possible. "(Miriam Neet, 2009)

Hassan Fathi, in his architectural school, he adopted a lot of methodologies that vernacular architecture had given birth to. But his ideologies changed and evolved by time and experiments. The methodology followed by Hassan Fathi in the beginning to avoid extreme climatic conditions was more oriented towards using thick walls made of mud bricks mixed with ache. Also, modifying the geometry of the window to become smaller to prevent light from entering the space, avoiding the solar gains resulted from it. The reasons behind using mud bricks were rooted in a variety of factors like low cost and convenience availability in the surrounding environment as well as some symbolic values.

And in order to increase the chances of achieving comfort in the hottest months of the year, he implemented domes, wind catchers and courtyards. However, using thick walls with smaller openings created dark and dim indoor spaces, reduced the amount of air interring the space and increased the gap of dysconnectivity between the indoor spaces and the outdoor environment. And this won't cope up with the communal life of the Egyptian society. (Howeidy, 2017)(fathi, 2000)

Fig 1.3.1 hassan fathi works, source (Seragaldin, 2007)

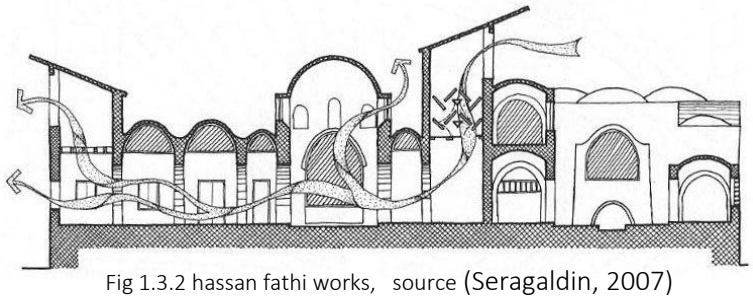


Fig 1.3.2 hassan fathi works, source (Seragaldin, 2007)

That's why this methodology was only implemented in southern Egypt where the average temperature is up to 8-10 degrees higher than the temperature in the northern part of Egypt. As for the northern part of Egypt Hassan Fathi applied a new methodology that managed to encourage the communal, social and cultural aspects of these regions. In this methodology he implemented courtyards and Mashrabiya.

"Being a scarce and expensive building material, wood was used very carefully and conservatively in his projects. The level of care in woodwork became especially apparent in opening treatments such as screens, shutters, railings, shades as well as casements, stairs, column, ceiling articulations and so on." (Miriam Neet, 2009)

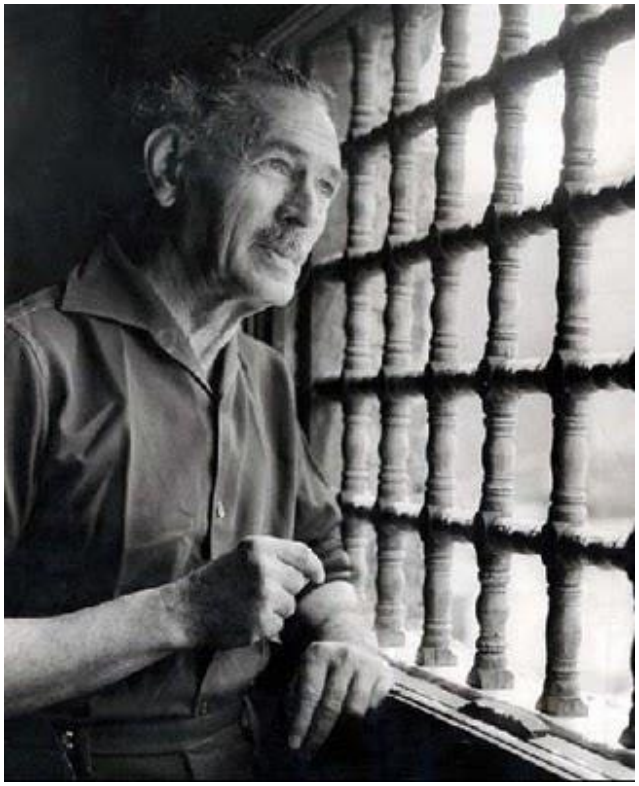


Fig 1.3.3 hassan fathi source (google)

For that Hassan Fathi studied the origin of Mashrabiya very well and selected the preferred typology according the function he needed to achieve and started providing detailed drawings of it in order to help the craftsmen and carpenters to create it in the closest form that he had imagined. However, Hassan Fathi in his works and books he provided only one typology of Mashrabiya with certain types of patterns, this model was the one he used the most and he only implemented it in a certain type of housing. That allowed to use Hight coast materials with very detailed decoration designs.

1.3.2. Hassan Fathi architecture and Mashrabiya.

Mashrabiya implementation as an idea fluctuates from a time period to another, and its implementation was very common in ottoman and Fatimc period in Egypt, which was during (641 to 1867). Where it was commonly used in any type of buildings, even low-cost housing. And then it faced a vast reduction in the 70s. Where the implementation of ideas from the heritage wasn't standard, and there was a typical move towards modernizing the facades by going far from historical components. Hassan Fathi managed to revive this art, and many buildings followed his lead after that. (waziri, 1999)(Hillenbrand, 2006)

One of the famous examples where Hassan Fathi introduced Mashrabiya for the first time in his contemporary designs, was Abdelrahman Nassef house. It was designed in 1973 in Jeddah -Saudi Arabia. Where the building was built by stone enhanced by the typology of traditional towers in the old cities of Saudi Arabia. Rather than using the familiar dome over the majlis (living and meeting room in Saudi Arabia's residential houses), Hassan Fathy thought that implementing an octagonal shape in the tower will be more regionally convincing. And in the top Mashrabiya was applied in its purest forms and geometry with no variation of patterns or establishing the ventilation approach out of it as it was covered with glass from inside, this didn't allow the use of the tower and the Mashrabiya in a more practical way. (Seragaldin, 2007)

After that Hassan Fathy started to implement Mashrabiya more in his buildings, in its simplest geometries as an adjacent lattice screen to the window geometry. Using one of the most commonly made patterns in Egypt "Arabesque " or what was named with "Maymooni pattern."

These buildings were:

I. Akil Sami house, a residential house located in Dahshur - Egypt, was built in 1978. In this building, he recalled the lattice work that was used in Motasrli residence 1950.

II. Alaa Al-din Mustafa house, a residential house located in Idfu - Egypt, was built in 1981.

III. Al Harini Vila, a private residential house located in Giza- Egypt, was built in 1938.

IV. Al Mashrabiya tourist centre, commercial building, located in Giza - Egypt, was built in 1976.

V. Casaroni house, a private residential house located in Giza- Egypt, was built in 1980.VI.Fares school, an educational high school for poverty located in Fares- Egypt, was built in 1957. Where mashrabiya screens were implemented latter according to the architect suggestions in 1974

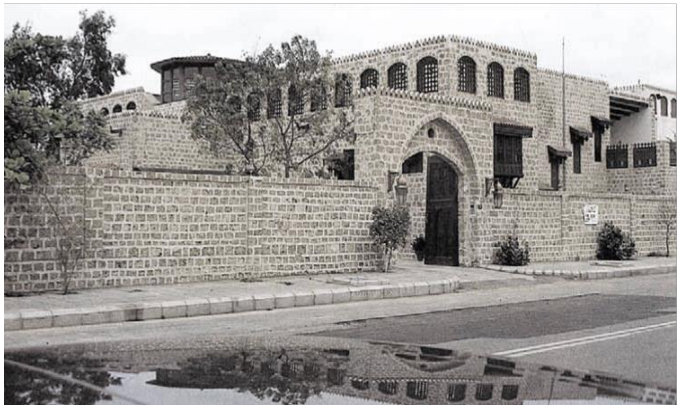


Fig 1.3.2.1 The Mashrabiyasketch made by Hassan fathi source (Seragaldin, 2007)

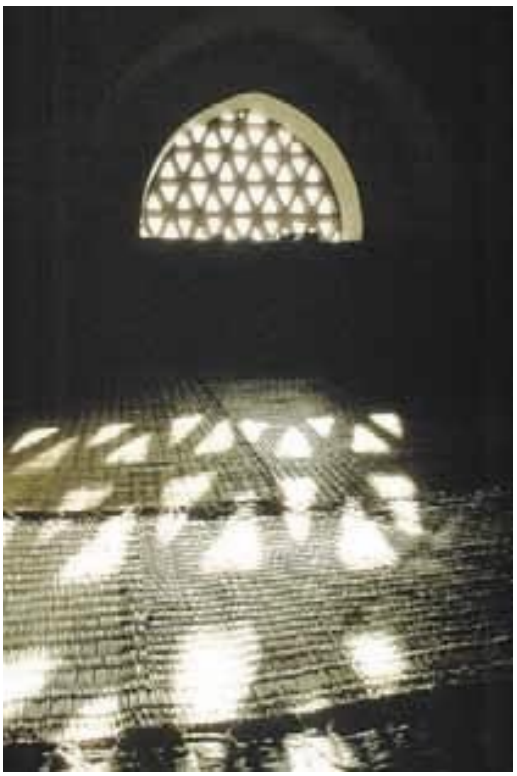


Fig 1.3.2.2 The Mashrabiyasketch made by Hassan fathi source (Seragaldin, 2007)

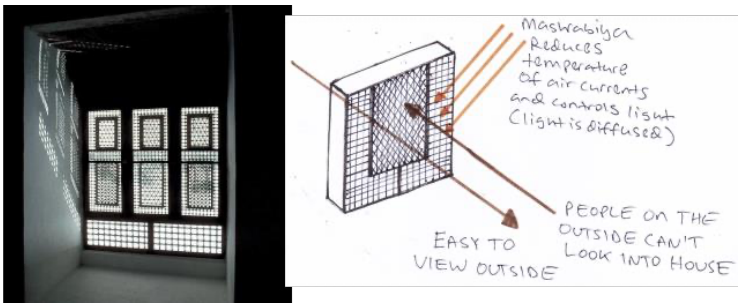


Fig 1.3.3 The Mashrabiyasketch made by Hassan fathi source (Seragaldin, 2007)

To provide a meaningful systematic study, it was important to understand the methodology of studying followed by previous architecture leader like Hassan Fathi, the paper works and research studies based on that topic. Despite the implementation of so many environmentally passive and heritage preserving ideas, Hassan Fathi was considered to be a legacy for all the new generations of architects where he was the most dominant representative figure of architecture in Egypt. The whole world widely acknowledged his impact. But this acknowledgement doesn't mean that he was easily discountable. His architecture ideology had sometimes been infuriated and disconcerted from its primary approach, as he always challenged the most influential characters in the building matters.

His strength was more generated from his ideas, not his buildings, as he only had built 30 buildings during his whole meaningful influencing career. Few of these buildings were known to the comprehensive public figures. Also, the governments demolished some of them to expand vertically. Yet his ideas managed to be widely acknowledged and survive through the time. (Seragaldin, 2007)

Hassan Fathi's books and papers that discuss his architecture was a very helpful step towards understanding a lot of points about Mashrabiya specially that Hassan Fathi provided a very detailed drawings for the model that he implemented and a lot of craftsmen use these drawings until now. (Viola, 2018) (Seragaldin, 2007)

Fig 1.3.4 old neglected work of Hassan fathi source (Seragaldin, 2007)

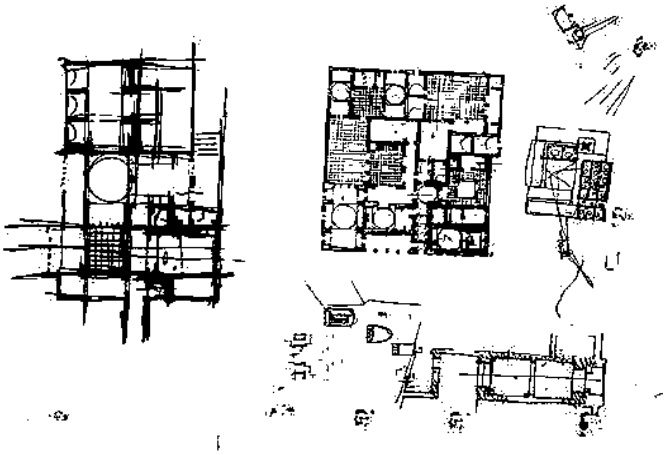


Fig 1.3.5 concept sketch made by Hassan fathi source (Seragaldin, 2007)

Fig 1.3.6 image of Hassan fathi source (google)

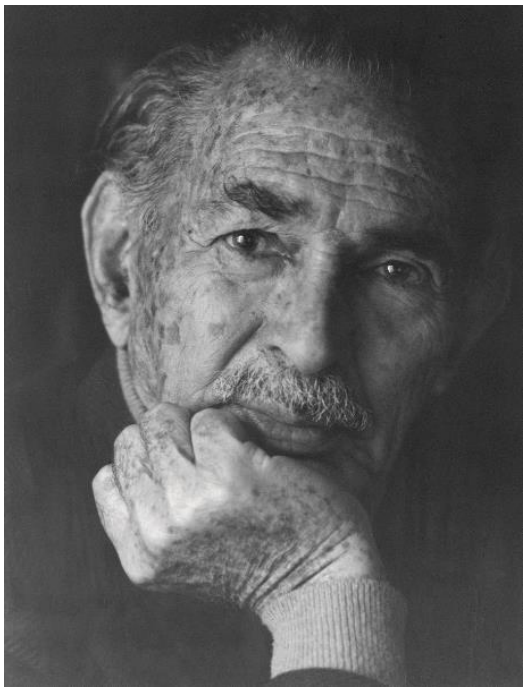
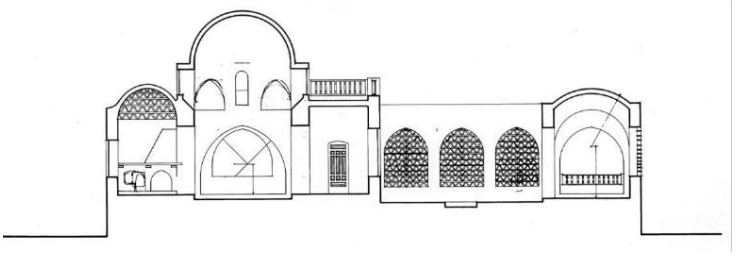
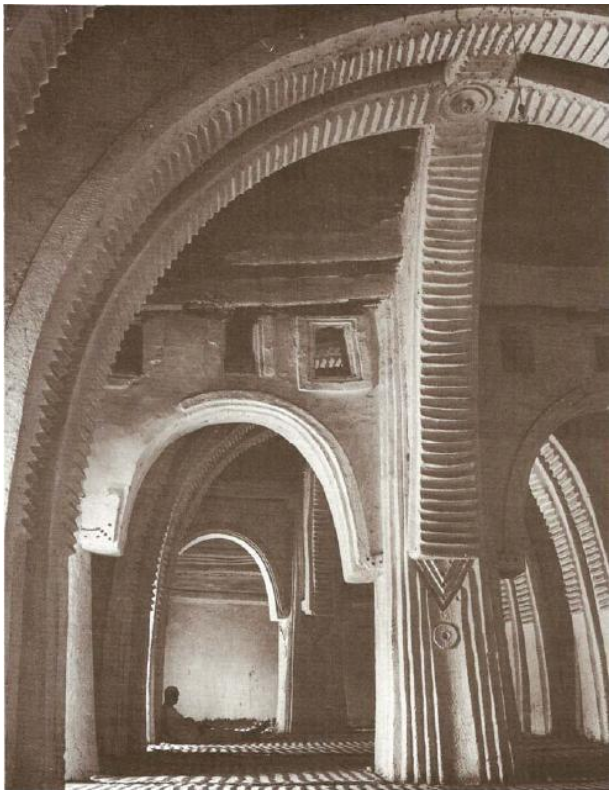
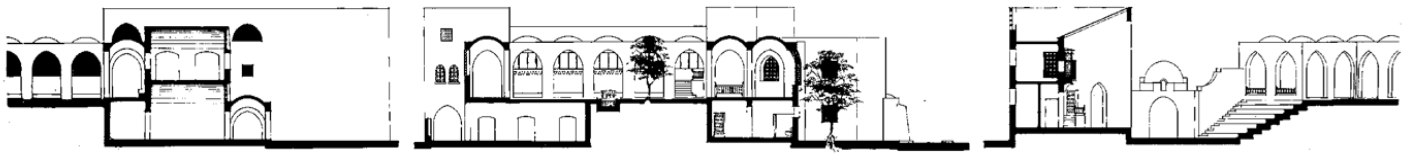


Fig 1.3.7 image of Hassan fathi source (google)





MAIN WEST ELEVATION



section A-A section B-B section C-C

Fig 1.3.8 collage image of hassan fathis work, source (Seragaldin, 2007)

4.Mashrabiya, ventilation and evaporative cooling.

The word "Mashrabiya" is driven from the word "Sharab" according to the Arabic dictionary. The word "Sharab" means "the drink "in English which is a location metaphor where they inhabitants used to place a ceramic pot filled with water inside the transitional space created by some types of Mashrabiya. Where the air flows at a proper velocity to achieve a multi-purpose process, where the water will get colder inside the pot for the inhabitants to drink and cool the temperature of the air entering the space. (KOUJAN, 2018)

Mashrabiya design was mainly generated from geometric shapes that were instituted from the traditional crafts and arts. Where the openings in the lattice screens used to vary according to the location, orientation and the function required from it. Where the smaller the openings in the screen, the less solar gains and glare penetrate from it, on the other hand, the wider the gaps in the lattice screens the more air will flow through it. (KOUJAN, 2018) (Schiano-Phan, 2004)

The environmental performance of Mashrabiya in terms of ventilation and shading also varied according to the typology, Where the geometry used to be different and the patterns used within this geometry changed as well. That's why Mashrabiya was considered to be portraying different single-sided ventilation strategies within its various forms. (Donald Watson, 1983)The variation in the ventilation performance was conducted to the different geometries and the combination of different patterns in the same component, and this accompanied by the huge variation in the shading performance as well, since it is significantly affected by the patterns shapes and perforation percentage. (KOUJAN, 2018)From the previous facts, it was essential to make a close understanding of how can the combination of different patterns affects the ventilation of the space and what types of patterns should be combined. Also, understanding how the indoor conditions will be affected if the evaporative cooling process was implemented within the Mashrabiya usual performances.

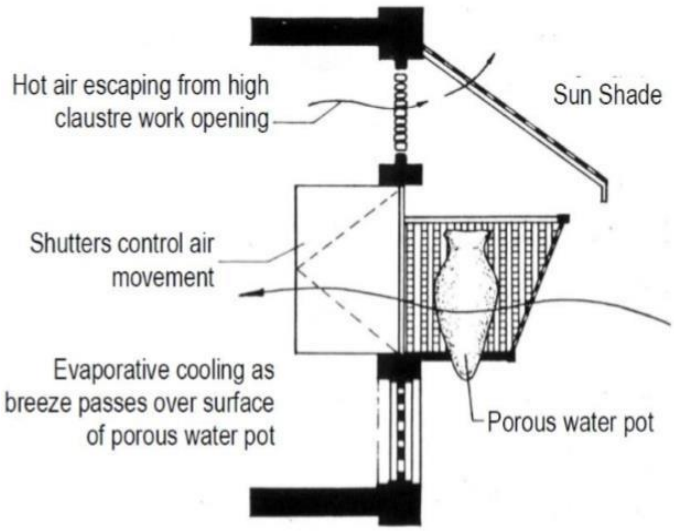


Fig 1.4.1 sectional sketch of evaporative cooling in mashrabiya source (Schiano-Phan, 2004)

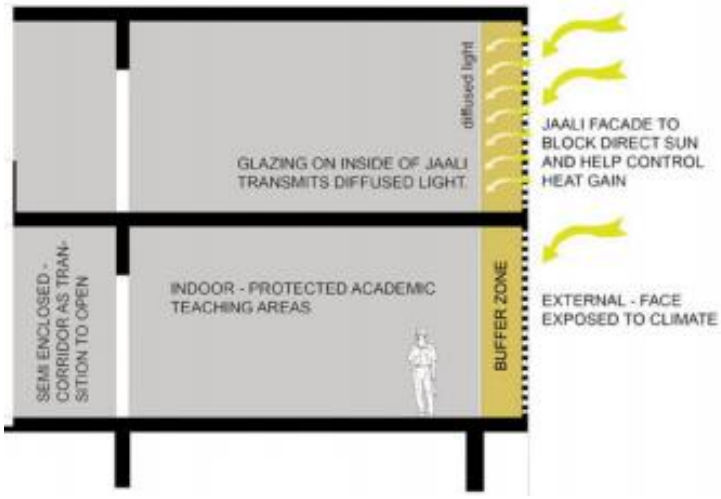
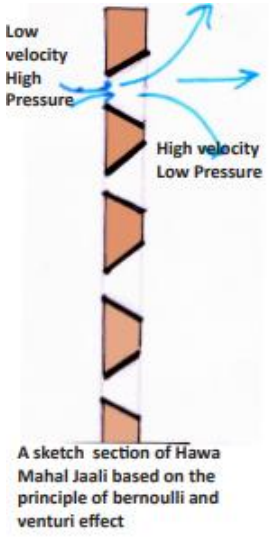


Fig 1.4.2 sectional sketch of performance of mashrabiya source (KOUJAN, 2018)



Loyola Women's Hostel, Trivandum

This is a building designed by Laurie Baker, as he was an architect who believed in cost effectiveness, he designed the complete wall with exposed brick jaali to minimise the use of windows to reduce cost. The Jaali designed by him is broader on the outside surface and narrower to inside ,mainly to avoid rainwater to penetrate

Fig 1.4.3 collage image of different types of of mashrabiya source (KOUJAN, 2018)

·Mashrabiya and evaporative cooling.

Evaporative cooling was first introduced to Mashrabiya when a ceramic pot was placed in front of the inlet, in order to achieve a multi proposal process, water was added to the ceramic pots in order to use the porosity of the ceramic and cool both the air temperature passing through and cool the water inside the pot to drink.

Thinking about this process, we can observe how sustainable and environmental it was. This was a significant approach back then. This process managed to achieve comfort and preserve the water used for drinking. This was a symbolic action that showed how was the appreciation of water as a source from nature back then especially that water was considered to be a rare natural resource in hot, dry climates.

"Ecooler is a ceramic product created by Studio Kahn in Israel, [7] with a promising passive design alternative to bulky, noisy, energy-sucking ACs. It is a concept that cools the air using a system of hollow ceramic tiles filled with water. The Ecooler system is integrating the two traditional elements, Jara Jugs, originally used to cool water, and traditional Mashrabiya tiles which were used to create perforated facades allowing airflow into homes, while keeping the sun out, to create a screen that serves as a natural, ecological cooling system " (Faggal, 2015) (Schiano-Phan, 2004)

Ecooler was mostly considering "Jaali" ceramic perforated screens passed in India that transformed the evaporative cooling occurrence as a process from the transitional space created indoor from the mashrabiya to be applied on the screen itself. This requested the implementation of wide smooth patterns to allow more air to flow through the patterns, eventually increasing its cooling effect. However, the efficiency of evaporative cooling as a process decreases noticeably when it is exposed to solar gains, as it accelerates the water evaporation reducing the time interval that within it the vapour droplets can cool down the air temperature. (Faggal, 2015)

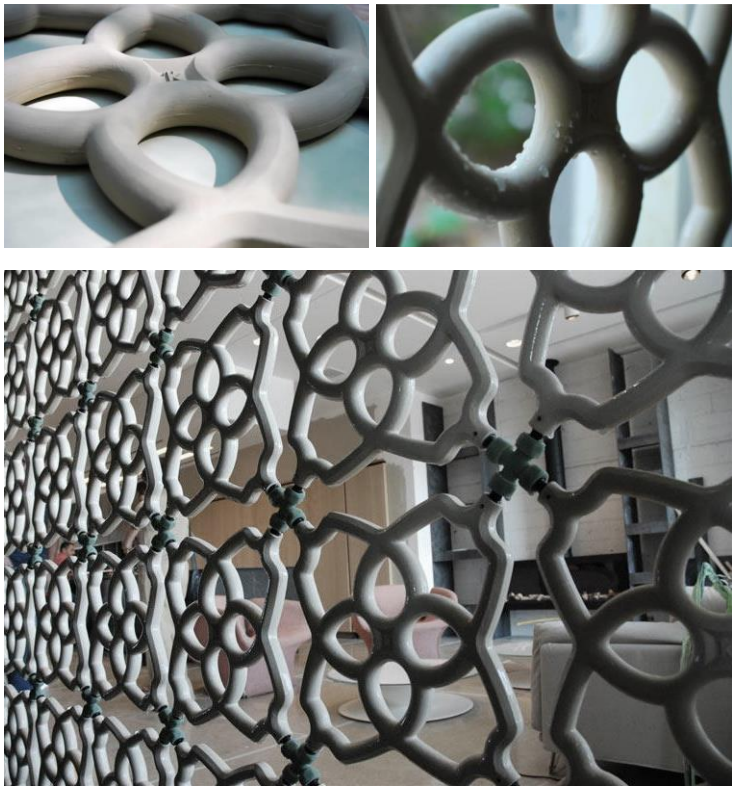


Fig 1.4.4 collage image of ecooler patterns
source (Faggal, 2015)

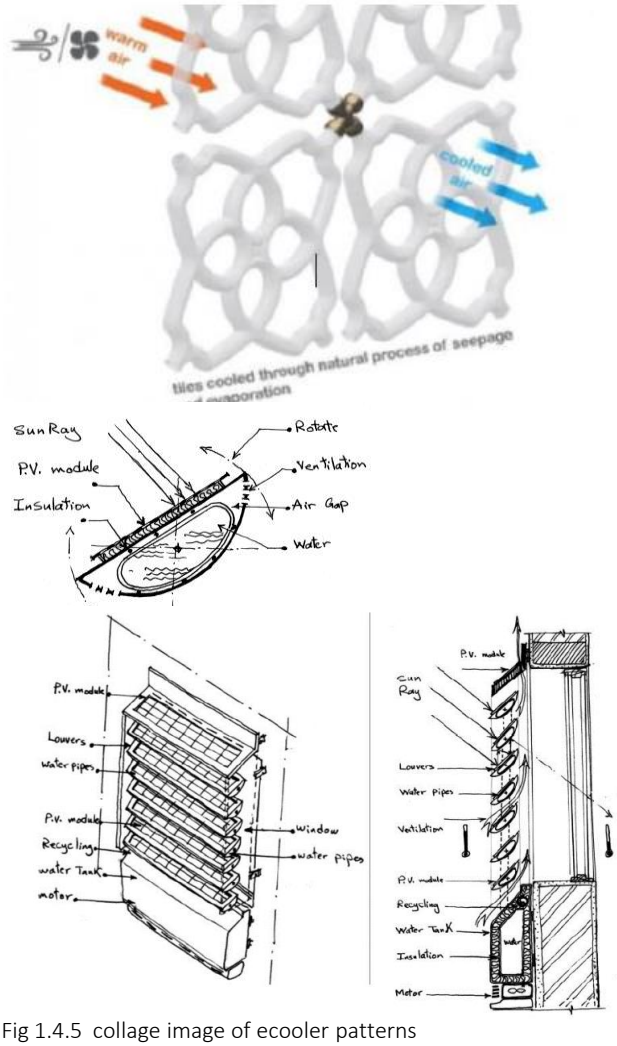


Fig 1.4.5 collage image of ecooler patterns
source (Faggal, 2015)

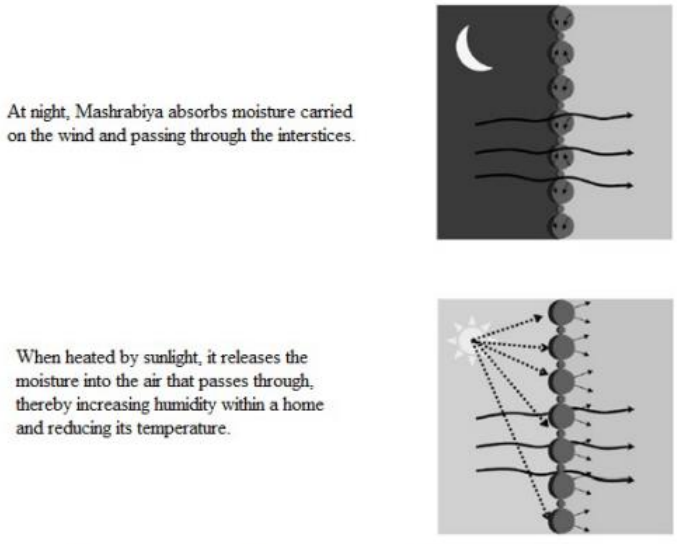
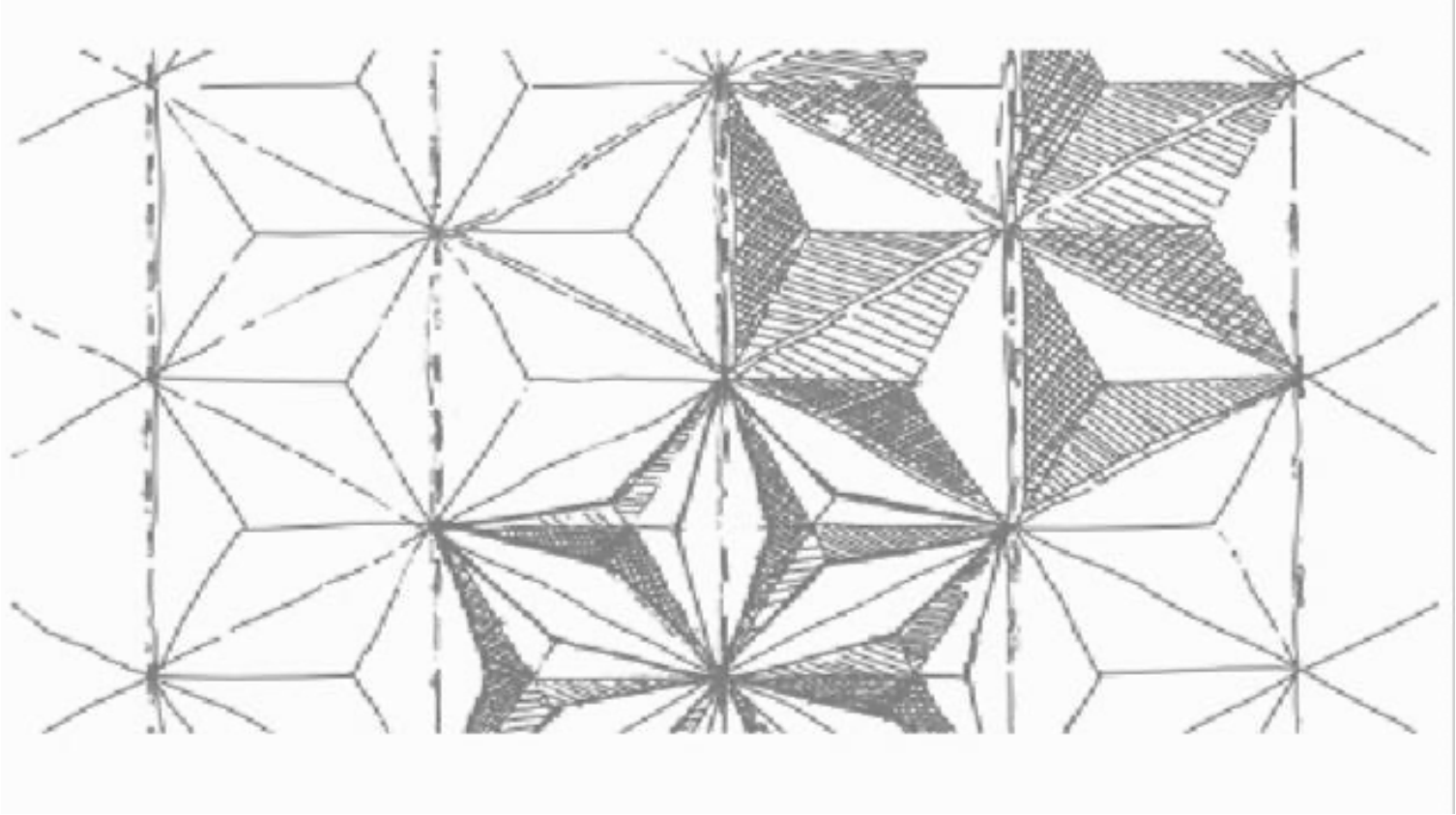


Fig 1.4.6: cooling effect of Mashrabiya through the evapo-transpiration process
(Source: Fathy, 1986, illustration by Gelil, 2014)



1.5 Contemporary Mashrabiya

from traditional Mashrabiya to modern double façade screens

1.5.1·Contemporary implementation of Mashrabiya

From studying how Mashrabiya was originally formed in vernacular architecture and how it evolved by time, a huge gap was found between the traditional forms and shapes of Mashrabiya and contemporary implementation. And it was a critical point to stop and try to illustrate more the reasons rooted in this. A lot of inquiries were made around this part, which was the motive towards applying more research and state the fundamentals of the fieldwork that should be made.

According to previous research papers the contemporary implementation of Mashrabiya only occurs in modern buildings, where Mashrabiya is redefined to be just perforated patterns that act as a shading element in order to prevent direct sunlight. To improve the performance in lower sun angles of incidence, it was modified to be dynamic and move according to the sun angle. Through time Mashrabiya was transformed to be a part of double facades. Where Mashrabiya was converted to be just an external texture that covers the building and prevent undesired solar radiations. Then technology was a stimulating element that evolved the Mashrabiya towards being a sensitive cover on a wider parameter that adapts to variable environmental conditions. (Sirryah, 2018)

But thinking critically about that, make us wonder, won't this method limit the use of Mashrabiya to be only applied in modern buildings. Where high cost and technology doesn't form any prior against new ideas. Then what will be the solution for low coast houses, the houses that represent that majority of houses in the developing countries, the countries that suffer the most from extreme climatic conditions? (Alothman, 2017) As for the contemporary low-cost houses that try to implement traditional Mashrabiya, It follows the visual tracing of the old models of Mashrabiya without paying attention to the orientation differences, and variation of performance from a pattern to another. This method of implementation is the main reason for the gradual disappearance of traditional Mashrabiya . Mashrabiya was lost in the transaction through time because no systematic studies provided enough data the emphasize the accuracy in implementation of Mashrabiya, since that the common implementation in basic designs is using Mashrabiya with inaccuracy where unsuitable patterns is placed for the unsuitable façade. This leads to impracticability that increases the reduction in the implementation of Mashrabiya now.

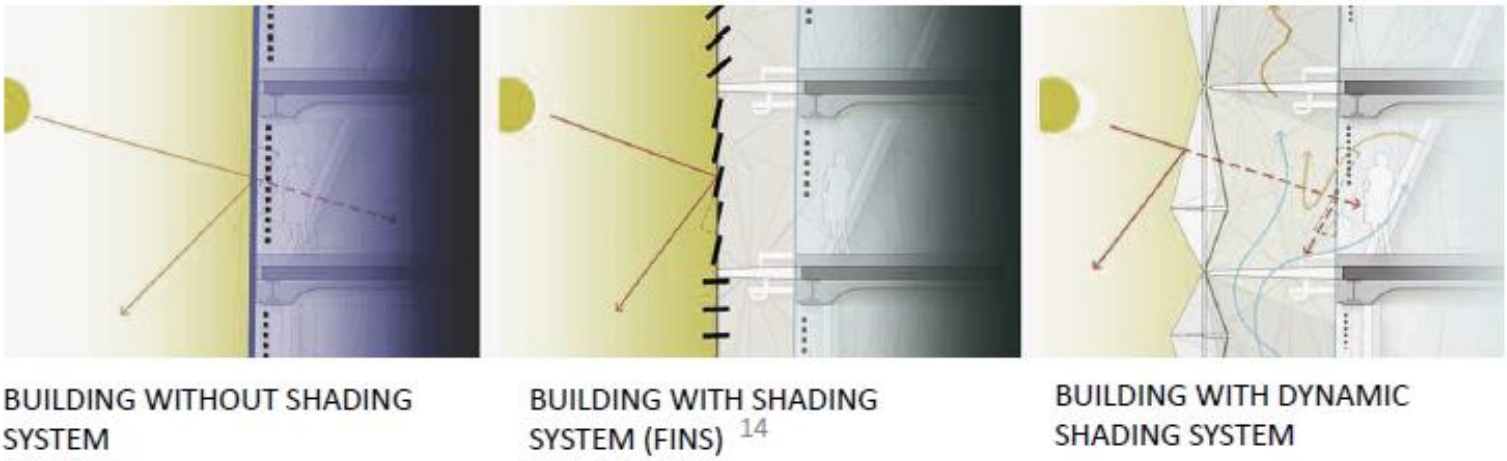


Fig 1.5.1 collage image of albhaar towers sections
(JYOTI AHLAWAT, 2015)

·Contemporary implementation of Mashrabiya

The eastern window technology in many of the new Middle East buildings has been transformed into double facades to increase the coolness inside. With the development of modern technology, the exterior form of the building has become a source of attraction for the users. Technology has opened up areas for light and shadow formation using Sensors and information technology.

The Sea Towers building in Abu Dhabi designed by Edis is considered to be a massive approach towards passive solutions implemented from traditional architecture. Where the towers were designed with a dynamic, sun-sensitive curtain that reduces its thermal gain, and has been placed two meters outside the building almost independently. It contains a large number of The gums, which were wrapped in glass fibres. These elements have been programmed on the movement of the sun to reduce its temperature inside the building, but in the evening all screens are closed. Also, with the beginning of the sunrise from the east, the element closes from the east. As the screens change throughout the day, the results from its performance improves as well. This reduces the building's need for industrial air conditioning and lighting by more than 50%. (JYOTI AHLAWAT, 2015)

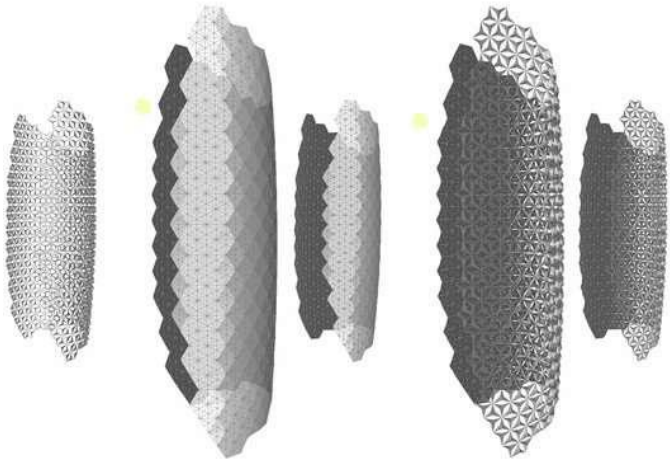


Fig 1.5.2 collage image of albhaar towers sections
(JYOTI AHLAWAT, 2015)

Fig 1.5.3 collage image of albhaar towers sections
(JYOTI AHLAWAT, 2015)

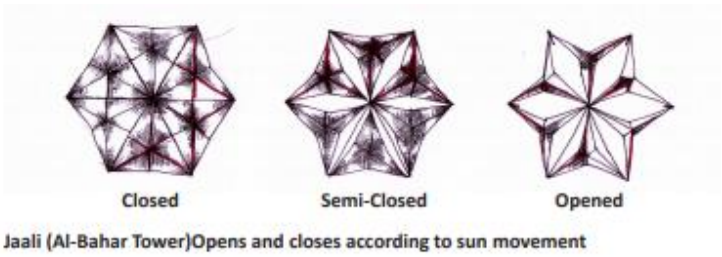


Fig 1.5.4 collage image of albhaar towers sections
(JYOTI AHLAWAT, 2015)

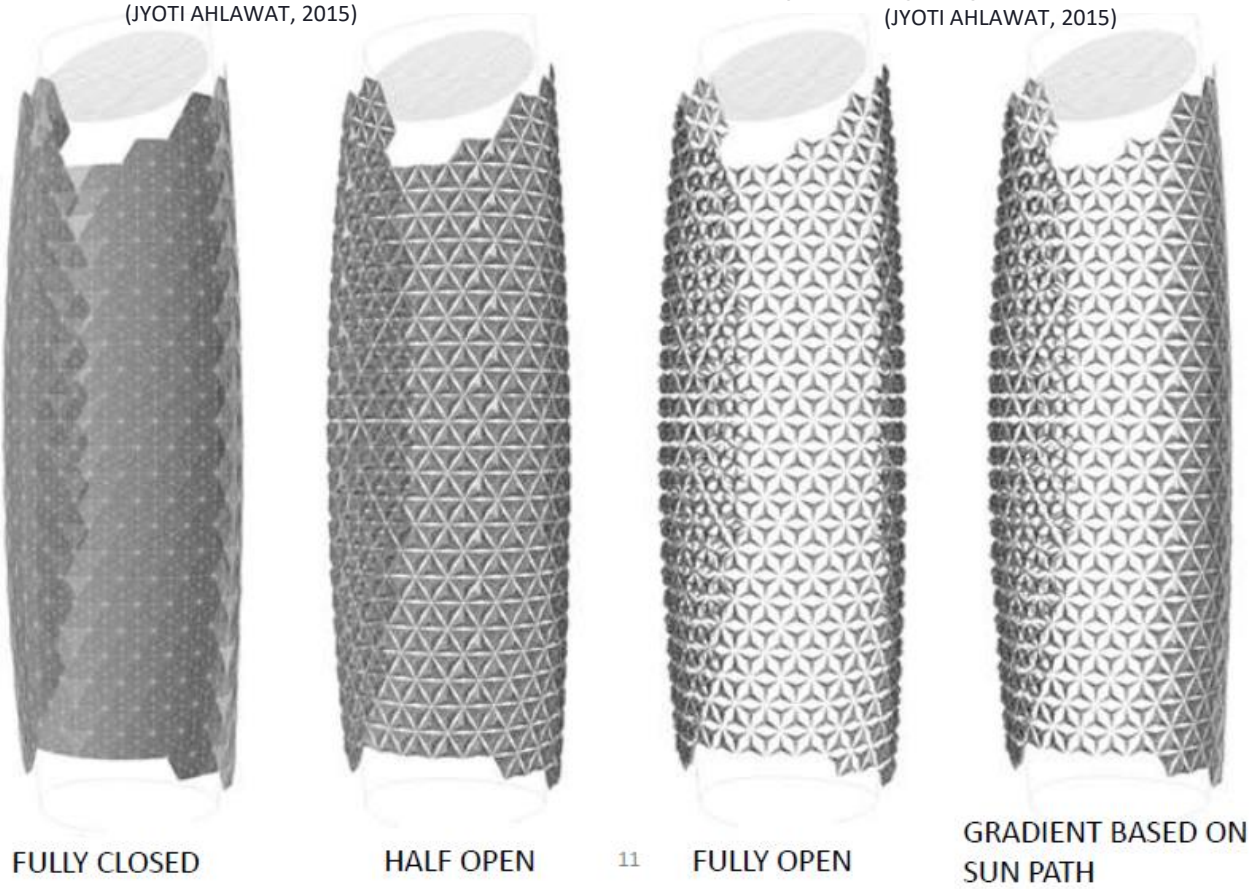


Fig 1.5.5 collage image of albhaar towers sections
(JYOTI AHLAWAT, 2015)

Fig 1.5.6 collage image of albhaar towers analytics
(JYOTI AHLAWAT, 2015)

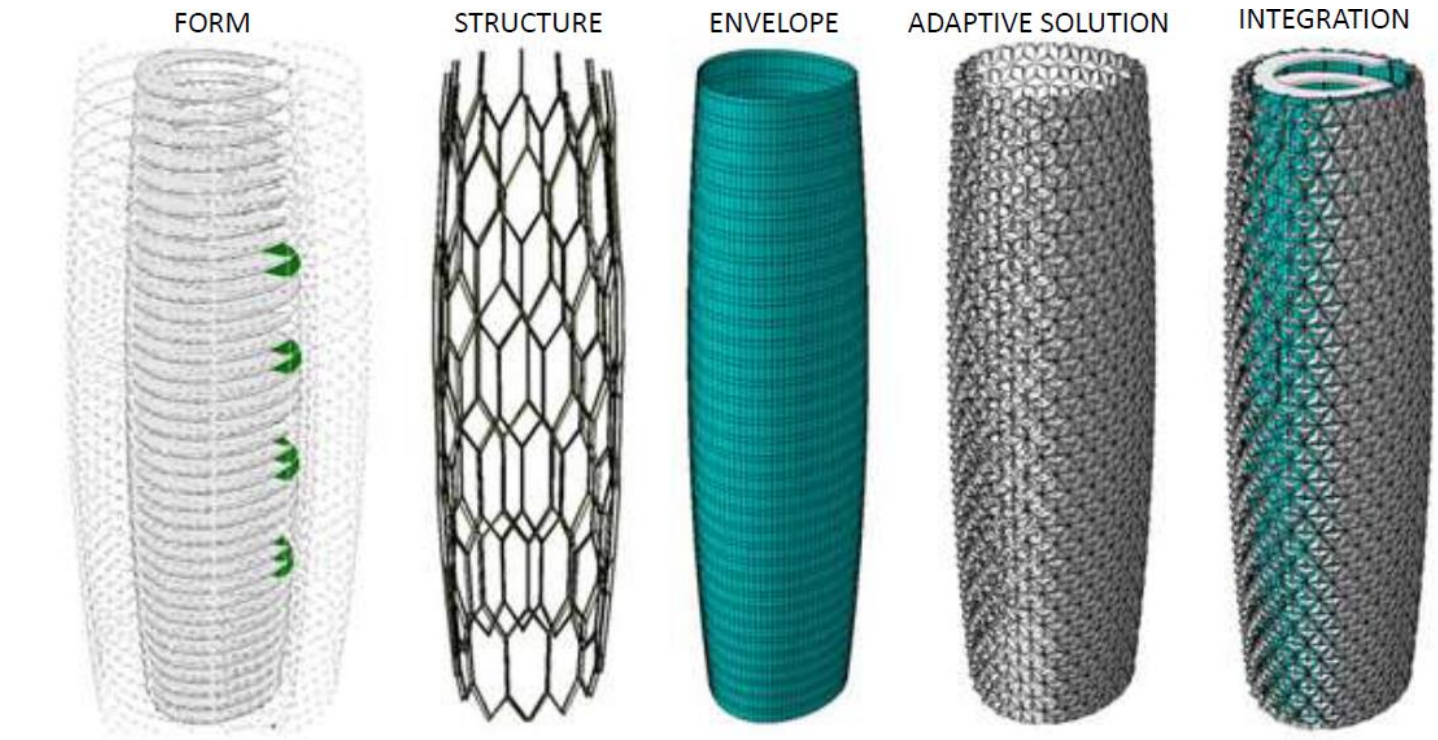


Fig 1.5.7 collage image of albhaar towers analytics
(JYOTI AHLAWAT, 2015)

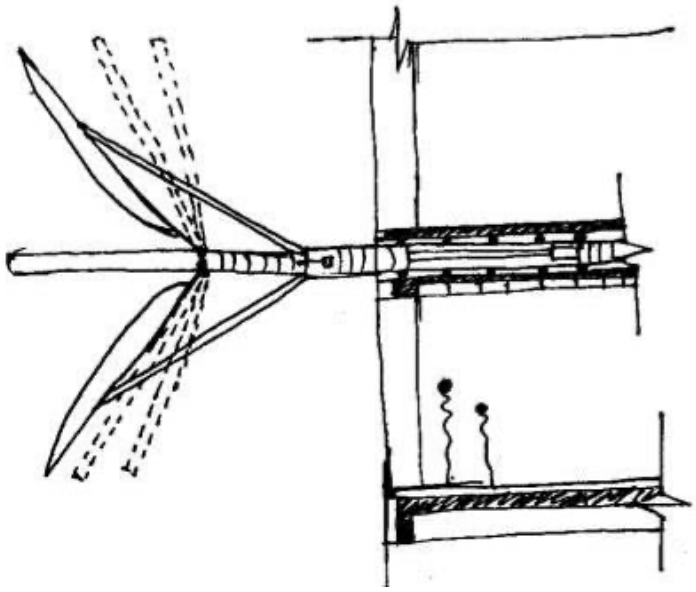


Fig 1.5.8 concept sketchs of Al Bahar toowers
(JYOTI AHLAWAT, 2015)

·Contemporary implementation of Mashrabiya

To understand the performance of these variable pattern configuration in terms of shading. Daylight and solar radiation simulations were made. Simulations of the building performance towards different environmental parameters were to estimate how much improvement these patterns had achieved. ”Theoretically, a shading screen should completely wrap the tower as direct solar rays hit the curtain-wall from all directions, especially during summer. The north face experiences direct solar rays only for a short time in the morning and later in the afternoon, i.e. before and after working hours. Shading units in the North zone was therefore unnecessary. “(JYOTI AHLAWAT, 2015)

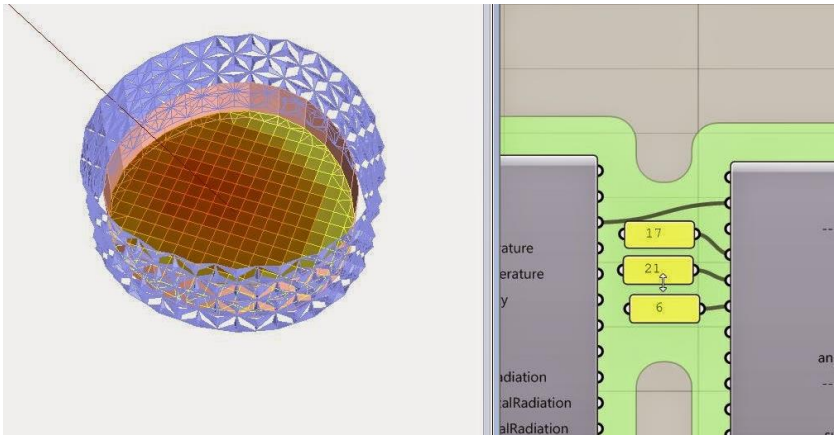


Fig 1.5.11 collage image of albhaar towers analytical sunlight hours simulation
(JYOTI AHLAWAT, 2015)

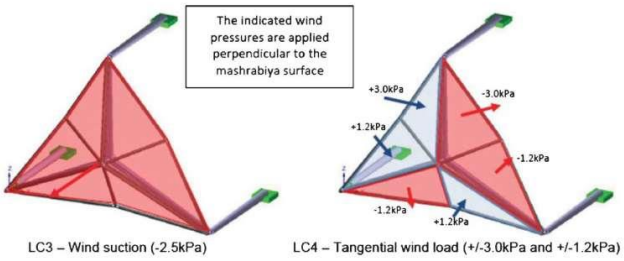


Fig 1.5.9 collage image of albhaar towers sections
(JYOTI AHLAWAT, 2015)

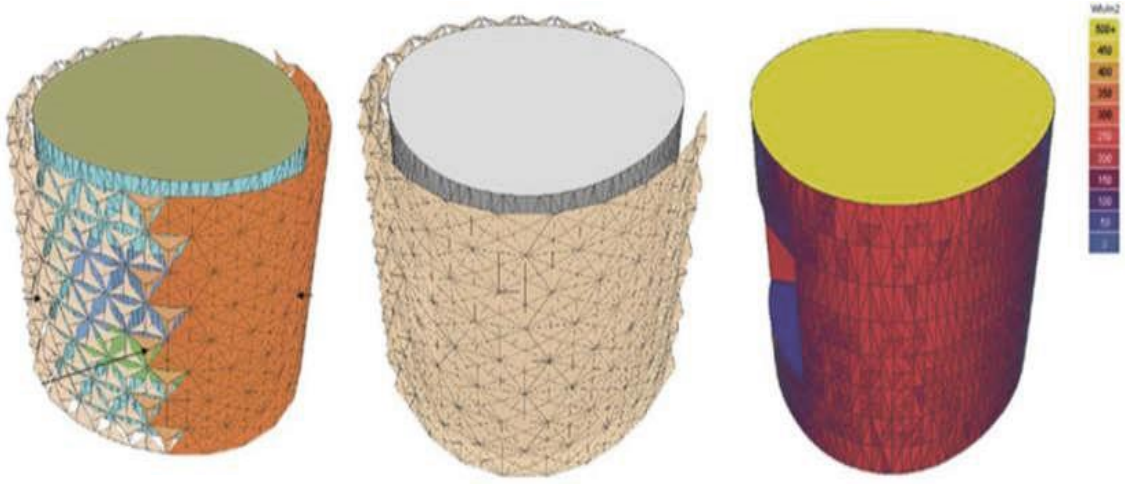
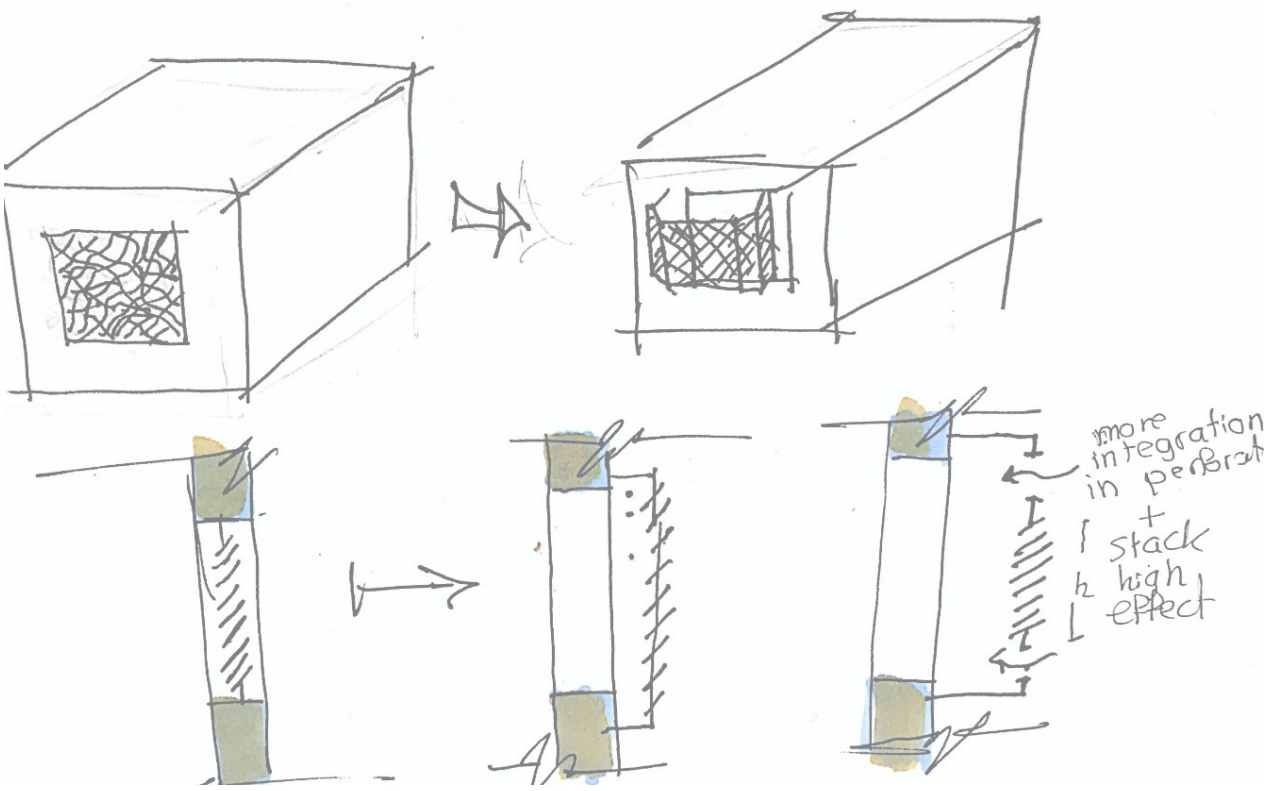


Fig 1.5.10 collage image of albhaar towers analytical sunlight hours simulation
(JYOTI AHLAWAT, 2015)

Hence, we can see the success of the application of sustainable approach through designing associated with the Islamic culture and heritage, mainly that the functional philosophy of Islamic architecture stems from the sustainable orientation, emphasizing the Arab identity and locality, where Mashrabiya always remains a renewed source of creativity. Therefore, the process of establishing the design style For Islamic architectural elements are not limited to a particular form or style but are the content and character. These elements are influenced by the original functional values. Thus, our goal is to revive the architecture with a modern and sophisticated way and restore the psychological balance of the contemporary human methods towards achieving goals and the purpose of life with the architectural values that define us.

Fig 1.5.12 Bahrain National Theatre- taken by Mostafa A Hadi
Source (google-paintress)



02. **Precedent of Mashrabiya**

typologies of mashrabiya and variation in the form according to the regional factor

2.0. Introduction:

Mashrabiya was developed in Islamic countries to provide privacy for the users inside the space and act multi purposely by performing towards different environmental parameters. The Mashrabiya was related to the Islamic architecture that was generated from Islamic political rule, that expanded through middle eastern countries, north Africa and Andalusia, which later become Spain and morocco. The expansion of such cultures through the land was accompanied by variation in climatic factors, which helped in modifying Mashrabiya into different typologies (سلمي, 1998 ;Jayyusi, 1994).

It was also known under many names according to the different regions, and the variation in the names used to represent the variation in the cultural aspects Mashrabiya was made to represent. Also, the variation of the function. The countries that mainly used the component for evaporative cooling named it Mashrabiya, driven from the word Sharab, which means drink. The countries that used to implement this name was mostly Egypt, Soudan, Libya and parts of turkey.

As for the countries that use it just for shading and ventilation called the component Rawshan or Shandal; this was common in the lands of Al Hijaz. The word Rawshan was driven from the Persian word "Rawza" which initially means the window or opening since it was used to block the possible amount of light and prevent solar gains. While Iraq and Turkish lands used the word the "shanashil" which means the word pavilion as it used to represent art, culture and cleverness in geometry and also was a feature of cooling in the middle eastern countries. Accordingly, it was widely applied in the middle eastern common houses, as it was very efficient in the spaces towards extreme climatic conditions, and merge it with social aspects required for each region.

- <u>Typologies of Mashrabiya:</u>

The different typologies of Mashrabiya showed how the Islamic architecture prioritize the environment, climatic factors and functionalism. Where the main goal was to achieve comfort indoors in hot climatic areas and balance it with social and religious aspects. For that, many typologies were driven from Mashrabiya that used different patterns and geometries. (waziri, 1999) These typologies were:

I. Andalusian Mashrabiya: was commonly used in Andalusia which eventually became Morocco, Spain, Tunisia

II. Ottoman Mashrabiya: the implementation of this type of Mashrabiya prospered during the Ottoman period, and the application went viral during this period where Mashrabiya was commonly used in all building's typologies.

III .Fathemic Mashrabiya: considered to be one of the oldest types of Mashrabiya where it was implemented during the Fatemeh rule, which was capitalized in Egypt-Cairo from 969 until 117. In this era, Mashrabiya was used only in luxurious houses and palaces as it used a specific type of wood that wasn't available with a proper amount in the countries

IV. Hijaz Rowshan: was commonly implemented in Hejaz lands that later become west of Saudi Arabia, a part of Yemen and Oman

From these different typologies, two main typologies were selected as they represented the vast variations in all the other typologies and were the most commonly used types tell nowadays. These typologies were:

A. Andalusian Mashrabiya
B. Ottoman Mashrabiya

Fig 2.0.1 image of Islamic architecture collage word ornament
source (google)

2.1 **Andalusian Mashrabiya**

lattice screens in lands of Andalusia

- Andalusian Mashrabiya

During the Islamic expansion that started from in the eastern countries, the eyes of the Islamic leaders and decision makers, back then, started to head towards south and north of Africa and Europe. The conquering of the African countries neighbouring Egypt, started in year 711where it started to expand to conquer Algeria, Tunisia, Morocco. The conquering armies and leaders stopped at Morocco to state a new approach for the future plans in order to insure the borders safety through land and sea to protect the colony that was formed there. Thereupon the marching was headed towards Spain in order to take control of strait of Gibraltar for taking control of any passing ships through it and insure the safety of the neighbouring lands. This formed a new era that fundamentally appeared in the history and culture of Spain generally and of Andalusia particularly. (El legado Andalusia, 2016)

After conquering Spain, the Islamic leaders and warriors become less worried about the danger that could approach them from Europe and the focus on the defence mechanism and war strategies became less. And the main attention was diverted towards culture reinforcement by encouraging art, poetry, science and architecture. Andalusian culture is well known for its symbolic art that promoted peace and balance through direct implementation of conceptual drawings and patterns driven from nature. And these patterns attracted so many architects and painters back in that era. Eventually affected the Mashrabiya to express balance and elegance. To finalize the characters of the Andalusian pattern a table was made to summarize such a category. (waziri, 1999) (سلمي, 1998)

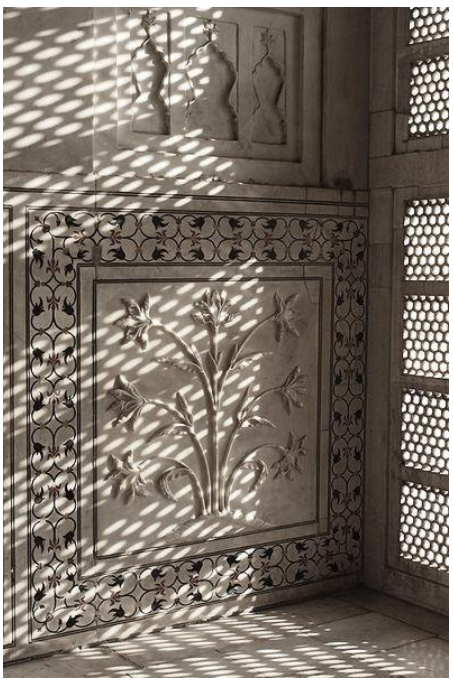
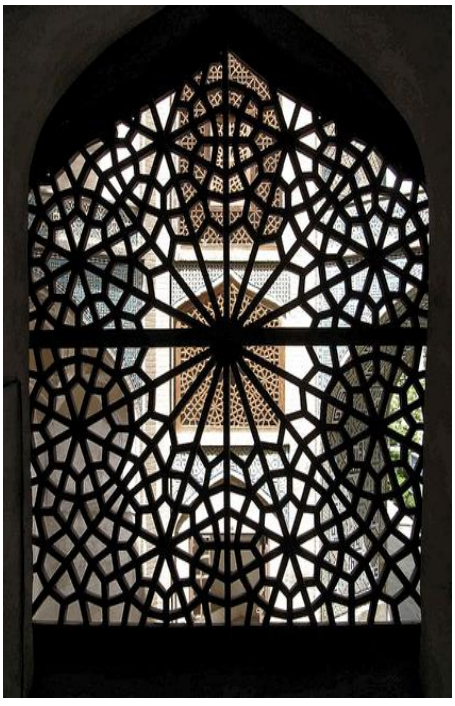
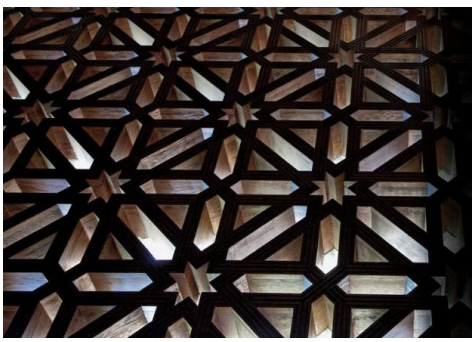
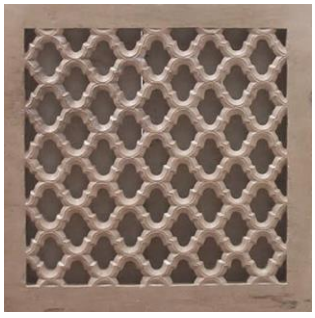
Fig 2.1.2painting for Cordoba in andlusia ,alfaragios
source (google)

Fig 2.1.1 image of morocco 1840
source (google)

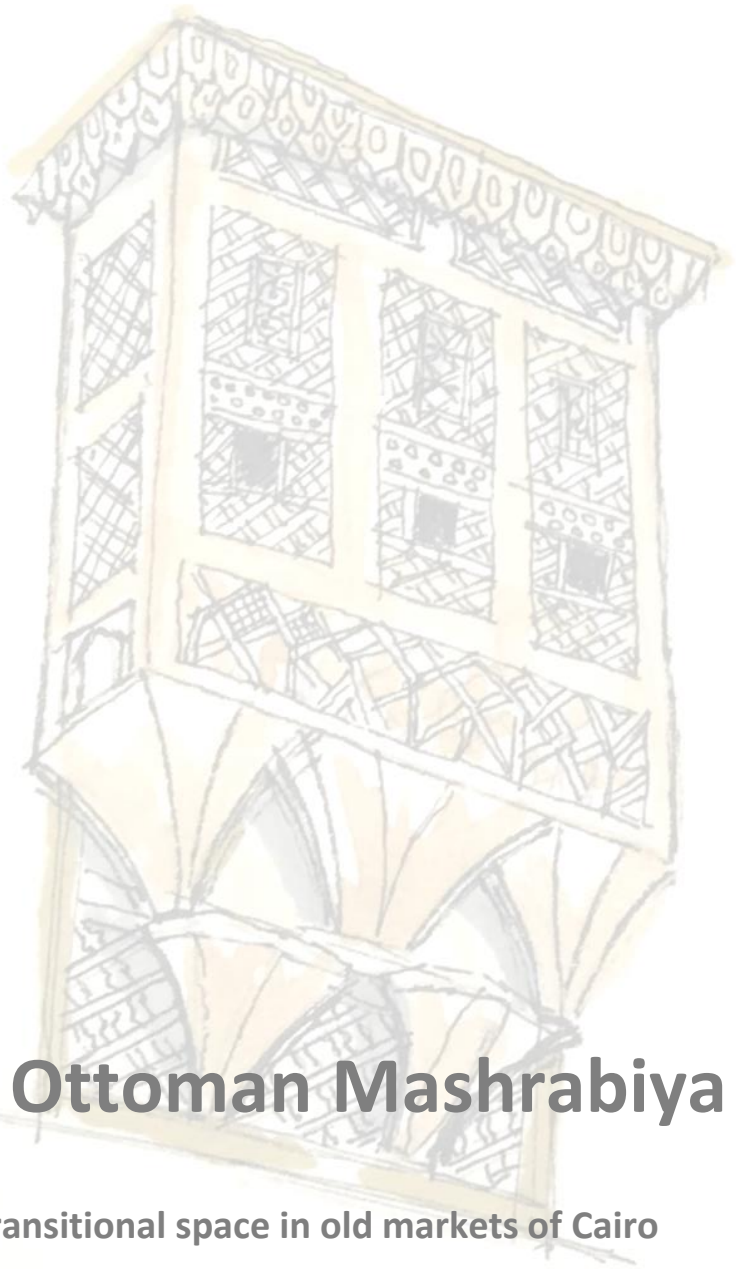
Fig 2.1.3painting for Caliph's reception of the monk John Gorze, ambassador of Emperor Otto I
source (. (El legado Andalusia, 2016)

Patterns forms	Floral patterns and patterns driven from plants were common, where the patterns were homogeneous and smooth, that represented peace, beauty and balance. This required a lot of effort in crafting it.
Perforation percentage range	Used to implement wider patterns with perforation percentage ranges from 24 to 47%
Geometry	Didn't have any complex geometries its beauty was in its simple forms. Where the patterns were applied in form of a screen adjacent to the window where only one pattern was used among the whole screen
location	Spain, Morocco and some part of Tunis.
typology of the spaces that commonly implemented it	Its elegant patterns required a lot of effort in crafting and designing, which made it more specified for luxurious houses, great mosques and palaces.
Material used for it	rods of different Woods or stone.

Fig 2.1.4 collage images of different shapes of Andalusian mashrabiya
source (El legado Andalusia, 2016)

Mainly when the screen will be made of stone or applied on a large scale less floral patterns is used and the patterns tend to be more geometrical with less curves, in order to facilitate its crafting and manufacture . (waziri, 1999) (سلمي, 1998)



2.2 **Ottoman Mashrabiya**

transitional space in old markets of Cairo

- Ottoman Mashrabiya

After the Ottoman Empire defeated the Mamluks in Syria, Egypt and the Hejaz and ruled these countries (1517 - 1802), they began a new architectural history that led to the change of the architectural system in these countries. In the beginning, Many of those who were studying Islamic architecture in Cairo believed that the Ottoman architecture not worthy of attention and their architecture models in Egypt were similar to their models in Istanbul. But the fact is that the architectural style used by the Ottomans in Egypt was a combination of their style in Istanbul and the Mamluk style in Egypt. After the Ottomans entered Egypt and defeated them, they decided to take the Egyptian craftsmen, artists and architects to Istanbul and leave Egypt empty without its main engine (building and architecture). Not long after, however, the Ottoman Sultan died and his son, Sultan Suleyman, took power and allowed the craftsmen and architects to return to Egypt again to spread Ottoman architecture in Egypt. After construction began again in Egypt, it was found that the architecture of most of the newly built buildings was an architectural mix between Ottoman and Mamluk architecture such as Khair Bek Mosque, and Emir Sultan Mosque. Cairo was full of architectural development as the Ottomans were interested in the application of the technology of large-scale interior building spaces that were confined to enormous weightless domes. But the Ottomans also decided to pay some attention to other things in architecture as they chose to care about developing not inventing. For example, they cared about increasing the number of minarets and domes and also cared about decorating, engraving, and using marble in the places of worship. They also decided to pay attention to the development of something that Egypt is already famous for, which is the Mashrabiya and wood crafting.

Fig 2.2.1 collage images of different paintings made to describe Cairo during the ottoman empire
source (google)

Fig 2.2.2 collage images of different paintings made to describe Cairo during the ottoman empire
source (google)

The main religious orientation in the ottoman civilization was Sufis, a version of Islamic religion that was popular on the frontier, that didn't require praying in mosques or reading from the Quran. Some of its worships were through a special dance, its practitioners were known as whirling dervishes. This was a practice more than a dance for the individual to forget himself to get connected to the universe and god and clear all the way to be just him and his god.

This art reflected how the Sufi as a worshiping method had a great impact on the culture and social aspects. This eventually affected everything and changed the prioritized thought of the society from fighting for independence from ottoman empire, to think more in nature, art and make their thoughts pure to god.

From that a lot of science and knowledge were added to architecture, this made the architecture in the ottoman period considered to be the one of the best most developed ,self-maintaining architecture in history. We are still learning from it in contemporary architecture.

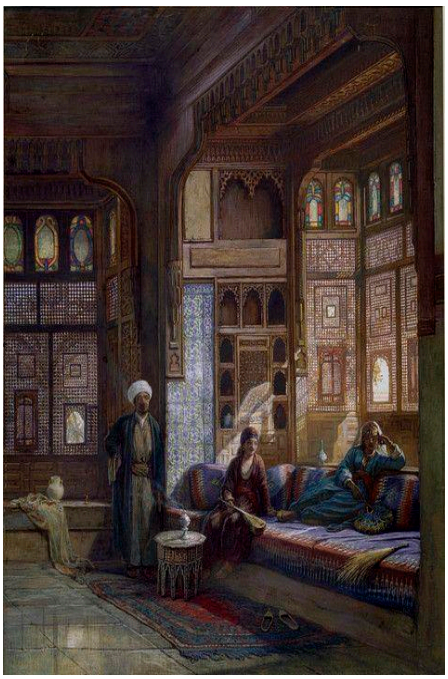
The cultural, social and religious aspects of the designers had a big impact on their architectural designs as

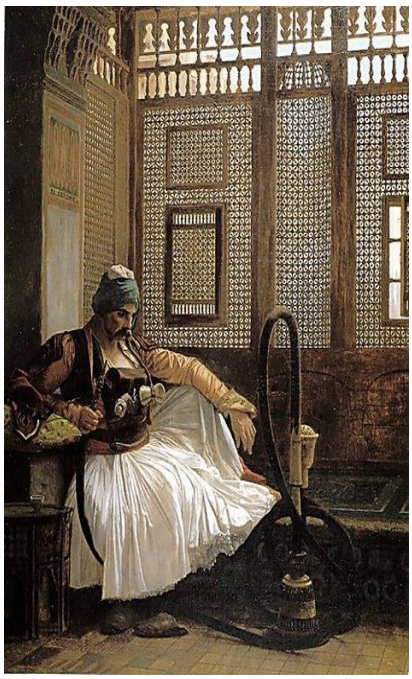
Fig 2.2.3 collage images for the common figure of the social layers in Cairo
source (google)

Fig 2.2.3 a painting made for a sufi man divirshes
source (google)

Dervishes dancer painting It stems from the sofa's sense of Islam, and it shows that the movement in the universe begins at a point and ends at the same point.

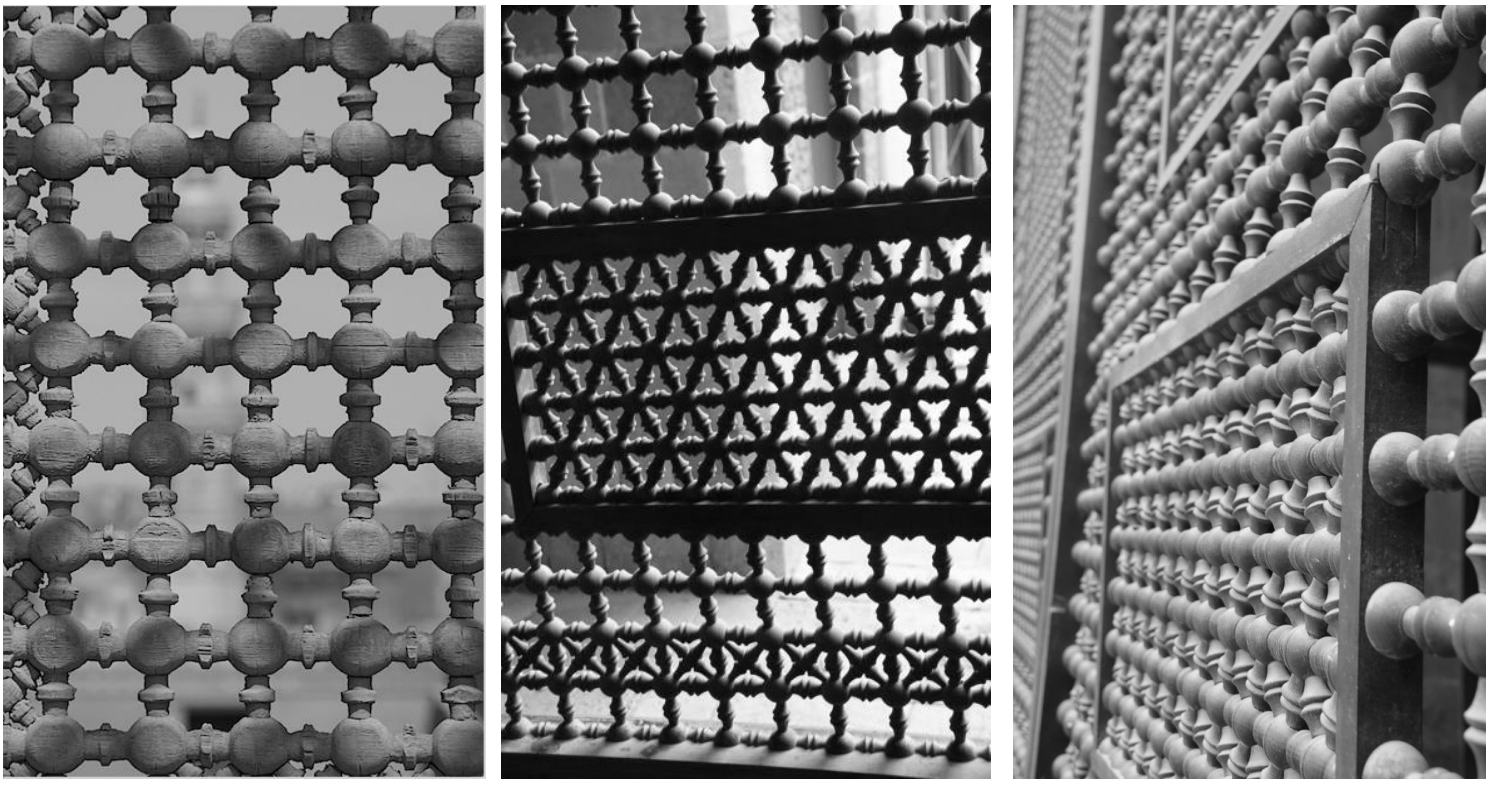
Fig 2.2.4 collage images of different paintings made to describe Cairo indoor spaces during the ottoman empire
source (google)





Patterns forms	Geometrical patterns and arabesque patterns driven from Lego and one of its most famous types was “Maymooni patter”, where the patterns were balanced and symmetrical. This facilitated its crafting which was eventually the main reason behind its common implementation in most of the buildings back then
Perforation percentage range	Used to implement small patterns with perforation percentage ranges from 13% to 25%
Geometry	Used to have complex geometry with mixing different patterns together in the same component according to the desired function. This was because of the competitive spirit in science and geometry back then and the huge desire towards creating an everlasting culture that doesn’t die through time.
location	Turkey, Egypt, Algeria, Syria and Iran
typology of the spaces that commonly implemented it	It was considered to be low cost and functional so it was implemented conventionally in all building’s typologies.
Material used for it	rods of different Woods or stone.

Fig 2.2.4 collage images of different ottoman maymooni patterns
source (google)



3. **Context**

the city , the climate and the architecture

3. Context

This study focused on the different typologies of Mashrabiya situated in the middle east countries , morocco and Spain. However, the main focus in the context and climatic conditions was on Egypt as it represented the center of the most common typologies of Mashrabiya.

3.1 History of Mashrabiya in Egyptian architecture :

Because Egypt was ruled by more than one kingdom, it contained many architectural marks for each kingdom. One of the most famous architectural marks were those of the Ottoman era, which ruled Egypt after the Mamluk period. Most of the architecture found in Egypt from the Mamluk and Ottoman rule appeared on the centers of religion (churches and mosques). It also appeared in the walls of schools, graves, castles, and Islamic museums.
The Mamluk era significantly changed Egypt's architecture, so it was called the Golden Age in the history of Islamic architecture (1250-1517 CE). This era was famous for its lighthouses, facades and domes that was mostly made of Stucco and marble.
Al-Zahir Bibars Mosque (1269-1267), famous for the origin of Mamluk architecture in its design, contains designs similar to Al-Hakimi Mosque in the Fatimid Period. This mosque is characterized by its main entrances and its four prominent pillars. As well as The arches that were designed in the form of horseshoe horses (Tilted and pointed). After the establishment of the mosque by the engineer Mohammed Ibn Belek Al-Mohseni, a school was established with the same name as the mosque. That school was damaged and Al-Sultan Hassan School (1356-1363) was built with the same architectural system. In the form of orthogonal shape on a huge scale, the school dish was up to 100 feet flat.
Architecture in Egypt then began to follow the same architectural system of the school and the mosque, especially domes and dishes of mosques and then after it the churches. But most of the things that have been built on the same system so far are the graves: Qaitbay Vault and Barsbay vault). Most of the Mamluk dwellings were destroyed and only parts of the façades and entrances were left. The Ottomans took the architectural system of gilded ceilings and wooden decorations of the Mamluk era after ruling Egypt and starting making changes on it and started their own Architectural era.

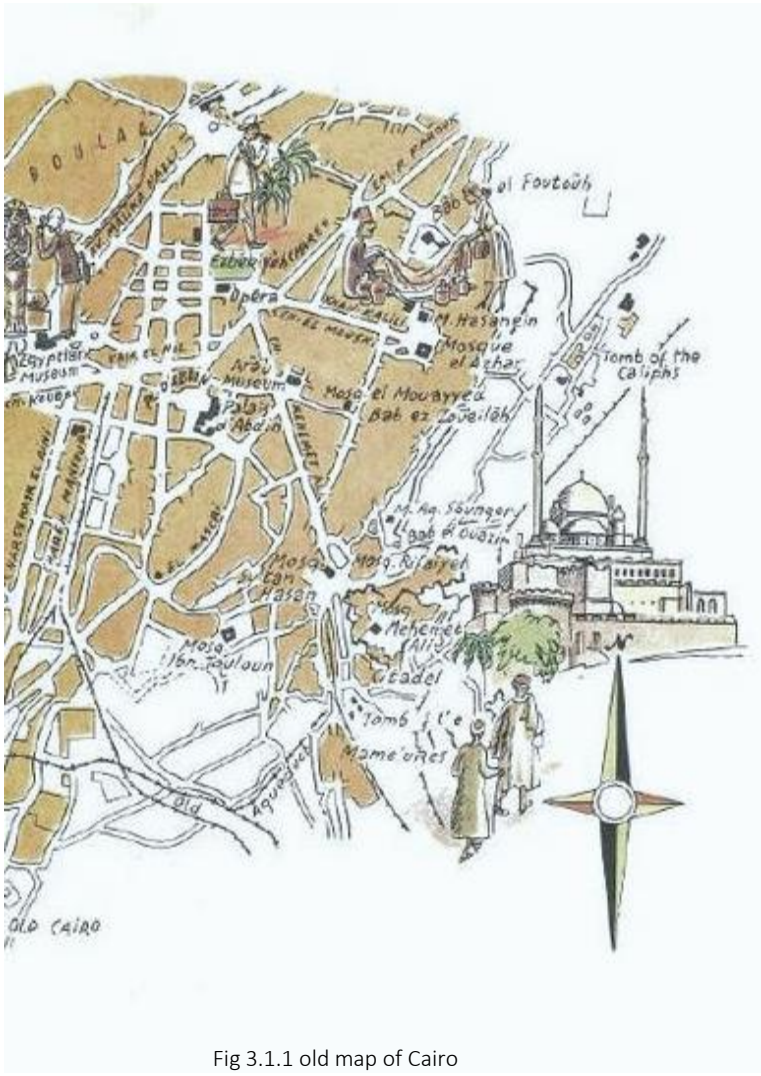


Fig 3.1.1 old map of Cairo
source (google)

3.2 Implementation of Mashrabiya through time in different space typologies:

The Ottomans were interested in developing Mashrabiya, which was initially built during the Abbasid era of ruling Egypt. They began to develop Mashrabiya spaces to achieve perfect harmony between internal and external spaces as well as light and shadow placed. The Ottomans made the Mashrabiya an essential part of Islamic architecture. While the Abbasids used Mashrabiya in palaces and public buildings. The Ottomans were the reason for its prosperity as they used it in most of the buildings like schools, hospitals, and offices. They spread it widely in Egypt, Sham, Iraq, and The Arabian Island; then they started publishing it in Istanbul.

The Ottomans changed many details in Mashrabiya. They designed Mashrabiya in different shapes according to the types of wood used and made a difference between house Mashrabiya, palaces Mashrabiya and public places Mashrabiya: They designed open Mashrabiya for houses and closed Mashrabiya for palaces and public places such as hospitals.

For the opened Mashrabiya, they cared about the privacy that the people needed; they used methods to mislead the eyes through the openings with Mashrabiya. They started using coloured glass to reflect the light to give a clear view of those inside the house to the street, but at the same time preventing those in the streets from seeing those inside the houses.

They highlighted the geometric shapes (triangles, pistols, octagons and round shapes) on the Mashrabiya to escape attention from looking inside and give a beautiful view of the short houses. Most of the famous Mashrebiyat (the plural of Mashrabiya) were those of the mosques such as Mohamed Ali Pasha Mosque which was built during Mohamed Ali ruling period in Egypt. But not only mosques stayed famous for their Mashrabiya but also houses like the nobles' houses. Like the Golden House that is still standing for notables. That house is full of expensive accessories that are important for the country. The house was famous for its numerous entrances and its big receptions and the long stair to the balcony that was fraught with decorations from everywhere. But all these were nothing compared to its Mashrabiya that shows the whole country yard while having amazing places with many Mashrabiya. Also, many houses till now contain Mashrabiya in Egypt, for example, Khan Al Khalil houses. Mashrabiya was one of the beautiful basics that the Ottomans took care of and used to attract the attention of the world so far. They cared about using it in Iraq and Egypt because of their desert climate, but they also used it in Istanbul and many other countries to show the world the creativity of its architecture so far.

Fig 3.1.2 3d max image to visualize Cairo streets of Cairo
source (previous papers by the author)

3.3 Mashrabiya in Egypt of 2019:

As the urban fabric of Egypt, and Cairo in specific, develops the architecture develops with it as well. And Mashrabiya as an art started to lose its high rated influence on the designers and users. A lot of the old buildings were demolished and rebuilt to satisfy the users and societies needs. Vertical expanding in the building process became more common, as well as the implementation of non-passive solutions to achieve indoor comfort.

Egypt managed to maintain its history and heritage through all the political, cultural and social changes. For many reasons, the urban transformation and the rapid growth led to dissolve this history. This factors contributed to restraining Mashrabiya as an art and craft in certain places in Cairo. During the unstable political and economic conditions, some people tend to burn the buildings to get governmental approval to demolish them. These actions lead to destroying beautiful arts and spectacular history, which eventually made Mashrabiya an endangered art that needs to be saved and revitalized.

The previous factor was the primary motive towards choosing Mashrabiya as an element to study and evaluate. They also helped in shaping and defining the steps of the study and the fieldwork. For that during the fieldwork, craftsmen, carpenters and inhabitants were asked about many questions that can help to understand the problem

The organization of the urban fabric is understandable only in terms of the historical context of the city. The three old areas are densely populated slums that almost surround the heart of the city centre relatively western. The largest of these cities is the city built in the Middle Ages during the Fatima dynasty (909-1171), with its pre-19th-century extensions (Aesthetic, Dar al-Omar, Bab al-Shaariya, Sayyida Zeinab to the east and the caliphs to the north). This densely populated area is located in most of Cairo's historic landmarks, including the First Bibars Mosque on its northern outskirts and Salah al-Din Castle in the south. Among the main bazaars within the central-walled city in Khan al-Khalili.

The Islamic historical architecture and most famous buildings that were mainly formed in Cairo were commonly located in:

1.Muizz Street
2.Saliba Street

Fig 3.1.5 old map of Cairo
source (google)

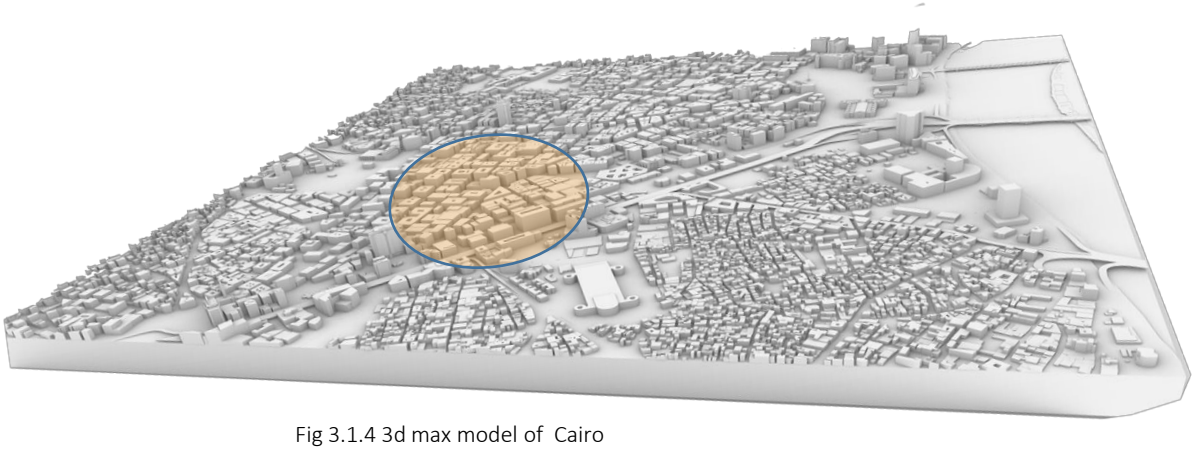


Fig 3.1.4 3d max model of Cairo
source (previous papers by the author)

Context

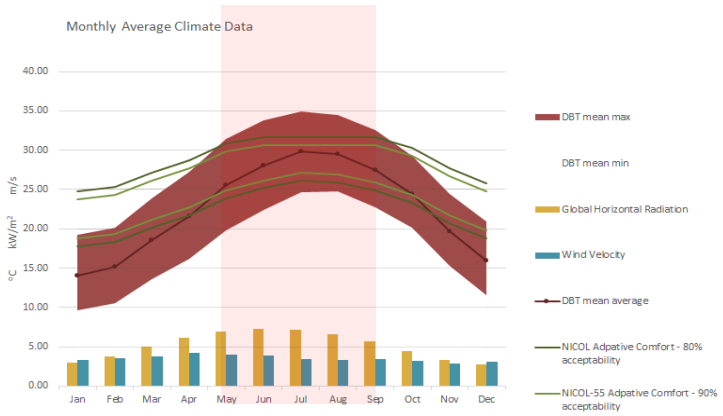
3.4 climatic factors of regions that implemented the Mashrabiya:

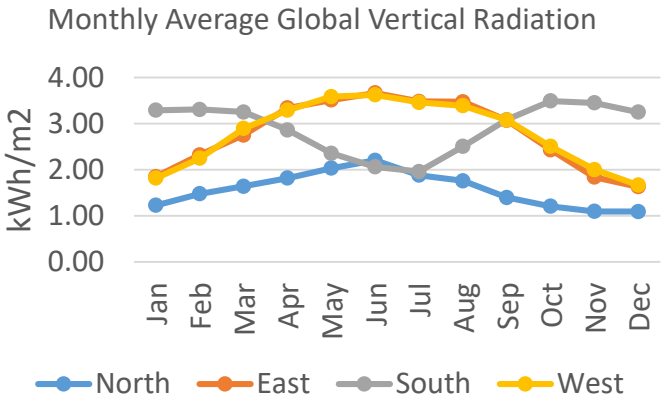
Egypt is considered to be the capital of Islamic architecture in general and mashrabiya in specific. For that it was importation to understand the climatic characteristics of the regions that were famous of its implementation of mashrabiya.
The climate in Egypt varies from a city to another according to the typology of the land, however, Cairo is considered to have hot dry climate. The temperature reaches extremes in summer days of June and July. The average temperature in summer days ranges from 34.8 to 38 degrees.
The winter in Egypt is very short, that it last for about 3 months only. The temperature in winters is considered to be moderate that's why the buildings and the urban fabric were made to adjust to extreme hot weather.

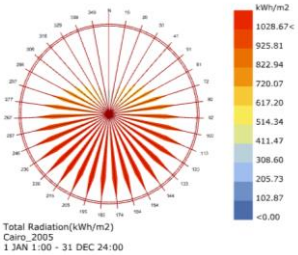
The capital, Cairo, is located about 180 km (110 miles) from the sea, but on the edge of the huge Nile Delta, so its climate is halfway between the Mediterranean Sea and desert climate: summer temperatures are lower than in the desert region but higher than in the desert region. On the coast, highs are about 35°C (95°F) in July and August, but humidity is very high, so the heat can be suffocating. Summer is long: Maximum temperatures exceed 30°C (86°F) from May to mid-October. Within the city, the unpleasant sensation of heat is increased by the so-called "urban heat island effect" that occurs in large cities, and also by pollution.

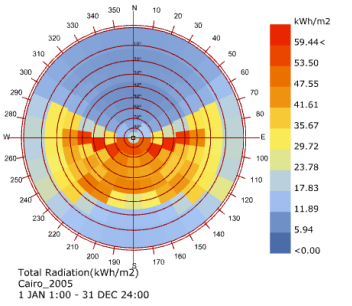
The radiation resulted from the sun is relatively high in both direct and defused radiations, this increased the challenge of designing an efficient shading element.

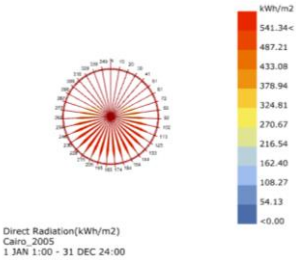
In hot summer days the east and west facades receive more radiation than southern facade which increased the challenge of shading them. Specially that the dominant wind most of the year comes from the east as well.

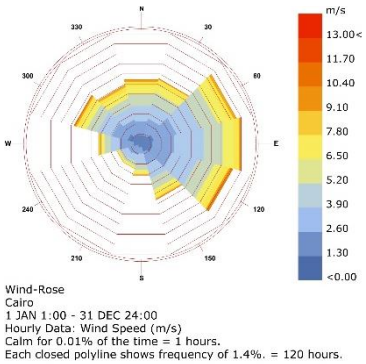




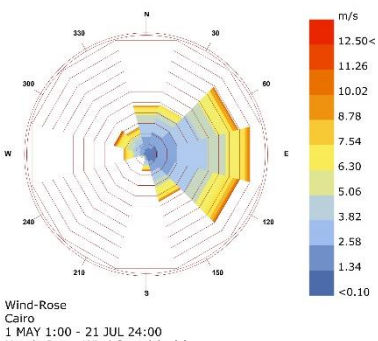






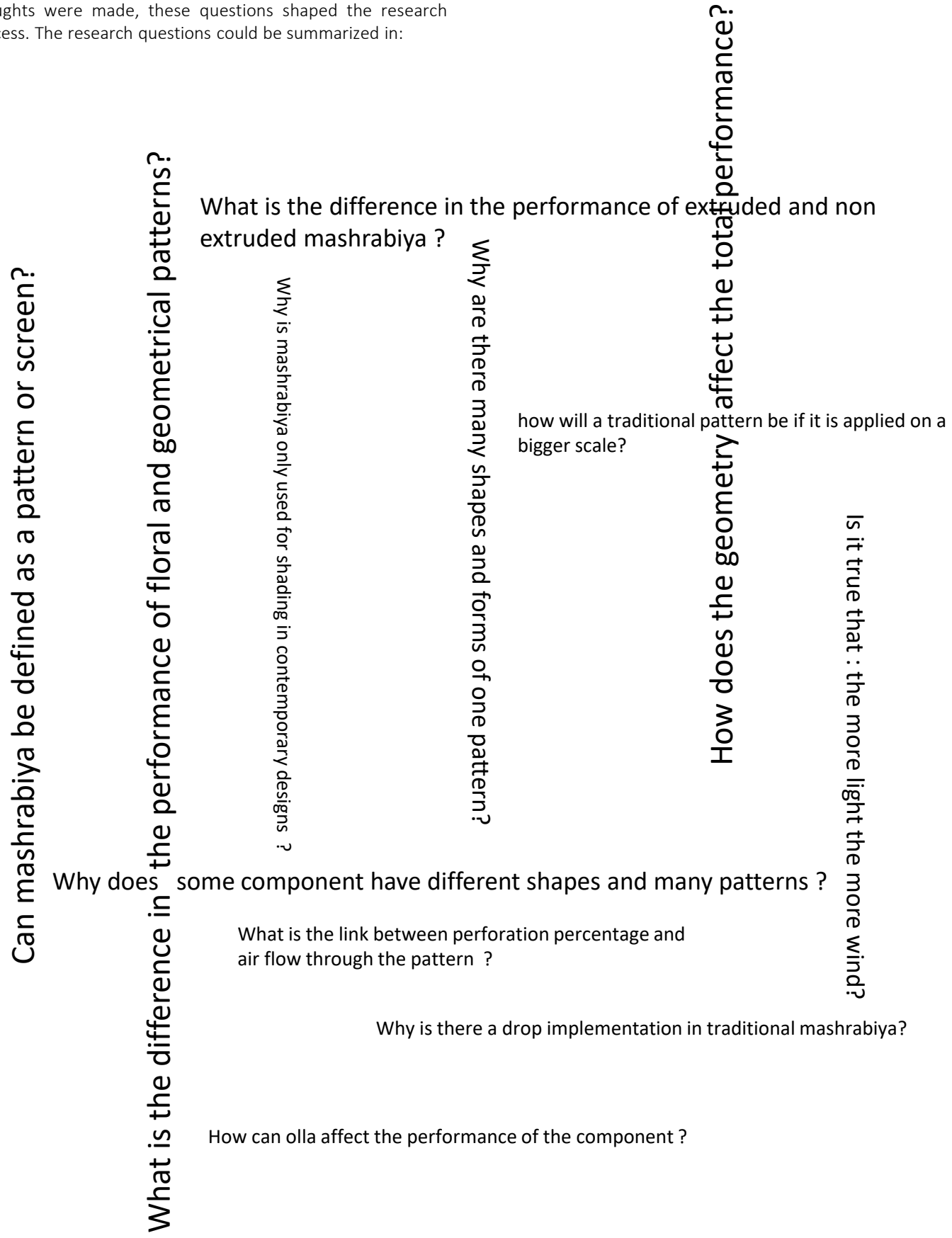


	January	February	March	April	May	June	July	August	September	October	November	December
Avg. Temperature (°C)	13.1	14.1	17.4	20.7	24.1	27	27.6	27.6	25.8	23.5	19.2	15.1
Min. Temperature (°C)	7	7.4	10.5	12.9	16.2	19.3	20.7	20.8	19.1	16.8	13.1	9.1
Max. Temperature (°C)	19.3	20.9	24.3	28.5	32.1	34.8	34.6	34.5	32.5	30.2	25.4	21.1
Precipitation / Rainfall (mm)	5	3	2	1	0	0	0	0	0	0	3	4



3.5 Research questions and objictives

Based on what was mention before a lot of questions and thoughts were made, these questions shaped the research process. The research questions could be summarized in:



4. Field work

Observing- measuring-hypothesizing-concluding

4.1 Objectives of the filed work

In this study, to make a good understanding of the performance of facade perforated screens, it was necessary to do the study on a wide parameter of sites in different climatic regions to understand different uses of Mashrabiya.
To have a full understanding of the environmental performance of Mashrabiya and be able to quantify its performance, it was necessary to run fieldwork studies to understand how the living examples preform.
The fieldwork was necessary to illustrate the reason behind the drop in the implementation of traditional Mashrabiya.
According to the literature review and illustrating the different historical examples of Mashrabiya, and how it evolved in terms of form and pattern, through time, from region to another. The literature review showed how different climatic conditions merged with different climatic factors had given birth to different typologies of Mashrabiya.
Various challenges formed prior against making a fully understanding of the total difference in typologies. This was conducted of the lake resources that could provide detailed information about the different forms of Mashrabiya, more so ever, no systematic studies went in a deep analyzation to the difference in the details of Mashrabiya. For that fieldwork was so important to provide a full understanding towards the particulars of Mashrabiya, by monitoring these differences through detailed drawing and measurements, this will facilitate providing the most accurate modelling of the component for further analysis.
The differences in the location of each selected city in both countries played an essential role in the climatic conditions.

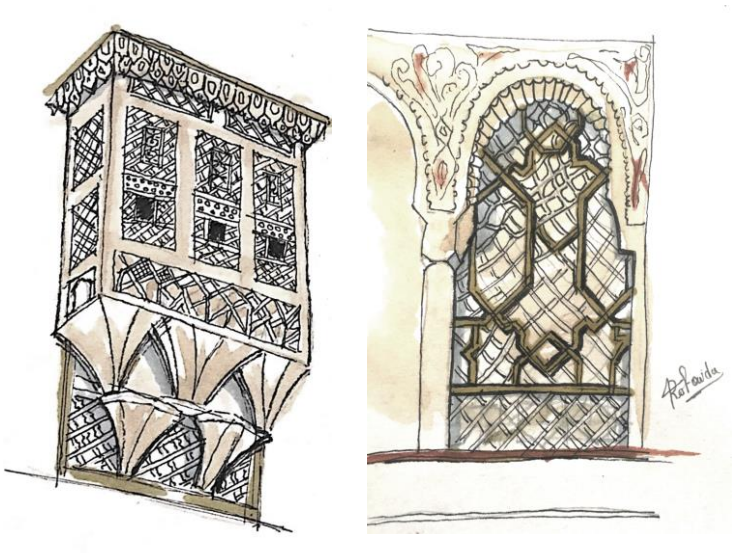


Fig 4.1.1 ottoman and Andalusian mashrabiya drawn sketches
source (sketches made by the author)

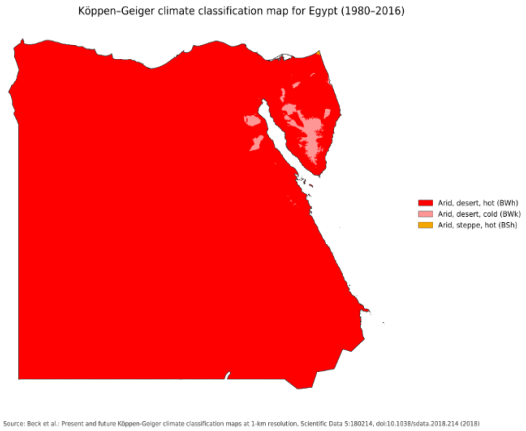


Fig 4.1.2 Egypt koppen climate map
source (Wikipedia)

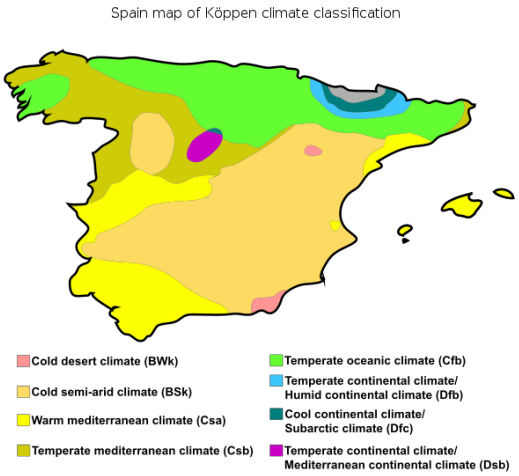


Fig 4.1.3 Spain koppen climate map
source (Wikipedia)

4.2.The methodology followed in fieldwork:

4.2.A: Setting the methodology of the fieldwork

It was essential to provide a standard methodology for fieldwork that was going to be made twice in two different countries with a different climate, with putting into consideration the issues inaccessibility to the selected spots and different languages.
For the fieldwork made, the following methodology was followed:
The main sites that were selected in Spain weren't considered to be domestic anymore and considered to be touristic sites. Where there was no possible way to establish weakly monitoring equipment for safety conditions and security issues. As a result of these factors, the most suitable solution to make a basic understanding to the performance of Andalusian Mashrabiya towards different environmental parameters was through motoring spot measurement.
The methodology followed for monitoring:

- Measuring the air velocity reaching the windows from outside by placing the monitoring device (wind meter) at a proper distance from the window.
- Calculating the shading efficiency of the patterns through measuring the illuminance outside (outdoor) and compare it with the illuminance inside, using lux meter.
- The illuminance inside was measured in the form of 3 spots that vary according to the distance from the pattern as shown in figure number (4.2.1) in order to calculate the average.
- The surface temperature was measured inside and outside to estimate the rough amount of solar gain prevented from the patterns through temperature differences.
- Room temperature was measured multiple times to be informed with any variation in the resultant temperature of the room.
- Afttab illuminance software was used to mainly evaluate the quantity and quality of light conditions resulted from the patterns by taking photos. Also, it allows determining the possibility of harmful glare existence near the windows.

4.2.B. Expected outcomes

Andalusian Mashrabiya is known for its floral, wide, smooth and fancy patterns. Its existence was for environmental function more than for cultural aspects. As it was mainly implemented in luxurious houses and royal palace. As for that, the perforation percentage of these patterns represented in the ration between the area of the vacuum gapes to the area of the solid parts of the pattern was large.
This was expected to affect the performance on the two main parameters performances.

I. In terms of Shading; Andalusian patterns were known for using floral shapes with high perforation percentage. Because of that, it is expected that it will not block most of the sunlight and it will allow more solar gains to enter the space
II.In terms of ventilation: it is expected that it will perform better in ventilating the space as it will allow more wind to enter.

Based on that, the trip field started in Spain and ended in Egypt where a lot of examples were tested, and new points were observed.

Fig 4.2.2 collage images of the tools used in the field work
source (google)

Fig 4.2.3 the app used in analysing the illuminance
source (Wikipedia)

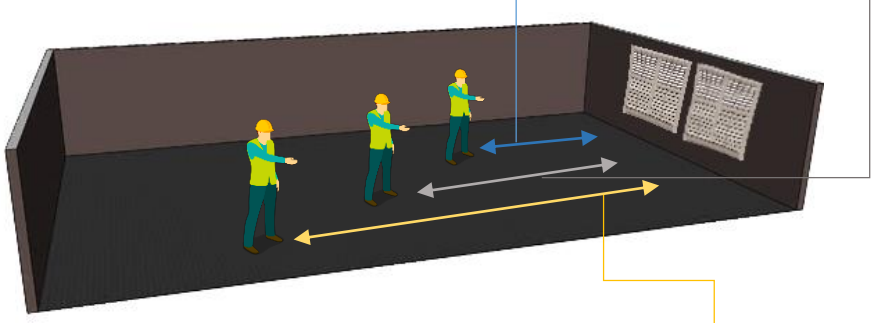
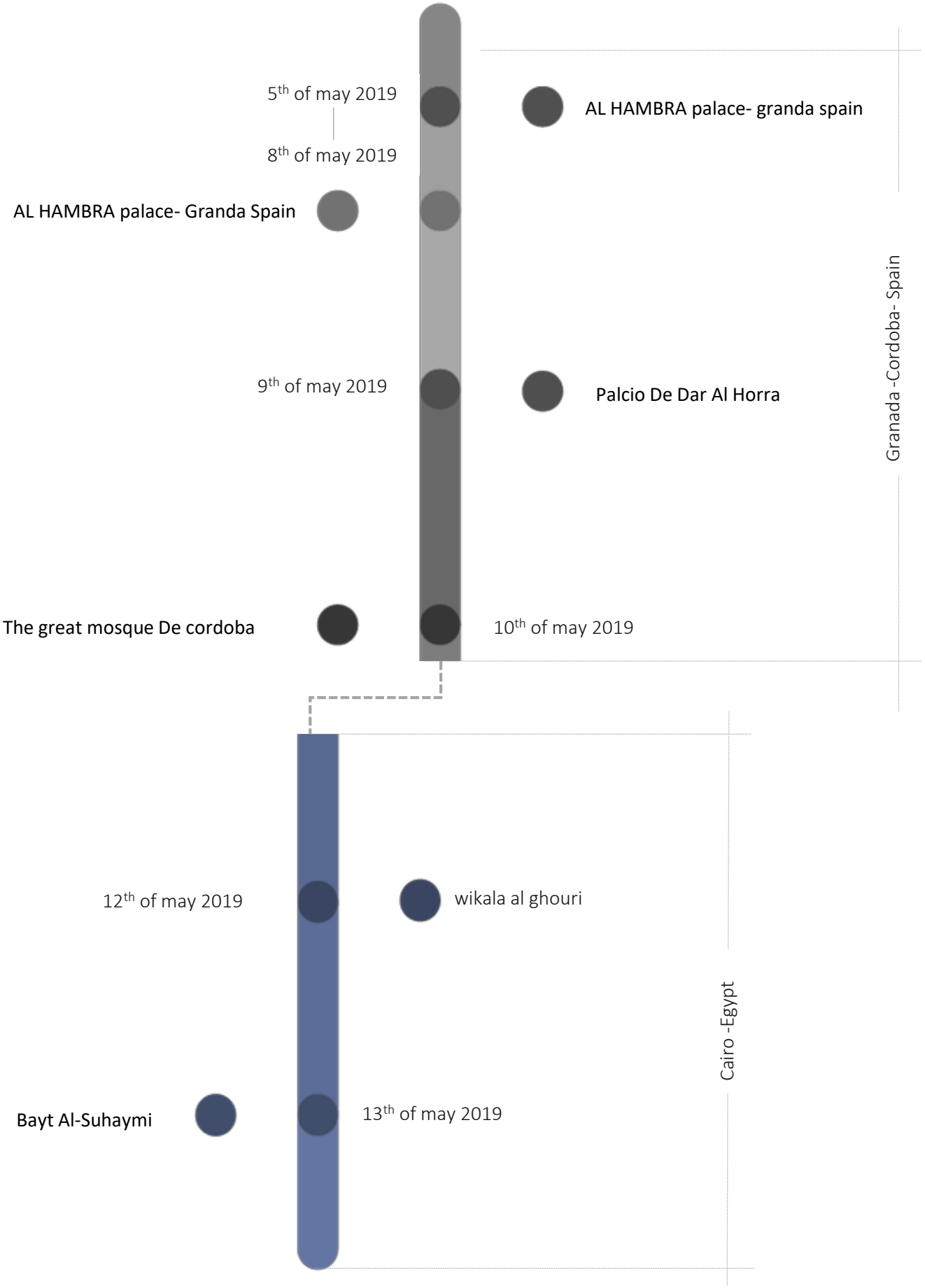


Fig 4.2.1 Spain koppen climate map
source (Wikipedia)

4.2.C. time table



GRANADA

SPAIN

4.3. **Andalusian Mashrabiya**

lattice screens in lands of Andalusia

1.Introduction

To have a full understanding of the environmental preform according to its climatic context, Spain was visited in order to see the different typologies and shapes of patterns.
The objective of the fieldwork wasn't only to understand the environmental performance of Andalusian or ottoman Mashrabiya but also to understand the difference between them in terms of details and patterns.
•Spot Measurements were taken
•Photos were taken to provide a full understanding of the details
•Measurements were made to know the dimensions of the components that formed the patterns Sketches were drawn to keep the details of the measures for the modelling stage to be used in the analytical work.
In Spain the following sites were selected:

1.Al hambra palace
2.Placio de dar al hora
3.The great mosque de Cordoba

These sites were selected according to the typologies of the perforated screens inside them. These buildings managed to maintain the screens implemented in it, which was considered to be the pure definition of Andalusian Mashrabiya.

For security issues, the long time continuous monitoring devices couldn't be applied. So spot measurements were taken on a three days interval in each site. Mesurments were taken more than once through the day in order to estimate the average performance of the patterns.
Estimating the average performance of the patterns was through comparing the outdoor conditions and indoor conditions at different depths of the spaces.
Some of the patterns in Alhambra palace were cover with glass in order to maintain the ornaments curved in the wall by protecting it from different climatic factors. But the performance was measured during the maintenance period where the glass was removed to be cleaned.
The journey to Spain started on 2nd of May 2019 and continued to 15 days after. Each site took an average of 5 to 3 day of monitoring and observing, except for the great mosque de Cordoba the fieldwork and observing occurred in one day as the patterns were totally covered with fixed glass that cant be removed during the maintenance.
The great mosque de Cordoba was an observing case to understand how the pattern was created in terms of geometry and dimensions if it is applied in large scale windows or arches. Also, to understand its performance in terms of shading in such a scale.

Fig 4.3.1 Spain geographic map
source (Wikipedia)

A-AL HAMBRA PALACE

Background of the place:

The Alhambra is an archaeological palace located in Granada, Spain, 430 km south of the capital Madrid and is a distinctive example of the palaces of Andalusia. The palace spans an area of 142,000 square meters with a length of 740 meters and a maximum width of 205 meters. Built during the second half of the 10th century, it is considered one of the most famous tourist attractions in Spain.

The features of Islamic architecture are evident in the buildings of the Alhambra through carpet decorations, the writing of Qur'anic verses and prayers, as well as the sabotages and descriptions of the poets' systems, surrounded by decorations of windows using beautifully detailed lattice screens. These factors contributed to the beauty and balance inside the space.

Fig 4.3.4 Alhambra palace image
source (Google)

The palace is divided into three sections! The first is the shura, in which the king holds his council, the second is the department of official receptions, the diwan and the throne hall, and the third is the harem section, which includes the residence of the kings. One of the most important landmarks in the palace is the Grand Basil Courtyard, the Ambassador's Lobby, the Saro Courtyard, the Sisters Hall, the Black Lobby, the Kings Hall, the Queen's Dress and others.

Fig 4.3.3 Alhambra palace 3D section- ambassadors hall
source (Google)

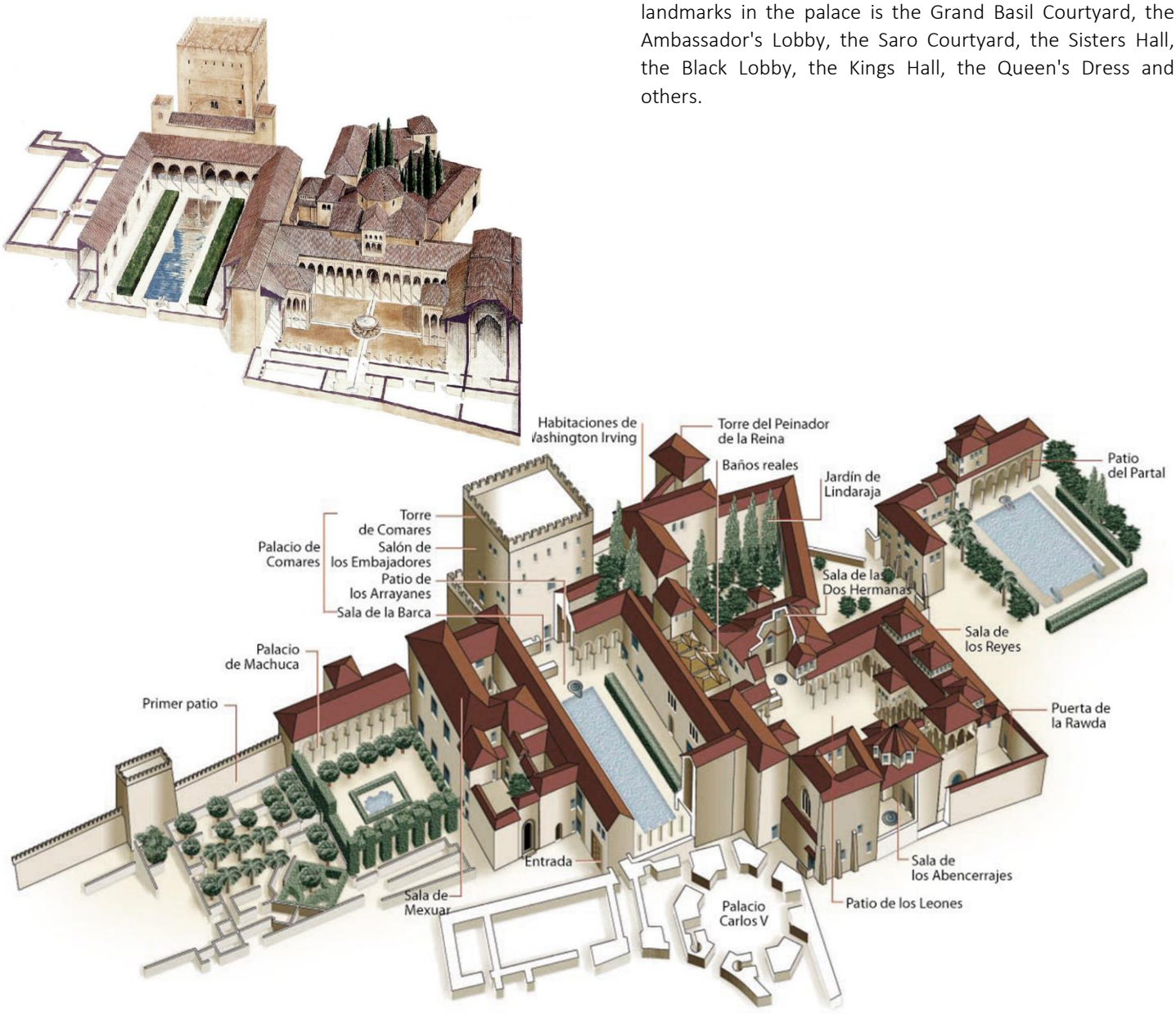


Fig 4.3.2 Alhambra palace 3D model
source (Google)

A. ALHAMBRA PALACE:

1. AMABSADORS HALL :
This hall implemented the Andalusian mashrabiya in a smart way as it used a wide perforated patterns in the small window and less perforated patterns in the big arches.
According to the measurement it was found that the amount of light passing from both the patterns was almost the same, despite the difference in the size and the perforation percentage of each pattern applied to them.
This led to creating a fair distribution of light and wind through the whole space.

This integration of patterns motivated the air to move through the whole depth of the space, as the wind was mainly entering from the big windows and goes out from the high small windows as it is a stack single side ventilation.

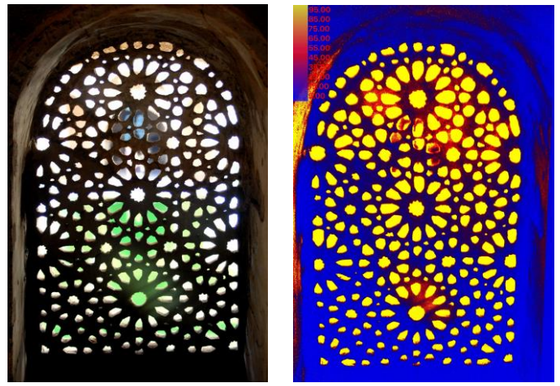
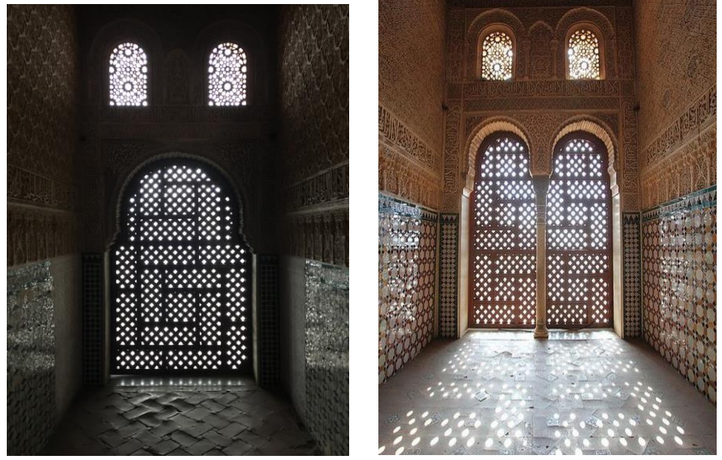


Fig 4.3.7 Alhambra palace-ambassadors hall small top windows
source (author)



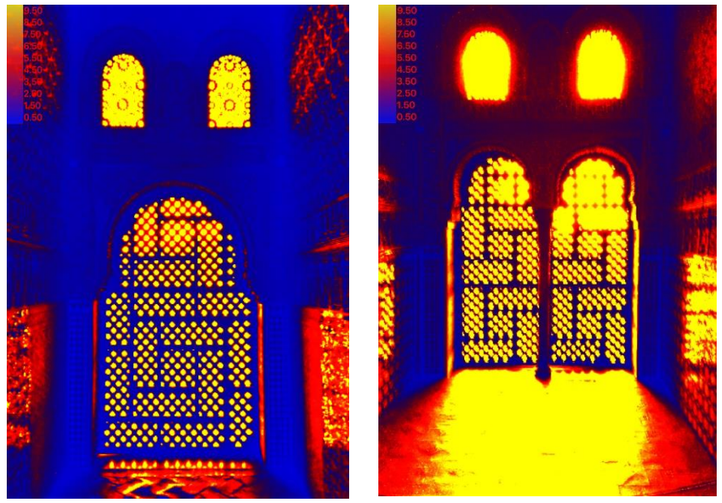


Fig 4.3.5 Alhambra palace-ambassador hall north facade
source (author)

Fig 4.3.6 Alhambra palace-sultan praying room
source (author)

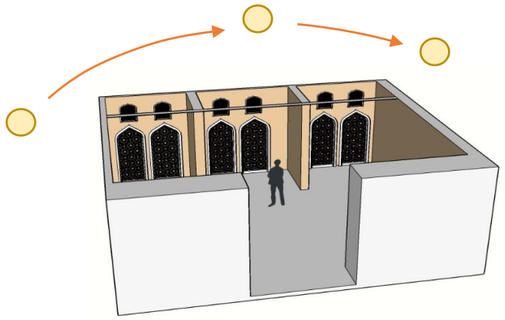


Fig 4.3.9 Alhambra palace-ambassadors hall analytical 3D sections
source (author)

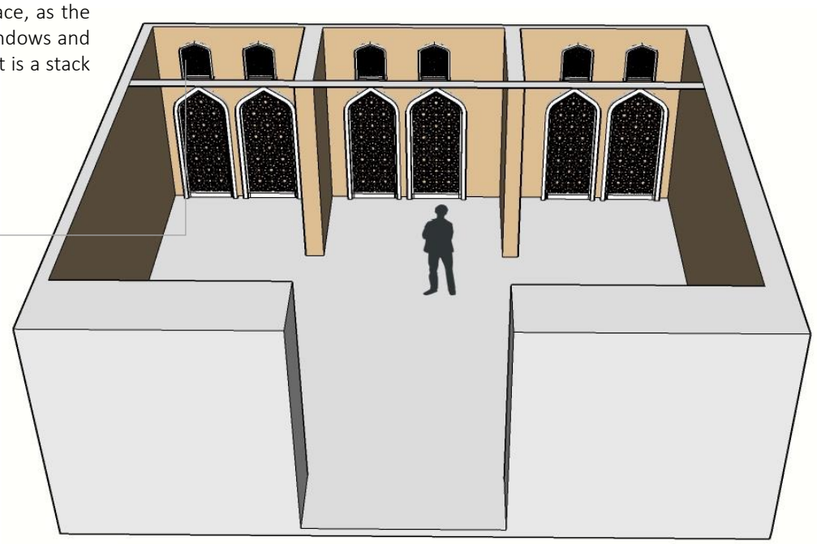


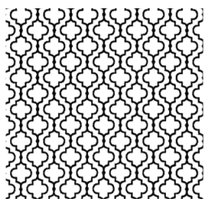
Fig 4.3.8 Alhambra palace-ambassadors hall analytical 3D sections
source (author)

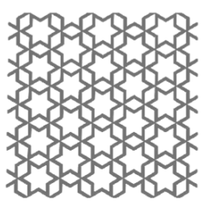
Outdoor temperature average : 25 ℃

indoor temperature average : 17.2 ℃

Outdoor wind speed average : 2.3m/s

Outdoor illuminance level average : 10,301 lux





Surface temperature of the pattern : 20.7℃	The patterns allowed 65% of the air to pass through	Indoor illuminance level average : 370 lux	Percentage of perforation is 29 %	The screens in top part of the Embajadores hall oriented to north
Surface temperature of the pattern : 22..2℃	The patterns allowed 65% of the air to pass through	indoor illuminance level average : 381 lux	Percentage of perforation is 47%	The screens in the big arches in the Embajadores hall oriented to north

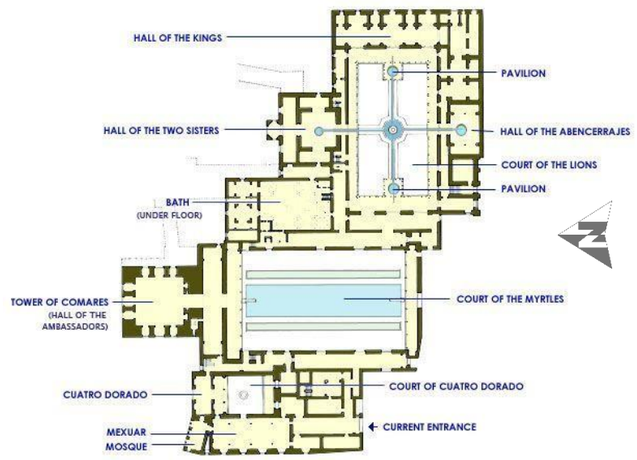


Fig 4.3.10 Alhambra palace plan
source (google)

A. ALHAMBRA PALACE:

1. AMABSADORS HALL :

The same small pattern was applied to the top slab part of the hall and if the user in the space tries to raised his hand an air flow can be sensed, even at the deepest part of the room where we are fare away from the inlets.

According to analysing the space and testing the air flow directions inside the space using basic methods, it was understood that:

- There are two levels of stack ventilation inside the room as shown in the figures.

The top parts allows more light to enter the hall depth of the room and delight the ceiling in order to show the beauty of the ornaments curved into it.

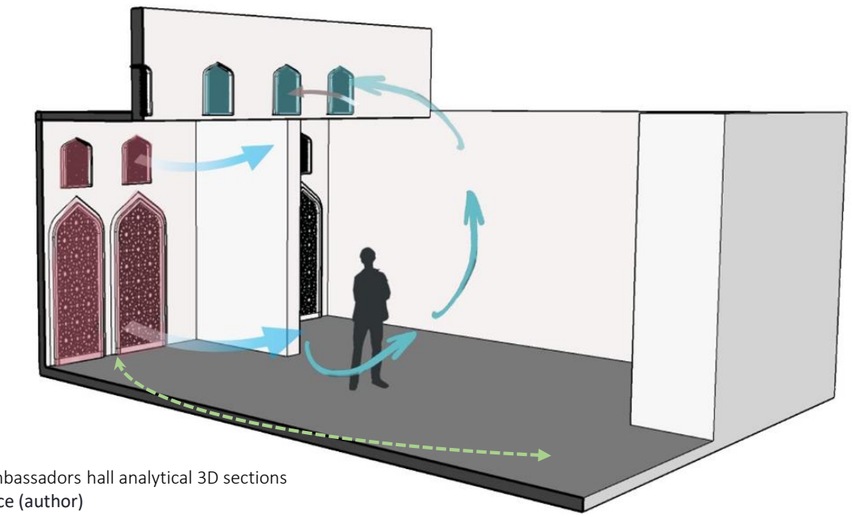


Fig 4.3.12 Alhambra palace-ambassadors hall analytical 3D sections
source (author)

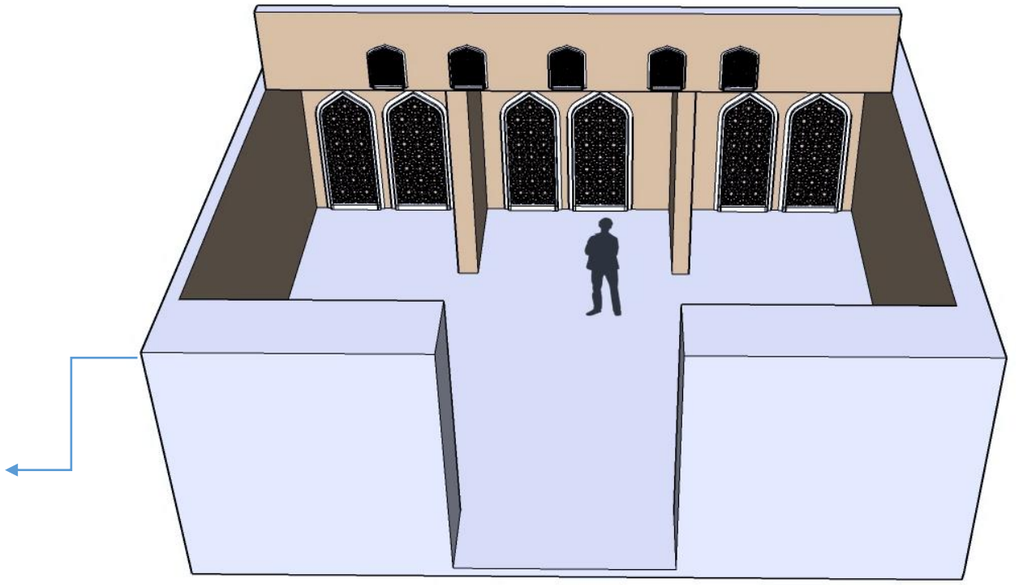


Fig 4.3.13 Alhambra palace-ambassadors hall analytical 3D sections
source (author)

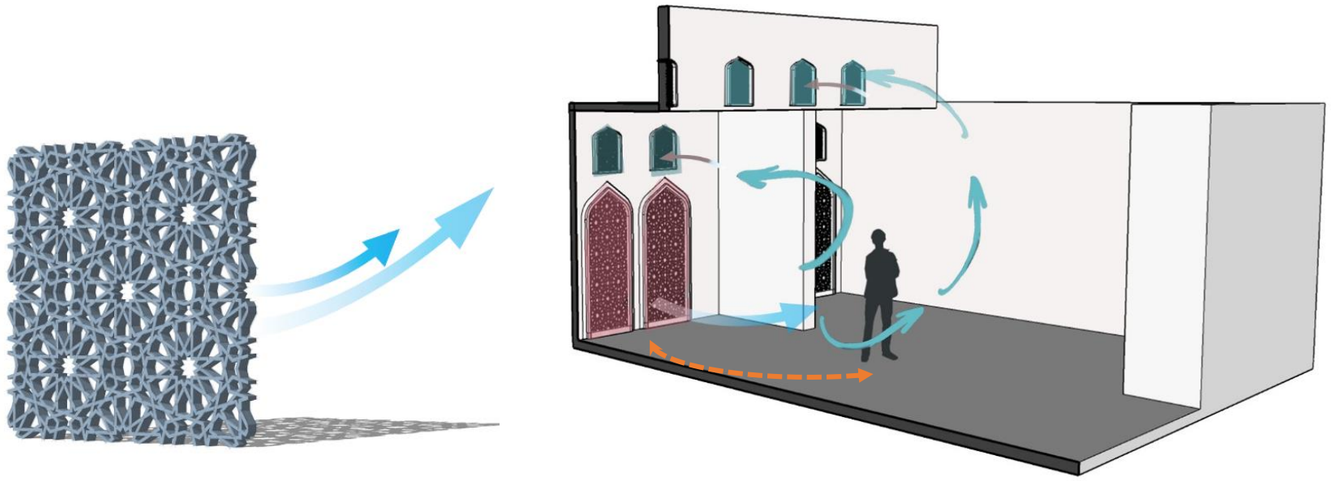


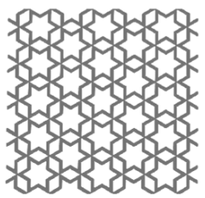
Fig 4.3.11 Alhambra palace-ambassadors hall analytical 3D sections
source (author)

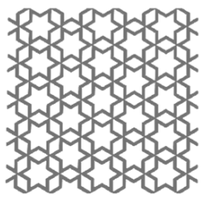
Outdoor temperature average : 25 ℃

indoor temperature average : 17.2 ℃

Outdoor wind speed average : 2.3m/s

Outdoor illuminance level average : 10,301 lux





Surface temperature of the pattern : 23℃	The patterns allowed 65% of the air to pass through. The air moves from the lower patterns to the patterns in the top	Indoor illuminance level average : 370 lux	Percentage of perforation is 47 %	The screens in top part of the Embajadores hall oriented to north
Surface temperature of the pattern : 24.2℃		indoor illuminance level average : 381 lux	Percentage of perforation is 47%	The screens in the top slap Embajadores hall oriented to north

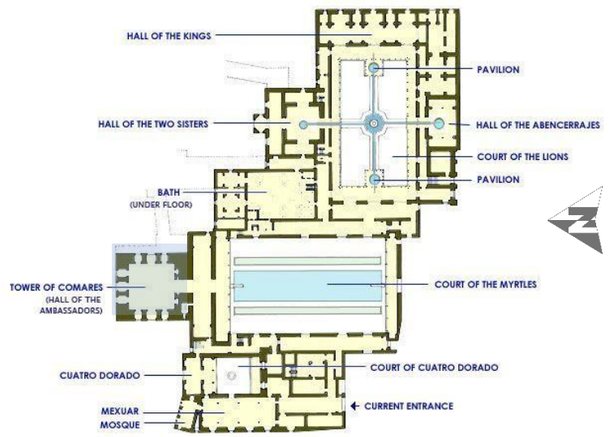
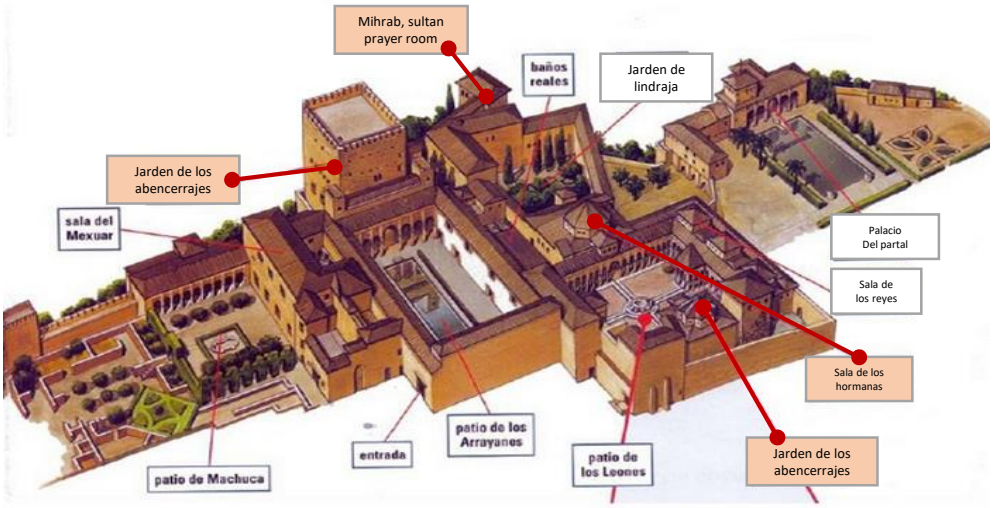


Fig 4.3.10 Alhambra palace plan
source (google)

A-AL HAMBRA PALACE

Four spots in the place x]were found to have mashrabiya and 2 of the spots had the Andalusian mashrabiya without the existence of glass adjacent to it. The glass was added to the patterns in the preservation and maintenance schemes in order to block the wind from entering the space with big volumes that can be accompanied with humidity and minerals that can spoil and affect the ornaments on the walls and ceilings.
Mashrabiya existed in other places as well inside the palace, but these spots were selected in order to compare how the patterns perform according to a different orientation.

The four spots are:
1.Muqarnas Chamber
2.Sultan's prayer room
3.sala de las dos hermanas
4.The Comares Palace, hall of the ambassadors



Muqarnas Chamber

Sultan's prayer room

sala de las dos hermanas

The Comares Palace, hall of the ambassadors

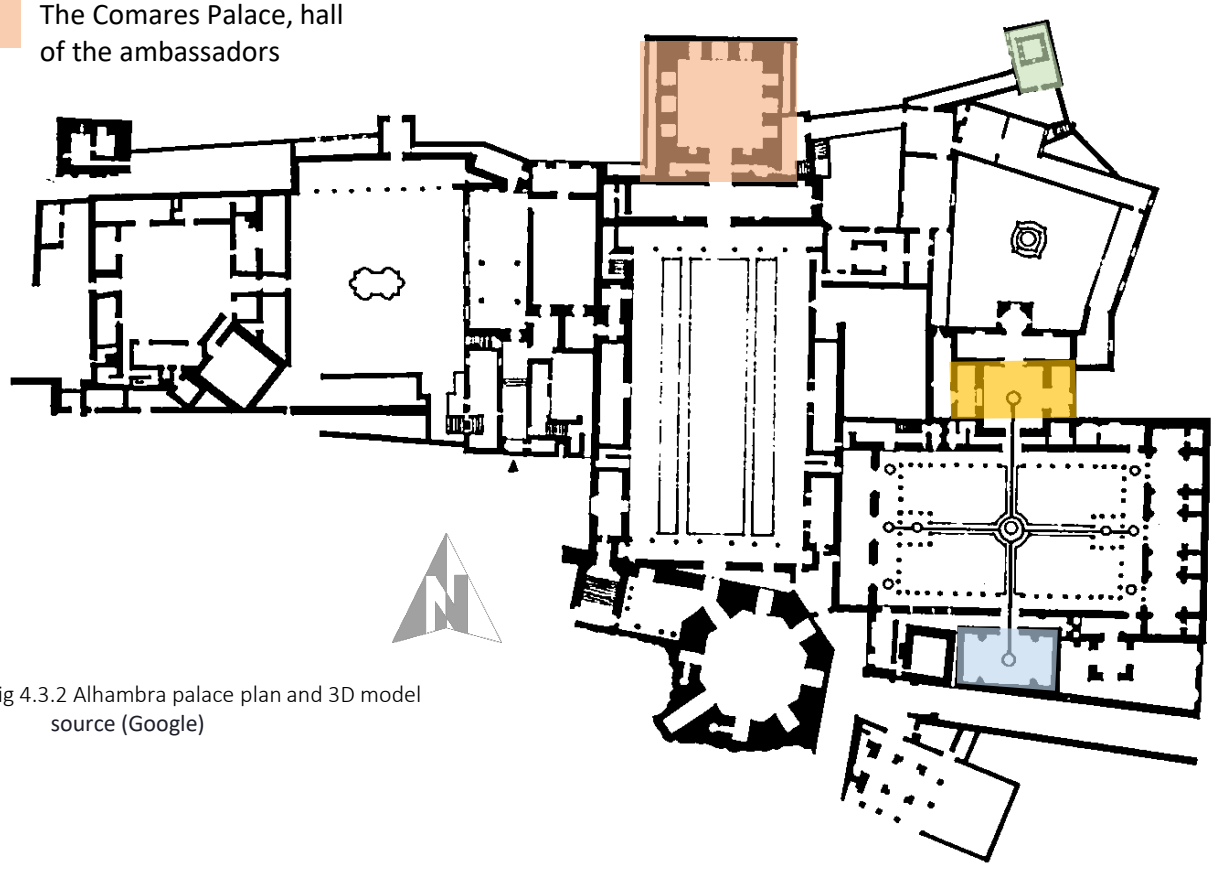


Fig 4.3.2 Alhambra palace plan and 3D model source (Google)

A. *ALHAMBRA PALACE:*

2.The sisters hall:
The dome in the sisters hall hade the same perforated patterns applied to all the diagonals and it was interesting to see how the same pattern will preform in different orientations at the same place.
the wind mete was raised as high as possible in order to estimate how the wind moves inside the space. The dominant wind direction in Granada in 5th of may was south west according to measurements and simulations.
the patterns had high perforations so the patterns oriented to south allowed more light to enter. The position of the patterns makes the entering light reflects on the wall to make the whole hall illuminated by natural light equally, creating a very beautiful view, also showing the contrast between the light and the shade projected in the ornaments curved on the ceiling and walls.

This case showed that the smart implementation of perforated screens is not limited on shading it can also be used in creating equal light distribution through the whole depth of the space.

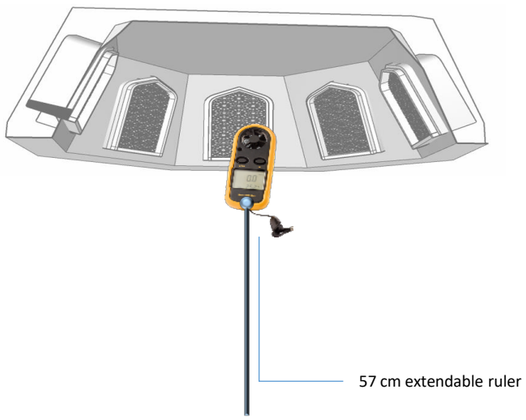


Fig 4.3.15 method followed in measuring source (author-grasshopper)

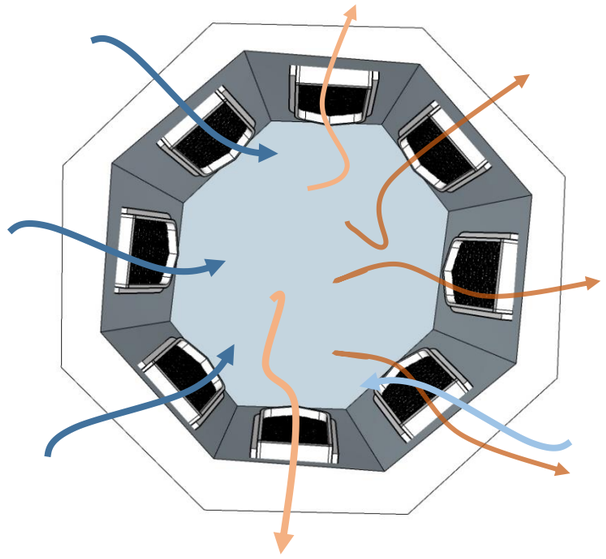


Fig 4.3.13 Alhambra palace-sisters hall hall analytical 3D sections source (author)

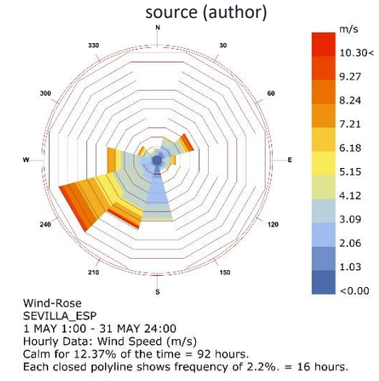


Fig 4.3.14 wind rose of Spain source (author-grasshopper)

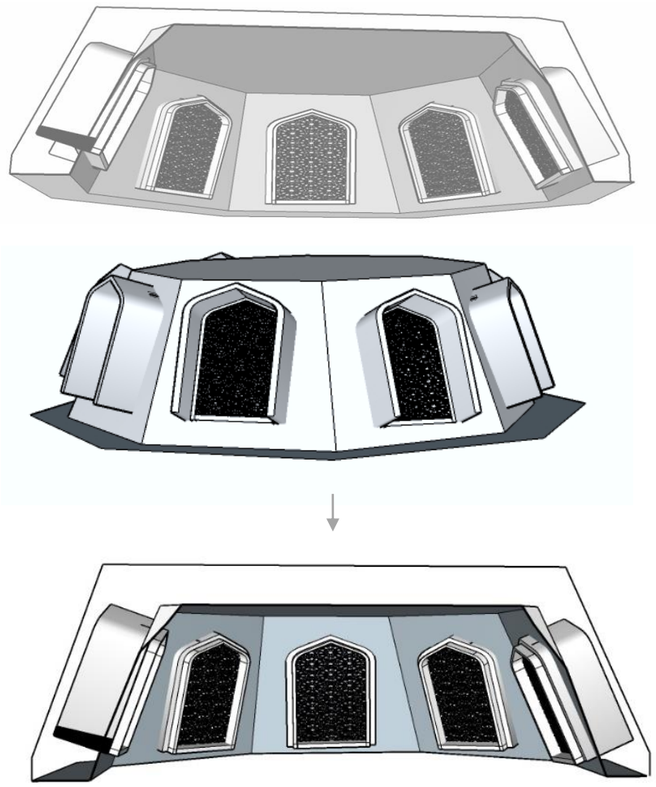


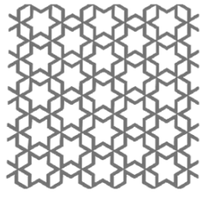
Fig 4.3.12 Alhambra palace-sisters hall hall analytical 3D sections source (author)

Fig 4.3.13 Alhambra palace-sisters hall analytical 3D sections source (author)

Outdoor temperature average : 25 ℃
Indoor temperature average : 17℃

Outdoor wind speed average : 2.3m/s

Outdoor illuminance level average : 10,301 lux



Surface temperature of the pattern : 27℃	The air is moving from outside to inside.	Indoor illuminance level average : 350 lux	Percentage of perforation is 31 %	The screens in Embajadores hall oriented to south
Surface temperature of the pattern : 25.6℃	The air is moving from inside to outside.	indoor illuminance level average : 172 lux	Percentage of perforation is 31%	The screens in two sisters hall s hall oriented to north

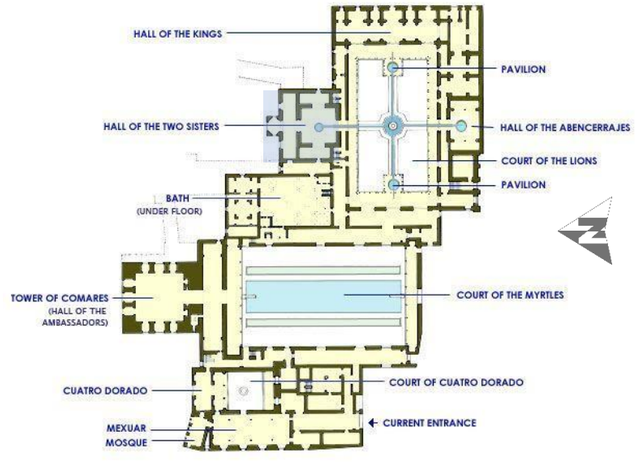


Fig 4.3.10 Alhambra palace plan source (google)

Fig 4.3.16 Alhambra palace-sisters hall at 10 AM
source (author)

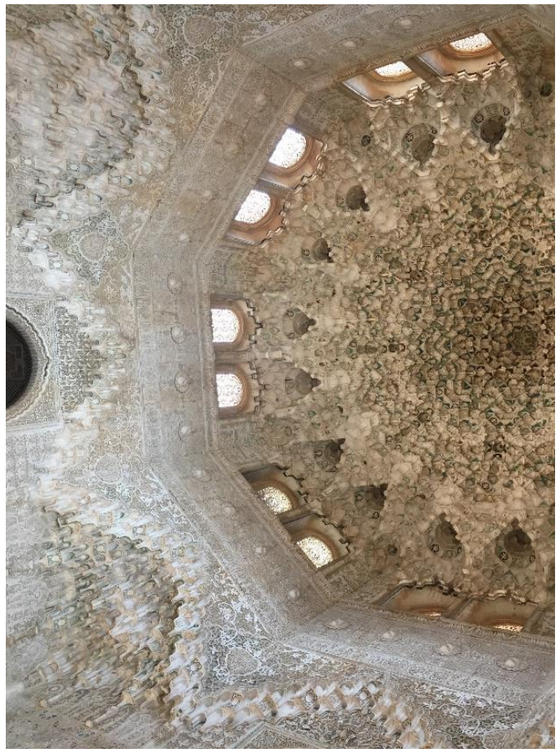


Fig 4.3.16 Alhambra palace-sisters hall at 12 AM
source (author)

Fig 4.3.16 Alhambra palace-sisters hall at 13:00
source (author)

Fig 4.3.16 Alhambra palace-sisters hall analytical illuminance image at 10AM
source (author-AfTTAB)

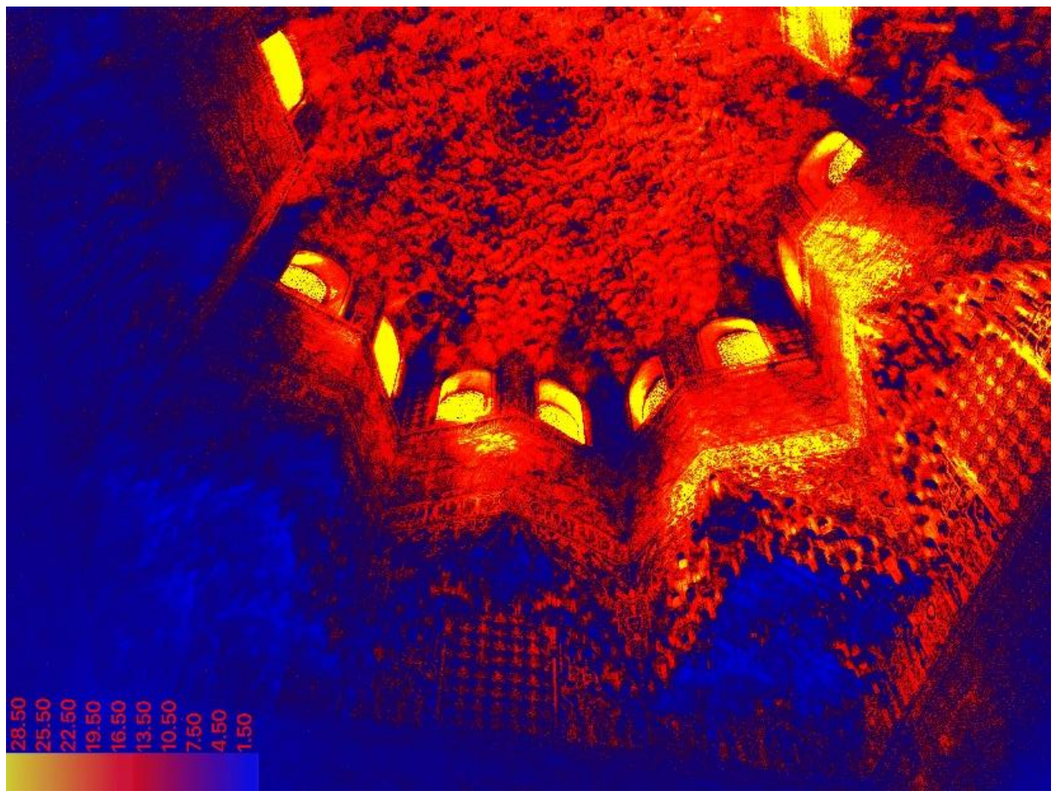


Fig 4.3.16 Alhambra palace-sisters hall analytical illuminance image at 12 AM
source (author-AfTTAB)

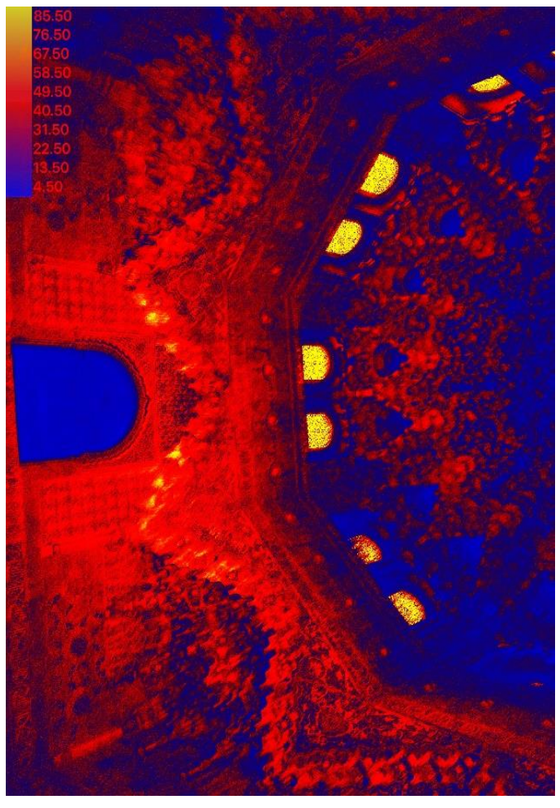
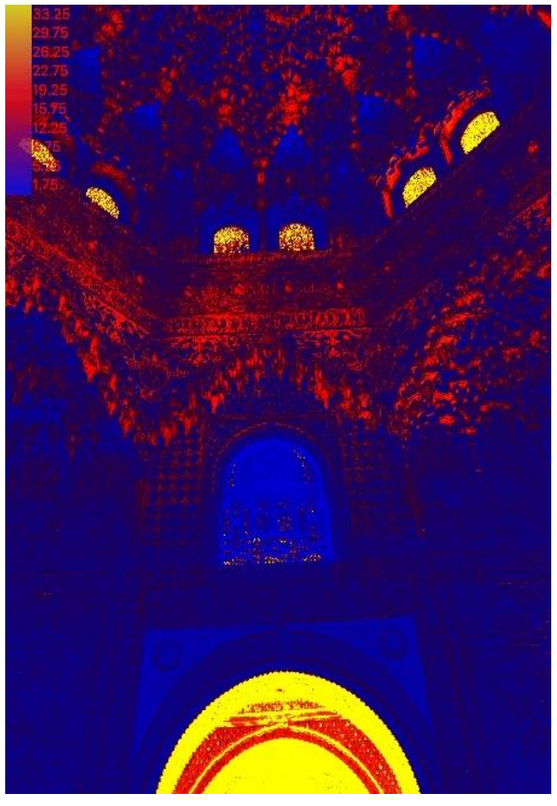


Fig 4.3.16 Alhambra palace-sisters hall analytical illuminance image at 13:00
source (author-AfTTAB)



B-PLACIO DE DAR AL HORA

T he palace is located on the heights of the old Alcazaba, and from its tower one has a magnificent view of the surroundings.
it was characterized by simple shapes of Andalusian mashrabiya that ventilated the terraces and provided shading for it

The field work was held in this building to understand how the performance of the same pattern varies from an orientation to another as this was the only available case that applies the same pattern in both northern and Sothern façade.

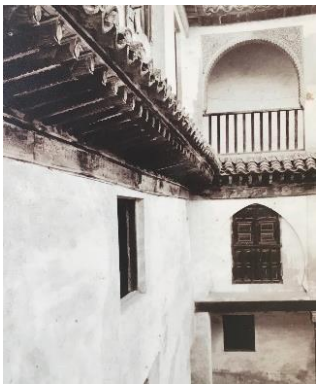
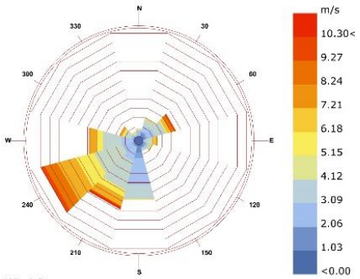


Fig 4.3. 17 collage images for palacio de dar al Hora
source (Google)

B-PLACIO DE DAR AL HORA

B.1: Setting the methodology of the field work

The patterns in the palace were highly perforated and the main material used in them is copper.
The same pattern was applied to both southern and north faced, we can see the difference in the performance of the Andalusian mashrabiya very clearly if the same pattern is applied to different facades.
Shutters were added to the patterns of the southern façade as it was a highly perforated pattern with less efficiency in terms of shading, specially in the southern façade.



Wind-Rose
SEVILLA_ESP
1 MAY 1:00 - 31 MAY 24:00
Hourly Data: Wind Speed (m/s)
Calm for 12.37% of the time = 92 hours.
Each closed polyline shows frequency of 2.2%. = 16 hours.

Fig 4.3.28 Spain wind rose
source (author, grasshopper)

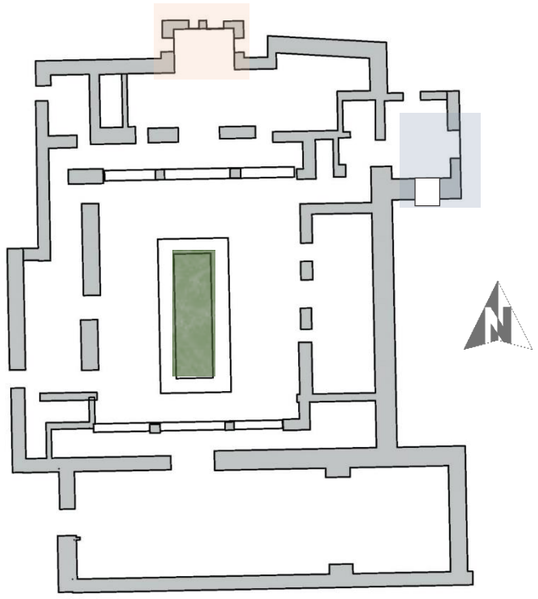


Fig 4.3.27 palace de dar al hora
source (author)

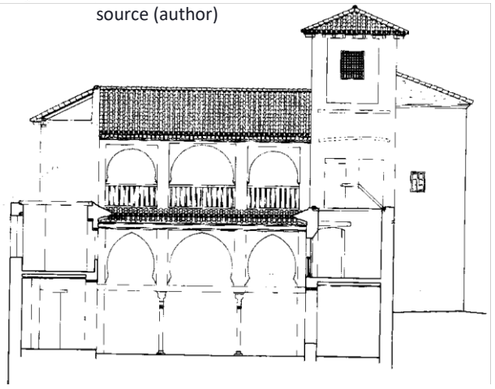
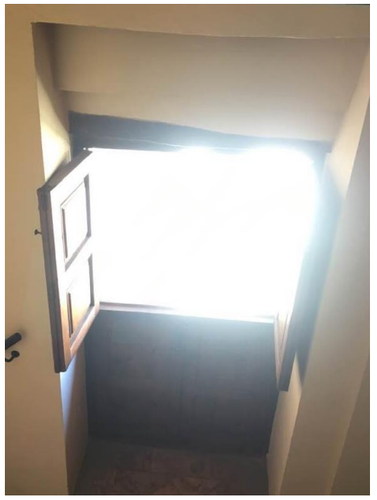


Fig 4.3.25 palace de dar al hora section
source (Wikipedia)



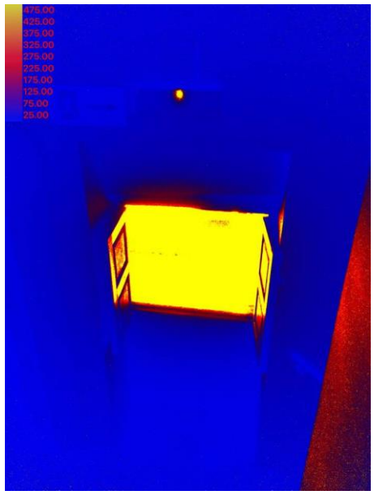


Fig 4.3.21 palace de dar al hora realistic and afttab illuminance image of south façade
source (author)

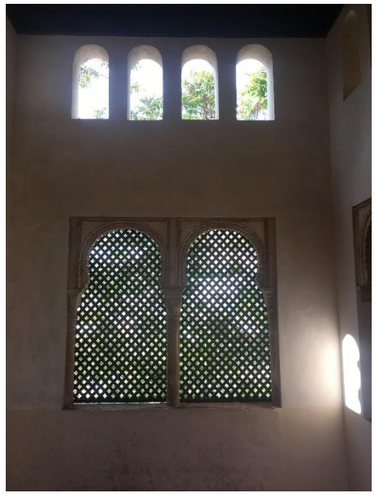
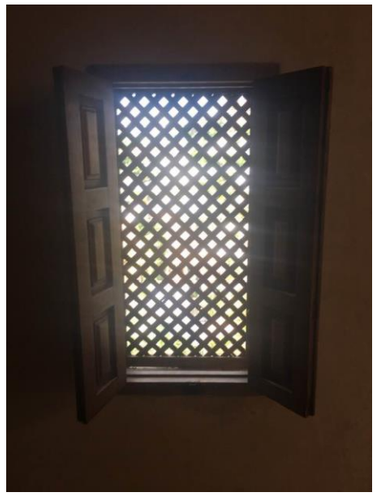


Fig 4.3.22 palace de dar al hora realistic and afttab illuminance image of north façade
source (author)



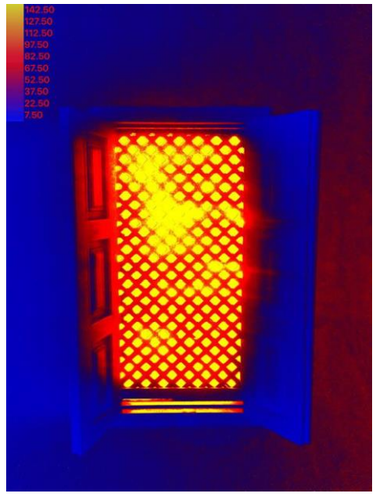
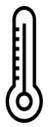


Fig 4.3.23 palace de dar al hora realistic and afttab illuminance image of south façade
source (author)

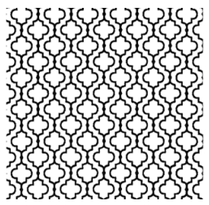


Outdoor temperature average
: 28 ℃

indoor temperature average :
19.7 ℃

Outdoor wind speed average :
2.1 m/s

Outdoor illuminance level average :
12,301 lux



Surface temperature of the pattern : 13℃	indoor wind speed average :0.1m/s	Indoor illuminance level average : 713 lux	Percentage of perforation is 42 %	The screens in th hall oriented to south
Surface temperature of the pattern : 15.6℃	indoor wind speed average : 1.1 m/s	indoor illuminance level average : 452 lux	Percentage of perforation is 42%	The screens in the hall oriented to north

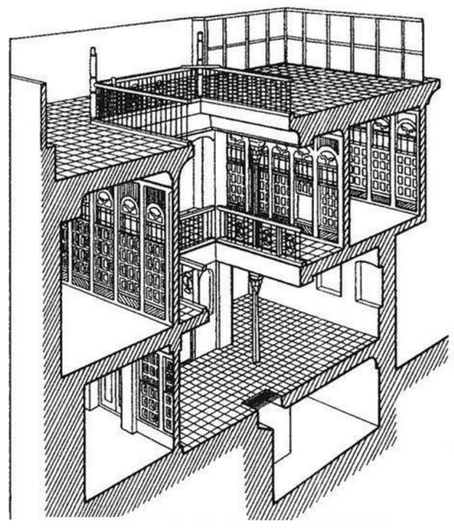


Fig 4.3.24 palace de dar al hora section
source (architag,2011)

C-THE GREAT MOSQUE DE CORDOBA

The great mosque de Cordoba is considered on of the masterpieces of Islamic architecture in Spain, however in this paper we wont go in details about the field work held there.
What needs to be mentioned about this sites is how the Andalusian mashrabiya was applied in a big scale like the arches that was almost 20 meters high.
It was found that the patterns with applied with same length to width ratio.
And the perforated patterns applied to the southern façade used to be small to prevent excessive solar gains.

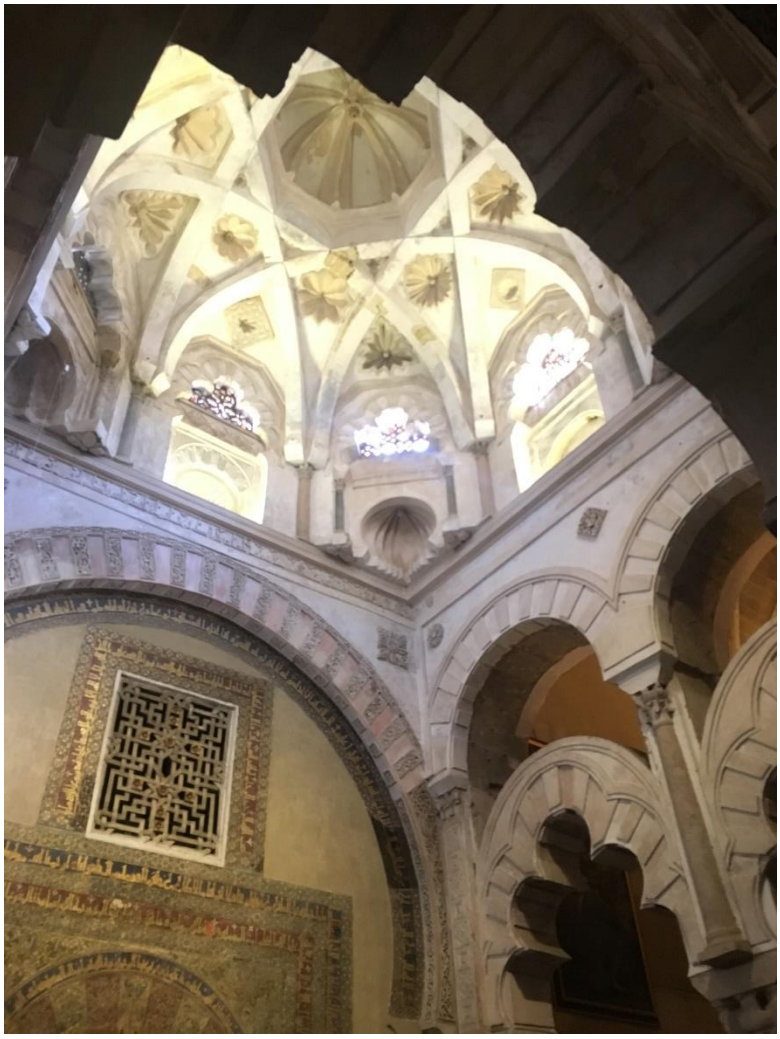
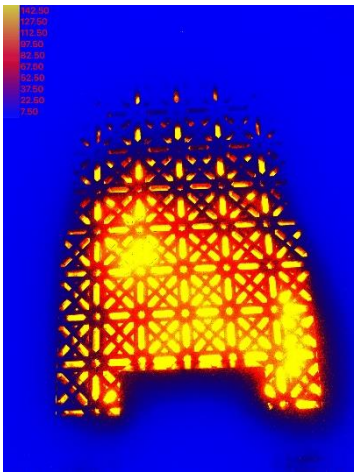


Fig 4.3. 19 collage images for the great mosque de Cordoba
source (author)



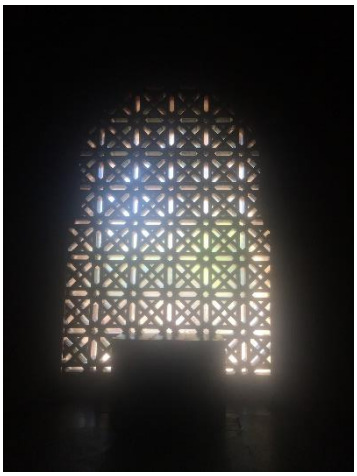


Fig 4.3. 19 collage images for the great mosque de Cordoba northern arches
source (author)

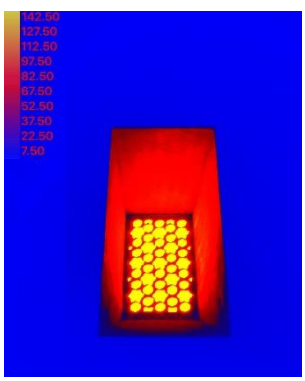


Fig 4.3. 19 collage images for the great mosque de Cordoba-southern windows
source (author)

4.4. **Ottoman Mashrabiya**

transitional space in old markets of Cairo

To understand the variation in performance and typologies of mashrabiya it was important to see living examples of ottoman mashrabiya.
For that Egypt – Cairo- old Cairo was selected to see and understand different typologies of mashrabiya, based on these 2 sites were monitored for two consecutive days.
The sites were:
1.wikala al ghouri
2.Bayt Al-Suhaymi

After that, there was a survey represented in friendly chatting with the carpenters and craftsmen who are still working in crafting mashrabiya until nowadays in Al Moez street. The questions asked to them was as following:

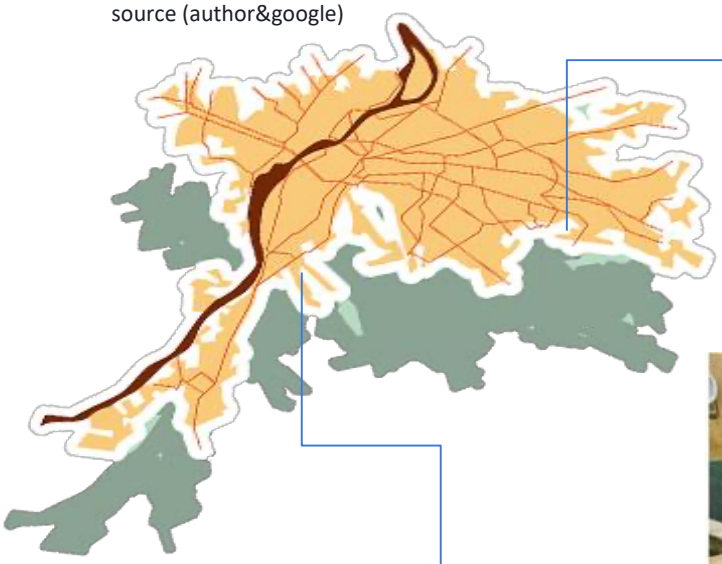
1. how often are you asked by a customer to make him a mashrabiya ?
2. How do you make it?
3. Do you have any guidelines to follow in crafting it?
4. With what do you start? Geometry or pattern?
5. What materials do you use?
6. What is the requirement needed to start modelling it?
7. Why there is a vast reduction in its implementation?

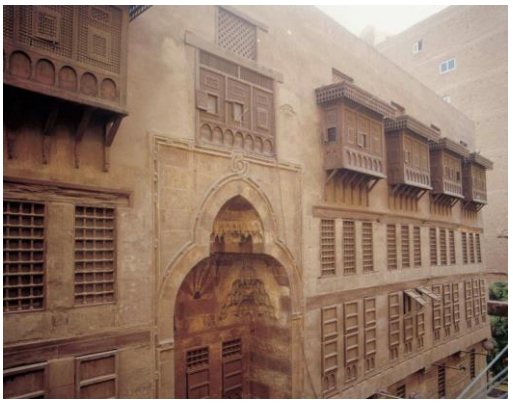
20 out of 22 gave spectacular answers to these questions, and some provided us with some of the data they used.
This data will be documented in the appendix. However, brief notes of the most common answers will be provided at the end of the chapter

Fig 4.4.1 collage images of field work
source (author&google)

wikala al ghouri

Fig 4.4. central old Cairo graphical map
source (author&google)





Bayt Al-Suhaymi

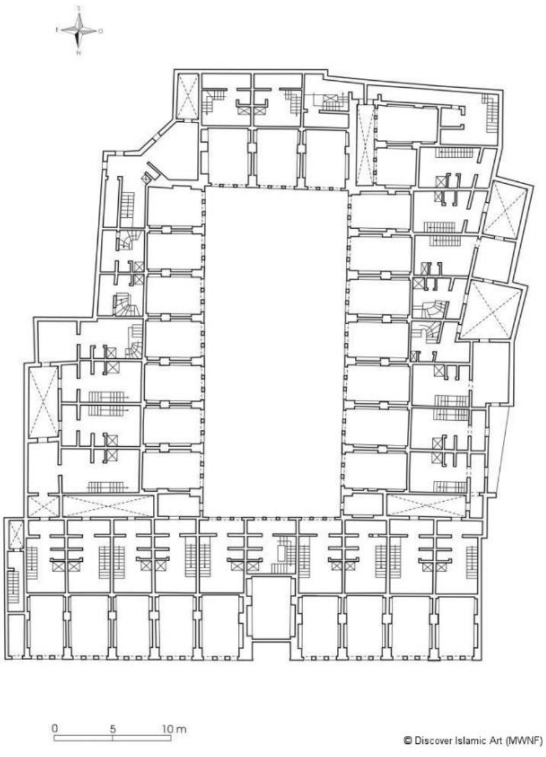
A. wikala al ghouri

•**The origin:** The sultan is the honourable King Abu al-Nasr, qassa al-Ghori of Berdi. He is the 24th sultan of the Mamluk clay of the jars.
•**Location**: Al-Ghori Agency is located on Mohammed Abdo Street (formerly Al-Tablita).“
•**The history of construction**: it was established in 909 Ah /1504 AD and ended in the year 1505 The era of the Goori sultan in the early 10th century was characterized by architectural activity. His buildings are at the top-ranked in the historical architecture list in this era, and the agency is a model of what it, as it represents the agencies shapes of this era, and the years rolled over the Goori agency and remained without care or maintenance until it got worse and cracked. Then it was bound to be protected and maintained. Before the Department of Conservation, the agency had a large courtyard where the deals were made. Where Commercial surrounded by the courtyard and preceded by the trumpets used to sell different goods and slaves.

The three levels of the view are oriented to the inside plate where the windows are solid. On the first level. Wide openings were designed to allow light to pass through space and the third level of the harem (women of the house). The screens applied to it let the user inside see the plate, and they can't be seen from the outside, to maintain privacy as well as ventilation and lighting for the void.
These houses and units inside are oriented from the four sides to the courtyard of the house. The designer of the building managed to maintain the privacy of the traders and their families by adding the element of mashrabiya on the windows that varied from a floor to another.
The screens in the mashrabiya that were situated in the third level (the highest level of the building) had narrow screens with low perforation percentage. As for the screens applied to the first floor, the patterns were wider with higher perforation percentage and supplied with an inclined shading screen that can be opened as a window.
The second floor, the openings covers were in the form of wooden grids that had gaps within it and can't be opened. The main goal from it was to ventilate the space.

Fig 4.4.A. 2 plan of the first floor
source (government documents)

Fig 4.4.A. 2 plan of the first floor
source (government documents)



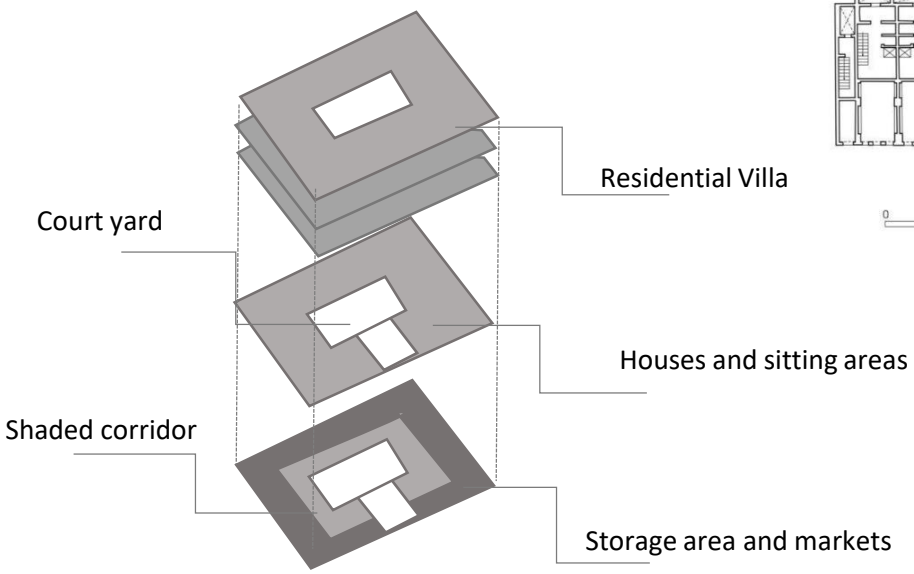


Fig 4.4.A. 3 analytical sketch of the building
source (author)

Fig 4.4.A. 4 shape of the courtyard during refurbishment
source (governmental document)

A. wikalat Al Ghouri

In the southern hall, which was used as a living space for the inhabiting families of the traders who worked in the market, the main façade is oriented to the hall (the courtyard) where the mashrabiya implemented there.

The geometry of the openings wasn't big in order to limit the light entering the space. The component had a combination of two patterns. Where a patterns with a big arched windows was applied to the bottom and a pattern with small perforation percentage supplied with a small window.

The interesting part. Is the existence of an extra opening added above the component. This opening was added during the construction process as it is a part of the wall with wide perforated arabesque pattern that is 70 % perforated. This formed a source of stack ventilation between the mashrabiya and the top window. But the room temperature was considered to be high because of the excessive solar gains coming from the top window, as there was no external shading provide for the top window.

Fig 4.4.A. 5 shape of mashrabiya of the third floor
source (author)

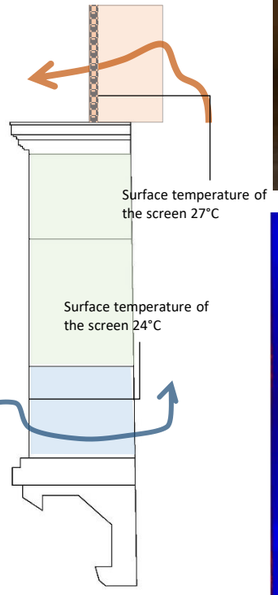


Fig 4.4.A.7 analytical section if the total performance of the component.
source (author)

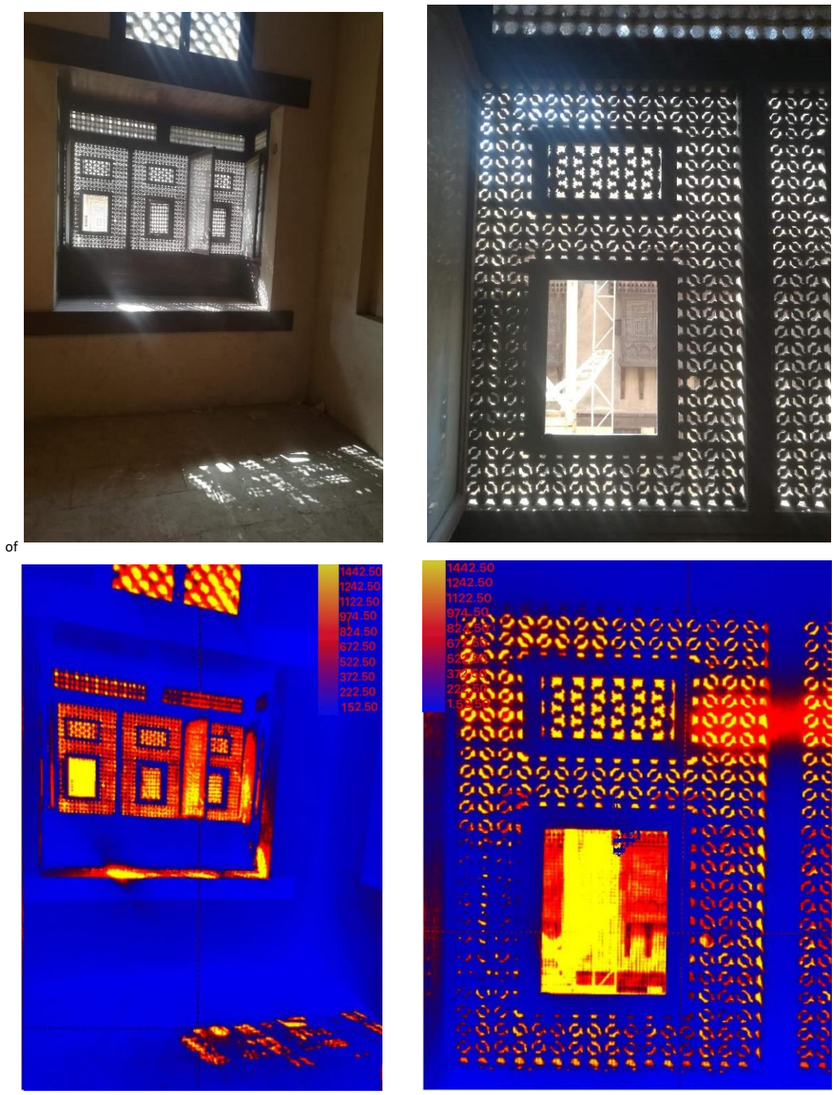


Fig 4.4.A. 6 shape of mashrabiya of the third floor from inside with aftab illuminance analysis
source (author)

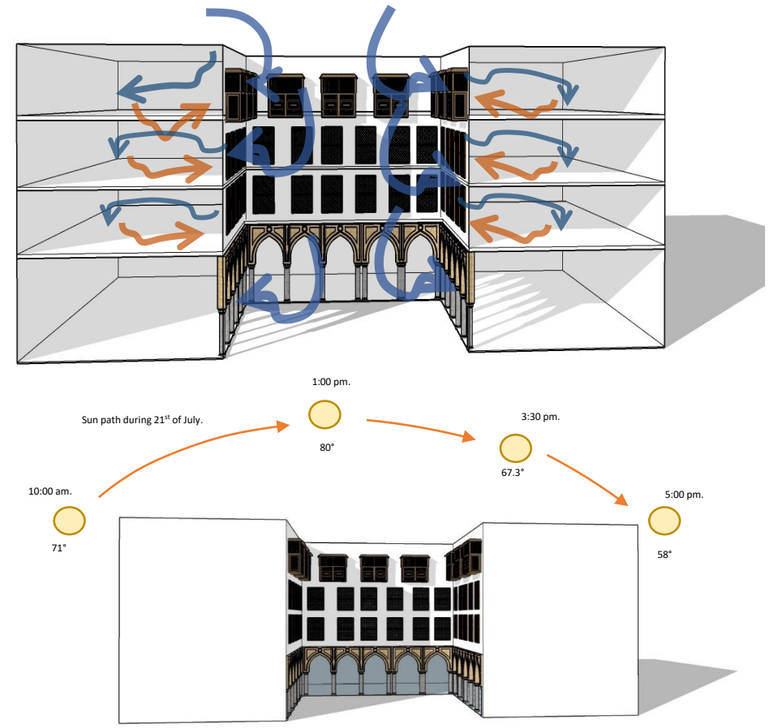


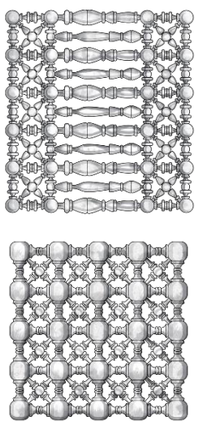
Fig 4.4.A.9 analytical sections
source (author)

Outdoor temperature average : 27.3° C

indoor temperature average : 19.2 ° C

Outdoor wind speed average : 0.1 m/s

Outdoor illuminance level average : 13,455 lux



Surface temperature of the pattern : 27.9° C	indoor wind speed average :0.1m/s	Indoor illuminance level average near the pattern : 850 lux	Percentage of perforation is 49 %	The screens in top part of the component
Surface temperature of the pattern : 24.3°C		indoor illuminance level average near the pattern : 579 lux	Percentage of perforation is 13%	The screens in middle part of the component

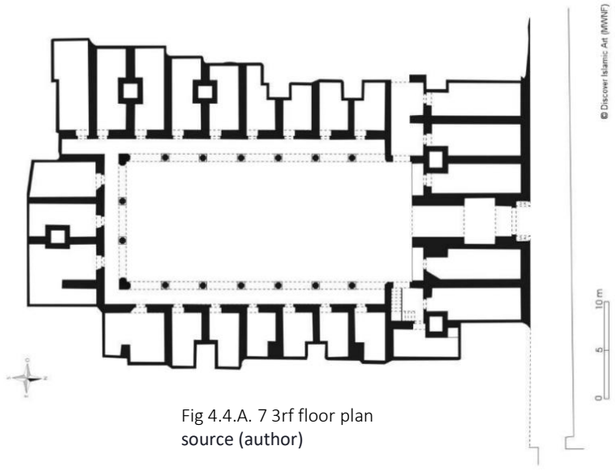


Fig 4.4.A. 7 3rf floor plan
source (author)

B. Bayt Al-Suhaymi

Bayt Al-Suhaimi is an example of traditional Arab houses and is one of the most popular heritage places in Egypt, located in Cairo city, located from Al-Mu'izz Street to the religion of God, where the house is named after the last person who lived, Sheikh Mohammed Amin Al-Suhaimi. The area of the house is estimated to be limited to (2000 square meters), and the house has two sections (tribal) which are the oldest and was built by Sheikh Abdul Wahab Al-Tablawi in 1648 A.D., and the section (Al-Bahri) was built by Haj Ismail Shalabi in 1699 AD and then connected to the first section.

The construction of the house began since the Fatimid period, and its planning was influenced by Ottoman architecture where the foundations of its designs at that era were based on three factors, Sufi religion, social conditions and climatic factors, so the ground floor was allocated to men and is called "Salamlak", and the upper level was for women, it used to be called (Haramlak).

Also, in the north section, there is another sitting space similar to the first one in the design and is considered larger. It has more and more architectural details and with it a basin of water of gilded marble. In addition to the presence of a drinking place made of wood Azizi, which is one of the finest types of wood and its most expensive types.
Also, there is a room to read the Qur'an next to the council with a large chair of busy wood with mashrabiya on all the openings of the room in order to provide equal light distribution in the whole room.

Fig 4.4.B. 1 Photo of bayt al suhaymi from the western street
source (Khalil,2017

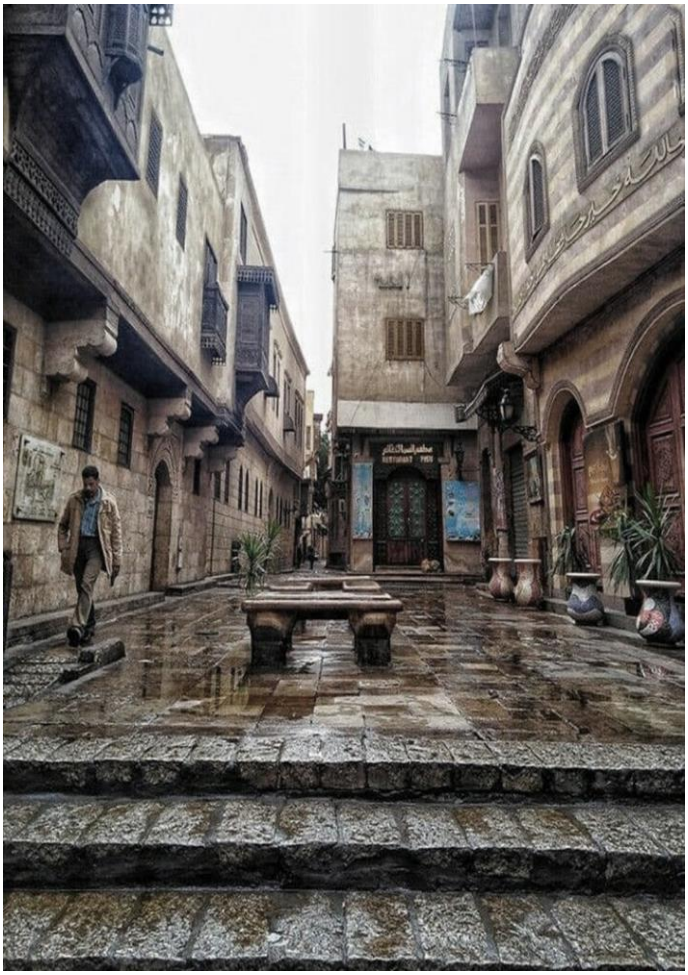


Fig 4.4.B. 2 collage image that summarize the scenes inside and outside the house
source (author, painters)

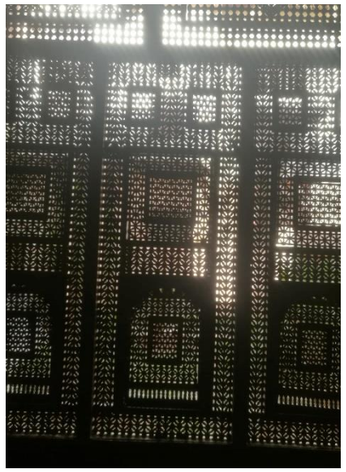
B. Bayt Al-Suhaymi

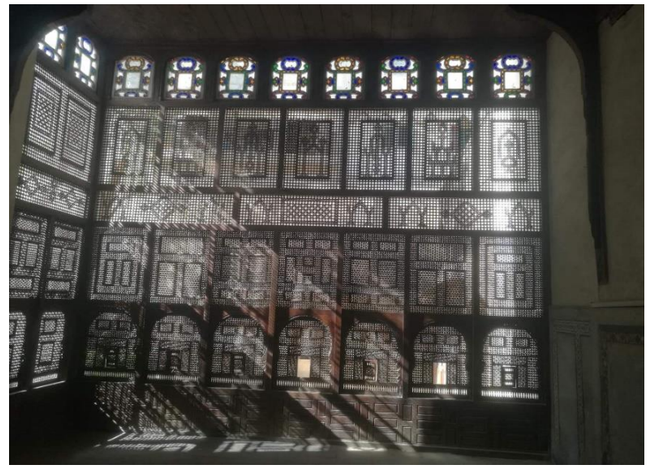
The typology of mashrabiya implemented in the southern room of bayt al suhaymi represent a more delicate and functional type of ottoman mashrabiya, that managed to achieve the same performance of the previous example with depending on the mashrabiya only.

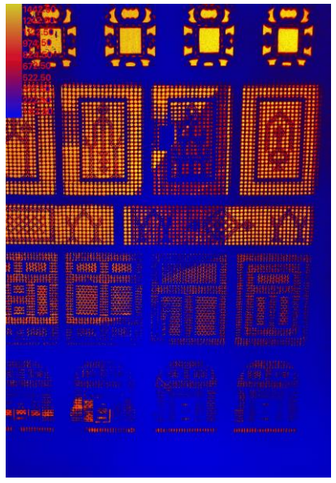
This managed to move the stack ventilation to be happening on the level of mashrabiya and no extra openings were needed. Where a wide perforated patterns are applied to the top and bottom parts of mashrabiya, the component is divided into 5 parts horizontally and 3 parts vertically, this facilitated the applying of different patterns on the same components.

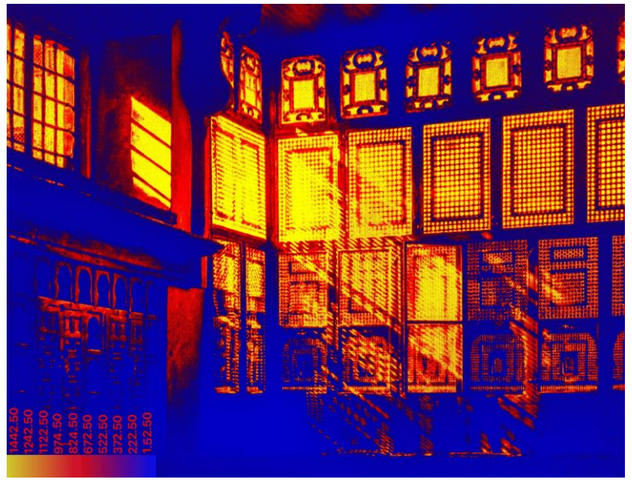
The air flows from the bottom where the surface temperature of the pattern is higher as it is exposed to the sun. Then the air moves through the space with average air velocity that can be sensed if the person is standing in the middle of the room moving to the top part of the screen, where the surface temperature of the screens in the top is almost 1 °C less than the bottom part. This might be because of the cap added to the component to shade the top part.

Fig 4.4.B.7 shape of the Mashrabiya implemented in the site.
source (author)









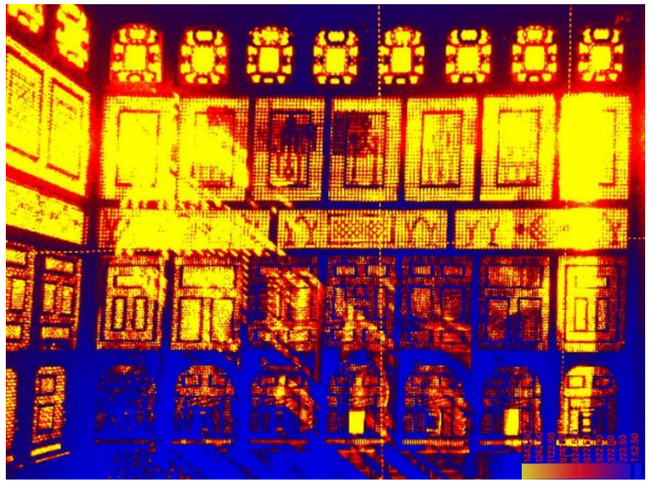


Fig 4.4.B. 3 collage image that summarize the scenes inside and outside the house with Aftab illuminance image
source (author)

Fig 4.4.B. 5 shape of mashrabiya of the 1st floor
source (author)

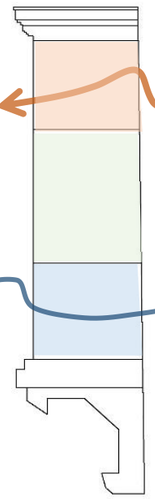


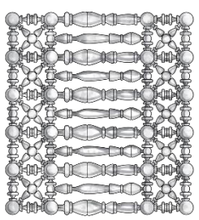
Fig 4.4.B.6 analytical section if the total performance of the component.
source (author)

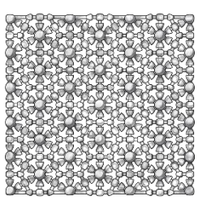
Outdoor temperature average : 25 °C

indoor temperature average : 17.2 °C

Outdoor wind speed average : 2.3m/s

Outdoor illuminance level average : 13,756 lux





Surface temperature of the pattern : 23.4° C	indoor wind speed average :0.3m/s	Indoor illuminance level average : 70 lux	Percentage of perforation is 33 %	The screens in the top
indoor temperature average : 17 °C	indoor wind speed average : 1.1 m/s	indoor illuminance level average : 81 lux	Percentage of perforation is 15 %	The screens in bottom

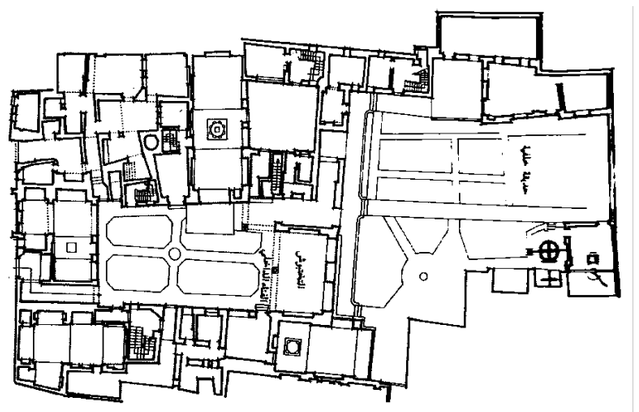


Fig 4.4.B. 7 1st floor plan
source (author)

Fig 4.4.B. 2 collage image that summarize the survey held among carpenters and workshops owners
source (author and previous papers)

Comments from the craftsmen and carpenters of Almoez street- Cairo who manufacture mashrabiya in Cairo nowadays

Khalil Emam:
Mashrabiya is a beautiful art, I love crafting it and see my efforts come to life in such a beautiful piece, but this effort takes time and this will coast the customer , no one wants to buy a proper coast on such a thing.

Safwat Ali:
I found easier ways to make the screens, all I have to do is draw a pattern I choose from a pattern magazine on the wood and cut it through. I don't make any hard patterns to save time, effort and money and I just choose the patterns I want.

Dawood Sami:
We do it just by looking at the examples that still exist in the street, but once these examples are gone we don't know who will remember this craft.

Mohamed Farouk:
I make it a lot, but only modern ones, just a decorative screens that people use inside the living space to separate the space into multiple space. No one asks me to make him the old traditional one, people use air conditioners and curtains, it is not functional anymore.

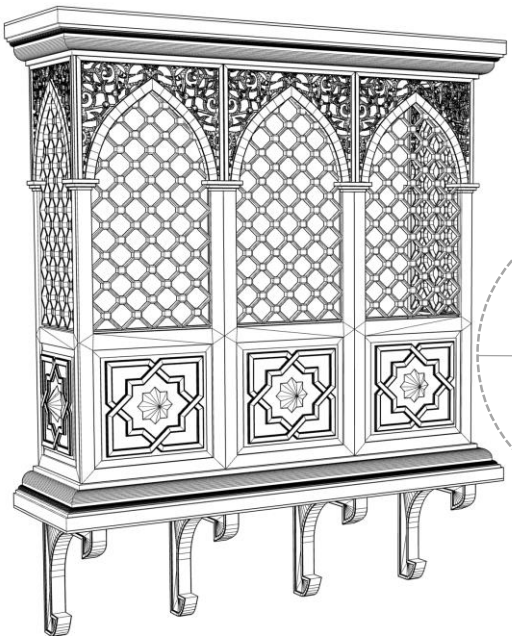
Sayed Mahmoud:
Wood in Egypt is getting more expensive every day , why would we waste it in on a screen, when we make the screen we use thin artificial fabrics and save the wood for furniture. I use iron to make these screens and apply it to any thing… windows… gates…fences.

4.5.conclusion:

Outcome of the field work:

1. The Andalusian mashrabiya can achieve the same approach of the ottoman mashrabiya through using the building geometry.
2. Widely perforated screens are recommended to be applied to small openings in order to maintain the light passing through it and allow more wind to pass through.
3. When a pattern is applied on a big scale, like the patterns applied to the arches of the great mosque de Cordoba, it must maintain its depth to length ratio. Which is the ratio between the length of the void and the thickness of the pattern itself.
4. Ottoman mashrabiya can contain variable patterns, and this variation in patterns create a stack single-sided ventilation. This can increase the efficiency of the total performance of mashrabiya in terms of ventilation.
5. It is not recommended to use wide perforated screens on the southern or western façade
6. Adding a cap to the top pattern of the ottoman mashrabiya has a significant impact in improving the indoor comfort as it was found in Bayt Al Suhaymi. While not adding it can have a negative impact on the indoor comfort as it was found in Waklit Al Ghori, where the top window added above of mashrabiya allowed excessive amount of solar radiation.
7. The main approach mashrabiya was created for is not limited to shading. Mashrabiya can be used in ventilation and providing equal light distribution inside the space.

- Outcomes of the fieldwork:

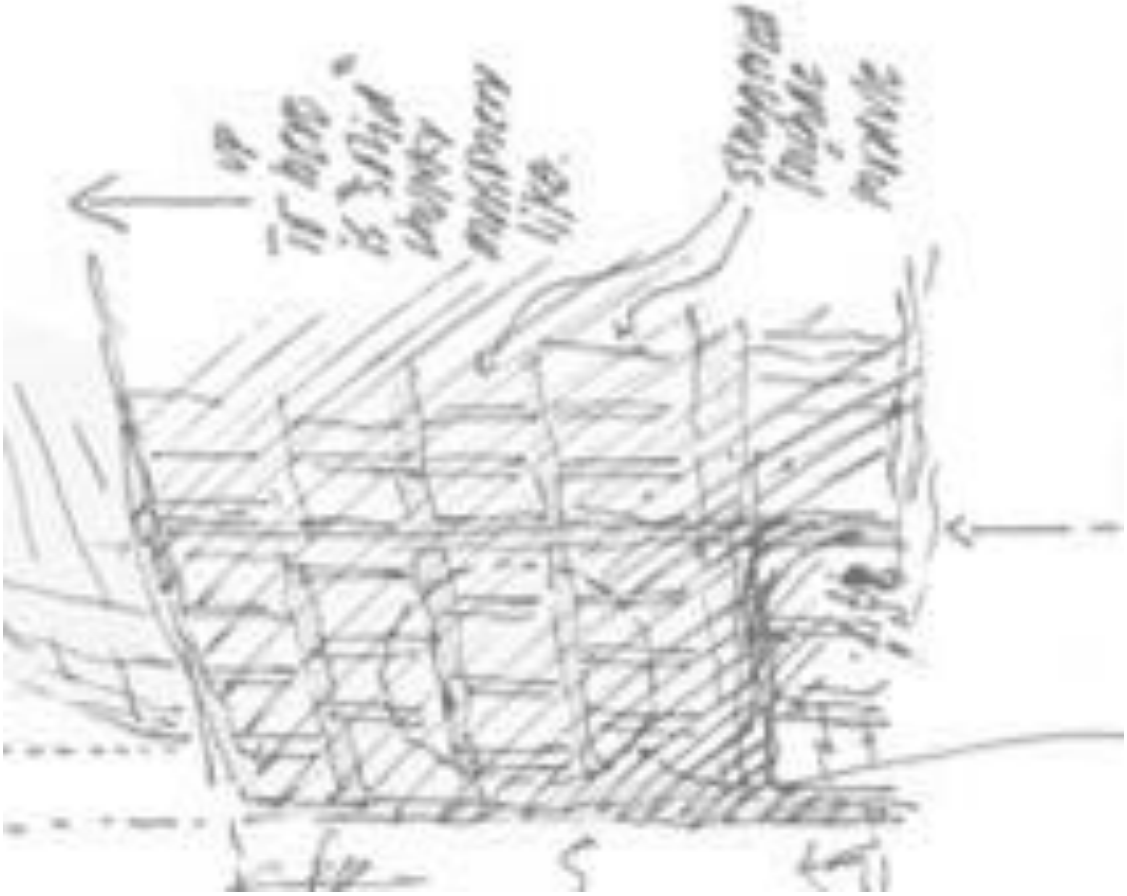


Advantages

Disadvantages

- Protection from solar radiation
- Fire and safety issues since the commonly used material is wood.
- Shading element that doesn't block the view from inside
- Lower resultant temperature indoors up to 7 degrees different

- Less adaptability in cold temperatures
- Fire and safety issues since the commonly used material is wood.
- Maintenance issues
- Noise allowance in the indoor spaces so its implementation in lower building levels had been reduced



5. Analytical work

Quantifying the environmental performance of Mashrabiya

A. Introduction:

According to the fieldwork and literature review, a lot of factors were found, and from that, it was concluded that the performance of Mashrabiya in terms of shading and ventilation varies according to:

- The perforation percentage of the pattern
- The type of the pattern, whether it could be floral or geometrical
- The thickness of the pattern, as it was found in the great mosque de Cordoba, where the patterns that were applied to a bigger scale maintained the same ratio between the depth and the length of gapes.
- The activity preformed in the transitional space created by Mashrabiya affects its performance significantly. Whether an evaporative cooling element is added or not

So the performance of the Mashrabiya varied in terms of shading and ventilation according to the patterns, the component style and whether it had more than one pattern combined on it as it was found in the ottoman Mashrabiya examples found in Egypt.

All these factors managed to shape the process of analysation and quantification of the environmental performance of Mashrabiya. According to that, the analytical method was formed and divided into two main steps:

A. The first step was the micro level:

Where the main focus was on the patterns and zooming on the smallest details of Mashrabiya. after that, start testing them under studied and fixed conditions to understand the difference of each pattern in terms of performance if they are exposed to the same climatic and physical surrounding conditions.

B. Macro (component) level:

After realizing how each pattern preforms in terms of light, solar radian and airflow, the next step scope is adjusted to zoom out and focus on the change in the performance according to the whole component. And understand how the geometry and combination of patterns can significantly change the performance.

Also, in the macro level, the effect of adding an evaporative cooling element in the transitional space created by the component will be roughly estimated.

C. site analysis:

After that a testing step will be made in order to visualize and quantify the environmental performance of a current existing site and see how it preforms in terms of shading with putting the surrounding context into consideration. To visualize a true case, it was important to visualize a surrounding context represented in climatic context and neighbouring buildings.

Micro level

1st step was to understand the performance of mashrabeya on the micro level and start understand and quantify the performance of different patterns on micro level

Macro level

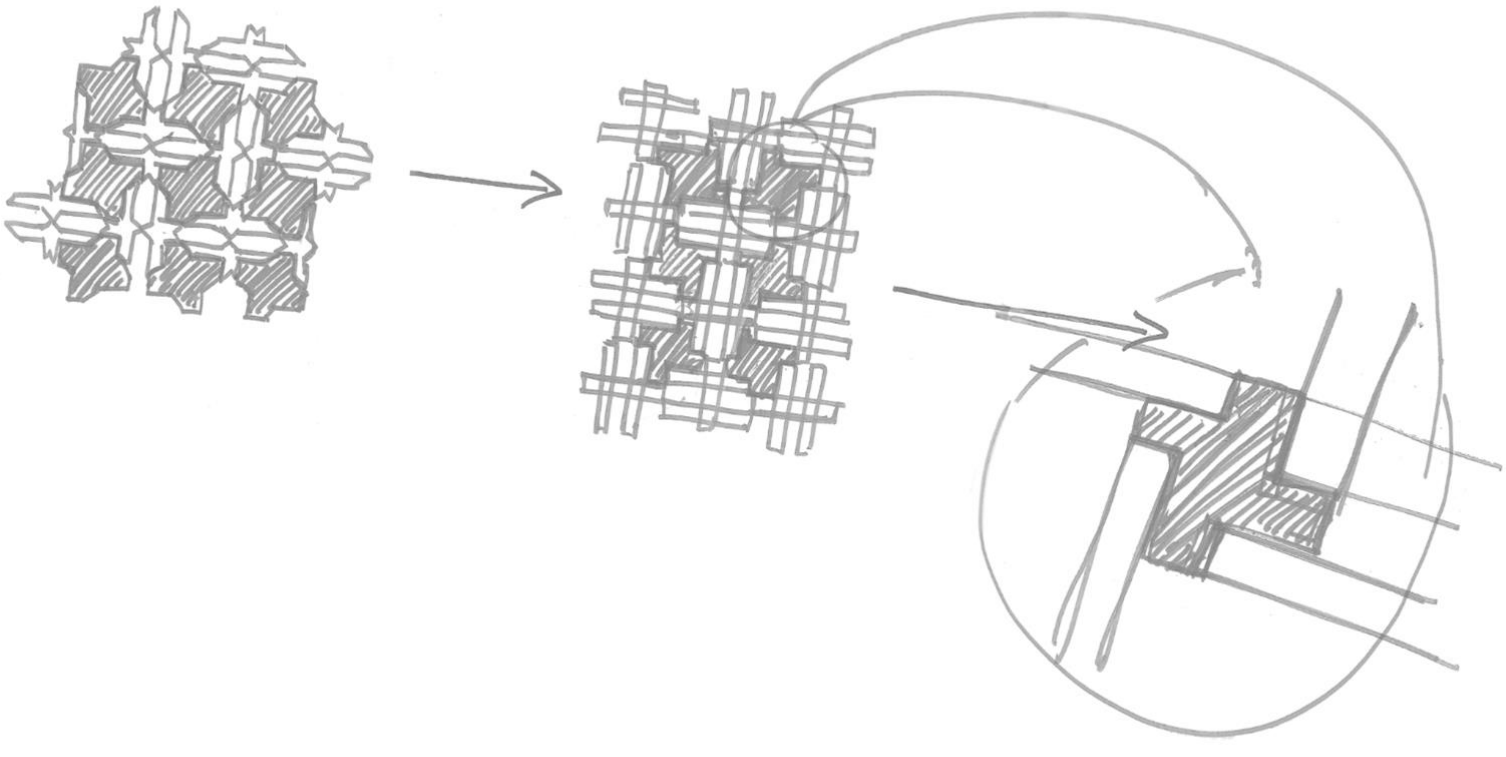
2nd step was to study the performance as a component and understand the variation of performance in different environmental parameters

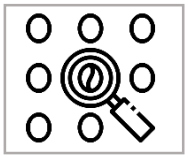
Site level

As a starting step to have a general look on the performance of the living cases examples through understanding its effect on the indoor thermal comfort with considering the context

5.2 Micro level

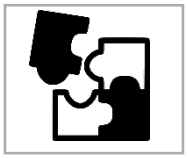
Understanding the patterns





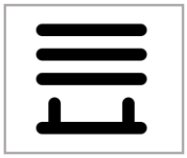
- Step 1

Identifying the most commonly used patterns that gather most of the patterns different typologies
Which are **7 patterns**



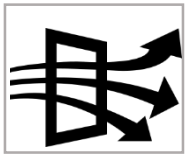
- Step 2

Defining the components that forms each pattern and how they are being repeated on a certain grid.



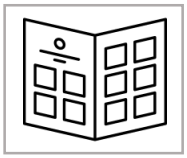
- Step 3

Analysing the shading performance in form of solar radiation, illuminance and length to depth ratio.



- Step 4

Analysing the environmental performance of each screen in terms of ventilation by calculating a conditional discharge coefficient.



- Step 5

Gather all the analysed pattern in the form of catalogue to be used later in the design guidelines.

A.2. Shading performance :

To quantify how the perforated screens performs as a shading element, it was important to understand each pattern and illustrate the component that forms each pattern and how it is being repeated among a certain module grid.

6-7 patterns were selected to analyse. The patterns were chosen in order to collect various Islamic pattern into these six main patterns which were found during the fieldwork in different typologies. The patterns that were selected were the most commonly used patterns in the mashrabeya as a component. The patterns were:

1.Pattern 1: a wooden standard element repeated on a typical grid 17 cm X 17 cm, this pattern was evolved from the simplification of floral patterns, perforation per cent: 21%

2.Pattern 2: almost the same as pattern one but repeated on a grid 10.5 cm X 10.5 cm and with the difference in the density of perforation and length to width ratio, perforation percent: 41%

3.Pattern 3: a total geometrical complex pattern that gathers various small details in the design of its component which created more integration in the perforation, perforation percent : 37%

4.Pattern 3-a: almost the same as pattern three but the component that forms it is totally different, perforation percent : 19%

5. Arabesque pattern: similar to Lego, very simple pattern in its grid and formation but has a complex component that has various styles, perforation percent : 49%

6.Pattern 5 : complex floral pattern inspired by symmetrical flowers form, perforation per cent: 30%

The patterns were modelled in different software and simulated with their realistic scale and dimensions in order to get precise quantification.

The depth or the thickness of the patterns was almost the same since the main material that was used in the manufacture of these patterns was wood, which had standard craft dimensions and manufacture methods.

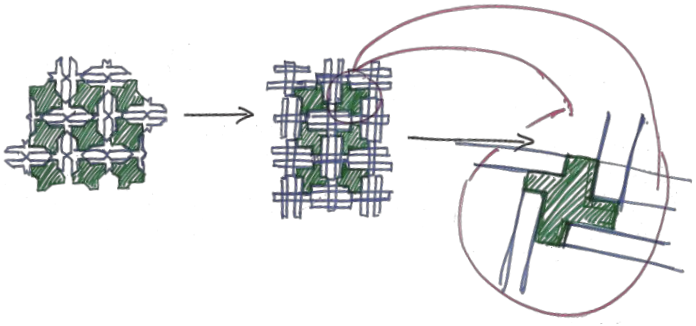


Fig 5.2.1 sketch of the concept of micro level
source (author)

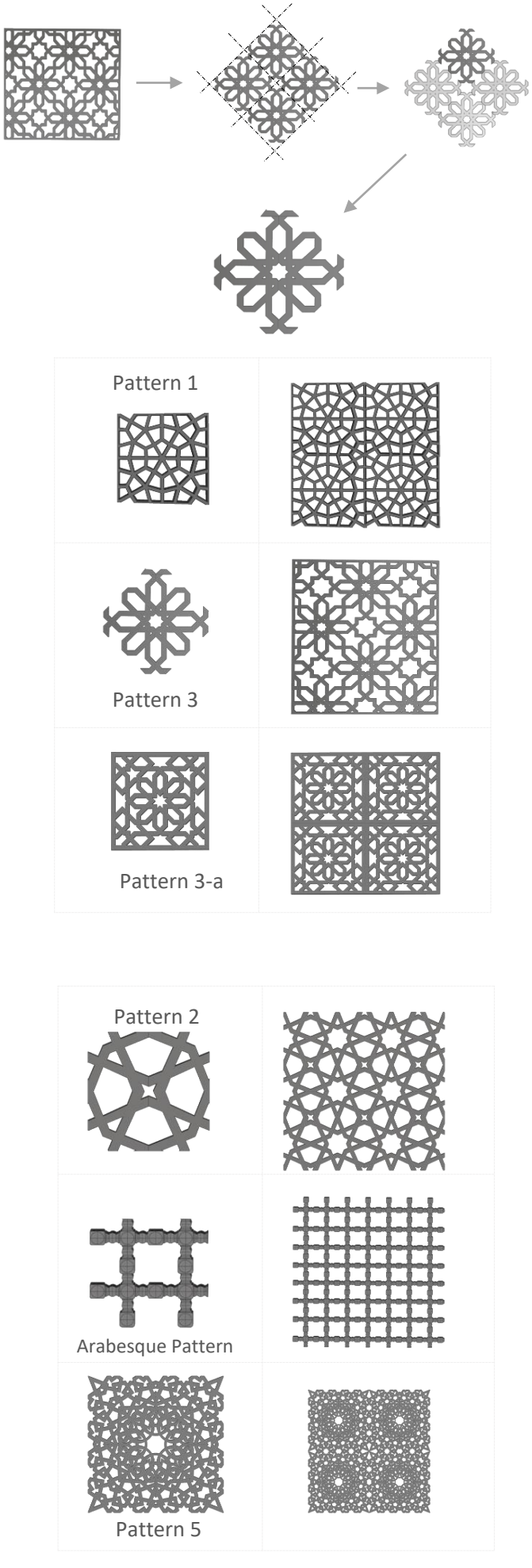
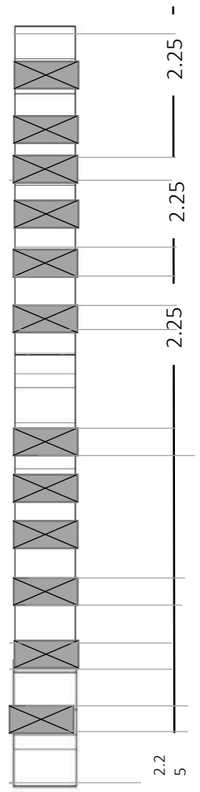
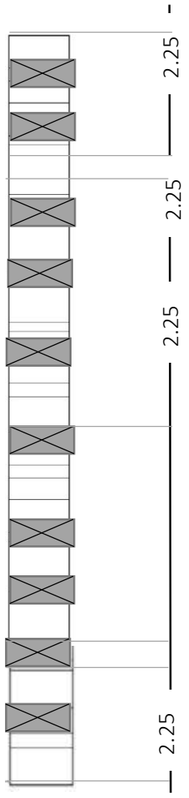
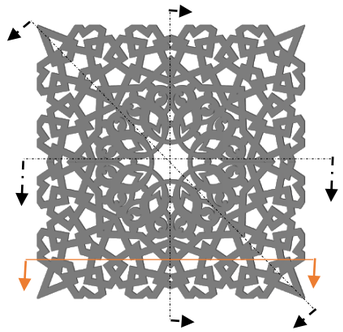
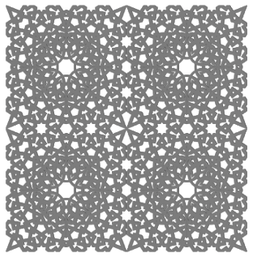


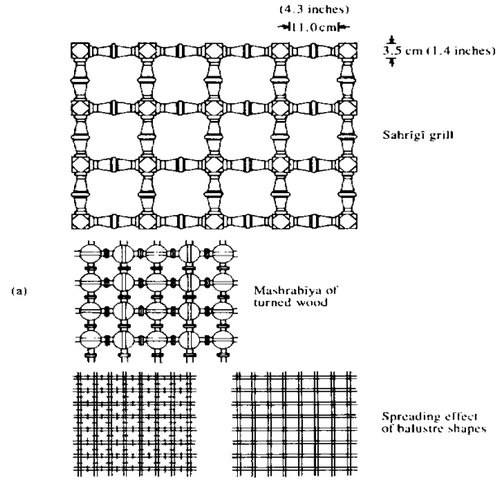
Fig 5.2.2 collection of the patterns selected to analyse
source (author)

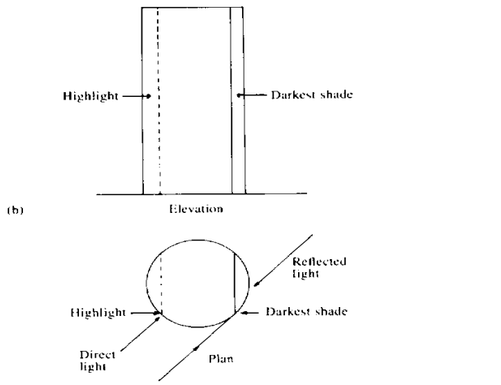


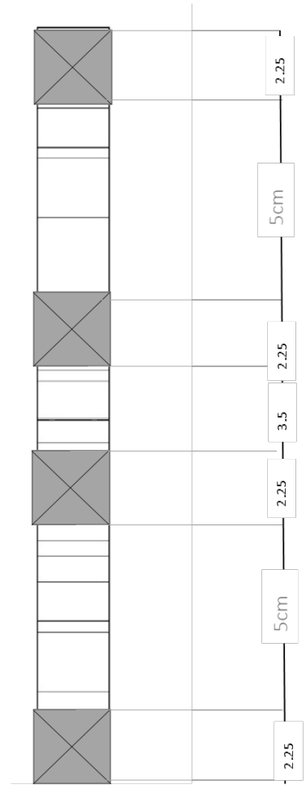


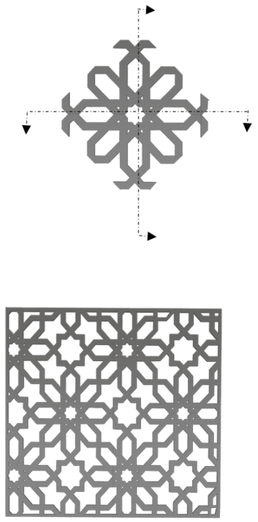


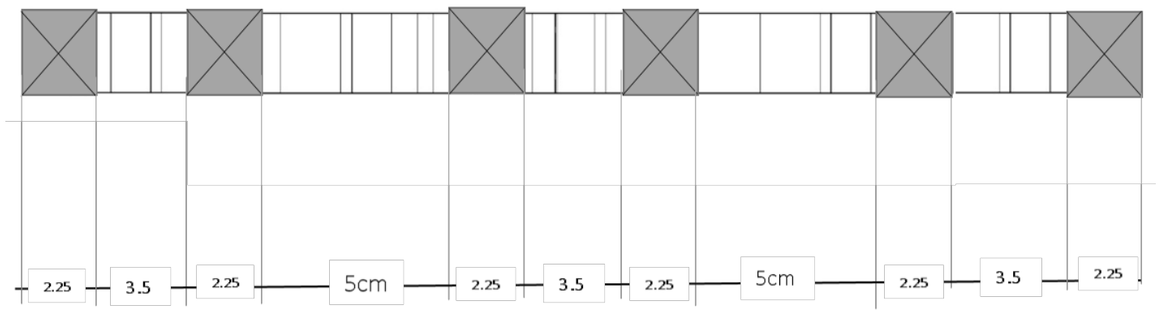


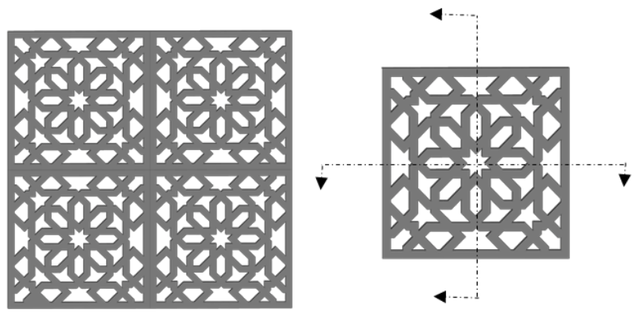


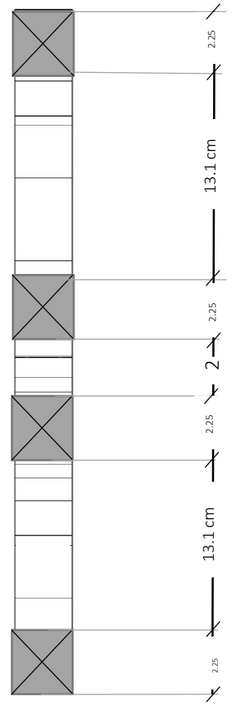


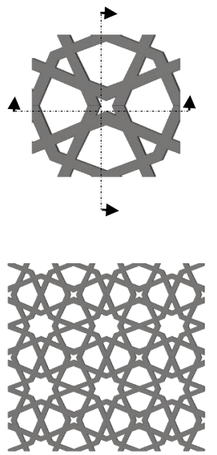


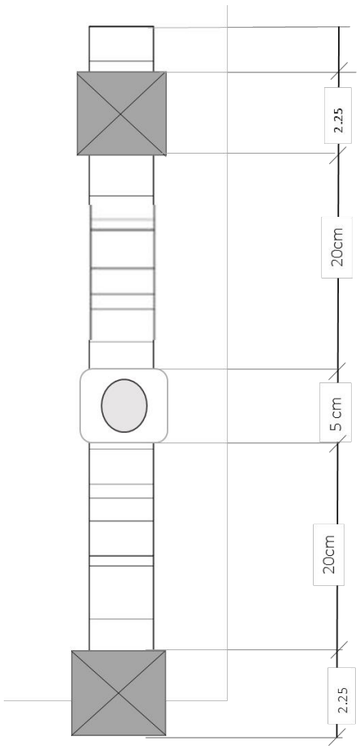


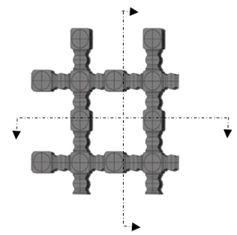












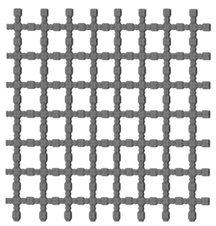


Fig 5.2.3.a. summary of some of the patterns sections
source (author)

5.1 Methodology of the shading performance quantification :

Understanding the dimensions of each patterns was based on the field work and some detailed drawings supplied from the carpenters and craftsmen interviewed in Egypt. Also, literature review of Hassan Fathi's works helped to understand more about arabesque patterns and Maymooni pattern (a pattern that will be mentioned later).

The dimensions of the patterns were the base stone of modeming the patterns in different softwares in order to quantify their performance towards different environmental parameters.

Quantifying the performance of different patterns that used to be applied on different facades with different geometries was quite challenging. For that a show box was created in order to provide the same fixed surrounding conditions for all the patterns. This was the method created to provide a fair comparisons between all the patterns by putting all the patterns under the same conditions.

For the quantification of the shading performance two analysis were made using the Rhino software in modelling and grasshopper-ladybug- honey bee. Two analysis were made to understand the patterns in terms of light and soar gains, these analyses were:

I. **Daylight analysis (illuminance):** to estimate the amount of light entering through the pattern component that plotted and projected on the fixed area of the box. The area of the plan receiving the light was fixed in each pattern in order to have a fair calculation of the average amount of lux entering the box. Since that the average amount of lux is affected magnificently by the area of the geometry receiving it.

II. **Vertical Solar radiation analysis:** the amount of light entering through the patterns component cannot provide a full understanding of the solar gains and heat resulted from it. As a percentage of the entering the pattern could be diffused with lower solar gains impact on the space. For that vertical solar radiations analysis were made in order to estimate the amount of radiation a fixed area of the pattern can allow. In this analysis the component of the pattern wasn't used but a fixed area of the whole pattern. Also, the geometry was represented in form of plan (testing geometry) placed adjacent to the patterns and from it average solar radiation projected was calculated.

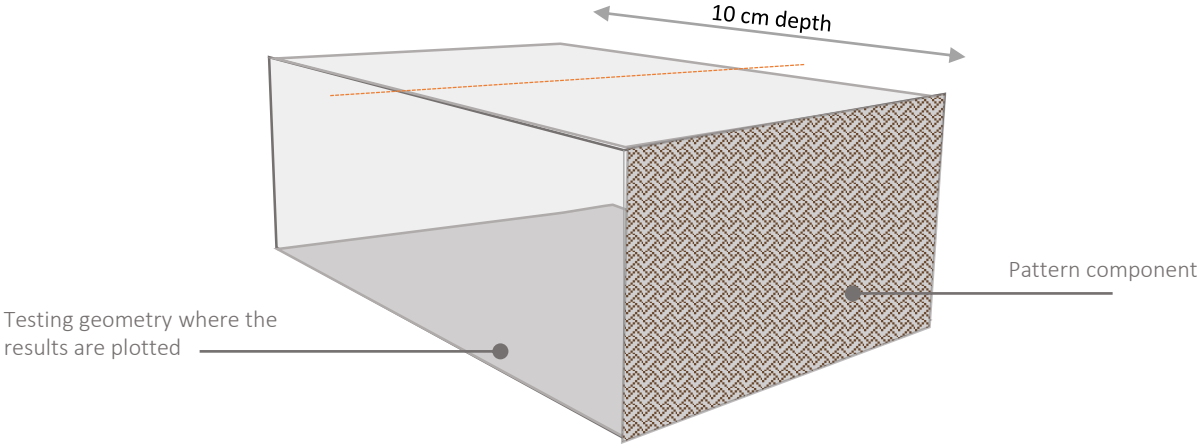


Fig 5.2.6 sketch of the method of daylight (illuminance) analysis
source (author)

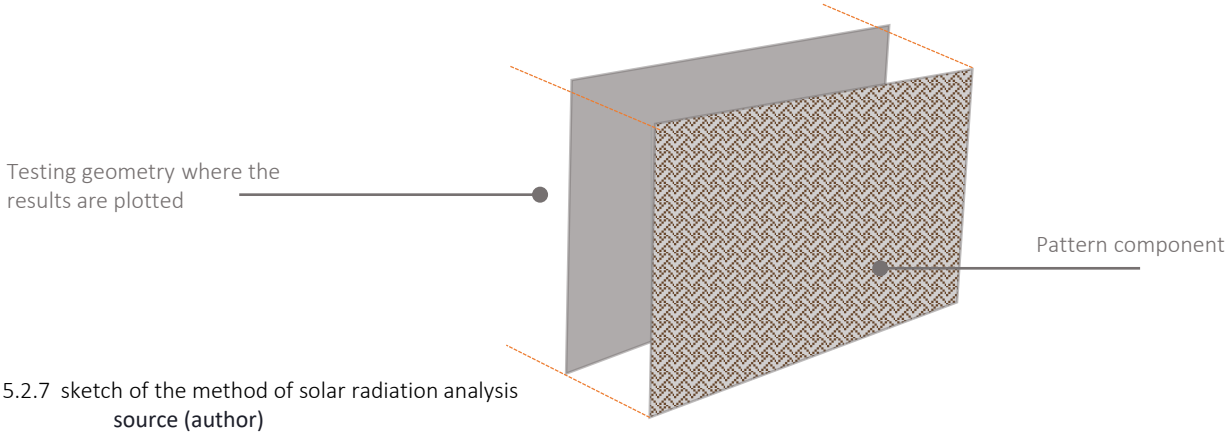


Fig 5.2.7 sketch of the method of solar radiation analysis
source (author)

A.2 shading performance quantification :

After identifying the component that forms each pattern and modelling it in Rhino software, the analytical studies took place and the following decisions were taken:

- **The patterns will be tested according to Egypt- Cairo climatic conditions:** since that a big part of the field work was held in Egypt. Also, Egypt gathered most of the typologies of patterns within the different forms of Mashrabiya, so it will be logical to test the patterns as if they were in Egypt.
- **The patterns will be tested in 21st of June:** as it is considered to be the hottest day in Egypt -Cairo recorded so far from the latest weather recording reports and Meteonorm.
- **The centre of the pattern will be located at height 9 meters:** as if it is on a window in the 3rd floor in a multiple story building.
- **No surrounding context was added:** in order to create a fair comparison between all the patterns.
- **The only context that was added is the ground:** a plan surface was added at zero level and assigned as a surrounding context to the pattern in order to estimate the defused and reflected light that may come in form of reflected ray from the ground.
- **The bouncing of the light in the grasshopper software was reduce to minimum:** both illuminance and solar radiation analysis took place in a very small scale considering the size of the components. So the light can keep bouncing inside the box an may cause inaccuracy in the results, for that the bouncy was reduced
- **The patterns were oriented to southern façade and western faced :**as these were the most critical façade in Egypt climate.

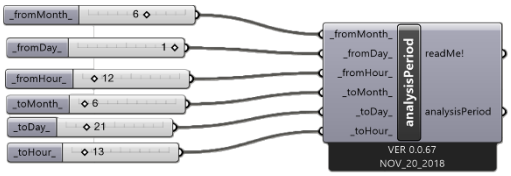


Fig 5.2.8inputs applied in grass hopper
source (author)

- **The patterns were tested in a sunny sky condition:** since the patterns were commonly implemented in hot countries with extreme climatic conditions, so it was important to test them in the most realistic common conditions.
- **The simulations made was finalized in form of an average result.** In order to facilitate the comparison between the different patterns. The average was taken by entering an average formula in the grasshopper simulation folders. This average was calculated through the Sumiton of all the results and dividing it with its number.

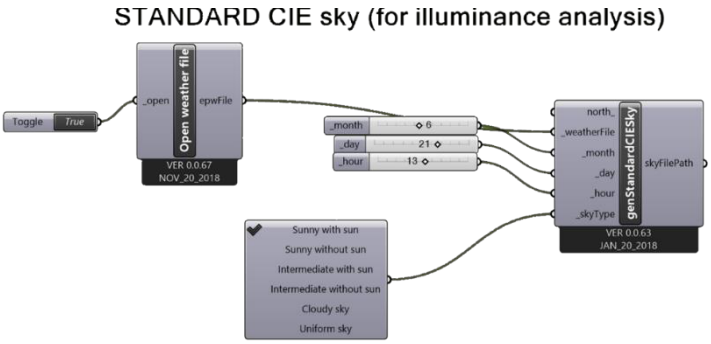
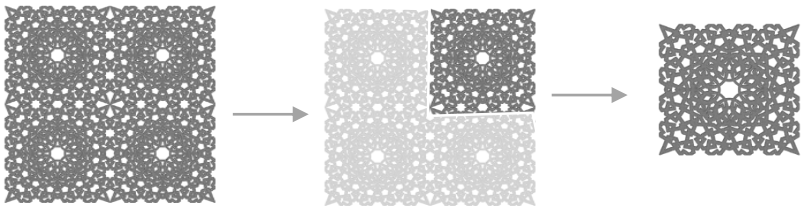


Fig 5.2.9 inputs applied in grass hopper
source (author)



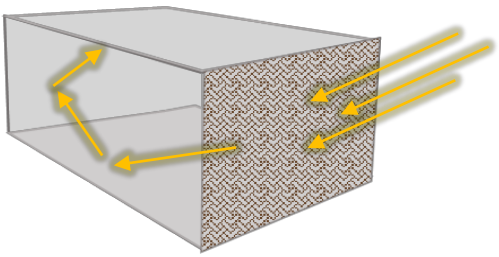


Fig 5.2.6 sketch of the method of daylight (illuminance) analysis
source (author)

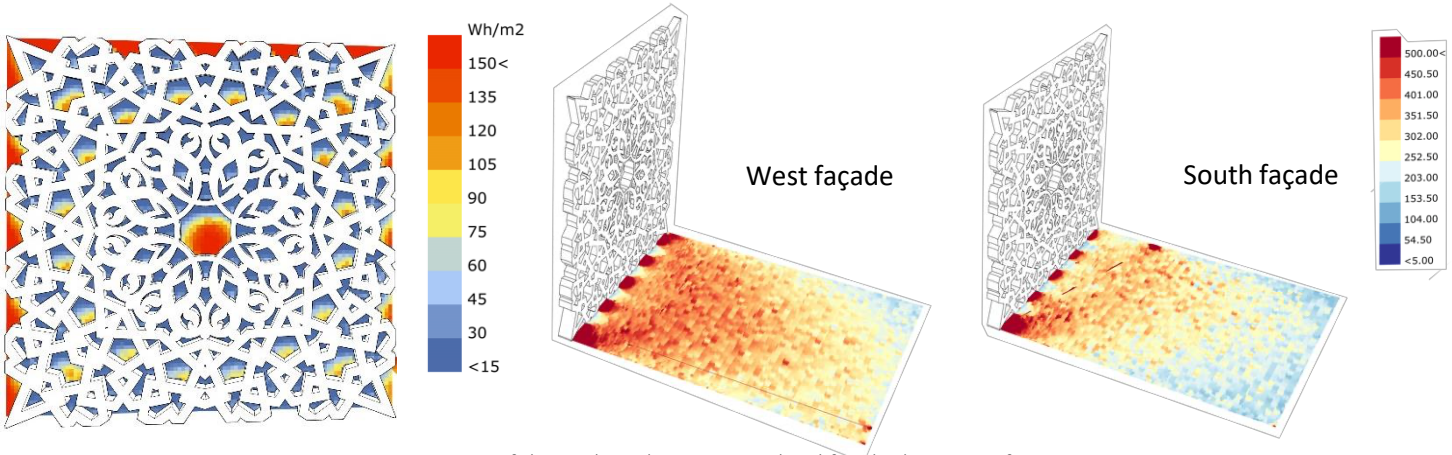
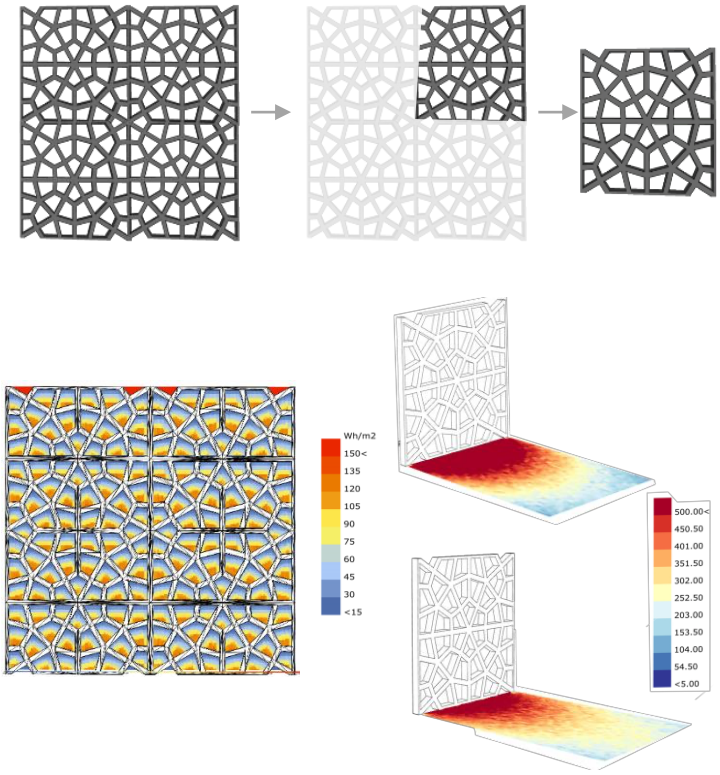
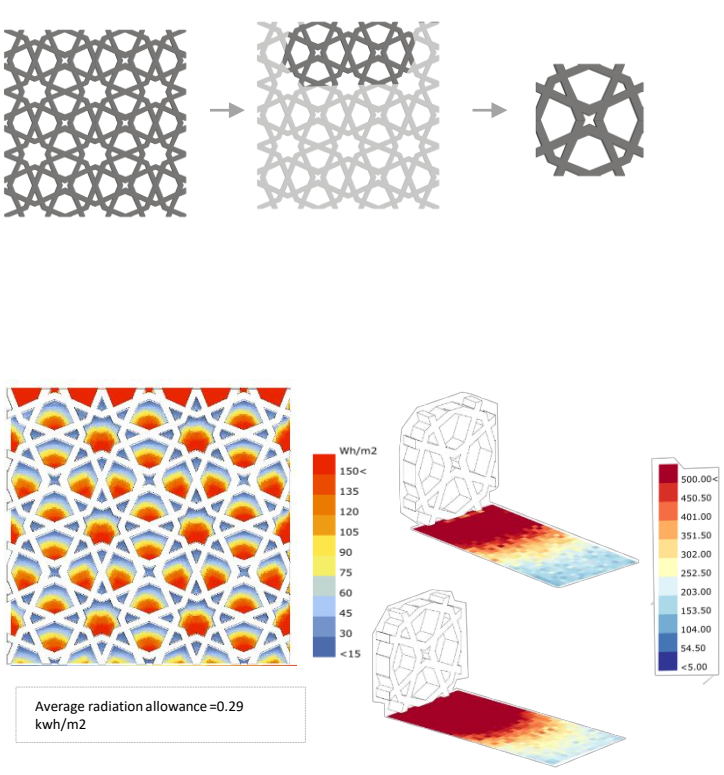
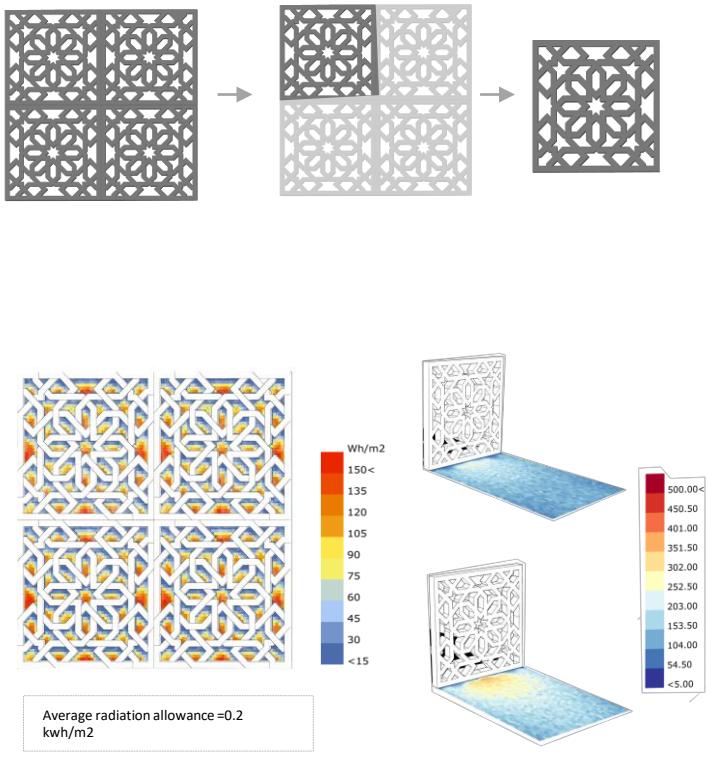


Fig 5.2.10 steps of the analysis done in micro level for shading quantification
source (author)

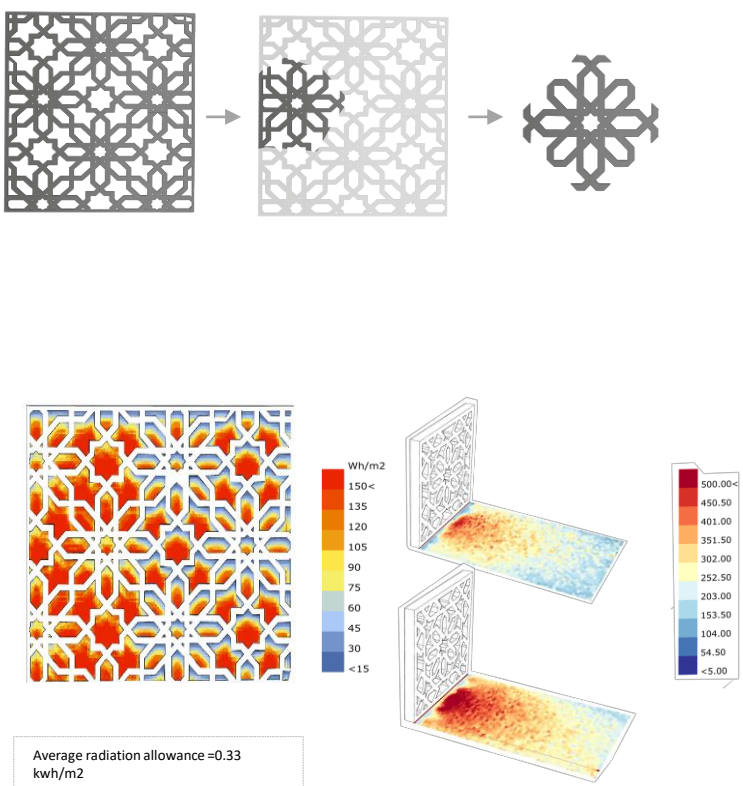




Average radiation allowance =0.29 kwh/m2



Average radiation allowance =0.2 kwh/m2



Average radiation allowance =0.33 kwh/m2

Fig 5.2.11 steps of the analysis done in micro level for shading quantification source (author)

A.2 shading performance quantification :

Some patterns like arabesque and Musky evolved from a shape to another according to the function desired from its implementation, the main shape and the main concept of the patterns remained the same where it used to be formed from Legos that can be removed and rebuilt. This pattern was widely implemented in Cairo-Egypt because of its efficiency and its maintenance and reconstructing is easy. The main variation between its shapes is in the component itself, as for the grid and the dimensions they remained the same.

This pattern portrays the main concept of the micro level analysis and raise its practicality, as it shows how the small differences in the component that form the pattern can majorly affect the performance of the pattern .

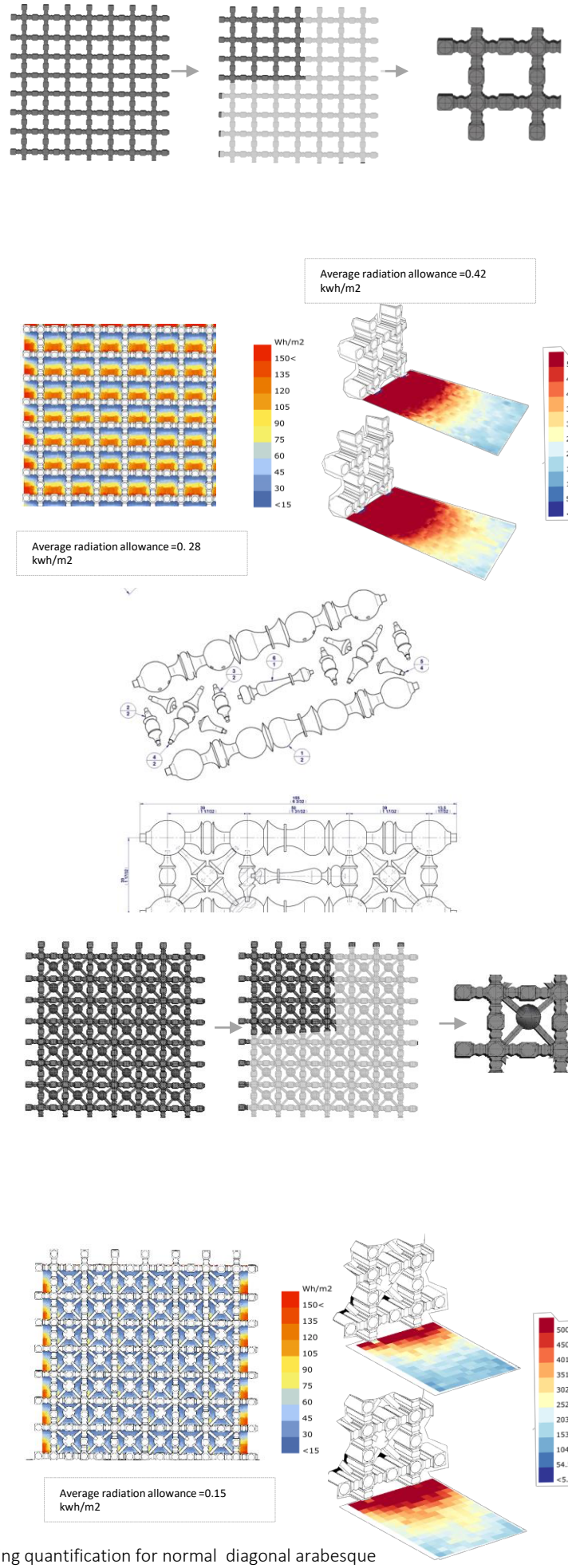
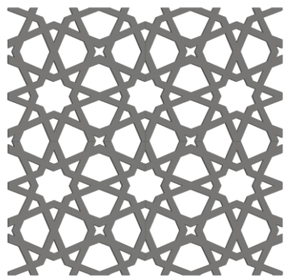
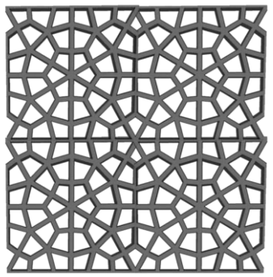
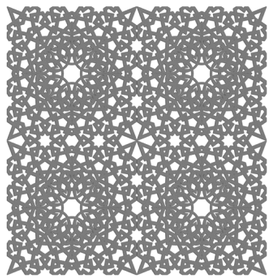
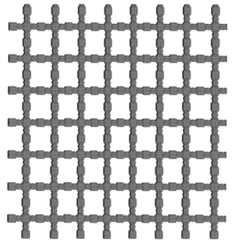


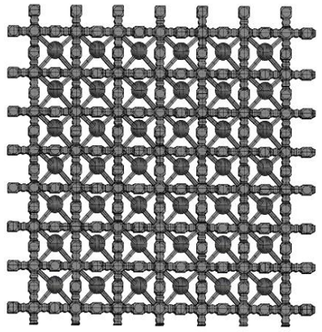
Fig 5.2.10 steps of the analysis done in micro level for shading quantification for normal diagonal arabesque source (author)

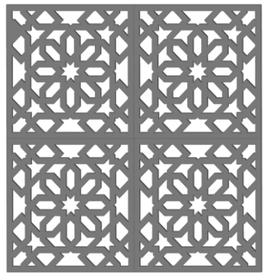


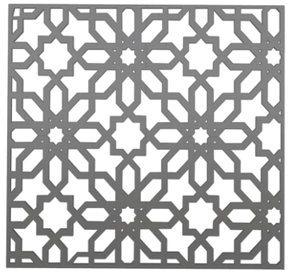


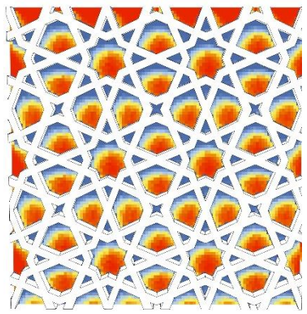


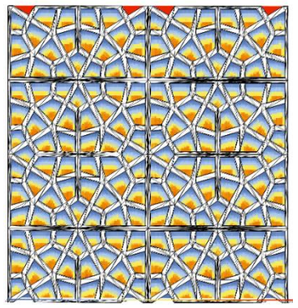


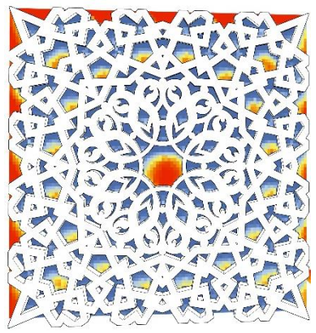


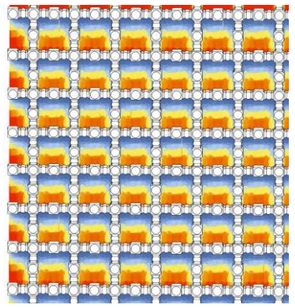


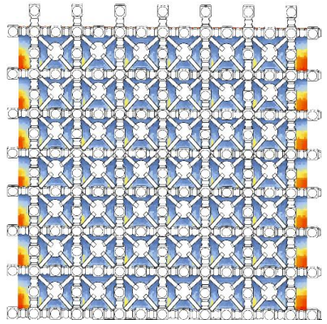


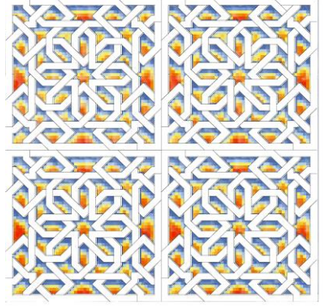


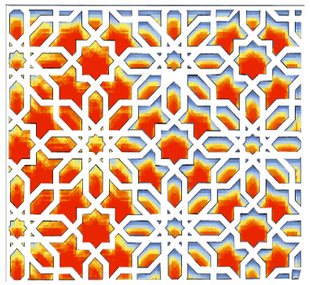












Average radiation :290 w/h

Average radiation :160 w/h

Average radiation :130 w/h

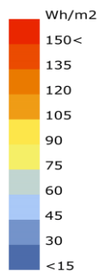
Average radiation :280 w/h

Average radiation :153 w/h

Average radiation : 201 w/h

Average radiation :330 w/h

Fig 5.2.3 summary table that gathers all the analysis and studies made (solar radiation for southern façade)
source (author)



A.2 Ventilation performance quantification :

Understanding the performance of each component that forms the pattern it was quite challenging because of software limitation. However, it was important to estimate a coefficient for airflow through patterns that can show the variation between different patterns in terms of how they allow the air to flow through them.

For analysing how the patterns components preform towards the air hitting the pattern it was important to model the patterns precisely with the exact realistic dimensions. Then each component was placed in a box with fixed diminutions and CFD analysis were made. CFD analysis was made **(Computational Fluid Dynamics)** by providing the same inlet air velocity at the inlet, same inputs and same box dimensions.

When the air flows through a pattern a variable percentage of the air defuses and hit the pattern causing small turbulences that happens on a very small micro level. These turbulences affect the pressure values closer to the patterns and decrease the amount of air entering through the patterns. CFD simulations was used to calculate the discharge coefficient created by each component. This calculations was made to link between the pattern shape, its perforation percentage and the amount of air passes through it. (K. Sheshagiri Hebbar, 2012)

The discharge coefficient was the only method that can help in comparing the different components. The reason for that is that it considered 3 main variables which are:

1. The difference in pressure
2. The area of the inlet that is mainly affected by the perforation percentage of the patterns
3. The airflow rate through the patterns

According to different studies and articles about discharge coefficient, the discharge coefficient can be defined as following: “The discharge coefficient is a dimensionless number used to characterise the flow and pressure loss behaviour if inlets and windows.” (neutrium, 2015) , perforated screens are typically and deliberately reduce pressure as they act as prior against the wind sometimes and allow only certain velocities to pass through.

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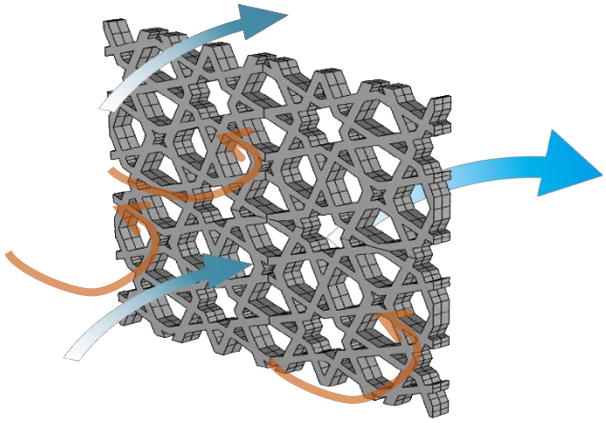
According to different studies and articles about discharge coefficient, the discharge coefficient can be defined as following: “The discharge coefficient is a dimensionless number used to characterise the flow and pressure loss behaviour if inlets and windows.” (neutrium, 2015) , perforated screens are typically and deliberately reduce pressure as they act as prior against the wind sometimes and allow only certain velocities to pass through.



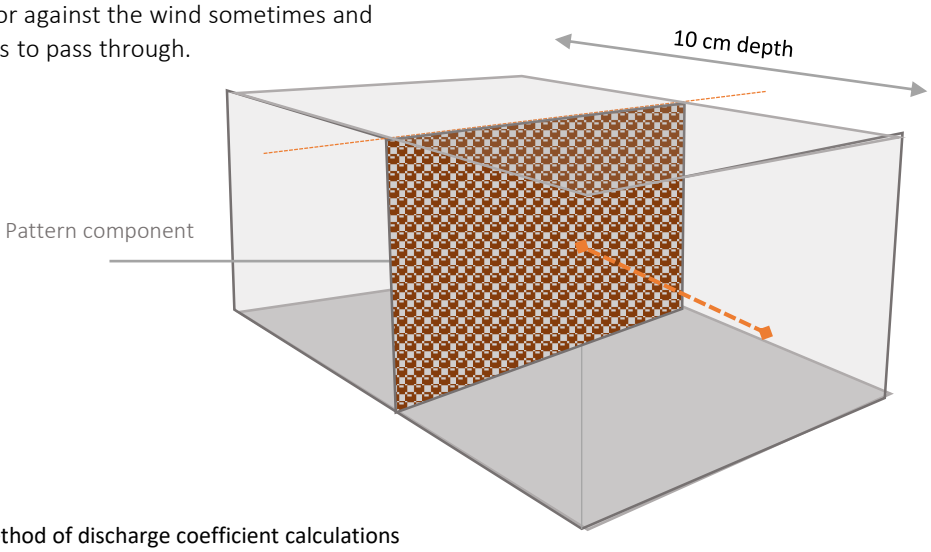


Fig 5.2.B.1 sketch of the method of discharge coefficient calculations source (author)

I. B. Calculating the discharge coefficient for each pattern:

$$C_d = \frac{\dot{m}}{\rho \dot{V}} = \frac{\dot{m}}{\rho A u} = \frac{\dot{m}}{\rho A \sqrt{\frac{2\Delta P}{\rho}}} = \frac{\dot{m}}{A\sqrt{2\rho\Delta P}}$$

The formula mentioned above was the formula documented to calculate the discharge coefficient in most of the previous research papers. Where the discharge coefficient = Mass flow rate at the outlet divided by the (area of the outlet * square root (density of fluid* difference in the pressure of the inlet and outlet))

For calculating these variable the patterns were modelled in the CFD as motioned before and two results planes were added. After that a bulk results was extruded for each plane to show the mass flow rate, pressure, area and density. Then the numbers taken was applied in the formula.

(sample of the results will be available in the appendix)

The CFD simulation was in form of the following main conditions:

- **The box**: the material assigned to the box was **fluid air**
- **The pattern:** the material assigned to the pattern was **solid soft wood**
- **The boundary conditions:** where all the faces were assigned to be **slip symmetry** except for the inlet and the outlet
- **The outlet**: was assigned to **pressure** as a boundary condition
- **The inlet:** was assigned to **velocity** as a boundary condition with velocity of 0.2 m/s

Calculating the discharge coefficient for PATTERN 5:

Calculating the discharge coefficient for each pattern:

1) Δp = P inlet - P outlet =15.4-(-0.1284)= 15.5 Pa
2)area of the outlet = 0.78m2
3)density of the air at this condition :
p=1.2041 kg/m3
4) mass flow rate at the outlet :
0.56 kg/s
5)Cd =
Cd=0.56/(0.78*√(2*1.2041*15.5) =**0.119**

The discharge coefficient doesn't represent the main performance of the patterns, it is not a constant number that can define the pattern. But it is a variable number that depends on the certain conditions of the moment that was simulated in the previous inputs.

However it can be used to create a fixed conclusion from comparing different patterns.

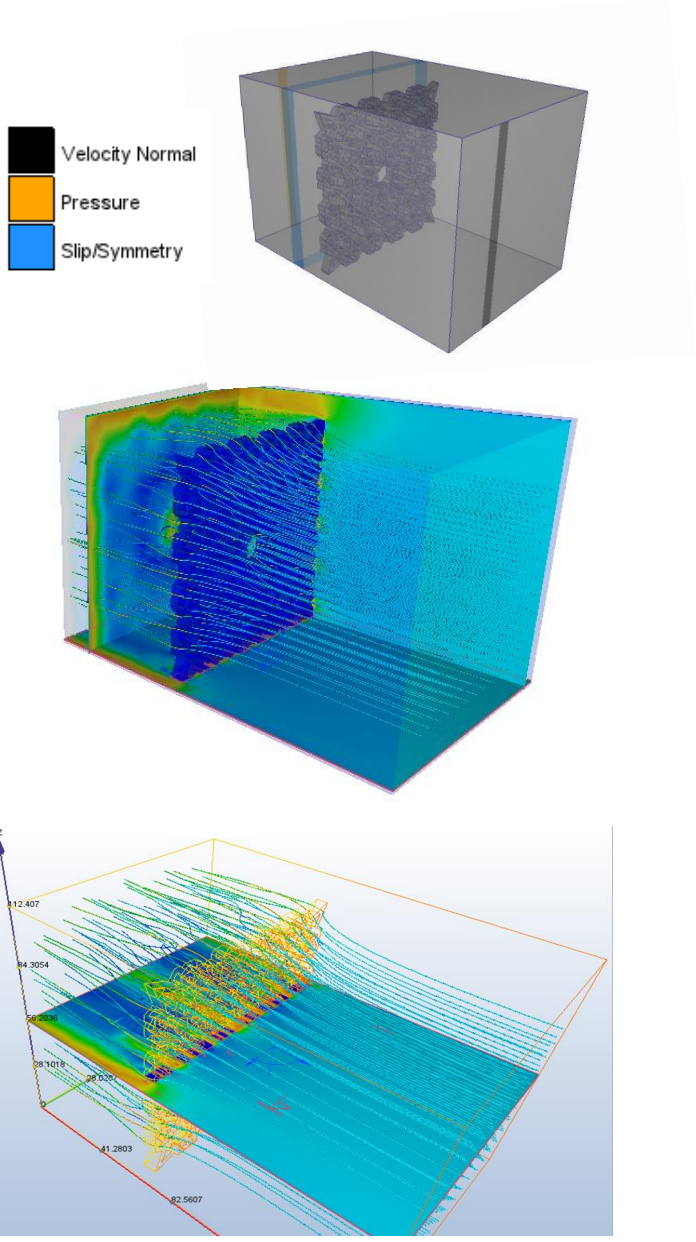


Fig 5.2.B.2 steps of calculating the discharge coefficient in CFD source (author)

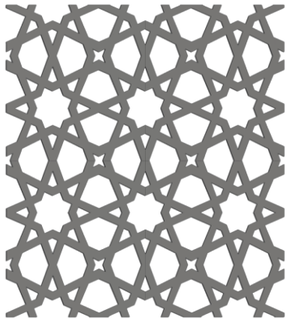
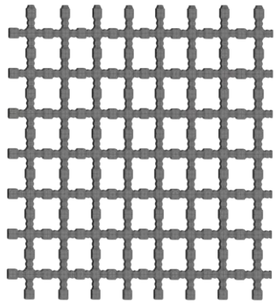
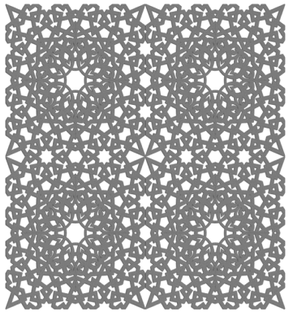
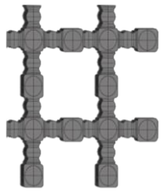
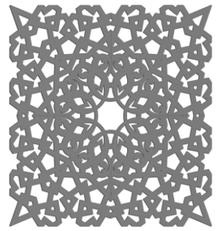
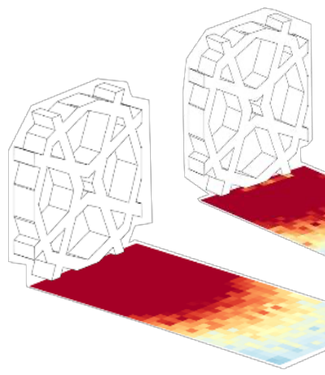
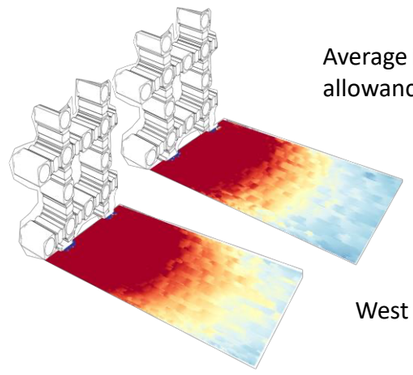
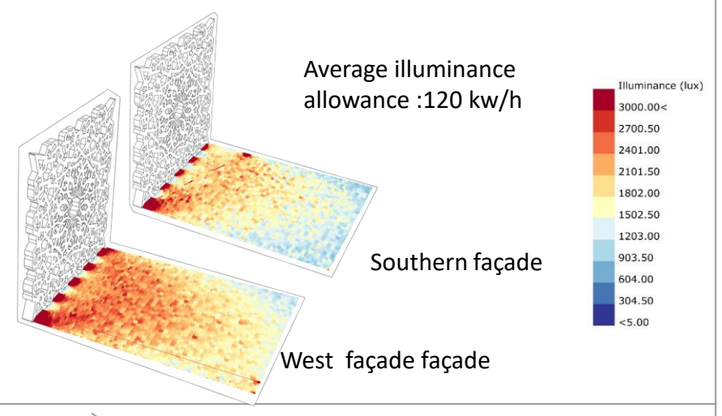
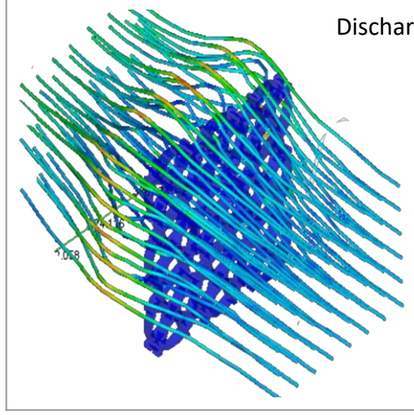
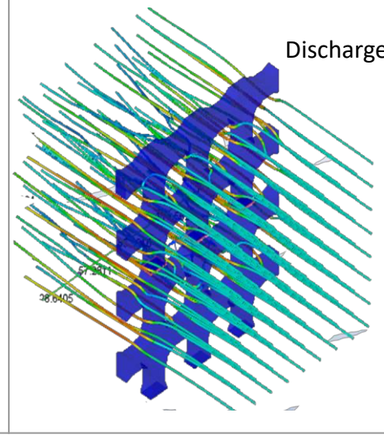
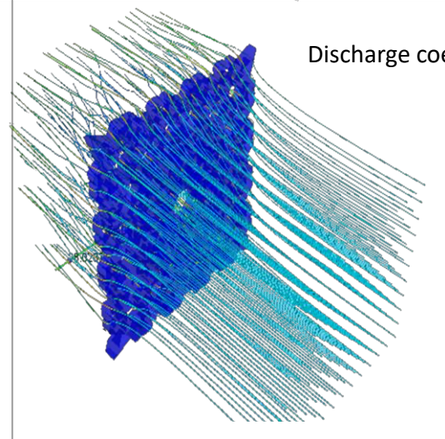
Pattern 3 	Arabsque 1 	Pattern 2 
Length to depth ratio= 1 : 2½	 Length to depth ratio= 1 : 4	 Length to depth ratio= 1 : 0.5
 Average illuminance allowance :110kw/h Southern façade West façade façade	 Average illuminance allowance :190 kw/h Southern façade West façade façade	 Average illuminance allowance :120 kw/h Southern façade West façade façade
 Discharge coefficient :0.2361	 Discharge coefficient :0.281	 Discharge coefficient :0.119

Fig 5.2.4 summary table that gathers all the analysis and studies made(daylight and discharge coefficient) source (author)

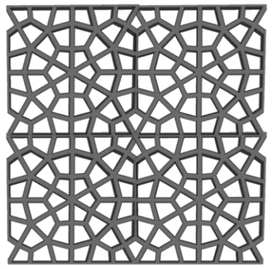
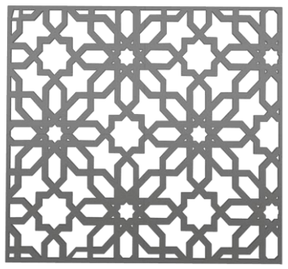
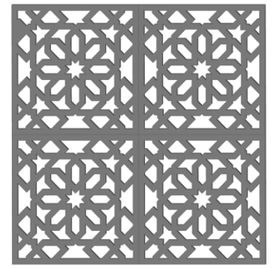
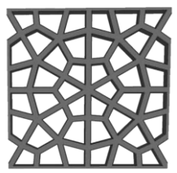
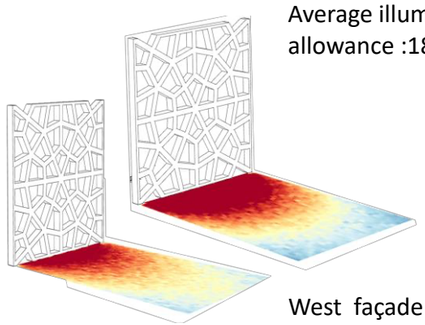
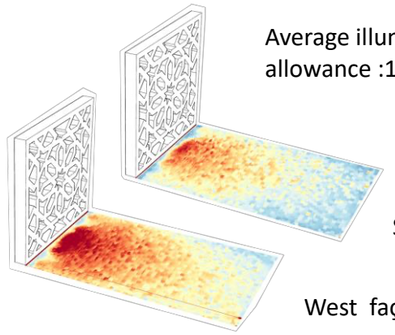
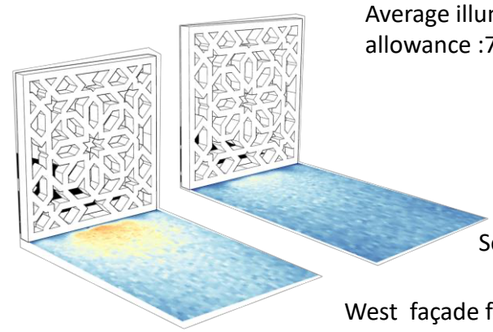
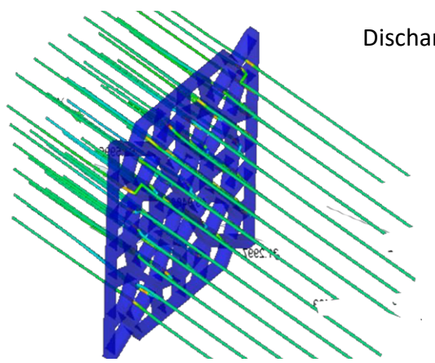
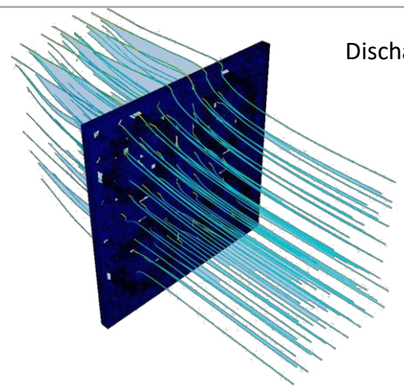
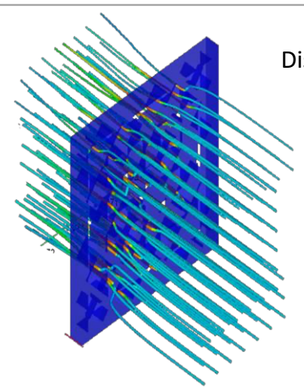
Pattern 1 	Pattern 4 	Pattern 5 
 Length to depth ratio= 1 : 2	Length to depth ratio= 1 : 2	Length to depth ratio= 1 : 1
 Average illuminance allowance :180 kw/h Southern façade West façade façade	 Average illuminance allowance :128kw/h Southern façade West façade façade	 Average illuminance allowance :79 kw/h Southern façade West façade façade
 Discharge coefficient :0.31	 Discharge coefficient :0.189	 Discharge coefficient :0.183

Fig 5.2.5 summary table that gathers all the analysis and studies made **(daylight and discharge coefficient)** source (author)

III. Outcome of micro level analysis

Based on the analysis made the following values was finalized in order to quantify and compare the environmental performance of the patterns:

1) Perforation percentage
2) Length to depth ratio
3) Average illuminance allowed by the pattern component
4) Average vertical solar radiation that can pass from the patterns
5) Discharge coefficient of the air flow through the pattern

Based on that a lot of conclusions were made that helped in understanding the relation between different environmental parameters and the typologies of the patterns

The micro level analysis provided a precise estimation of each patterns and its details , this helped in the analysis and modelling in the macro level in order to understand the performance of different patterns when they are combined together.

IV. Conclusion

In terms of shade and light:

- The amount of light that passes through the patterns (represented in the average illuminance) is directly proportional to the perforation percentage.
- The shading performance efficiency of a patterns is inversely proportional to the amount of light and solar radiation that passes from the pattern
- Floral pattern that have a close perforation percentage to a geometrical pattern tend to allow more light allowance to the space.
- Sometime the shading performance is not related to perforation percentage of the patterns, so even some patterns have closer perforation percentage their solar transmittance varies a lot. This might be conducted to the mesh density that increases the light reflectivity. Regarding this the solar transmittance can be a consequence of the patterns geometry.
- Most of the components performance is reduced at lower sun angles (as the patterns were tested in different days, times and orientation)

In terms of ventilation:

- There is a relation between the discharge coefficient and the aspect ratios of the patterns' components, where the aspect ratio is the ratio between the length and the width of the openings.
- The value of the discharge coefficient varies significantly at small pressure differences, also, it becomes almost constant at small pressure differences. That might be important to take into consideration that natural ventilation often operates at very small pressures.
- The discharge coefficient can vary in the same opening at small pressure differences.
- The highest discharge coefficient comes from arabesque pattern as it has a smooth spherical components the forms less obstruction against the air flowing through it
- The discharge coefficient of the patterns varies according to peroration percentage of the patterns, where the value of the discharge coefficient is directly proportional to the percentage of perforation in the patterns.
- The patterns that have small dimensional grids with smaller components tend to have less discharge coefficient and vice versa.
- Floral patterns tends to have height discharge coefficient than geometrical patterns

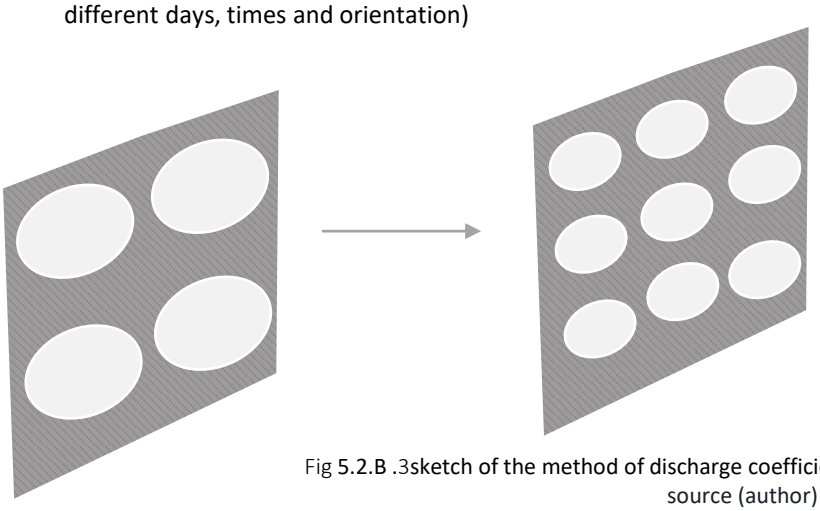
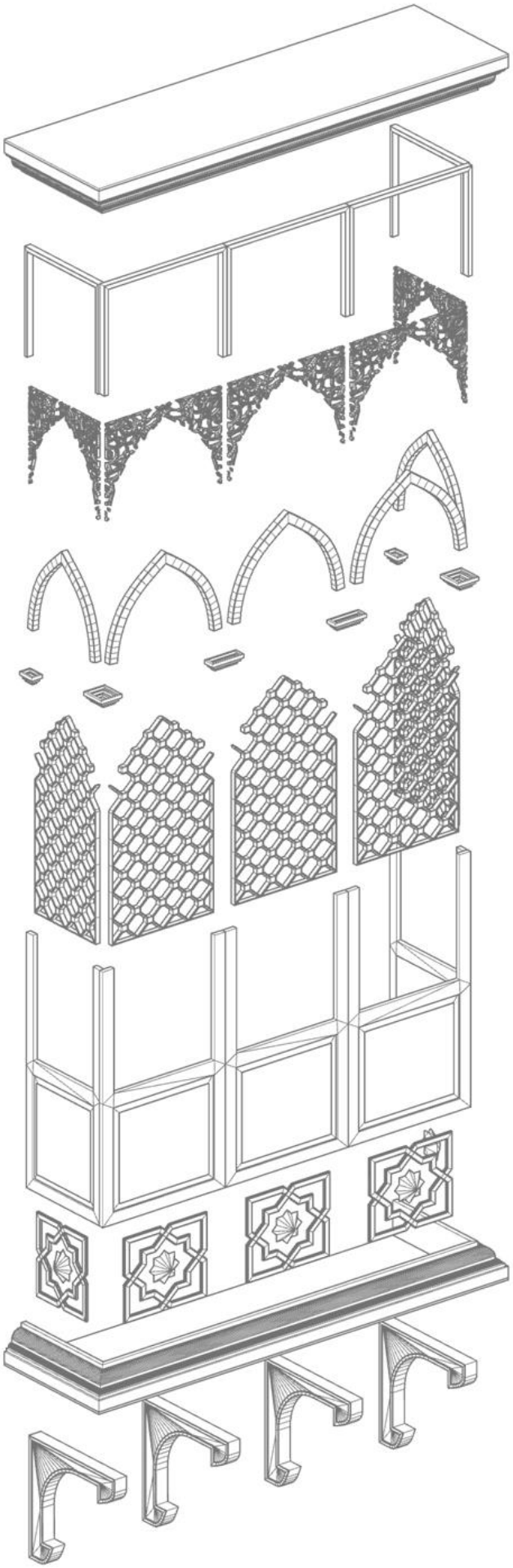


Fig 5.2.B .3sketch of the method of discharge coefficient calculations
source (author)

5.3 Macro level

Analysing the whole component



3-MACROLEVEL:

From the field work it was found that the performance of mashrabiya varies spectacularly according to the change in the component level. This change can be in:

1. **Geometry**:
   which is the main difference between Andalusian mashrabiya based in Spain and morocco and ottoman mashrabiya based In Egypt. Where the Andalusian mashrabiya is adjacent to the window using floral patterns that have more perforation percentage. While ottoman mashrabiya is an extruded component that forms a transitional space inside with using geometrical patterns with less perforation percentage. Also,
1. **Combination of patterns**:
   ottoman mashrabiya have different typologies where it can have one pattern or a combination of different patterns. This combination can create a sort of stack single sided ventilation where the patterns used in the top and bottom are comparatively wider and more perforated than the patterns used in the middle of the component.
1. **Evaporative cooling process interaction with the component performance**: where a ceramic pot is placed in front of the inlet in order to perform a multi proposal process, so it can cool down the air entering to the space by evaporative cooling process and providing cold water for drinking. Also, some modern approaches created a hollow ceramic patterns where water was ducted inside.

In macro level we are going to analyse and understand the performance of the component in different scenarios which are:

1. Adjacent patterns (non-extruded)- Andalusian mashrabiya
2. Extruded patterns -ottoman mashrabiya
3 Extruded component with combination of different patterns
4. Extruded component with combination of different patterns and evaporative cooling
   4.a) Ceramic pot situated in front of the inlet (CASE 1)
   4.b) Ceramic patterns with water ducted through it (CASE2)

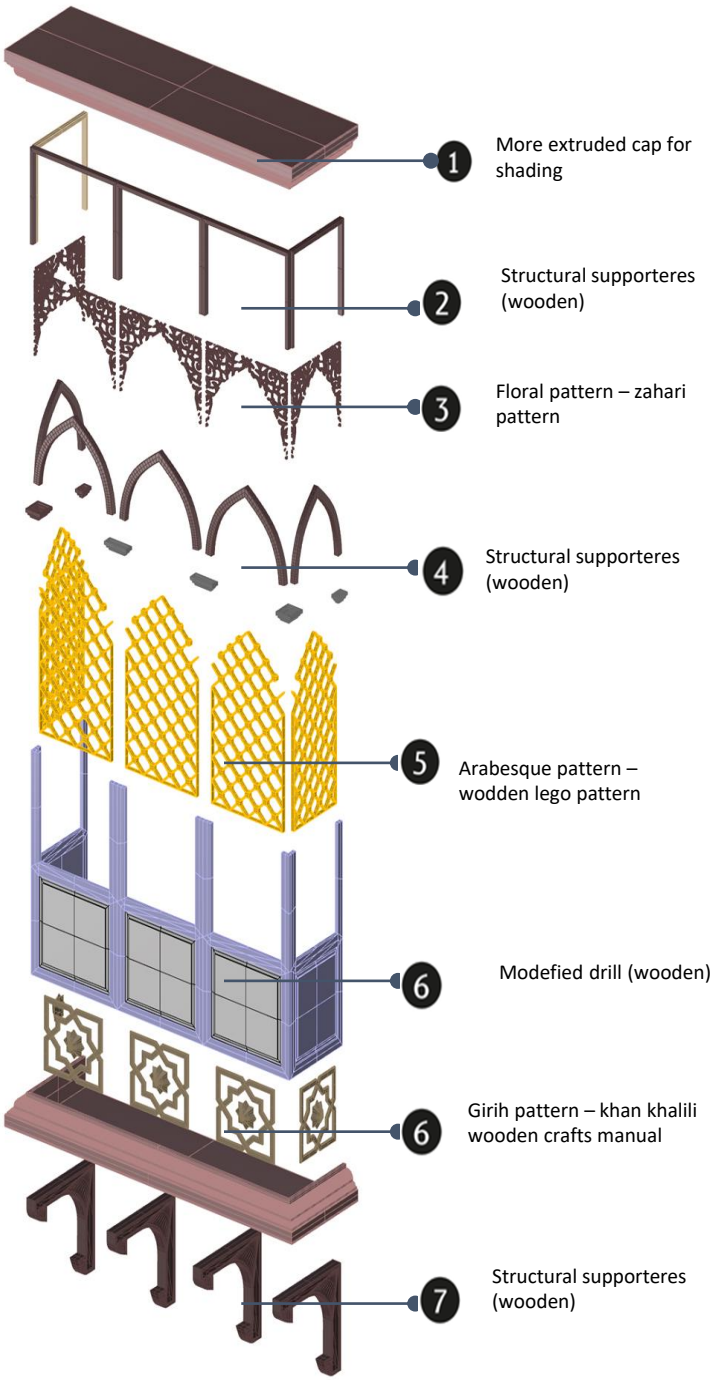


Fig 5.3.1 Analysing mashrabiya as a component source (author)

From the field work a lot of notes and sketches were made in order to document the different integrations in the patterns, and model these patterns combination in its existing geometry, in order to understand its performance.

The components were modelled in Rhino 2017 software in order to simulate their environmental performance in different analytical software.

From bayt al suhaymi it was concluded how the integration in the perforations of patters in one mashrabiya according to the height

The upper part of the mashrabiya has more perforation that reaches from 47~ 63 %

However by adding the side open the effective aperture area increases more.

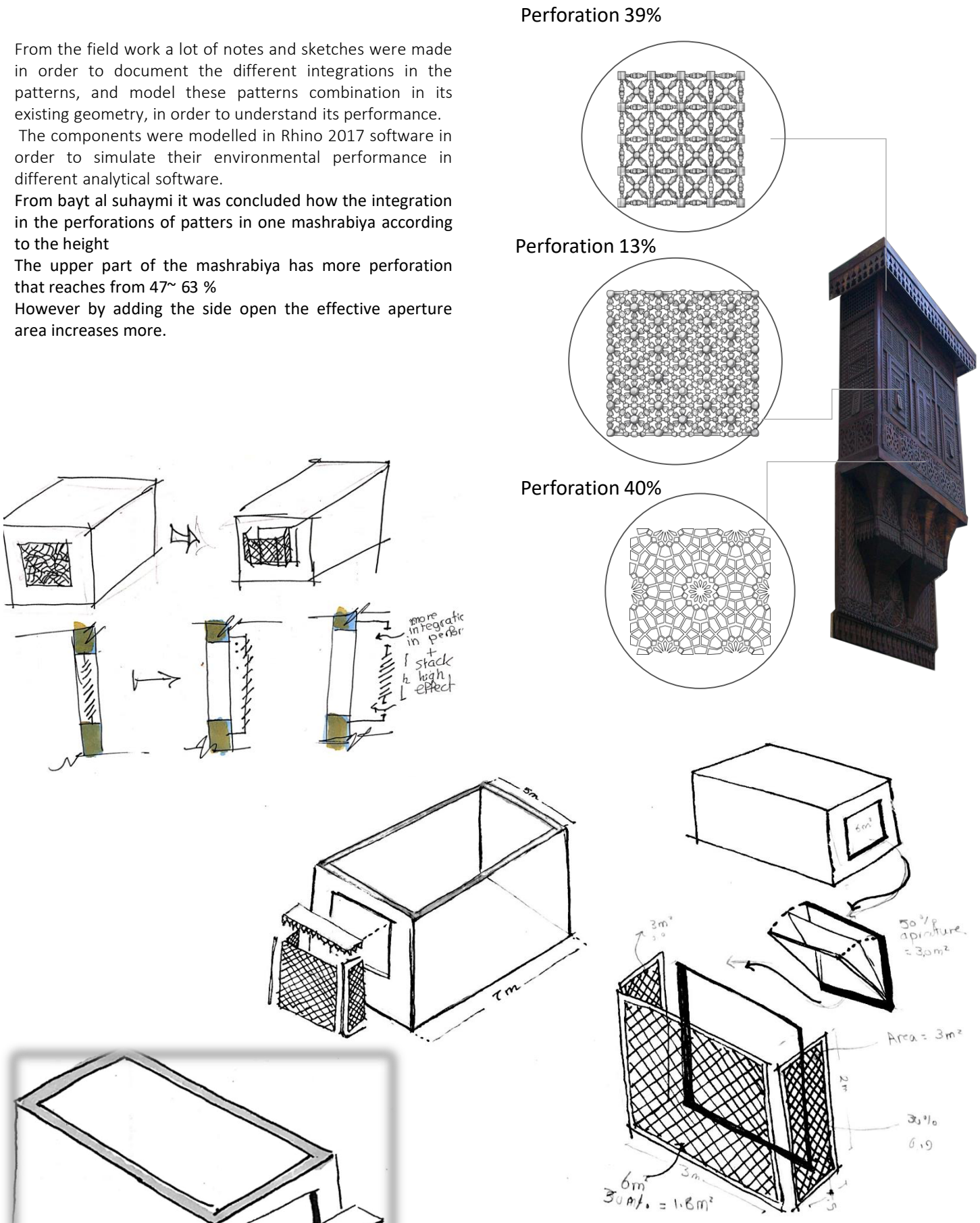


Fig 5.3.2 analytical sketches to understand the combinations of different patterns and different typologies
source (author)

II-MACROLEVEL:

II.1 Methodology:

There is no argument that patterns with high perforation percentage allows more air flow through the space since that the effective aperture area increases. But will that achieve comfort? that's why analytical tools were needed to make a fully understanding of the performance on various environmental analysis. The tools used for this analysis are:

A. **Optivent:** to form a basic understanding of how the patterns preforms through calculating the effective aperture area on the whole component using the following formula:

Effective aperture area = percentage of perforation (%) X total area of the window (m2)

This formula provides rough calculations of the effective aperture area which is not a precise constant number as it changes according to the conditional discharge coefficient and the turbulences cased from the air hitting the patterns, since that the patterns act as wind prior at some point.

A. **CFD (Computational Fluid Dynamics):** to provide a full understanding of how the air preforms through the pattern inside the space and how is this affect the mass flow rate at the end of the room

B. **Grass hopper-honeybee:** where illuminance analysis to estimate the amount of light the enters through the pattern as well as the amount of solar gains resulted from it.

C. **Testing room:** For testing the performance of different typologies of mashrabiya as a component a space was selected similar to typical living spaces in Egypt-Cairo with dimensions 5x7 meters (floor area of 35m2) , supplied with an opening on one façade only which is the southern façade and the reason behind these inputs are as following:

i. **Orientation:** The exterior façade was selected to be southern because according to the climatic data of Egypt as well as most of the middle eastern countries it

ii. **Dimensions of the room:** according to the typical houses architectural design standards in Egypt 2012 -the living space in atypical residential house designed for a family that contains from 3-4 members a suitable living room should be supplied with area that ranges from 25 to 40 m2. (committee, 2009)

iii. **Dimensions of the room:** according to the typical **the window dimensions:** according to the Optivent analysis many tests were made to reach to a base point towards stating the most suitable dimensions for a window that can achieve the required amount of air and closer as possible to the required for cooling, considering the internal heat gains resulted from equipment and occupants' sensible gains. Where the window dimensions were finalized to be (2 X 3) meters

D. **Dynamic Thermal analysis (TAS):**
Where a rough image of mashrabiya patterns was formed using inclined shading element which intersect using different directions and angles of inclination.
The inclination of the shading fins was what provided the thickness to it, since there were software limitations in modelling the patterns in a more realistic way.
Also, it was important to model the pattern in TAS and not simplifying it by calculating the effective armature area provided by the pattern and assign it as a window. Since that will decrease the accuracy of estimating the amount of solar gains resulted from the patterns. Specially that TAS was the only software that manages to calculate it.

the inputs:

All the analytical simulations (tools of understanding the performance of mashrabiya as a component) had the same input with no surrounding context added into consideration.

- **The surface temperatures** assigned to the objects was through the surface temperatures documented during the field work in Bayt al suhaimi house.
- **The climatic context** was the same climatic context assigned in the micro level analysis which is *Cairo- Egypt latest weather data files*.

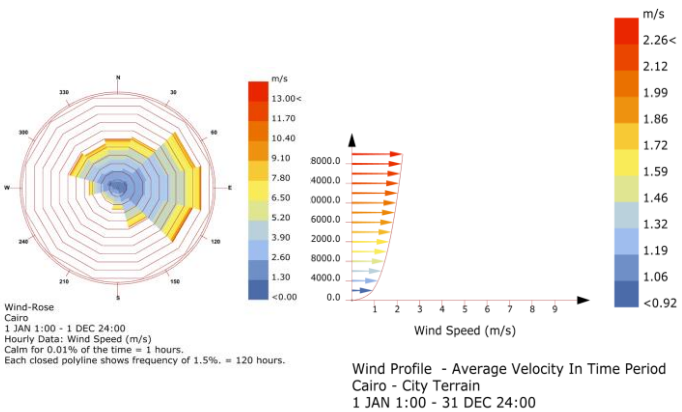


Fig 5.3.3 wind rose and wind profile of Egypt-Cairo source (author)

1. Adjacent patterns (non-extruded)- Andalusian mashrabiya

Andalusian mashrabiya is considered the basic case of mashrabiya where floral patterns with higher perforation percentage and higher discharge coefficient which eventually increases the effective aperture area and allows more air to flow through the space. But this study aims to evaluate the performance of mashrabiya not only as a source of ventilation but also as shading element. And the more the perforation percentage will be the shading performance efficiency will become less.

A. Optivent analysis:

A.1-inputs:

I. **Aperture area:** By assuming the effective aperture area which equals the percentage of perforation multiplied by the window area which was 6 m2, the effective aperture area was found to be 2.5 m2.

II. **Percentage of opening:** since that the area calculated was the area considered for the openings and perforation of the screens the percentage of opening was considered to be 100% opened

III. **The internal gains added to the space was the occupancy heat gains of 2 sitting persons (low metabolic activity)**

A.2-outcomes:

From optivent analysis it was found that

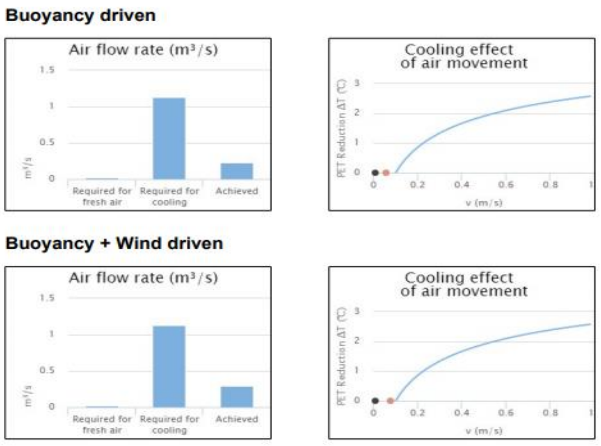


Fig 5.3.4 optivint result
source (author)

A. CFD analysis:

B.1- inputs:

To simulate a model of this pattern closer enough to reality calculations were made in order to provide rough estimation of the effective aperture area, as there were some software limitations in modelling such complex floral patterns in a bigger scale in CFD.

B.2- outcomes:

According to the simulations made in CFD the following results was found:

- The air flow at the end of the room was considered to be sufficient in terms of ventilation, however, the air speed closer to the pattern was more than 50 %higher than the air speed at the end of the room. The average air velocity at the middle of the room ranges between 0.07 to 0.1 but reaches zero at the end of the room
- This shows that the pattern may allow appropriate amount of air to flow through it but there is no good distribution for the air inside the space.
- Some turbulences were caused near the patterns due to the amount of air that passes through it and hit the surrounding context of the openings (the wall).

- Aperture percentage = 42%
- and windows area = 6m2
- which gives aperture area of 2.52m2

Average temperature : 23.1c°

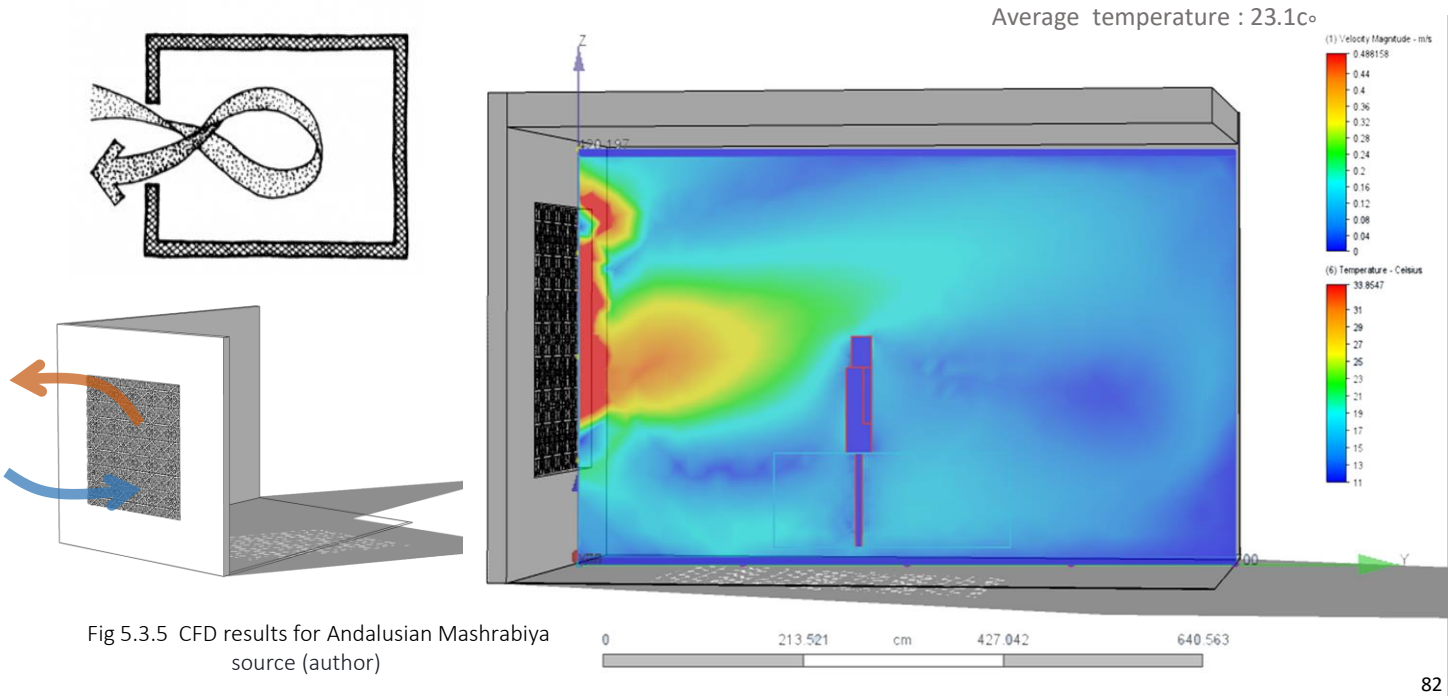


Fig 5.3.5 CFD results for Andalusian Mashrabiya
source (author)

2. Extruded patterns -ottoman mashrabiya

During the field work some typologies of the ottoman and Fatimic mashrabiya were found. Where there was no patterns integration. In most of its types it was only one pattern that was applied to the whole geometry.

This typology 80% of the patterns applied to its geometry is a Maymooni pattern with very small perforation percentage that ranges from 12% to 17%. From that the outcomes of its performance in terms of shading were expected. However during estimating the effective aperture area of the openings of the patterns, to be used in both CFD and optivent the following was found:

- The extrusions in the geometry of the component that have_patterns added to them as well add compensate a lot to the effective aperture area of the component.
- This addition in the effective aperture area can make this component with small perforated patterns equivalent to the Andalusian Mashrabiya performance in terms of ventilation.
- With having almost the same performance as the Andalusian Mashrabiya, this typology managed to achieve a better performance in terms of shading. As the main used patterns is less perforated than Andalusian pattern with less solar access.

- Perforation percentage = 17%
- windows area = 6m2 + (2m2)*2 extruded area
- which gives aperture area of:

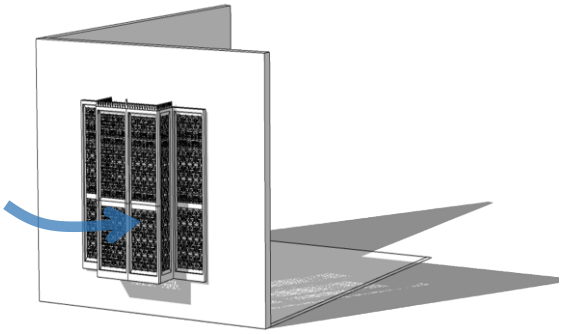
(17% X 6) + (17% X 2 X 2) = 1.02 + 0.68 =1.7 m2

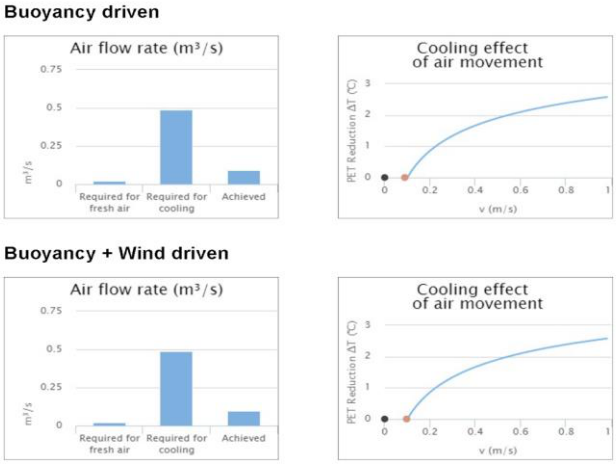
- **Inputs:**

The same inputs were applied here as well as the surface temperatures

- **Outcomes:**

1. The average temperature of the room was almost the same as the Andalusian Mashrabiya, however the CFD doesn't put the solar gains into consideration. So, we can estimate that this case preforms better in terms of shading, as it prevent the solar access more efficiently.
2. The spot measurements applied to the CFD model was the same spot measurements documented in Cairo-Egypt 15th of May where the outdoor temperature was 32°c. despite that ,the indoor temperature was considered to be within comfort.





Northern facade single sided ventilation

Fig 5.3.6 optivent results for ottoman Mashrabiya source (author)

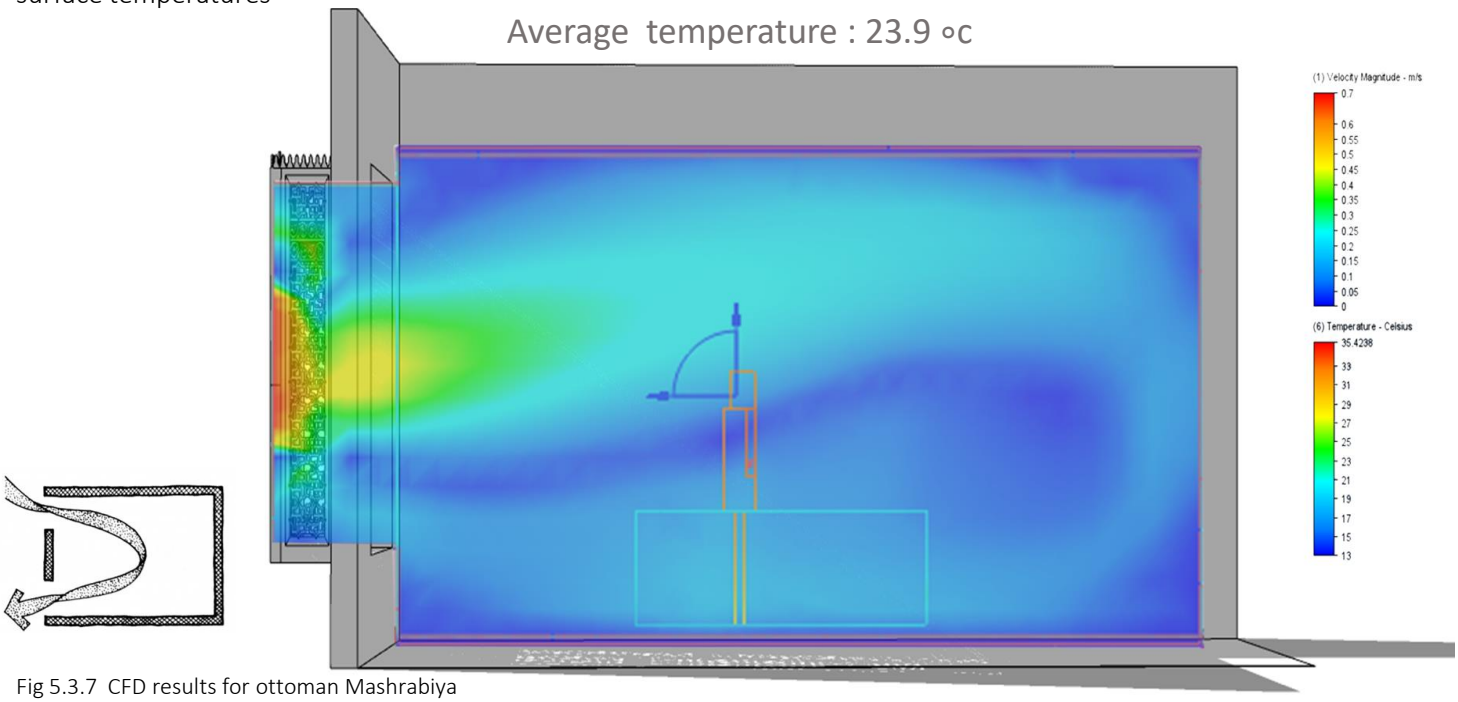


Fig 5.3.7 CFD results for ottoman Mashrabiya source (author)

3. Extruded component with combination of different patterns:

This case is almost the same as the cases before except for:

- The extrusions in the geometry of the component that have patterns added to them as well add compensate a lot to the effective aperture area of the component.
- This addition in the effective aperture area can make this component with small perforated patterns equivalent to the Andalusian Mashrabiya performance in terms of ventilation.
- With having almost the same performance as the Andalusian Mashrabiya, this typology managed to achieve a better performance in terms of shading. As the main used patterns is less perforated than Andalusian pattern with less solar access.

- **Inputs:**

The same inputs were applied here as well as the surface temperatures . The only difference was in the effective aperture area.

Effective aperture area :
Perforation percentage * (area)
Area = 4.5+ (0.75*2)
Effective aperture area =
[17% x 4.5] + [0.75 x 42%] + [0.75 x47%]+ [$\frac{1}{3}$ x 0.75 x 42%]
+ [$\frac{1}{3}$ x 0.75 x 47%] + [$\frac{1}{3}$ x 4.5 x 17%]

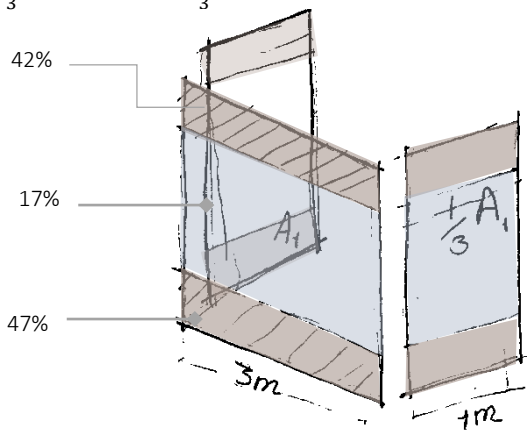
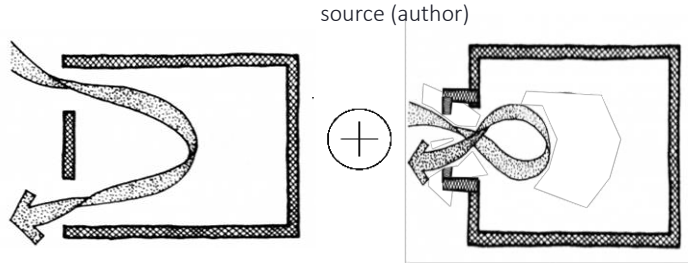


Fig 5.3.10 CFD analytical sketch for showing pattern comb.
source (author)

- **Outcomes:**

1. This method improved the ventilation performance of Mashrabiya up to 50 %and improved its cooling effect up to 40% according to both optivent and CFD analysis.
2. Adding stack height effect increased the air flow through the whole depth of the space making the average airflow velocity at the end of the room 0.002 m/s
3. The average temperature was 21 degrees which is almost 1.5 degrees lower than the cases analysed before .

Fig 5.3.11 CFD analytical sketch for showing geometry combination.
source (author)



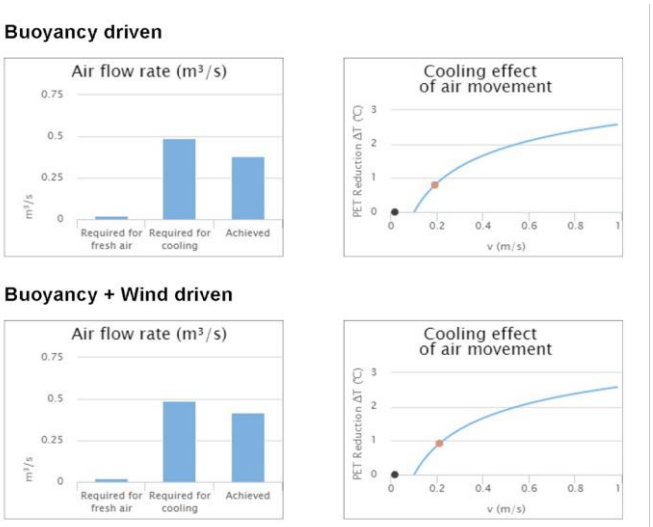


Fig 5.3.8 optivent results for ottoman Mashrabiya
source (author)

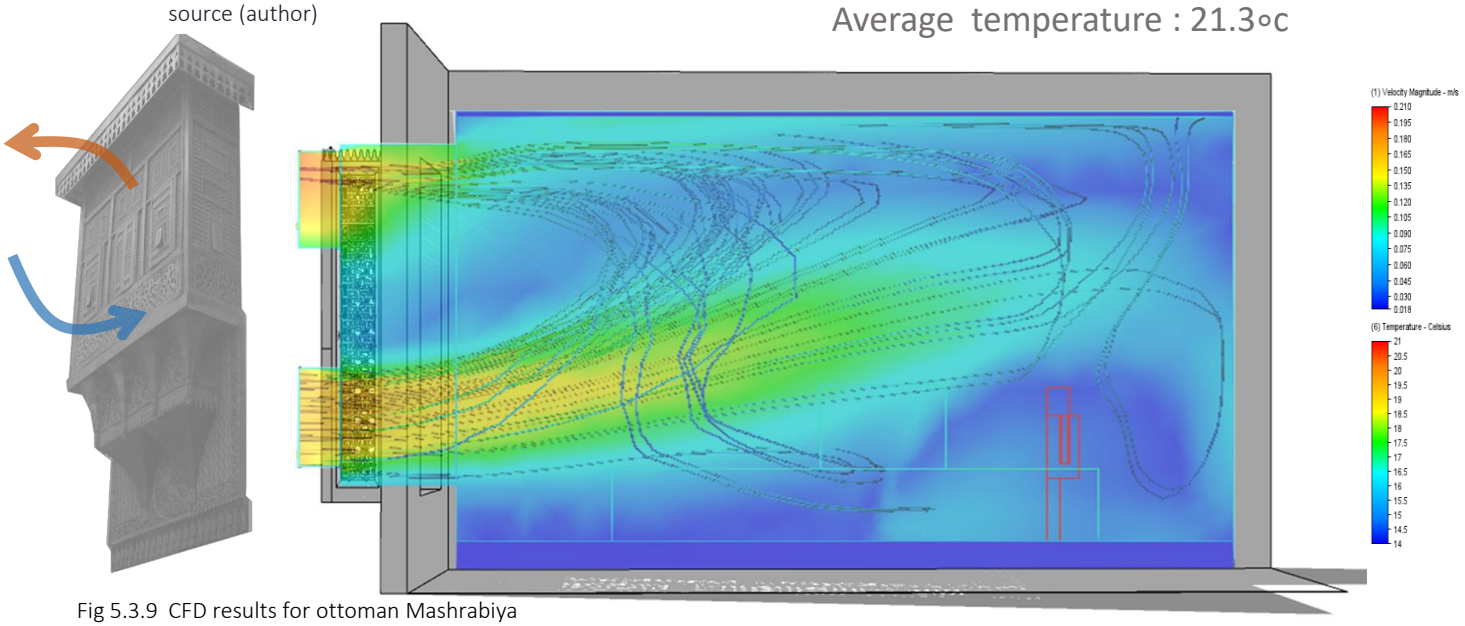


Fig 5.3.9 CFD results for ottoman Mashrabiya
source (author)

C. Illuminance

Each component used to have different shading performance that varies according to the patterns applied on it and the geometry represented in each form. For that, all the cases that were mentioned before were tested in terms of shading through using grasshopper, ladybug and honey bee software to estimate the amount of light that penetrates the patterns.
For that illuminance analysis were made and the average illuminance projected on the floor of the room was calculated

- **Outcomes:**

The top part of the component is the part that receives light and solar radiation the most. As a consequence, there is a very negative impact that occurs from using a wide high perforated screens in the top part of the component in southern and western facades.

But from the previous CFD analysis it was shown that to achieve an efficient stack single sided ventilation wide patterns must be used in both top and bottom part of the component. This showed that cap part added to the component in most of mashrabiya components is considered to be an effective solution towards shading the pattern and allowing air follow with proper amount.

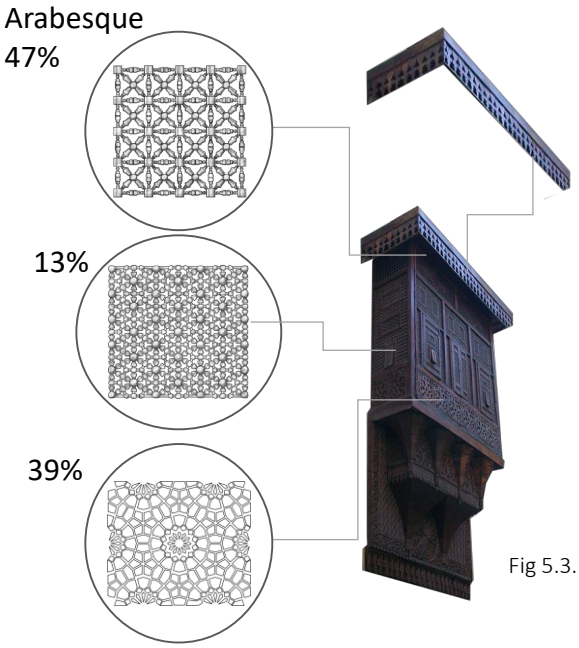


Fig 5.3.14 figure of analysing the complex component of ottoman M.
source (author)

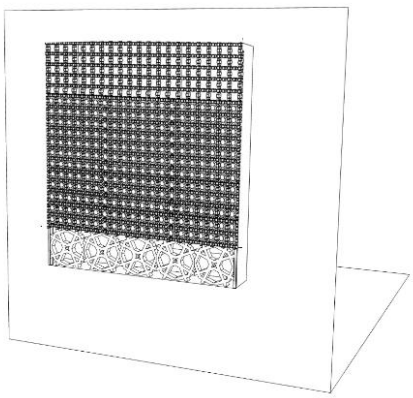
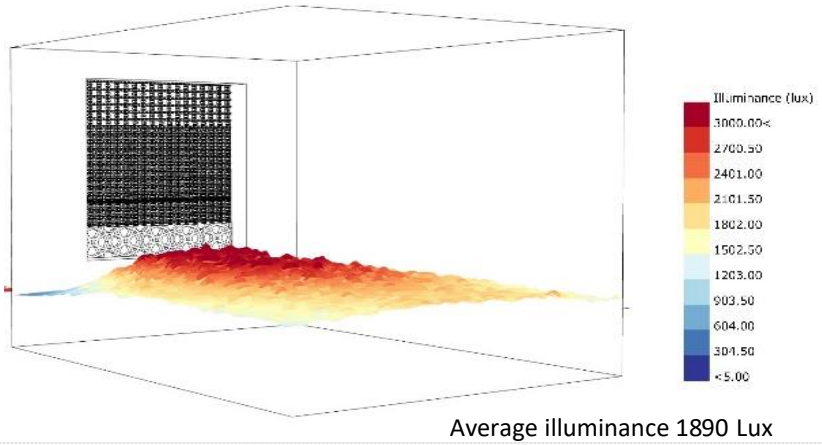


Fig 5.3.12 illuminance of the component without the cap added
source (author)



Average illuminance 1890 Lux

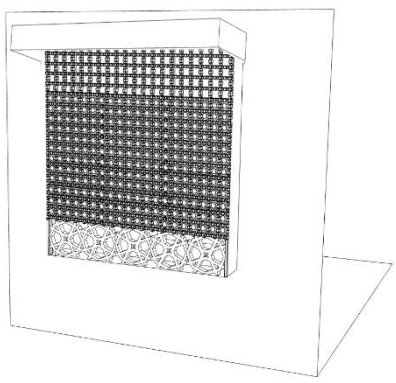
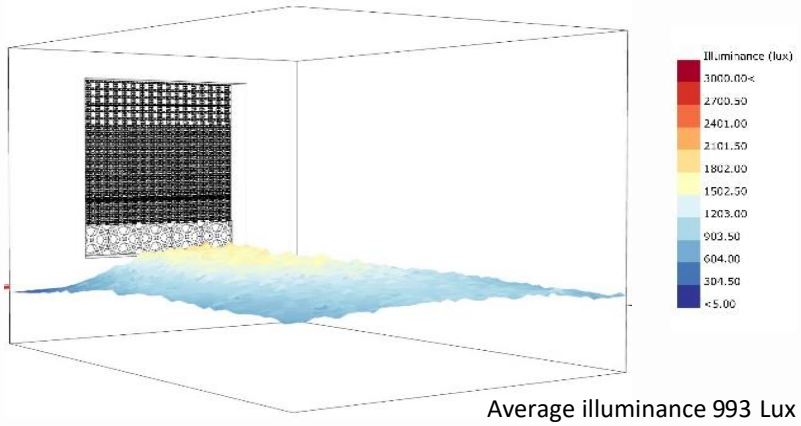


Fig 5.3.13 illuminance of the component with the cap added
source (author)



Average illuminance 993 Lux

4-Extruded component with combination of different patterns and evaporative cooling:

Case 1(Evaporative cooling and Mashrabiya):

In Arabic language the word “Mashrabiya” is driven from the word “Sharab” which means in English drink. This shows the mean usage of Mashrabiya on a different environmental parameter which is: evaporative cooling. Where a ceramic pot filled with water is situated in front of the inlet of Mashrabiya, this ceramic pot “Olla” used to be round in order to increase the surface area exposed to the air coming from the inlet. In this case a wider perforated pattern is used in the inlet which is situated in the bottom part of the component in order to allow more wind and avoid undesired solar gains which is received mostly in the top part of the component.

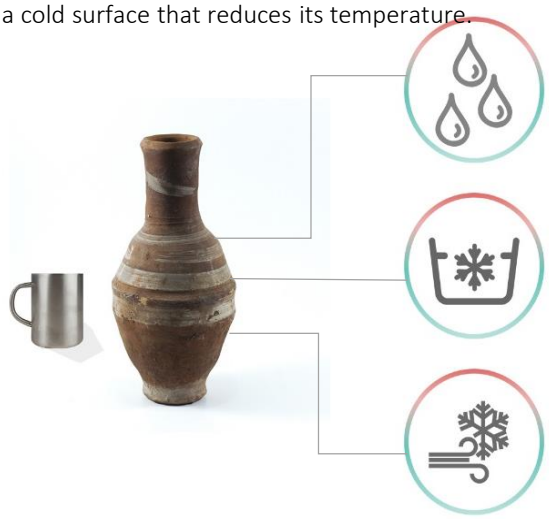
It was important to understand and illustrate how the component preforms towards the space if an evaporative cooling element was situated in front of the inlet.

To test evaporative cooling in terms of indoor comfort and air movement using analytical tools there was a lot of limitations. However, it was necessary how the aerodynamics around the patterns and through the room will be when adding a cold humid surface in front of the inlet.

For testing that CFD software were used where a hollow pot was modelled in front of the inlet with additional volume inside it. The model was simulated in CFD with brick as a material assigned to the ceramic pot as it has closer porosity to ceramic used in traditional pots. As for the volume inside the pot water was assigned to it and the temperature was edited to be 4 degrees lower than the room temperature.

The CFD simulations ran according to the measurements took in the field work in order to provide realistic results and full understanding of the parameters. And based on that surface temperatures were assigned.

From the CFD results despite the limitation the requested more searching and deep literature review it was hard to simulate evaporative cooling as a process but the simulations managed to show how the air flow preforms when it hits a cold surface that reduces its temperature.



- Evaporation process

Where the droplets on the exterior surface of the pot help to moisturize the air

- Cooling the water inside the pot

Where the air passing by the pot cool the surface which helps in cooling the water inside, so mashrabiya acts as a passive refrigerator to cool the drinking water.

- Cooling the air coming inside the space

Where the air temperature reduces significantly by the evaporative cooling effect or by hitting the cold surface of the ceramic pot

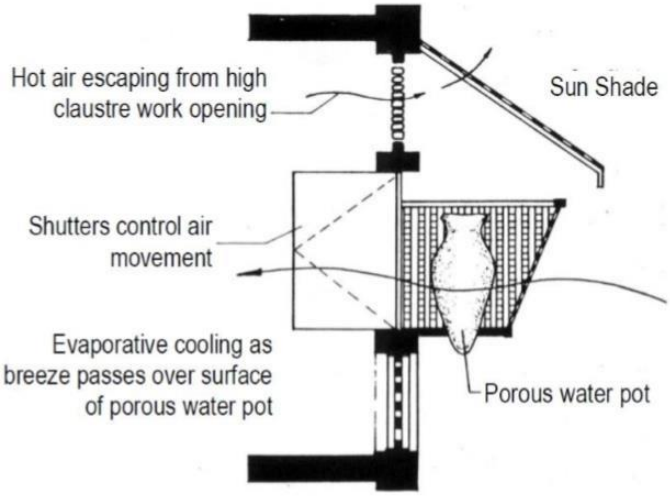


Fig 5.3.15 figure of analysing the complex component of ottoman M. source (schiano,2004)

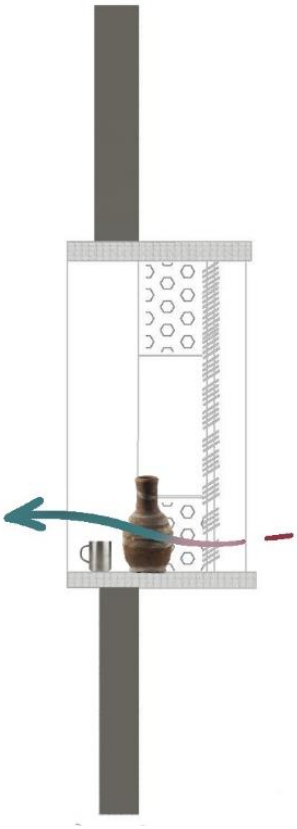


Fig 5.3.16 sketch showing functional use of the transitional space . source (author)

Outcomes:

- The air becomes colder in the inlet which reduces the room temperature up to 1.85 degrees less as an average. But the air doesn't penetrate the whole depth of the room since it become heavier it tends to settle down at a certain depth of the room. Where the mass flow rate at the end of the room was 31% less than the case where there was no evaporative cooling element added even though it was the same conditions.
- Also, another assumption was made to illustrate the reason of the low air flow rate, where the evaporative cooling element (ceramic pot) obstruct the air coming from the inlet and cool the air that manages to pass contacting its surfaces.

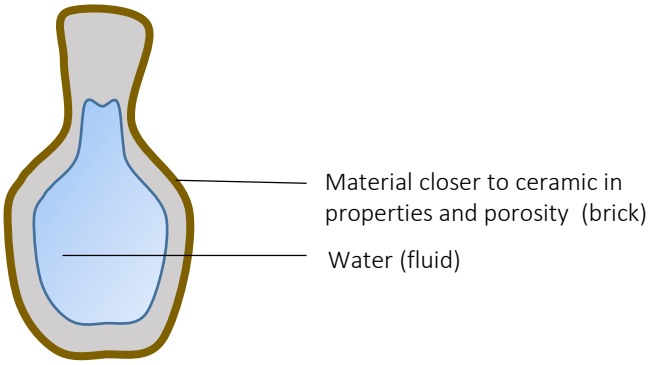
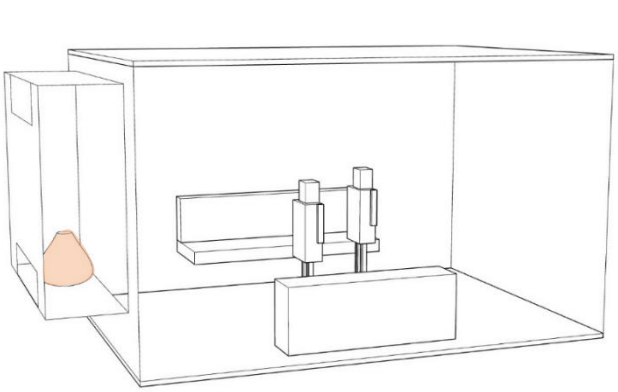
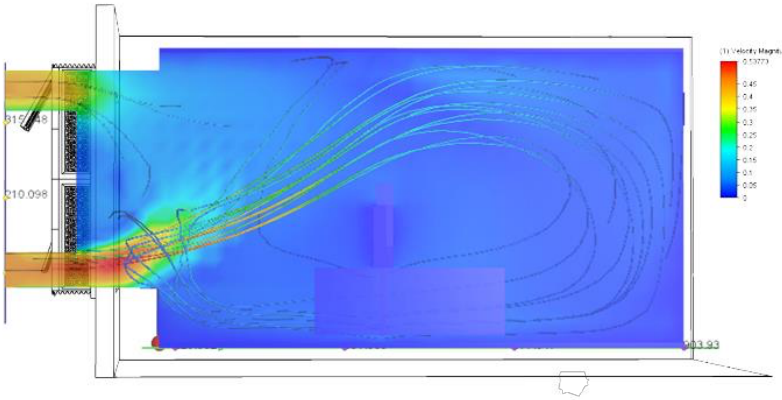


Fig 5.3.17 sketch introducing how the ceramic pot was modelled in CFD . source (author)





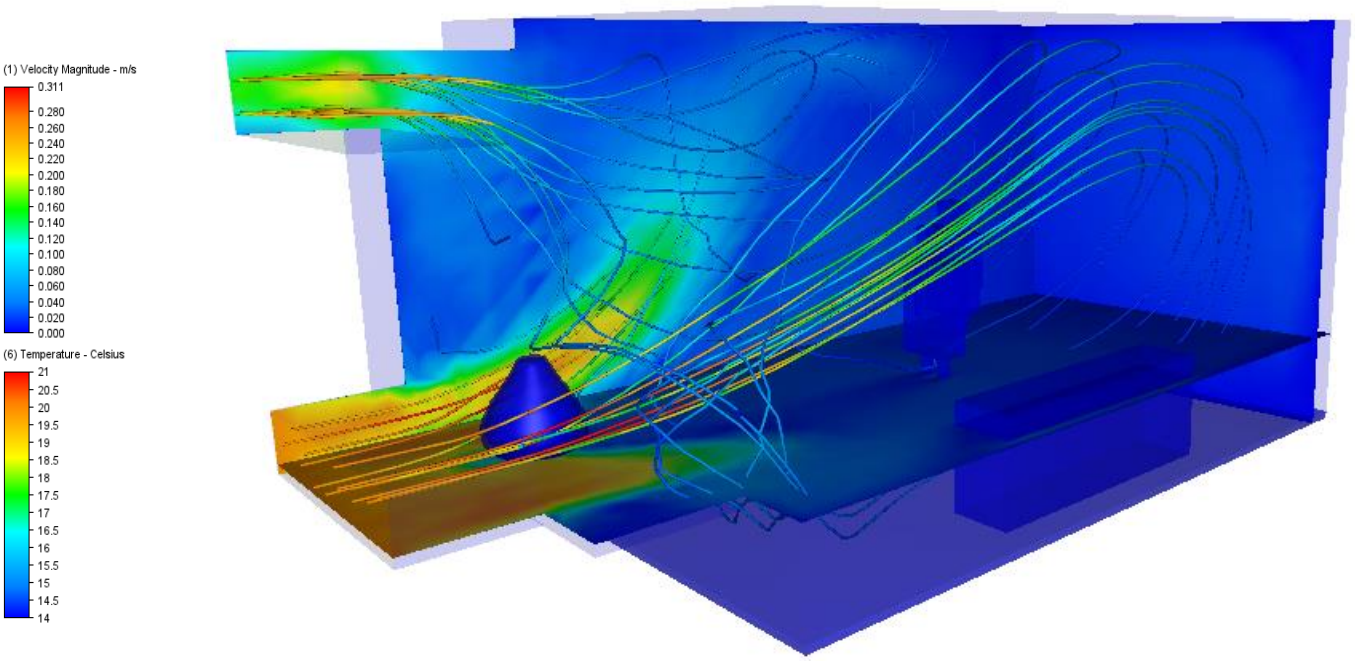


Fig 5.3.18 figure of analysing the complex component of ottoman Mashrabiya with ceramic pot in CFD source (author)

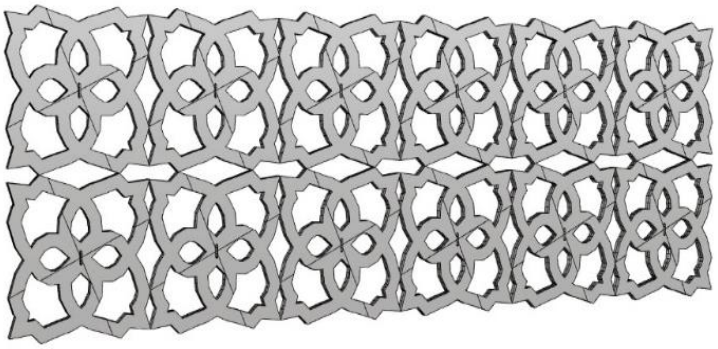
II-MACROLEVEL:

5. Case2(Ecooler (evaporative ceramic patterns))

As motioned in the literature review it was useful to understand how the modern approaches preforms, where the evaporative cooling as a process occurs on the patterns itself.

From that an example was selected to be tested and examined to evaluate it performance in terms of shading, ventilation and cooling.

The reason behind that was conducted to the literature review and reading other papers approaches where an example about evaporative cooling patterns was studied. One of these patterns is “Cooler” 5 cooling ceramic screens. Where the water is being ducted to the ceramic patterns to keep it saturated as much as possible in order to cool the air entering through the space, (SURNAME OF AUTHER, YEAR) .



By simulating the same exact patterns used in Ecooler which was characterized with higher percentage of perforation than normal patterns in order to allow more air flow through it. It was expected to allow more wind to flow to the room.

The pattern was simulated the same way the ceramic pot was formed where a fluid volume was modelled inside the vacuum of the pattern, And CFD showed the expected results from the pattern and reduce the room temperature up to 3.4 degrees.

However, the CFD doesn’t put into consideration the solar radiations allowed to penetrate the patterns. That’s why it was important to understand the solar radiations allowed through the pattern by using grasshopper software as Ladybug and honey-bee. To estimate the amount of solar radiations entered through the component daylight analysis was made to estimate the amount of lux that go through the pattern.

However, based on the literature review evaporative cooling efficiency in cooling the indoor temperature decreases noticeably when the evaporative cooling is exposed to solar radiation as it accelerates the evaporation of the ceramic. And mashrabiya is a shading element, that’s why when evaporative patterns are used it suggested to be in a shaded place.

And this explains why the main implementation of evaporative cooling in vernacular architecture was through adding the ceramic pot not through using evaporative patterns.

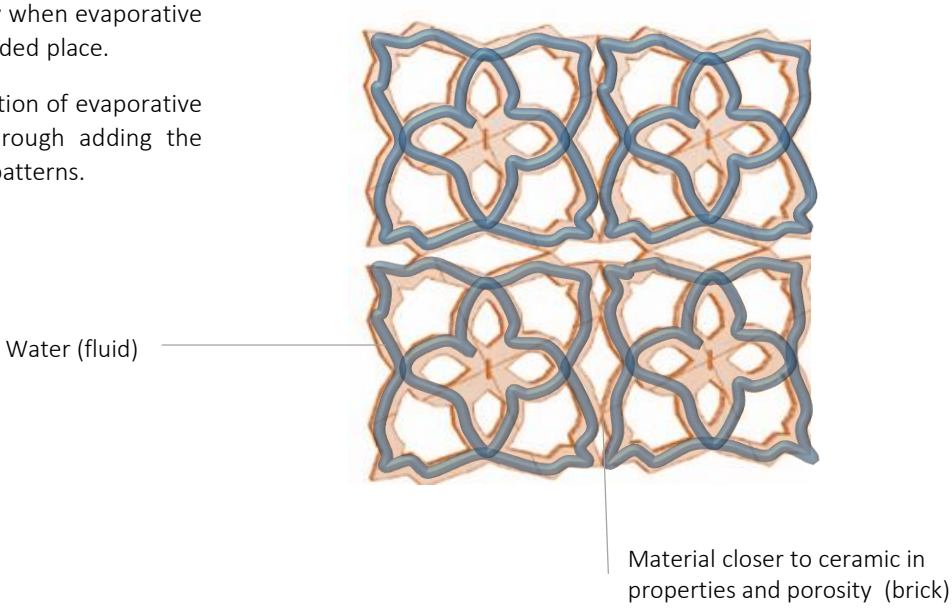


Fig 5.3.19 sketch introducing how the ceramic pattern was modelled in CFD .
source (author)

Outcomes:

Despite the software limitations that was a prior against making a full understanding of how the evaporative cooling can affect the total performance of Mashrabiya as a component. The CFD software managed to understand and simulate how the air will perform through the space whenever it was an evaporative cooling element placed in front of the main inlet or the pattern itself acts as an evaporative cooling element.

From CASE 1: it was found that it cools down the place up to 2 degrees as an average but the ceramic pot tends to act as an obstruction against the air coming from the main inlet.

And from CASE 2: it was found that evaporative ceramic patterns have much higher cooling effect since the area of the exposed surfaces of the evaporative element to air flow is much more than CASE1.

Its efficiency managed to reduce the average room temperature 2 degrees lower than the first case. Despite the software limitations in simulating the evaporative cooling process, the efficiency of the wide perforated patterns with lower surface temperature managed to affect the indoor conditions significantly. This might be conducted to the perforation percentage of the patterns as it allows more air to flow through it.

But adding water inside the pattern affected the density of the air entering the space and reduce the air flow rate the the end of the room to 0.003 m/s which is less than the case before.

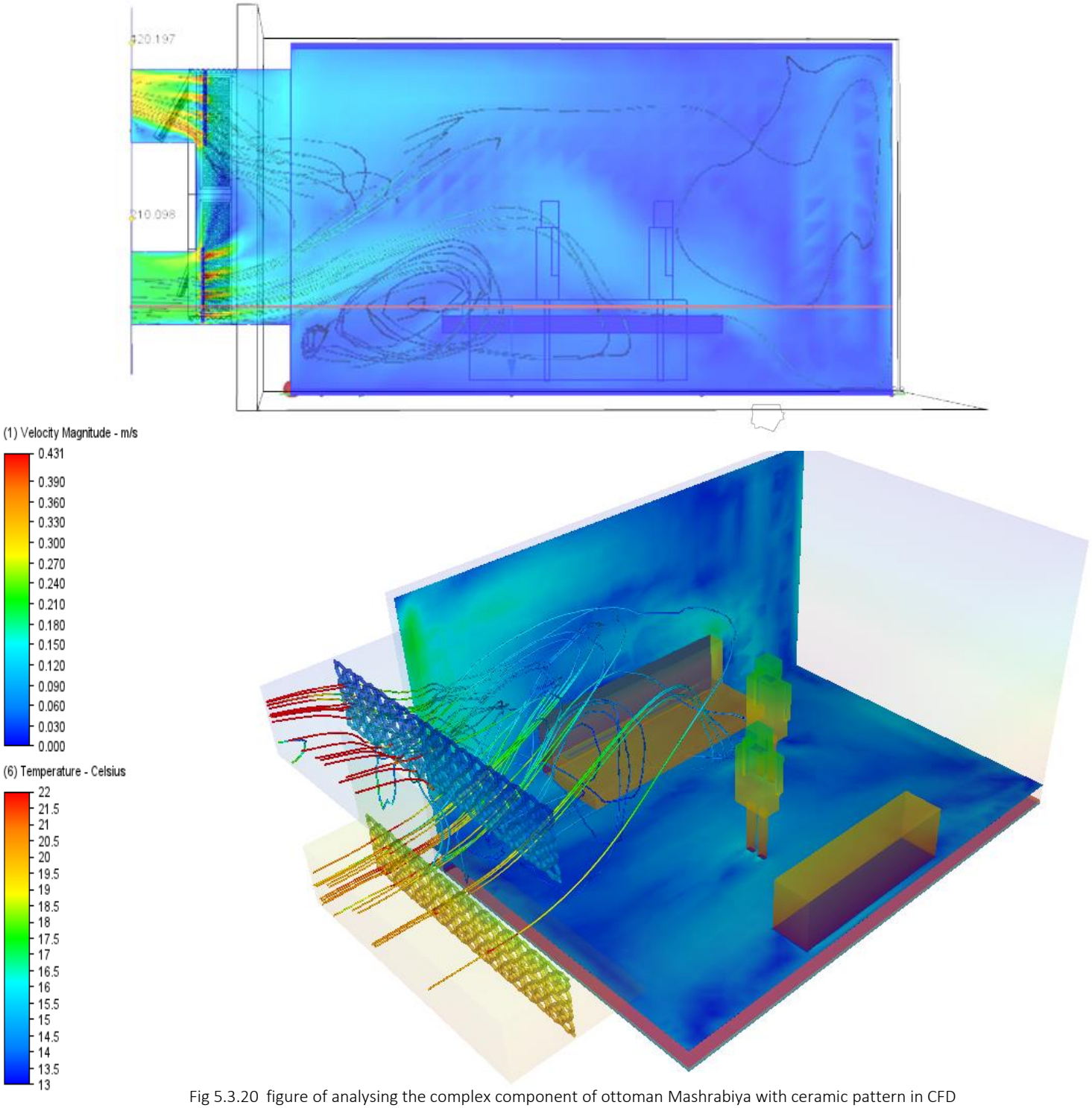


Fig 5.3.20 figure of analysing the complex component of ottoman Mashrabiya with ceramic pattern in CFD source (author)

5.3..2 The performance of case 2 in terms of shading:

Adding a high perforated patterns in the top and bottom of the component geometry increases the efficiency of the performance of mashrabiya in terms ventilation. This can improve its impact up to 50% from its original performance.

For that it was important to understand and estimate the amount of solar gains resulted from such combination of patterns, however according to the literature review this method is considered to be applied on its own or combined to other patterns bur no cap or any external part is added.

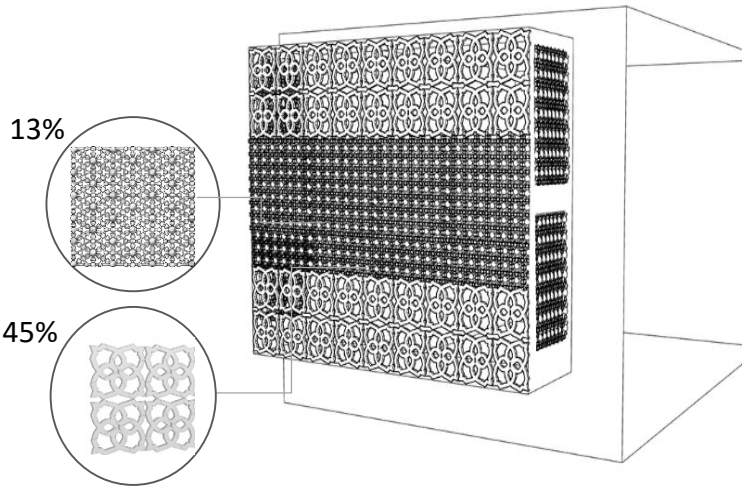
This was the reason that limited its implementation to be used only in indoors and semi indoor (transitional) space only. As the efficiency of the evaporative cooling reduce majorly in the presence of solar gains as it accelerates the evaporation process without giving enough time interval for cooling the space. (KOUJAN, 2018)

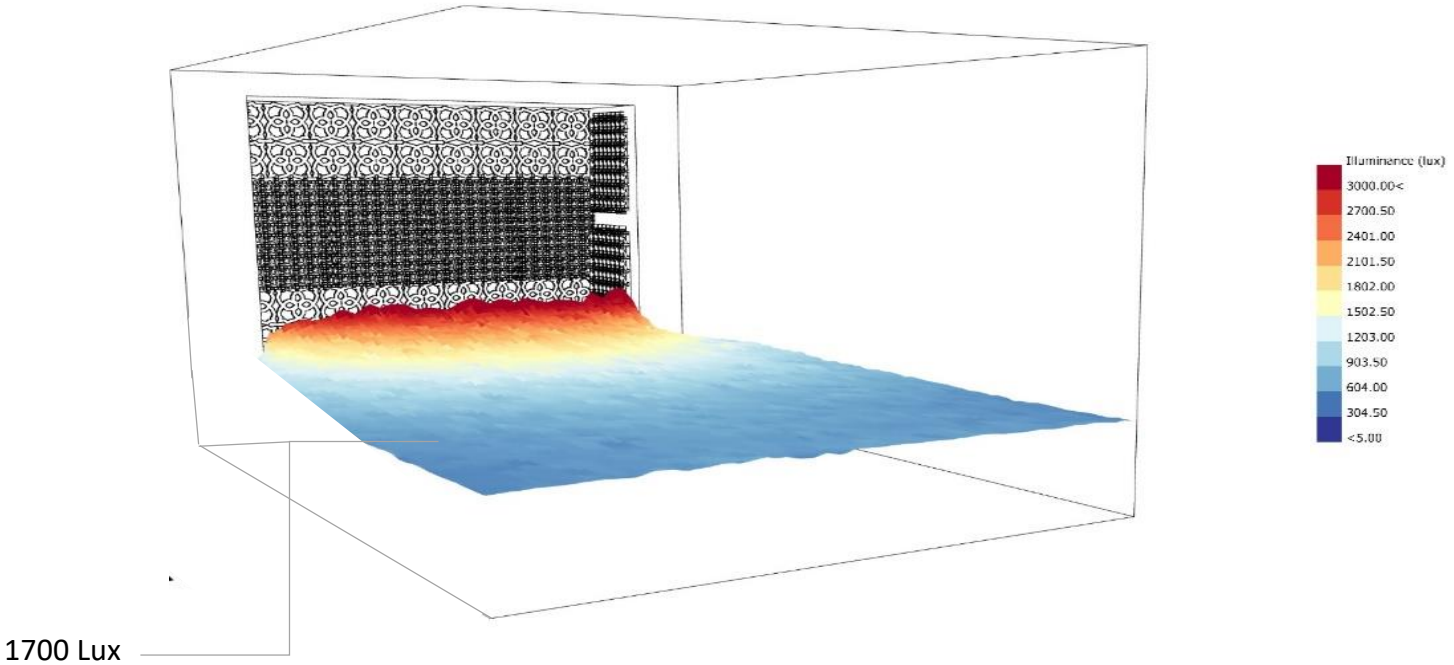
For that the ecooler pattern was modelled in rhino and applied to an ottoman mashrabiya component. The pattern was added to the bottom and the top parts of the component. The dimensions of the patterns was preserved in order to provide realistic results. For that the pattern was added to the top and the bottom and then the whole component was scaled to adjust to its dimensions.

- **Outcomes:**

The outcomes of testing this case were quite expected, as it was concluded before from testing the arabesque patterns in the top part of ottoman mashrabiya without adding external shading to it.

The daylight (illuminance) analysis showed that this combination of patterns allows excessive amount of light to penetrate through the whole component, the resulted average illuminance from this combination was 1700 lux which is considered undesirable in extreme hot climates like Egypt.





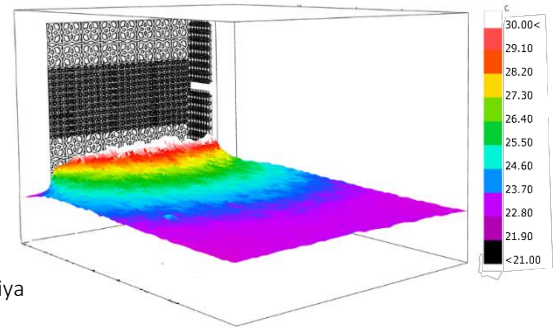
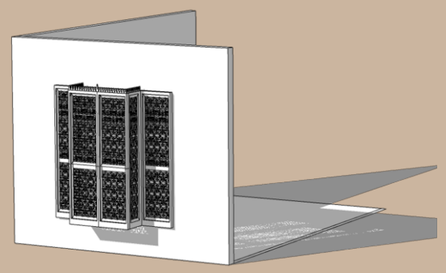
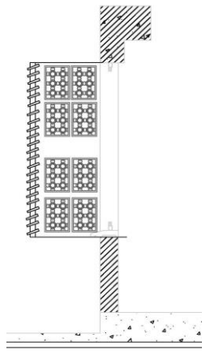
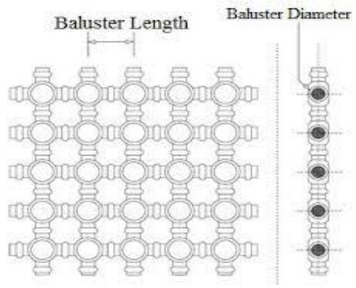
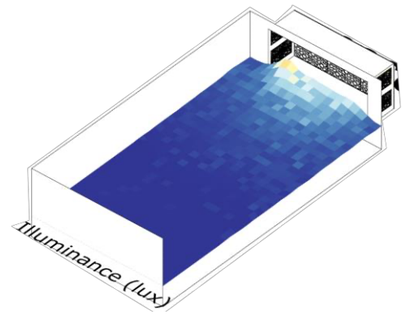
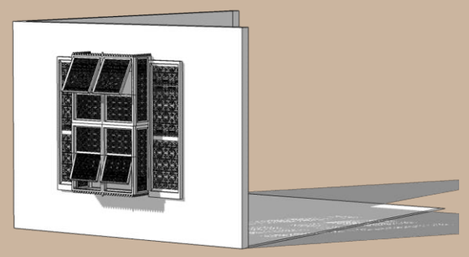
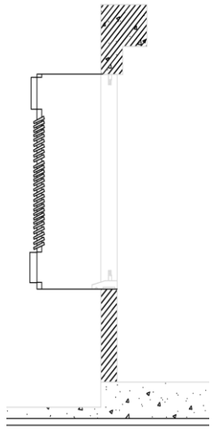
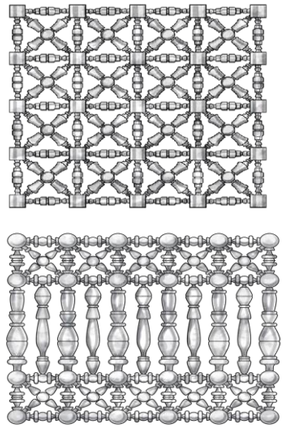
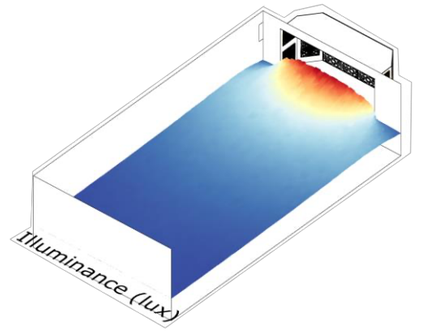
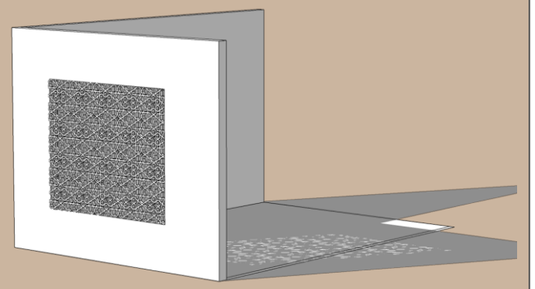
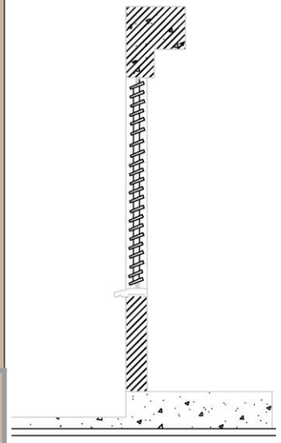
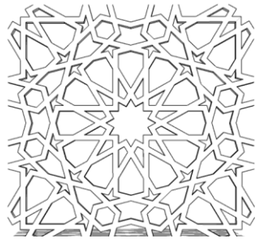
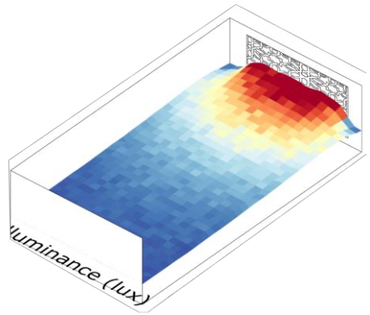
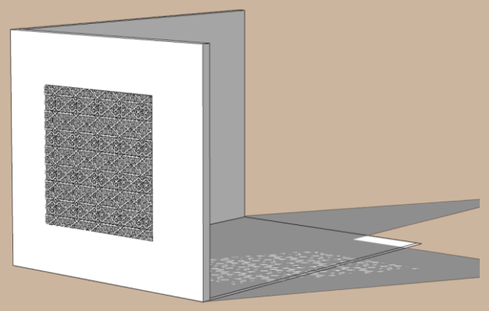
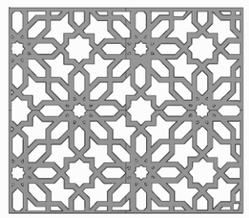
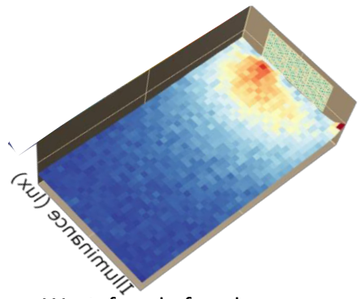
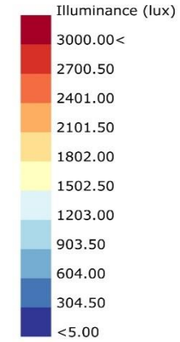


Fig 5.3.18 figure of analysing the complex component of ottoman Mashrabiya with ceramic pot in Grasshopper (illuminance and thermal analysis)
source (author)

Fig 5.3.19 summary table that gathers all the analysis and studies made
source (author)

			Percentage of perforation is integrating from 17 to 47 % on the whole screens	 West façade façade
			Percentage of perforation is integrating from 90% on the top 70% in the bottom and 14~20 % in the middle	 West façade façade
			Percentage of perforation is 42% on the whole window which gives area of aperture of 1.8 m2	 West façade façade
			Percentage of perforation is 31% on the whole window which gives area of aperture of 2.82m2	 West façade façade



A. Dynamic Thermal analysis (TAS):

This analytical tool was used to estimate how the performance of the main typologies of Mashrabiya contributes to indoor comfort. AS was the most suitable tool for such analysis as it estimates the amount of solar gains generated from the patterns and how they affect the average comfort hours inside the space. Modelling Mashrabiya in TAS was quite challenging, because of the software limitations the patterns couldn't be modelled in its exact form. And there were two methods to model the mashrabiya is such a software:

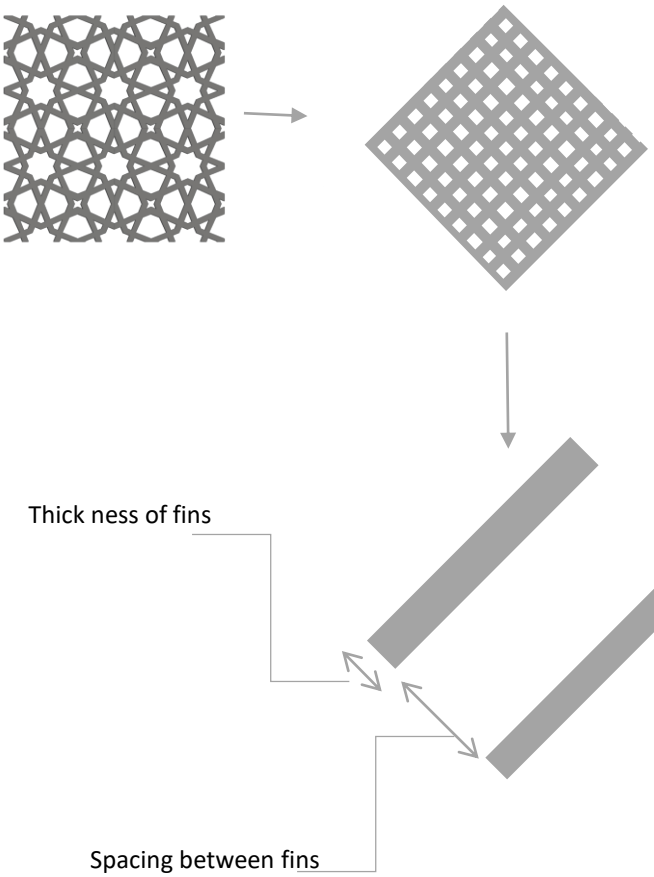
1. Calculating the rough effective aperture area resulted from a pattern and model it as an opening that will be fully (100 % opened all the time. But this will neglect the true performance of mashrabiya as a shading element and provide inaccurate estimation of its solar gains.

2. Modelling mashrabiya as shading element that have - opposite direction-inclined fins intersecting with each other, this can give a rough image of the realistic form of mashrabiya.

The second method was more accurate to be followed; after that, only two scenarios were modelled in TAS.

Case 1: Andalusian mashrabiya with perforation of 47%

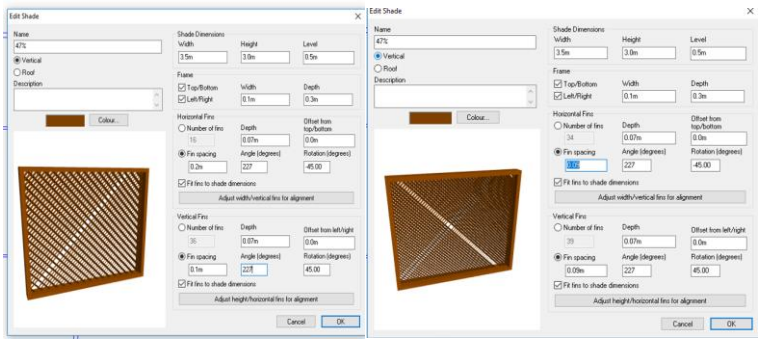
Case 2: ottoman mashrabiya with the combination of patterns These cases were compared with a normal shading element added to the same size of openings.

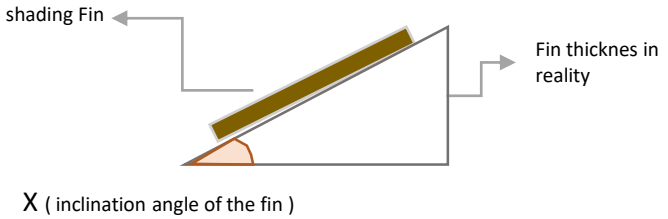


F.1: The modelling process and inputs:

The patterns was modelled from shading fins according to the following steps:

1.Calculating the effective aperture area created by the perforated pattern and start to estimate the number of fins required to cover the window to give a close perforation percentage
2. The shading fins in the TAS software doesn't have a thickness, so the thickness of the fins was defined through its inclination. By using the sin of the inclination angle to know what angle the fin should be inclined with to give the needed thickness.
3. After giving thickness to the fins, the spacing between the fins was defined.





The precentage of perforation in thermal dynamic analysis modeling depends on the following variables:

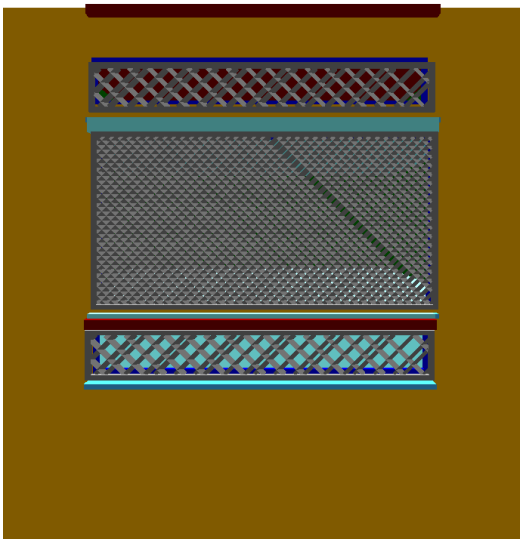
1-space between each fin
2- the angle of the inclination of the fin which can identify the thickness of the fins
3- rotation angle of each fin

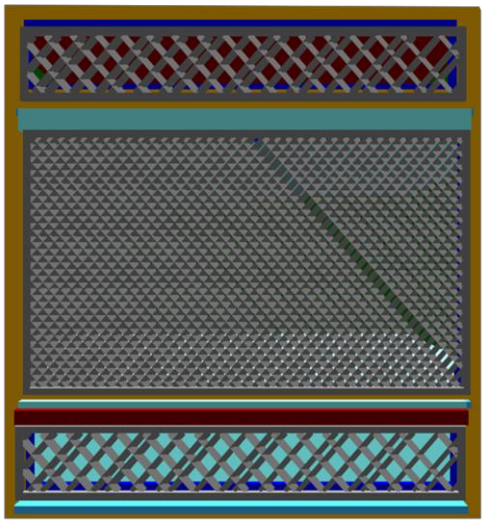
The sine of the angle = $\frac{\text{the length of the opposite side}}{\text{the length of the hypotenuse}}$

The cosine of the angle = $\frac{\text{the length of the adjacent side}}{\text{the length of the hypotenuse}}$

Fig 5.3. A.1 steps of modelling perforated screens in TAS
source (author)

4. The room mashrabiya was added too was the same example followed in all the previous cases with same dimension of opening.
5. No internal conditions was considered, as the main purpose was understanding the difference in solar gains in each room and how they affect the indoor resultant temperature.
6. The weather file applied to this analysis was the same weather file applied to all the cases before (Cairo, Egypt).
7. For modelling stack single sided ventilation (case 2) 3 floors were created. The height of each floor represents the height of each part in the component. The patterns applied to the top and the bottom had perforation percentage of 50 %and the pattern in the middle have perforation of 19 % The height of the top and bottom parts was 0.5 meters. The height of the middle part was 2 meter.
8. The floor and ceiling in case 2 for the middle part were assigned as null in order to make the software realize the stack ventilation process.
9. The same thing occurred in the floor of the top part and the ceiling of the bottom part of the component.
10. The calculation of the air flow rate in case 2 was a Sumiton of the air flow rates in the three zones .
11. All these shading element were applied to a window that was assigned in the TAS to be 100 %opened all the time and oriented to the south.





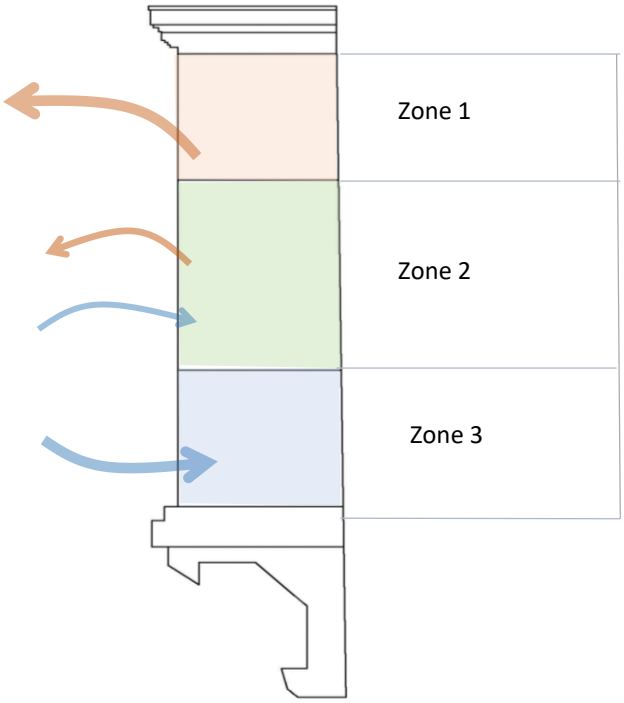


Fig 5.3. A.3 sketch that shows the concept of modelling the screens in TAS source (author)

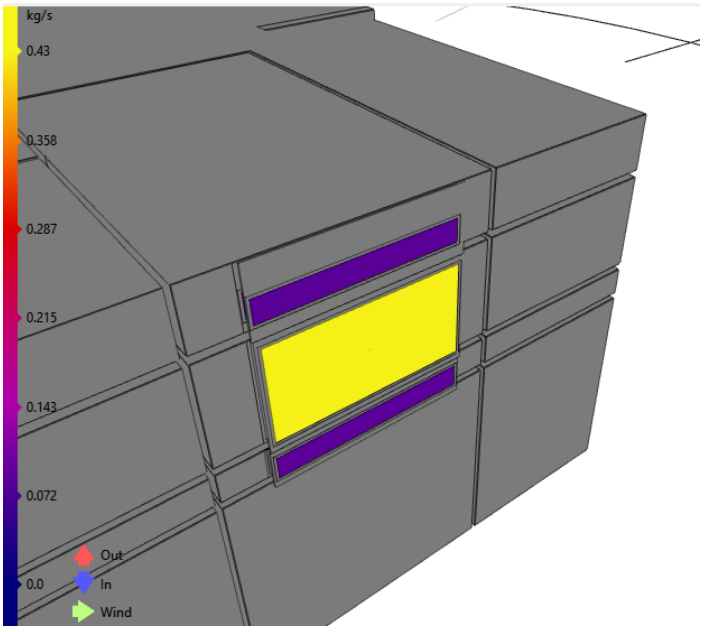


Fig 5.3. A.2 steps of modelling simplified ottoman mashrabiya in TAS source (author)

The aperture flow in (kg/s) from the TAS results according to the previus inputs in each case 21st of june, comparing them together in a graph in order to understand what scenario preforms better.

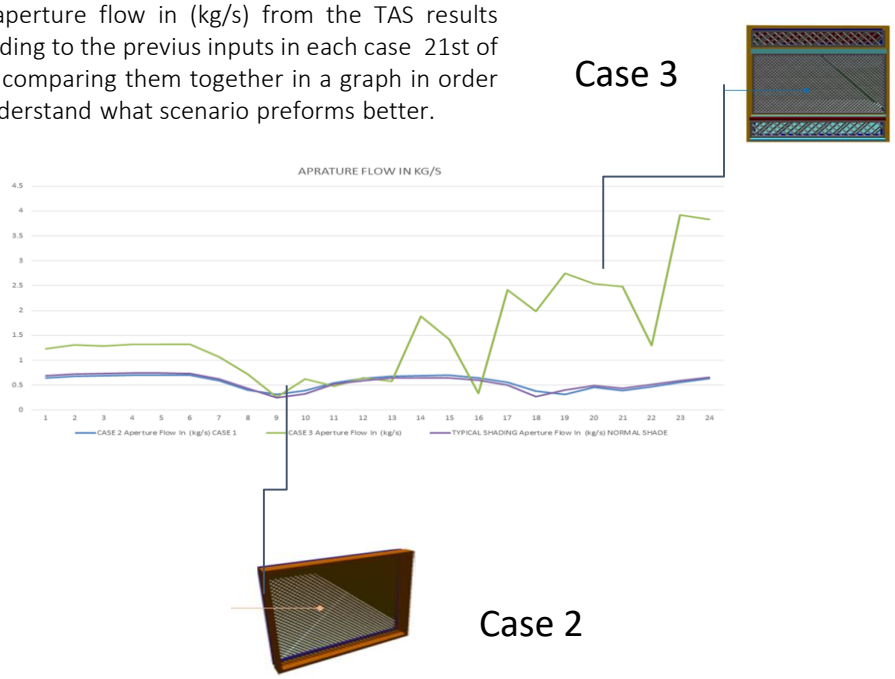
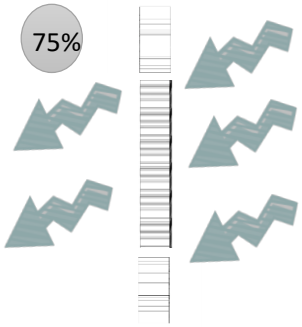
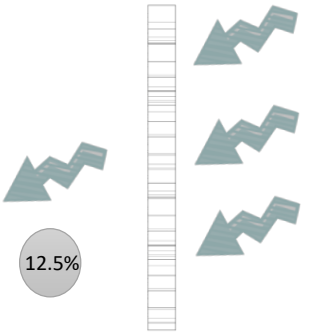


Fig 5.3. A.3 final result of aperture flow in TAS for the three cases
source (author)

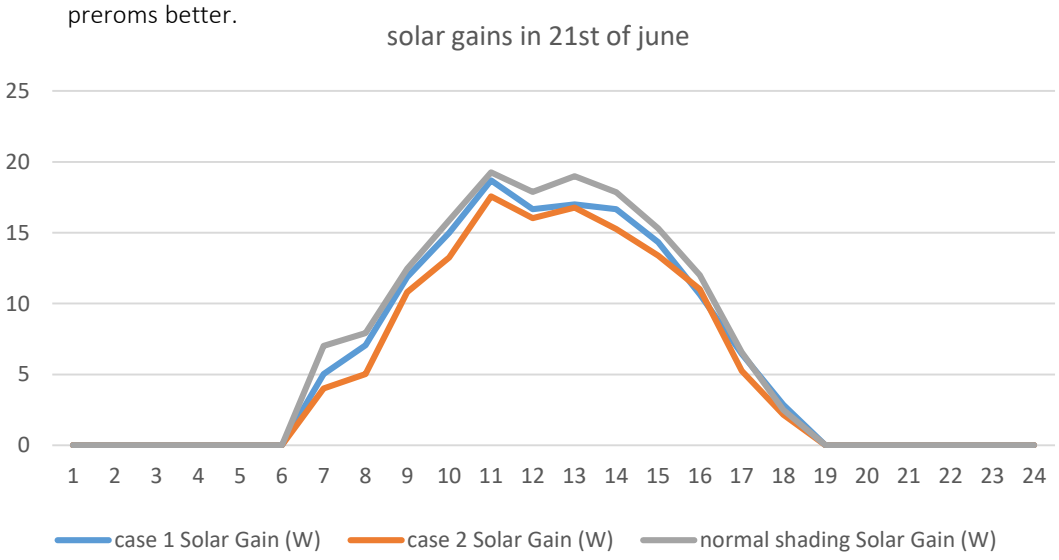


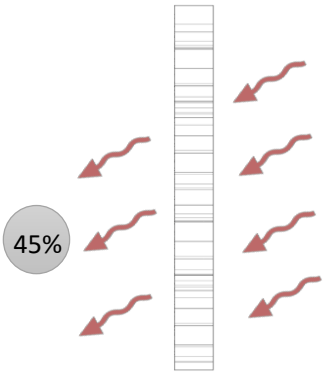
The ammount of solar gains pass throught case 3



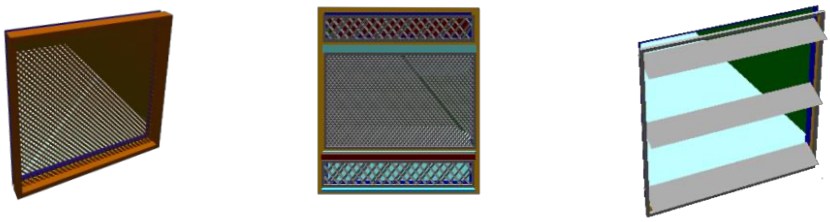
The ammount of solar gains pass throught case 2

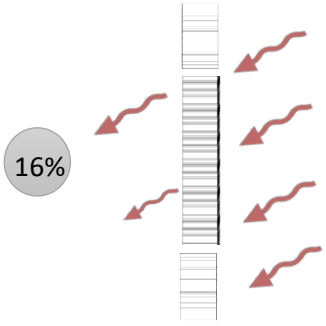
The solar gains resulted from each scenario, comparing them togother in a graph in order to understand what secnario preroms better.





The ammount of solar gains pass throught case 2





The ammount of solar gains pass throught case 3

Fig 5.3. A.4 final result of solar gains TAS for the three cases
source (author)

- **Outcomes:**

1. Case 2 performance in terms of ventilation is up to 70% better than case 1.
2. The average resultant temperature of the room in case 2 was 2 degrees above the comfort according to NICOLE comfort band calculated for Egypt.
3. The solar gains in case 1 and case 2 are almost the same.
4. Although case 2 had a less percentage perforated patterns in the middle, but the high perforated patterns applied to the top and the middle managed to reduce the efficiency of the component in terms of shading. Because of software limitation no external shading was added to the top part. It was expected that adding it could have improved its efficiency a lot.
5. Both cases of mashrabiya preforms better in terms of shading than using standard shading elements, this might be conducted to the efficiency of mashrabiya in shading even at lower sun angles.

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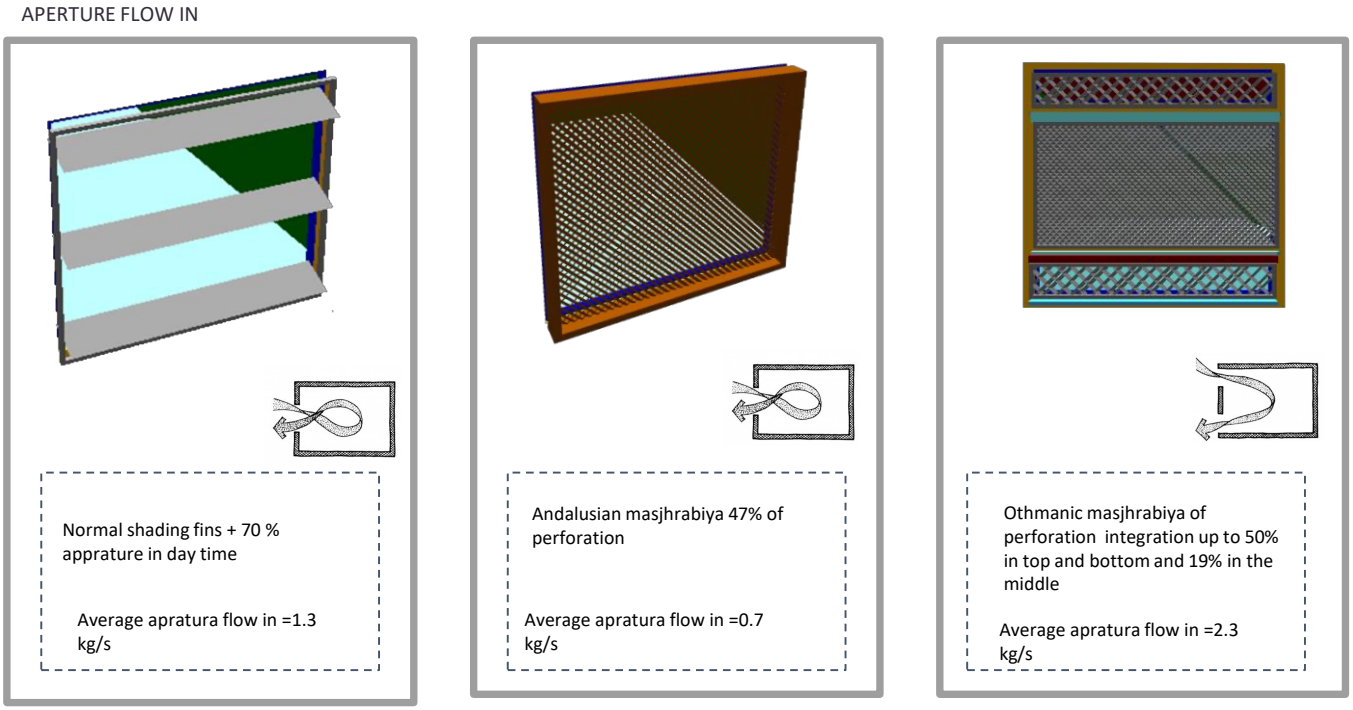


Fig 5.3. A.5 final result of aperture flow in TAS for the three cases source (author)

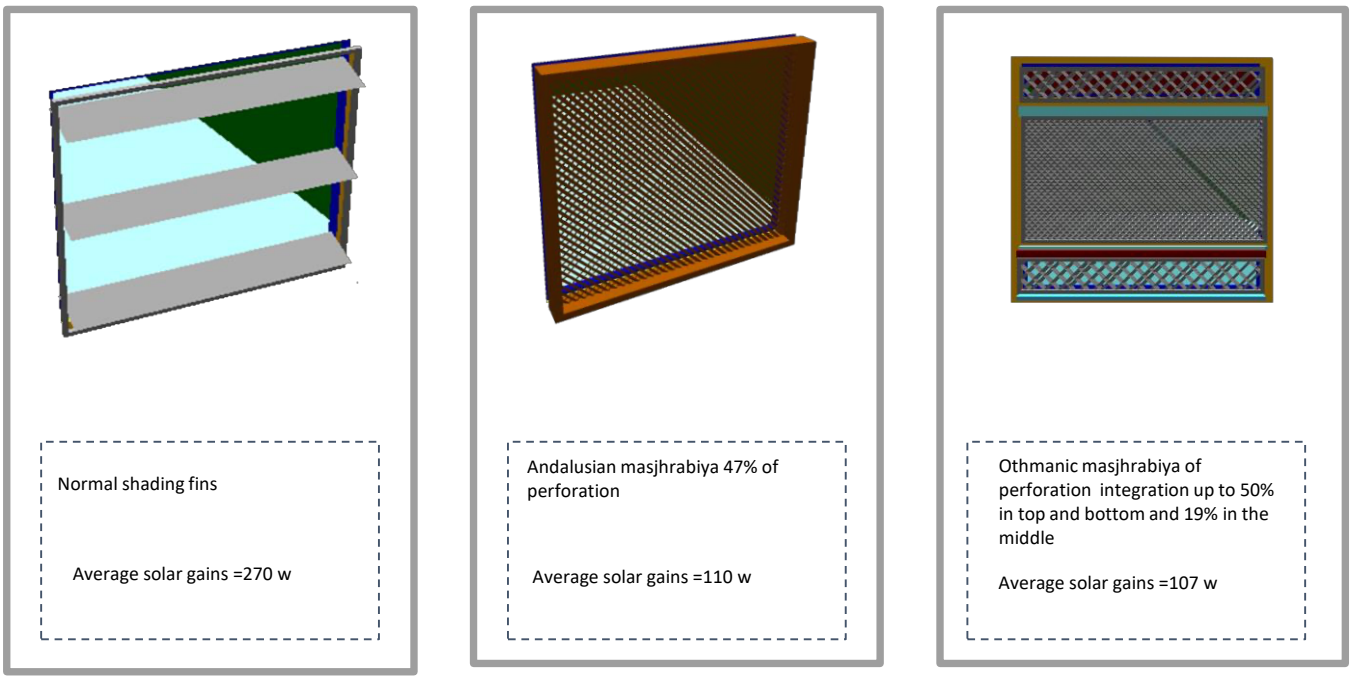


Fig 5.3. A.6 final result of solar gains TAS for the three cases source (author)

5.4.Site level

As a testing step to have a general look on the performance of the living cases examples through understanding its effect on the indoor thermal comfort with considering the context

3. Site level

4.1 Introduction

After analysing the Meshrabiya as a component and studying the impact of its details. It was important to understand how the Meshrabiya preforms in terms of site, with putting into consideration the surrounding context that have a major impact on its performance. For that a site was selected from the field work and ones space in this site was chosen to be analysed.

The site and the space that were selected :

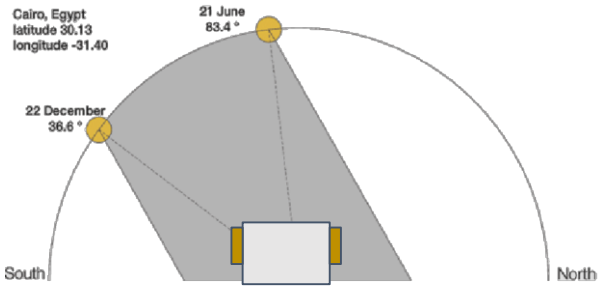
location	Al Moeez street-Cairo-Egypt- latitude 30.0444° N, 31.2357° E
Description of the space	Living space in an extended family residential block that was built in 1709 and refurbished in 1994, the house is still used by the same family as heritage inheritance between the family members.
Area of the space	70 m2 (7m X 10.5m)
Function of the space	Living space
Typology of Mashrabiya applied to it	Ottoman semi rectangular octagon.
Orientation	Western facade oriented space

The main challenge is this site is that the main space, which is the living space is oriented to the west. This site was found during the fieldwork, but it wasn't accessible, there were a lot and critical hypothesis made on the existence of a big opening on the western façade. So it will be useful to understand how mashrabiya performs in this case. The space selected for the analysis is a living space in a big family residential house. The space was mainly used for praying reading and discussing holly Quran and religious books, for that I was required to achieve a precise balance between light and shade, so the light can be distributed equally inside the space but doesn't exceed the proper amount to avoid solar gains resulted from it.

Neighbouring parts of the building

The space selected for the analysis

The ground floor and the entrance of the site



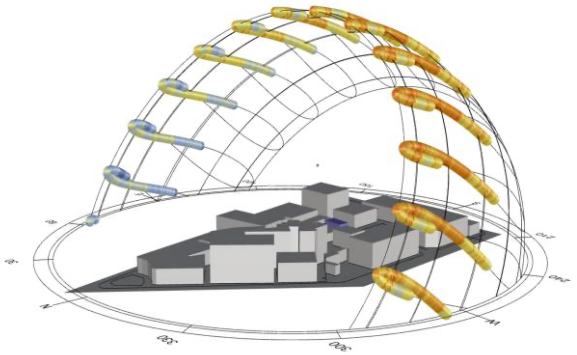


Fig 5.4..1 site analysis-cairo-Egypt
source (author)

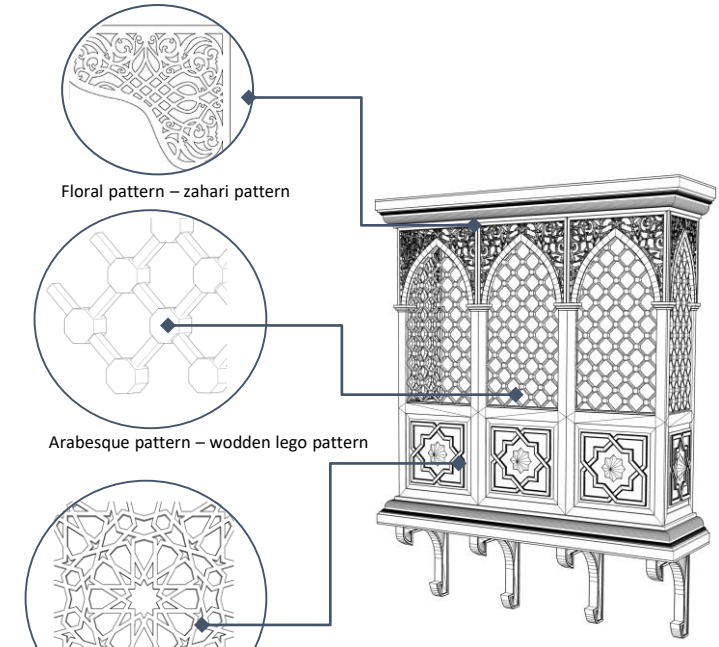


Fig 5.4..1 component analysis
source (author)

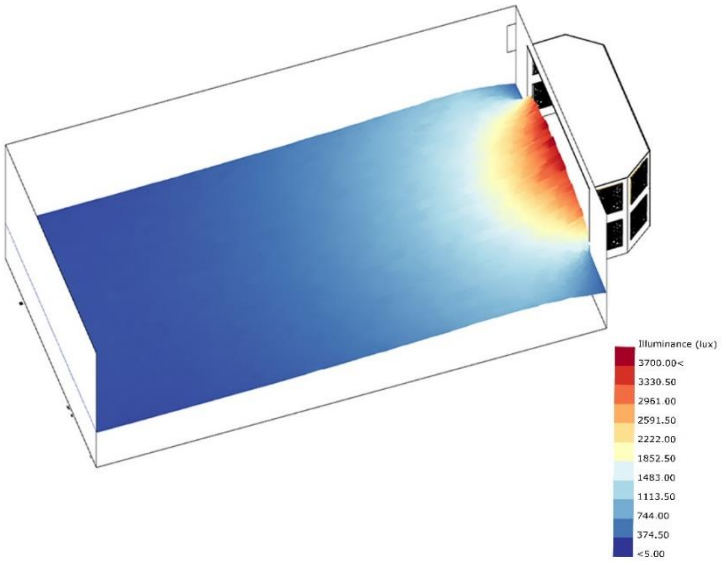
Fig 5.4..1 site analysis-cairo-Egypt
source (author)

4.2 Outcomes:

Space was tested on a typical summer day (21st of June 13:00) The living space was analysed with and without mashrabiya, and based on that the following results were found: The average illuminance that enters the space without mashrabiya added to it, putting into consideration the surrounding context and buildings, was approximately 3070 lux, which is not desired in such a hot climate like Egypt. Although the average illuminance was high, some parts received less than 50 lux. This shows that there is a poor and insufficient light distribution inside the space.Adding the ottoman mashrabiya that was originally applied to the site managed to reduce the average illuminance inside the space to become 1790 lux, which is acceptable in such space typologies according to the Egyptian architectural code. Although the average illuminance was not high, the lowest illuminance at the end of the room was 307 lux.

4.3 Conclusion:

From this case, it was concluded that Mashrabiya performance in terms of light is not limited to shading and protecting the space from excessive solar gains. Mashrabiya can also provide fair and equal light distribution through space. That's why mashrabiya was widely implemented in mosques as mosques spaces typology required equal light distribution through the whole space.



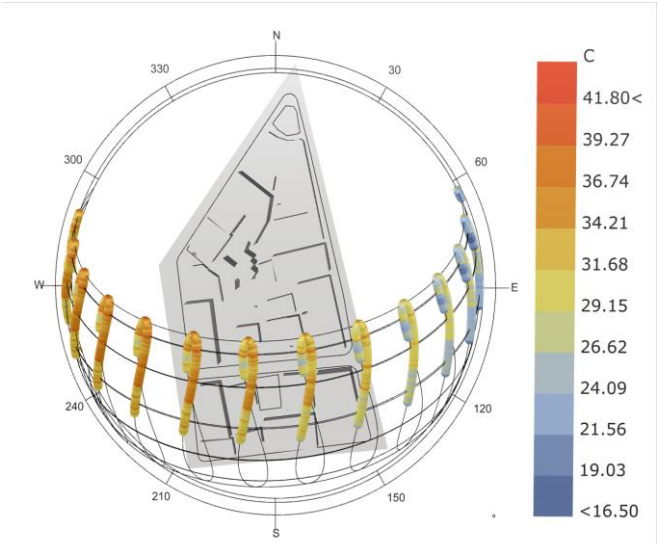


Fig 5.4..5 site analysis-sunpathe
source (author)

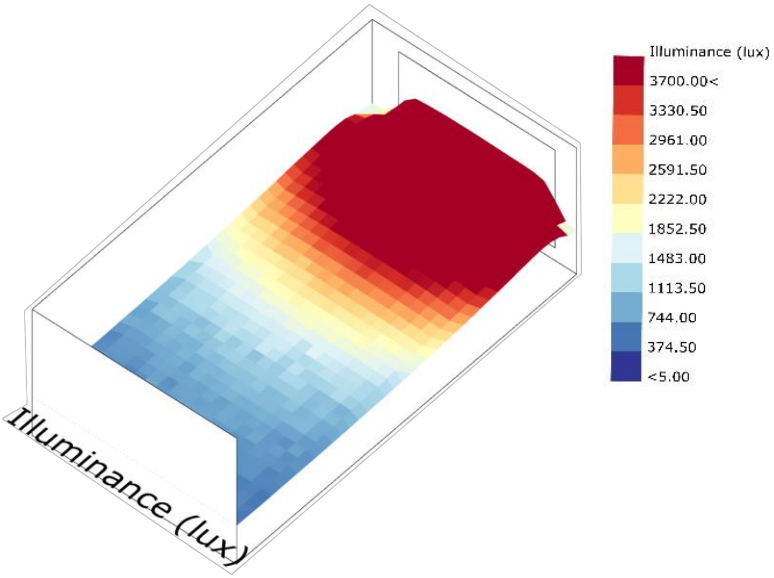


Fig 5.4..3 Illuminance analysis for a typical window with no shading
source (author)

Sun position digram for summer

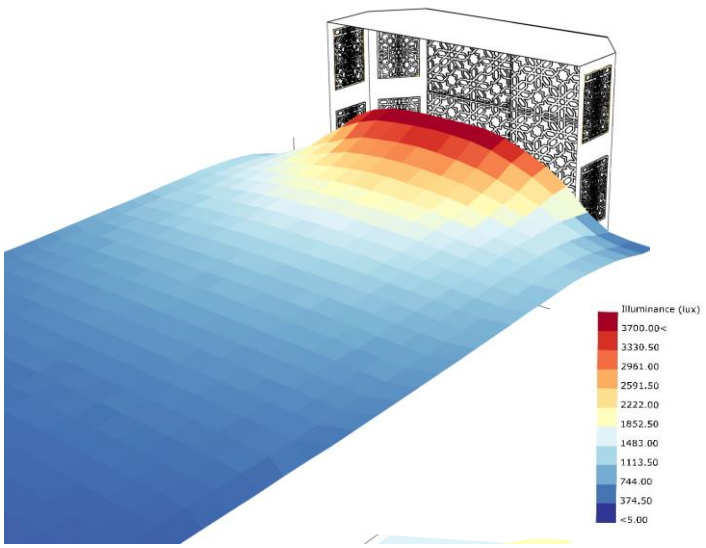


Fig 5.4..4 Illuminance analysis for a typical window with mashrabiya
source (author)

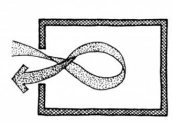
5.5.Conclusion

From all the parts of the analytical work, it was found that mashrabiya within its simplicity as a solution there is a lot of complexity in its performance. To understand and quantify its performance a lot of analytical tool were used, and despite software limitations, a total figure was formed about the performance of mashrabiya. This analytical work managed to show us how mashrabiya varies significantly in its environmental performance from the smallest change in any of its details.

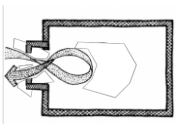
That's why it was important to make a full understanding towards the details of the mashrabiya. Despite the complications in analysing the details of the components that form each pattern, this step was essential to understand and compare the main details of mashrabiya.

Regarding, the macro level analysis it was found that combining patterns is significantly influencing the total performance of mashrabiya in both: ventilation and shading. This can be considered as a double edged tool , that can improve and strengthen the performance of Mashrabiya, or drain and demolish the strength points that exist in mashrabiya as a passive solution.

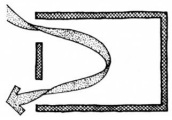
For that, design guidelines and recommendations must be provided to encourage the implementation of mashrabiya in contemporary designs with the most accurate and precise methods.



- Andalusian mashrabiya (un extruded patterns) allows more light inside the space and its performance in terms of ventilation is sufficient since it uses more perforated patterns.



- Ottoman mashrabiya (extruded patterns) performance in terms of ventilation is sufficient despite using less perforated patterns but adding patterns in the extruded geometry increases the effective aperture area , provide self shading and surface temperature variation which motivate the air to move more inside the space.



- Adding stack height effect to the component improves the air flow ratio through the space up to 45%

- Adding a ceramic pot in front of the inlet of the component reduces the resultant temperature till 3 degrees but the air movement through the space becomes less

- Using ceramic perforated pattern is un desired since the pattern is exposed to solar radiation

5.Conclusion

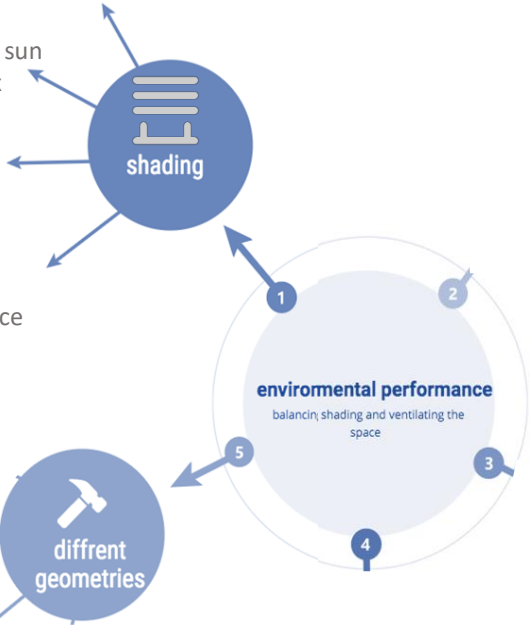
Based on the analysis made in both micro and macro level the following was concluded

- Depend on the geometry of the component and the percentage of perforation

- The performance of the patterns decreases at low sun angles except for some patterns the have complex components

  - Some patterns are not preferred for some facades unless additional element is added

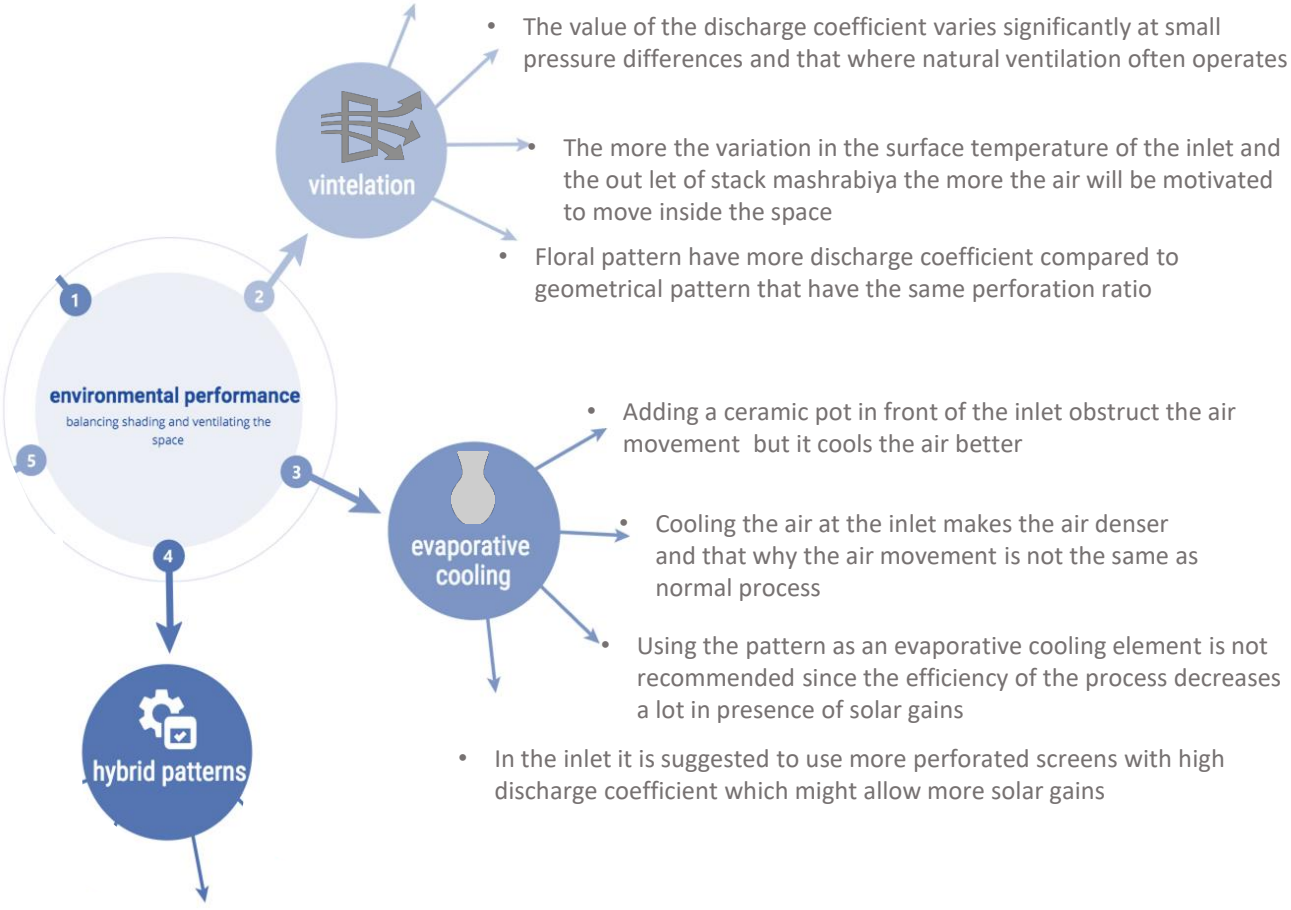
  - Some patterns have the same perforation percentage but different shading performance



- Extruded components provide self shading and indoor transitional spaces that can be used for various functions

  - Cap can be added in the top of the component to provide shade on the wide perforations in the top

- Discharge coefficient is variable from a case to another according to the influencing condition but it compared the performance of different patterns in the same conditions



- The value of the discharge coefficient varies significantly at small pressure differences and that where natural ventilation often operates

- The more the variation in the surface temperature of the inlet and the out let of stack mashrabiya the more the air will be motivated to move inside the space

- Floral pattern have more discharge coefficient compared to geometrical pattern that have the same perforation ratio

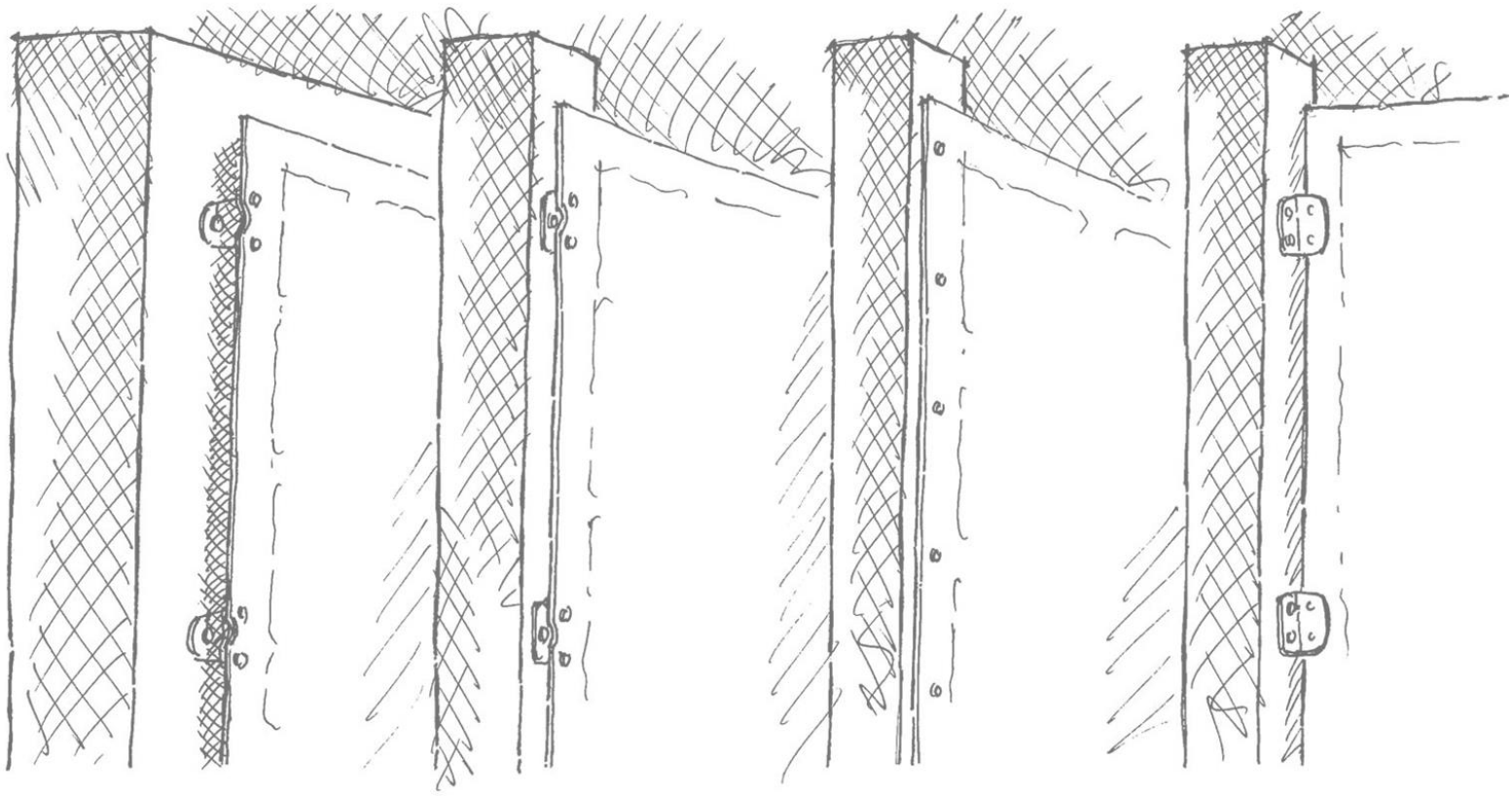
- Adding a ceramic pot in front of the inlet obstruct the air movement but it cools the air better

- Cooling the air at the inlet makes the air denser and that why the air movement is not the same as normal process

- Using the pattern as an evaporative cooling element is not recommended since the efficiency of the process decreases a lot in presence of solar gains

- In the inlet it is suggested to use more perforated screens with high discharge coefficient which might allow more solar gains

- Wide perforated patterns are commonly used in the inlet and the out let to allow more air flow where the middle of the window very narrow perforated patterns are used to prevent the undesired light and solar gains



6. **Design guidelines**

Design recommendations based on the analytical study of Mashrabiya

6.1 Introduction

In this chapter some recommendations and suggestions will be provided, this analysis is based on the analytical work analysis made to quantify the performance of different patterns and understanding the performance of these patterns in different geometries and combination.
From the literature review, fieldwork and analytical work, it was found that the perforated screens as an idea are generated from transforming a one fixed horizontal shading element. For shading a window from high sun angles that are mostly accruing in hot summer days, a fixed horizontal shading is applied to the window. The length of the horizontal shading element depends on two main variables, which are:
1.The sun angle incidence on the façade
2.The height of the window
To give the same performance with less l horizontal shading depth, the horizontal angle can be divided into 2 parts above each other, and the depth will be divided into two as shown in the figure. For decreasing the depth more, the number of shading fins can increase, and the depth will be divided on the number and so on. The number of fins can increase, and the more the number of fins increase, the less depth is required for the fins.
This horizontal shading shade the window only for the sun angle it is designed for, a horizontal shading element can shade the sun angle it is designed for and higher sun angles, but it allows the lower sun angles.

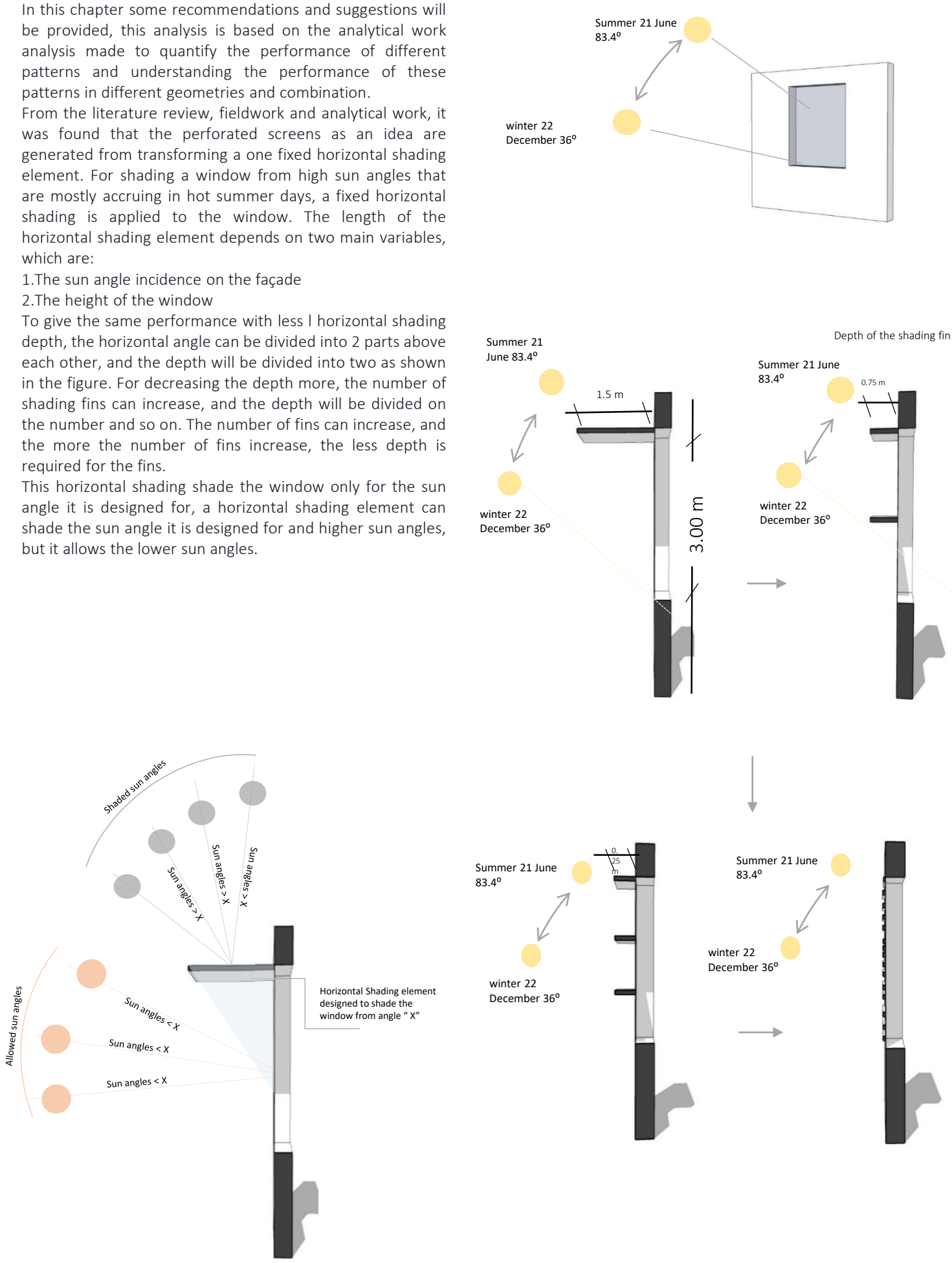
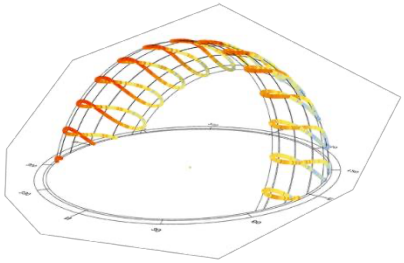


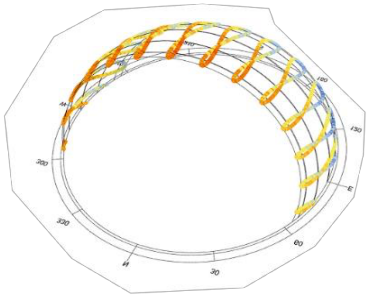
Fig 6.1.1 sketches that explain how the idea of mashrabiya was formed as a shading element source (author)

6.2 why perforated screens:

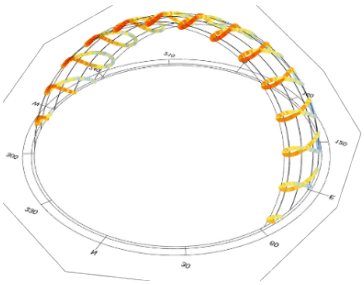
After stating how does the horizontal shading fins preforms, it was found that this typology of shading is desired in cold climatic countries as it allows the sun to enter the space in winter days, by just blocking high sun angles and allowing lower ones. But in hot climatic countries, the solar gains resulted from both low and high sun angles are not desired. For that, it was required to design a shading element that shades the window most of the year from high and low sun angles. According to the sun path diagrams shown in the figures, it is clear that the variation in the sun position, as well as the sun angles, is very small in the Middle Eastern countries. The main variations are in the sun temperatures. The similar sun positions generated similar shading solutions, which evolved through time to give birth to mashrabiya.



Saudi Arabia



Dubai



Algieri

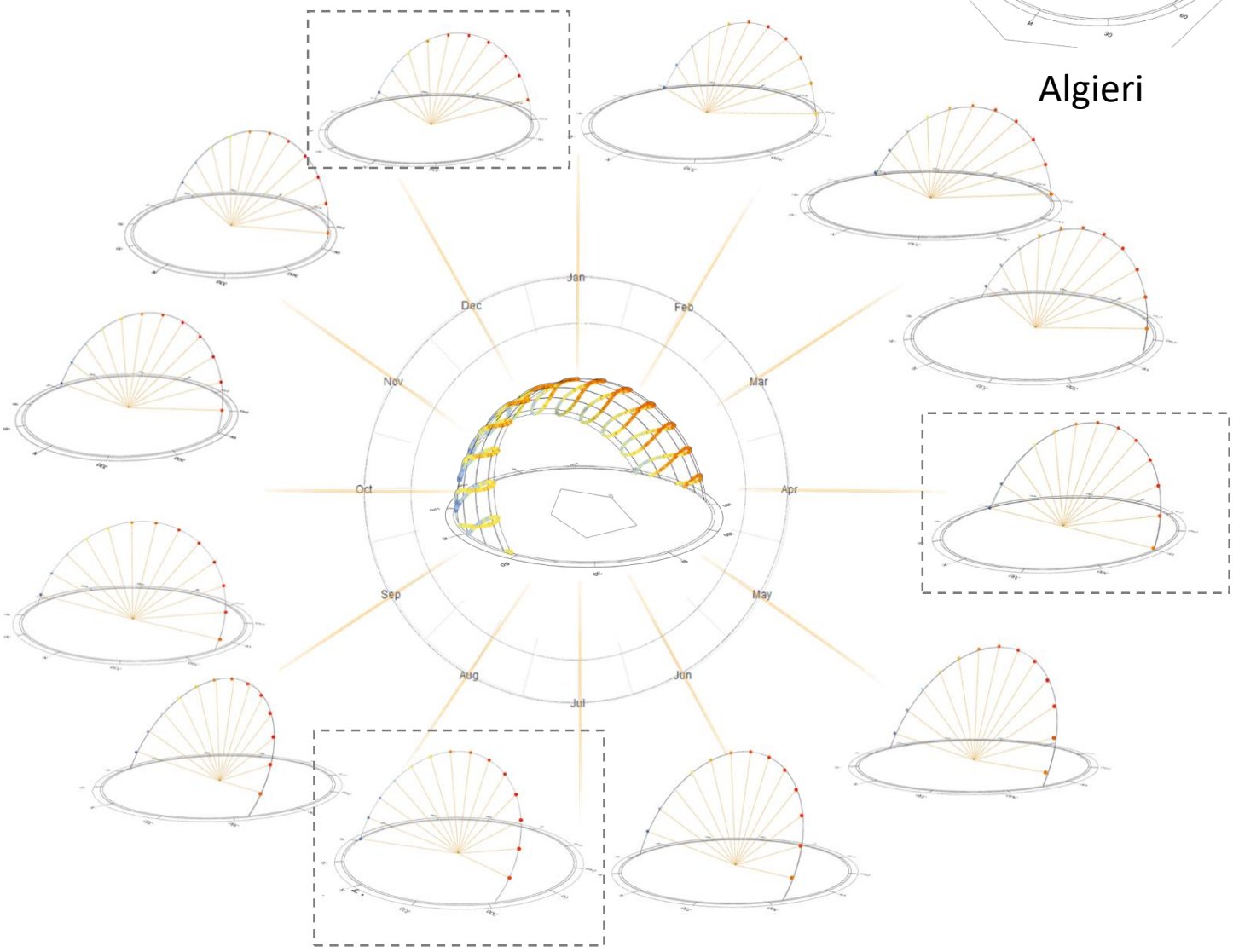


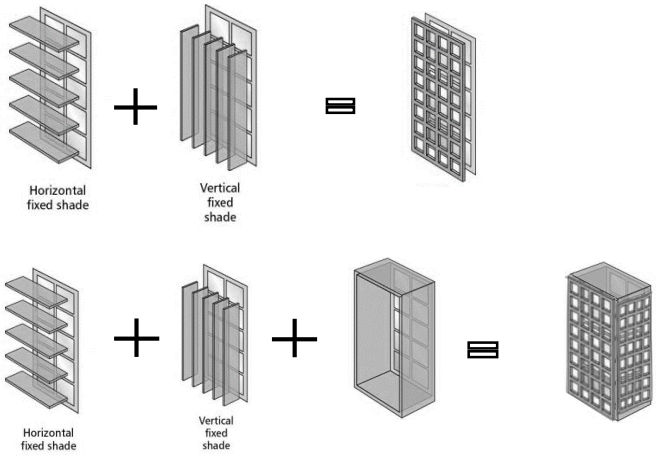
Fig 6.2.1 sun path of the whole year in Egypt+ on the right sun path of different middle eastern countries source (author)

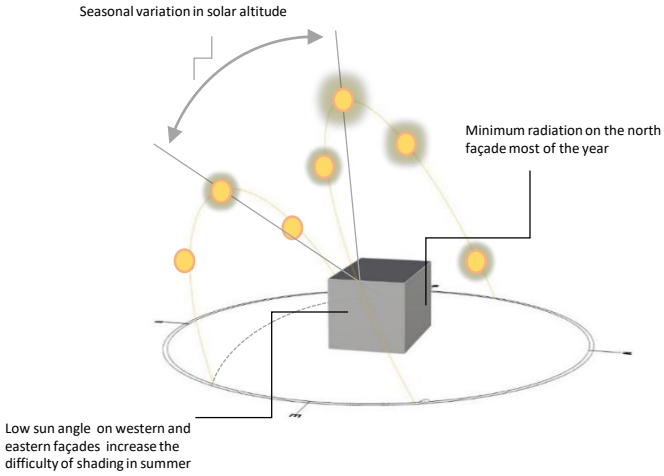
6.3 shading the western faced :

Every façade orientation requires a particular type of shading, either horizontal or vertical and sometimes both, this variation depends on the location, orientation and the function of the space. Shading a façade like the western façade is quite challenging as it requires both vertical and horizontal shadings that preform efficiently towards low sun angles. Mashrabiya is considered to be a perfect solution for shading the western façade as it is considered to be a shading element that gathers both horizontal and vertical shading and a better performance towards lower sun incidence.

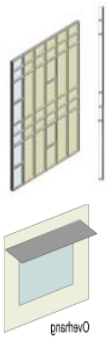
FENESTRATION SHADING DEVICE DESIGN For west facade

- EXAMPLE: Taking a window of width 1500mm and height 1800mm Shading by single vertical shade is not feasible as the shade depth becomes prohibitive.
- Hence the width is subdivided and several louvers are designed. If 6 divisions are created, i.e. effective width to be shaded is 250mm





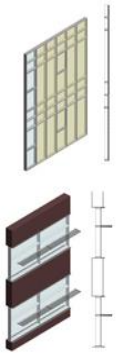
North façade shading recommendations

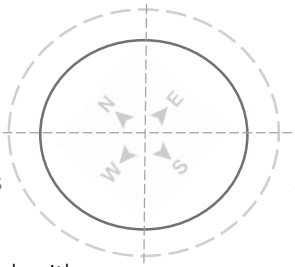


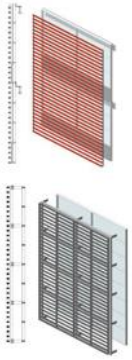
Generally not required except from low evening sun in peak summer days . when the sun path is long and hits the North.

East façade shading recommendations

Facades are challenging to be shaded with fixed shades as the sun is very low. Hence only small windows are recommended, external moveable. Shades/ rollable blinds are more effective than fixed shades. These . also help preserve the view from the windows.





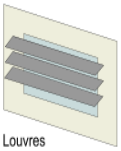


West façade shading recommendations

Facades are challenging to be shaded with fixed shades as the sun is very low. Hence only small windows are recommended, external moveable. Shades/ rollable blinds are more effective than fixed shades. These . also help preserve the view from the windows.

South façade shading recommendations

fixed horizontal device or window recessing.



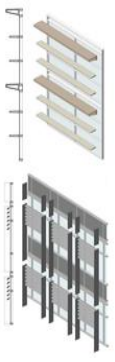


Fig 6.3.1 sketches that explain what is the required typology of shading for each façade
source (author)

6.4 Formula of perforated screens: :

To design a perforated shading screen that preforms efficiently towards sun angles instead of just implementing typical horizontal shading fins, a mathematical formula based on steps was driven.

This formula was made to link between the depth of the horizontal shading fin and the perforation percentage required in the perforated screen that will replace it.

Once the designer knows the required perforation percentage, the following matrix can be used in order to allow the designer to see the following in order:

1. Which pattern should be chosen?
2. How the patterns perform comparatively with the other suggested patterns in different environmental parameters.
3. What orientation is this pattern preferred to
4. In case of the combination of different patterns in one component, what position is preferred for each pattern?
5. What type of space is pattern preferred to?

- Relation between the depth of the horizontal shading element required to shade and the perforation percentage in the patterns.

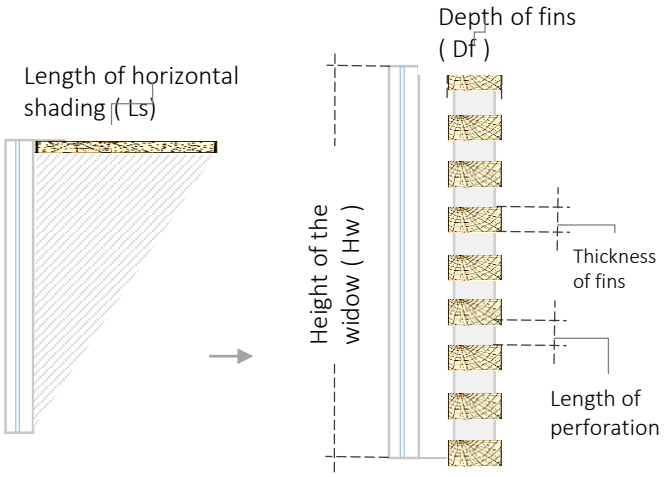


Fig 6.4.1 link between horizontal shading and perforated screens source (author)

(usually most of the cases the thickness of fins doesn't varies as much as the length between the gaps, as the length of the gaps is what varies the most between different patterns)

- Linking the depth of the horizontal shading and the percentage of perforation can be represented in the following formulas:

Depth or length of the horizontal shading element

(Ls) = depth of fins (Df) X No. of fins

- Perforation percentage of a screen =

[Total Length of the gaps / total length (height of window)] X 100

- Total length of gaps =

Height of window (Hw) –(thickness of fins X No. of fins)

- Depth of fins is usually constant between range of thickness standards according to the material
- So the dominant variable and the most effective factor is the no. of fins.
- Thickness of fins can be determined through this relation :

- $[\frac{1}{x}$ X height of the window] ÷ No. of fins

- Where X is a variable No. that depends how much allowance of other environmental parameters :

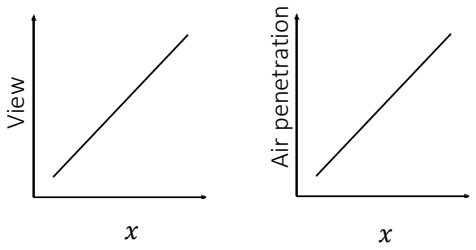
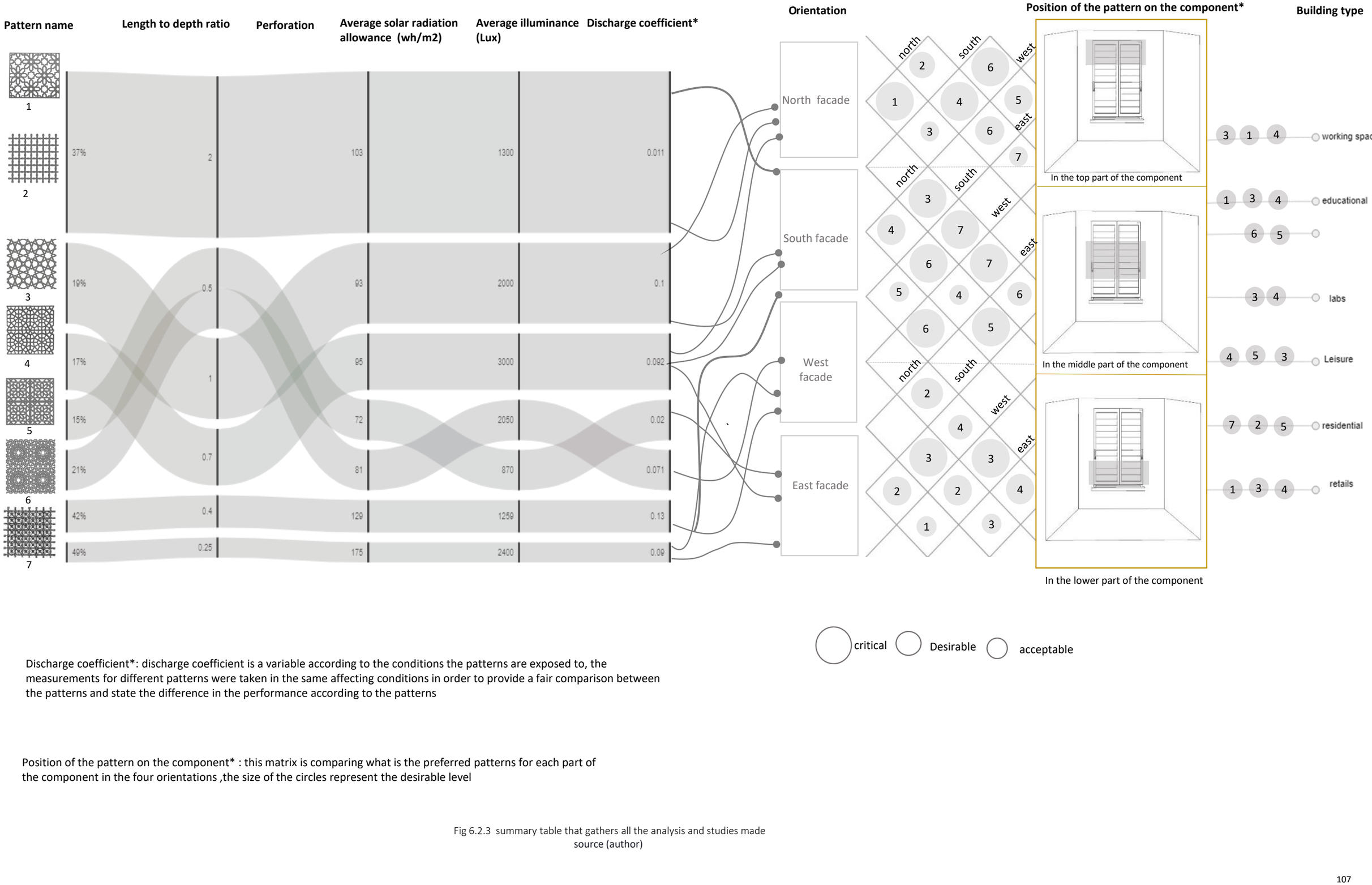


Fig 6.4.2 link between variable “x” and different parameters source (author)



Discharge coefficient*: discharge coefficient is a variable according to the conditions the patterns are exposed to, the measurements for different patterns were taken in the same affecting conditions in order to provide a fair comparison between the patterns and state the difference in the performance according to the patterns

Position of the pattern on the component* : this matrix is comparing what is the preferred patterns for each part of the component in the four orientations ,the size of the circles represent the desirable level

Fig 6.2.3 summary table that gathers all the analysis and studies made
source (author)

6.5 recommendations based on the geometry and micro level:

After that, some test were made in order to provide design recommendations in terms of geometry and patterns micro level. This is a test that was made in order to understand and based on that, recommend the suggested approaches form similar papers that went into analysing the JAALI. (Sanathkumar, 2014)

- The outcomes:

Inclining the whole component with using widely perforated screens can magnificently affect the performance of traditional perforated screens in terms of shading. However, it requires early consideration in terms of the construction process and cannot be applied to some buildings typologies with different geometries. .

Fig 6.5.1 analysing different component typologies for both north and southern facades source (author)

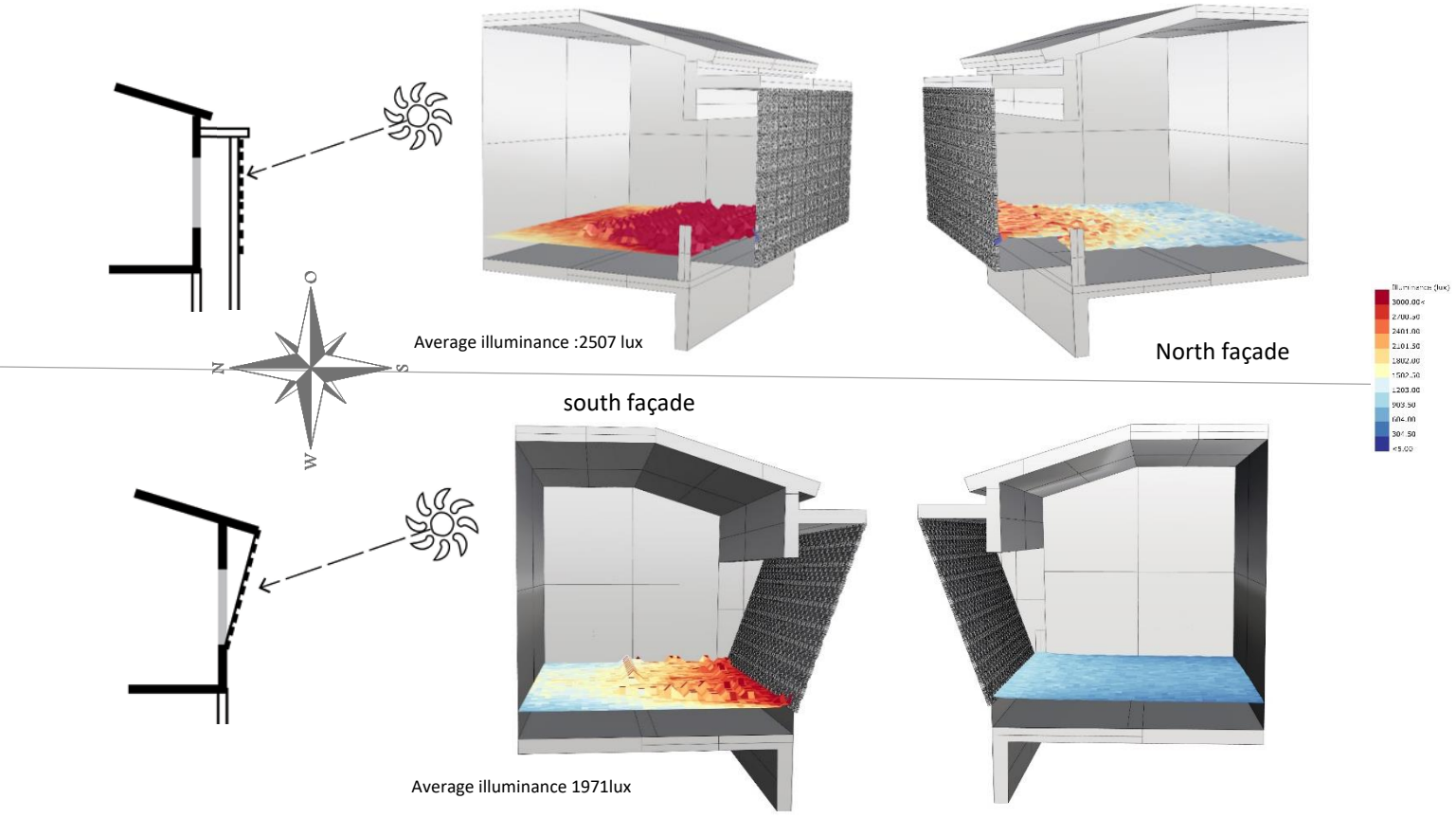
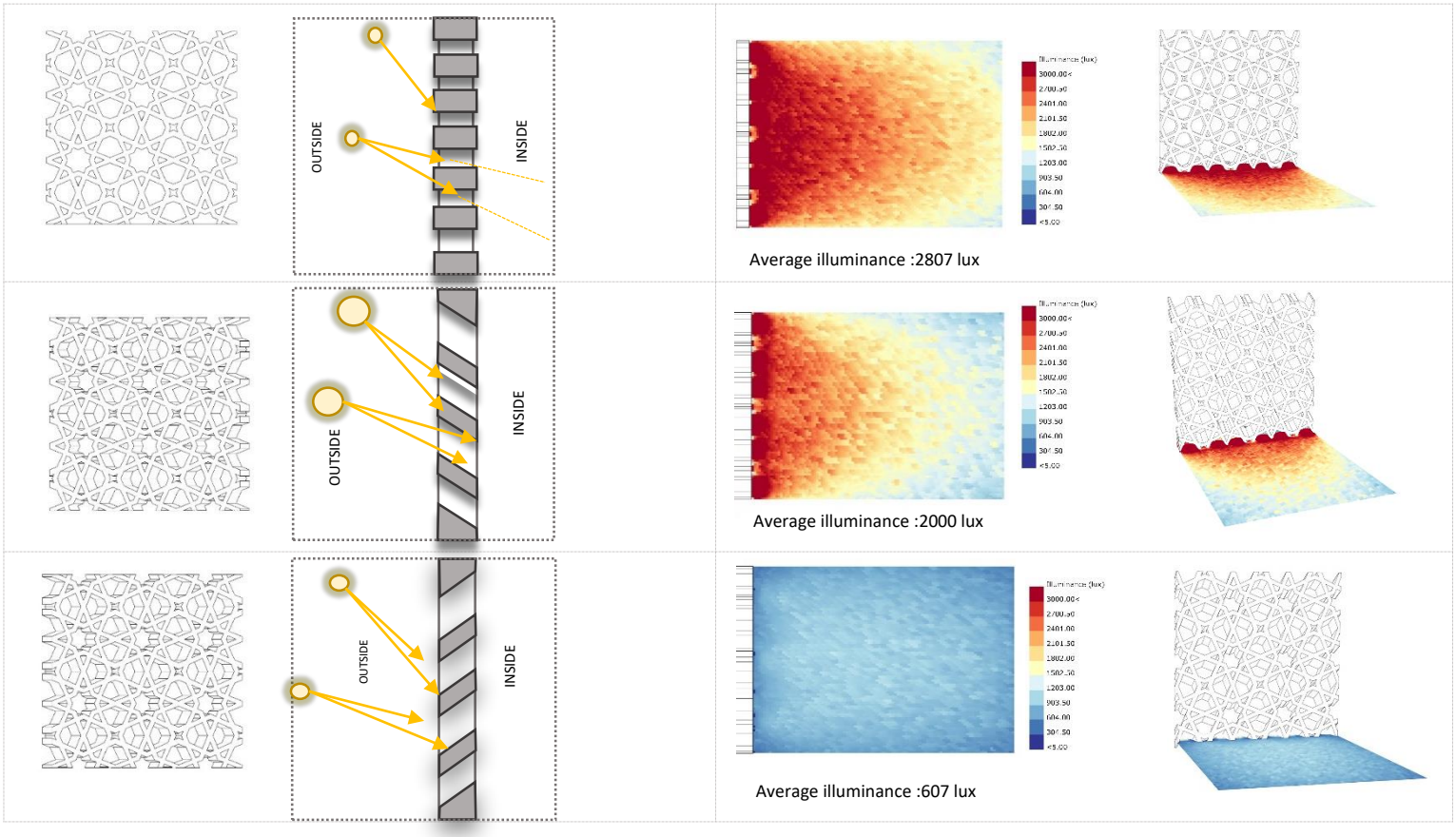


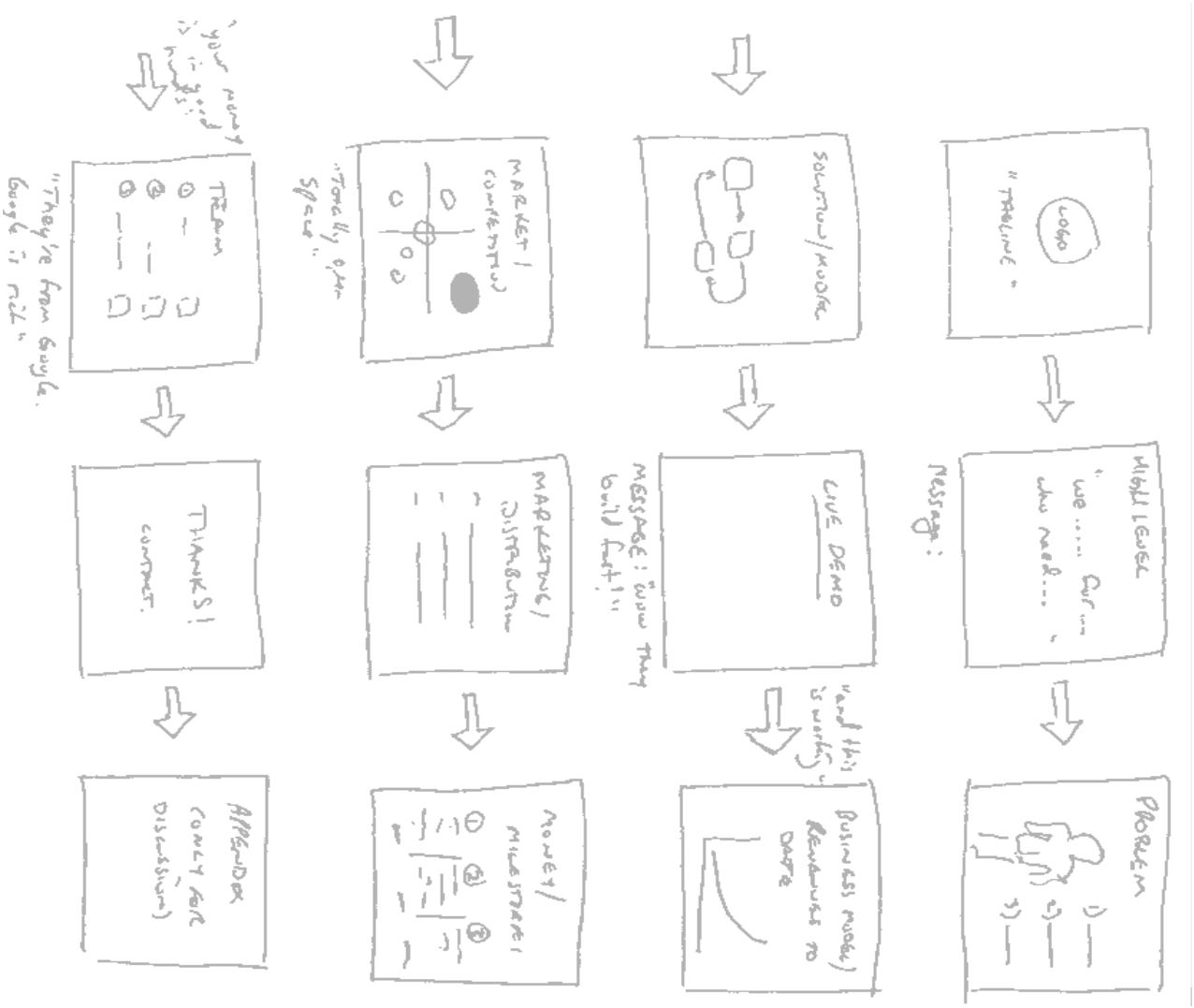
Fig 6.5.2 analysing different component typologies for both north and southern facades source (author)

Since the main study focus was on how the patterns implemented in different typologies of mashrabiya preforms. It was found that sometimes the environmental solutions lie within the smallest details.
Based on that, one of the highly perforated patterns were adjusted to be inclined within its perforations by 45 degrees (visual angle). Subsequently, it was tested how it will perform if the pattern itself was carved into the wood with an inclination angle of 45° to the inside and another case to the outside.

•Outcomes
inclining the pattern with 45° can reduce the amount of light entering through it up to 50 %.

Fig 6.5.3 comparing the shading performance of different forms of the same pattern
source (author)





7. **Conclusion**

Mashrabiya is a component not just a perforated screen

7.Conclusion

The previous studies and remarkable architectural books conducted about this topic was beneficial in forming a total background of Mashrabiya. It was found that Mashrabiya is a passive facade component that was created for shading in extremely hot climates. Based on the literature review of different books and analysing the documented works of Hassan Fathi, it was found that Mashrabiya as a component was mainly created to balance the religious aspects and cultural beliefs of the society to maintain the privacy of the house. The idea of Mashrabiya evolved from the evolution of the building's typology. The building in the past was formed of thick walls and small openings to maintain privacy and prevent excessive solar gains resulted from the sunlight entering the space. The Islamic aspect and environmental factors forced the building to increase the dimensions of the openings. So it can allow more wind to enter and a proper amount of light that can allow the users to read books. Also, to enable the inhabitants to hear the prayers sound "Azan" coming from the mosque.As the typology of the openings evolved to meet the user's requirements and achieve the desired functions, Mashrabiya was formed to achieve the balance between these desired functions and the cultural aspects. Mashrabiya also evolved through time from a region to another to meet the requirements and climatic conditions of each country. The typologies of Mashrabiya varied in geometry, patterns and materials according to the restrictions of the regions that implemented it. Based on that, Mashrabiya was apportioned to different forms and categories. Two of the most common typologies were: Andalusian screen and ottoman Mashrabiya. Most of the existing Islamic Mashrabiya examples were categorised under these two typologies. Analysing and forming a complete understanding of ottoman and Andalusian Mashrabiya can be documentation for more than 70 %of the different shapes of traditional Mashrabiya and perforated screens. Mashrabiya is a complex component with a variety of patterns and geometries, it can combine different patterns in one component. Therefore, it was important to have a full understanding of these variations and how they affect the total performance of the whole component. Consequently, fieldwork was held to comprehend the performance of different forms of Andalusian and ottoman Mashrabiya. Egypt and Spain were selected to observe the different typologies located in their old sites. Also, some of the main objectives of the fieldwork is documenting and understanding the patterns in terms of combination, scale and dimensions and what is the aspect considered in modelling the patterns as well as asking the craftsmen and carpenter who used to model such a component about the prioritised steps followed in the modelling process. The pursuit of these objectives proved a lot of points and introduced new idea and types that were not documented before. In the fieldwork it was found Mashrabiya within its typologies cannot be treated as a perforated screen. There were a lot of passive and effective environmental solutions that lie within the complexity of these components.

During the fieldwork it was found that even the Andalusian Mashrabiya used to have simple geometry and wide floral perforated patterns, it is capable of achieving stack single-sided ventilation through adding top windows with smaller areas, this depended mainly on the geometry and building design itself.

The practicality of these methods, controlled the amount of air entering the space at different depths inside the room. As for ottoman Mashrabiya, the complexity of its geometry and variation of the patterns shaped the total performance of the component and created a balanced harmony between the different environmental parameters and the cultural aspects. Furthermore, it was found that Mashrabiya's function as a façade component is not limited on shading the space and allowing air penetration through the space. It can also allow equal light distribution inside the space. Mashrabiya can be a very effective tool towards achieving indoor comfort and balancing the user's requirements in different environmental parameters. From observing the environmental performance of some Mashrabiya typologies it was found Mashrabiya can block up to 72 %of the daylight that can enter to the space, this is accompanied by more efficiency in protecting from solar gains resulted from the radiations which contribute a lot in reducing the indoor temperature up to 10 degrees less. Also, some other typologies were made to allow more daylight to enter the space with a proper amount, where the component reduces the daylight entering the space with 40 %less. Mashrabiya's role in improving indoor conditions is not limited to preventing excessive solar radiation from entering the space. It is also through orienting the air passing through it to flow at different depths of the room. In the fieldwork, it was found Mashrabiya can reduce the indoor temperature up to 10 °c compared to the same neighbouring rooms that didn't implement any shading element.The complexity found in the details Mashrabiya could gather, shaped the analytical work to focus on the quantification of Mashrabiya on different levels and scales. For that, the patterns were analysed on a small scale, and it was found that the performance of the patterns in terms of shading and ventilation is not always related to the perforation percentage of the patterns, it is more related to the shape and forms of the pattern's component. It was also found that Floral patterns tend to allow more uniform distribution of light inside the space. Analysing Mashrabiya as a component showed how can the total performance of the Mashrabiya vary significantly if patterns were combined in a way that can create stack ventilation. The performance of Mashrabiya can be improved through applying the accurate pattern in the suitable place for the right orientation. Within the experimental steps of the analytical study it was found Mashrabiya can have inoperative impact by having impractical performance, this can happen through applying un recommended patterns to the wrong façade. For that, it was essential to supply the users and decision-makers with recommendation and guidelines that orient

the total performance of Mashrabiya towards the maximum efficiency.
Based on the analytical work, the conclusions that shaped the design recommendations are as follows:

•To balance the need for shading and the ventilation of the space, it is better to implement Mashrabiya as a total component, not just a perforated screen, where a combination of patterns can be used.

•Applying simple perforated screens on the openings is linked with the fact that the more perforation, the more airflow to space, also the more solar access. However, the analytical tool managed to prove the inaccuracy of this fact in terms of complex Mashrabiya. Where complex Mashrabiya can be modified in order to allow more airflow and minimise the solar access, by applying three different patterns two of them are highly perforated, and one has very small perforations. This combination can create a stack single-sided ventilation that is considered to be more than the airflow resulted from using simple wide perforated patterns. Also, it has the capacity of moving the air through the whole depth of the space.

•Adding a cooling in front of the patterns can significantly drop the room temperature up to 4°c. In the end, Mashrabiya is a component that merged art and passive environmental solutions, for that it was widely implemented in the past. Now the contemporary implementation of Mashrabiya is lamented on façade perforated screens in modern building and some few crafts for low-cost buildings and common houses. Going through these examples showed the fact of how Mashrabiya got so disconnected from its original functional forms. After testing Mashrabiya with its different forms with analytical tools, it was found that Mashrabiya can be a double-edged weapon, and its impact can be useless if it is applied in the wrong way. For that this paper aims to revitalise Mashrabiya and reintroduce it for all buildings' types. This can be through offering systematic quantification that can show the performance of each part and type of mashrabiya individual or combined, and consequently suggesting them to different facade and space functions. This paper can be the starting step towards so many future studies that can document all the other typologies that this paper didn't mention and using environmental design as a tool to preserve the heritage and traditional architecture.

One of the contemporary models of Mashrabiya crafted in Cairo; this model represents the rough tracing of the traditional forms without knowing the achievable environmental performance of the component and the consequences of choosing the patterns.

Cairo- El Mosque street

Fig 7.1.1 a component that was being crafted during the field work in Egypt source (author)

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Appendix

- Contemporary implementation of Mashrabiya

The technology of the oriental window in many buildings in the Middle East as there is the Doha Tower in Qatar, which was designed and implemented by John Nouvel 2002-2012 at a height of 231 meters. The main designing concept was combining elements ,Islamic decorative units inspired by the aesthetic and functional philosophy of the barbeque. Merging this traditional arts with contemporary technology in a Cylindrical building, covered the building with a series of decorative layers suspended with Aluminium ore in order to shade the building from the sun and provide natural light in addition to the presence of layers of reflective glass that adds the concept of sun protection as the building is illuminated in the evening.

John Nouvel 's creativity did not stand in this field, but he is about to open his latest innovation, the Louvre Abu Dhabi Museum, which he started in 2007. As he was influenced by the philosophy of the aesthetic and functional barista and presented in a horizontal position through the formation of the dome of several metal layers to improve the thermal climate of the interior vacuum of the building. (Keazor, 2008)

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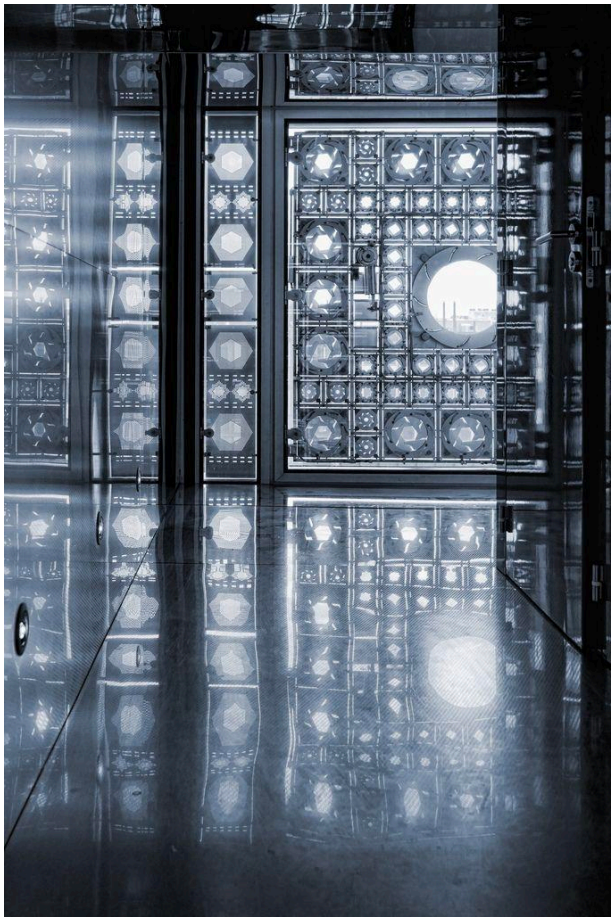

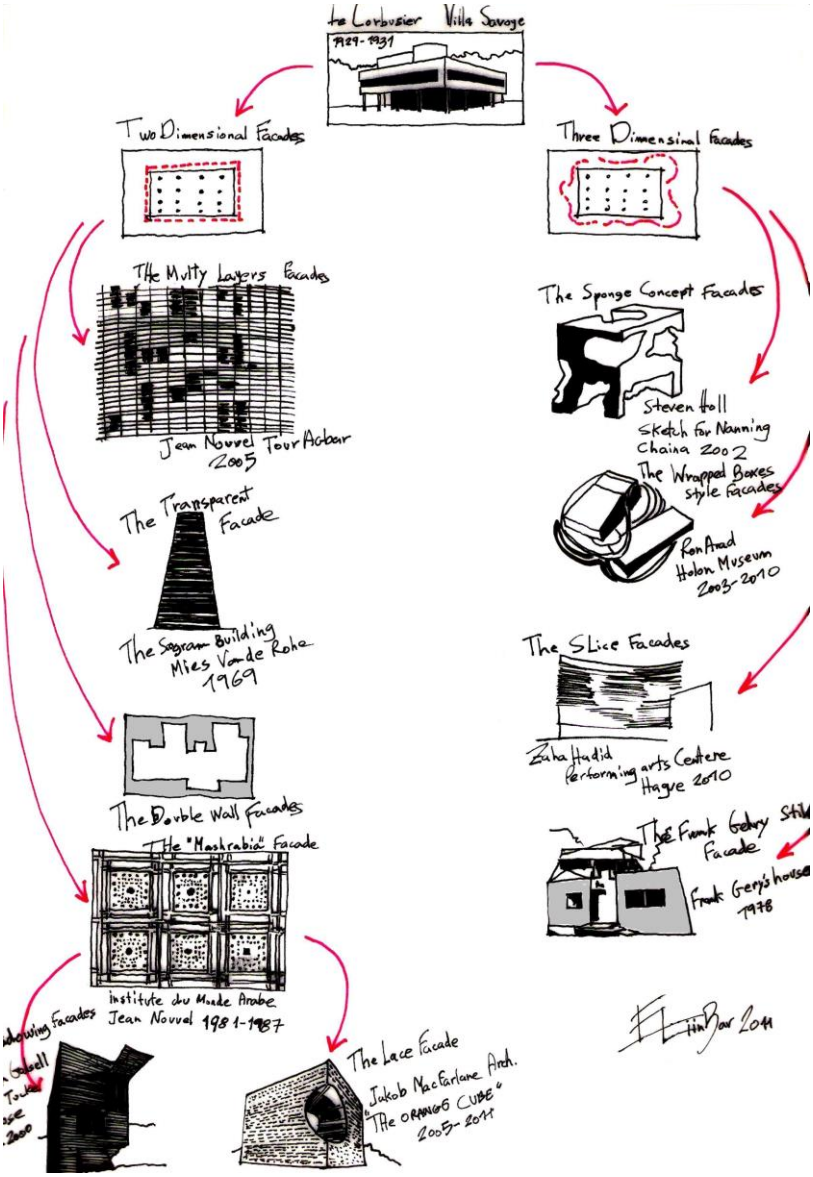

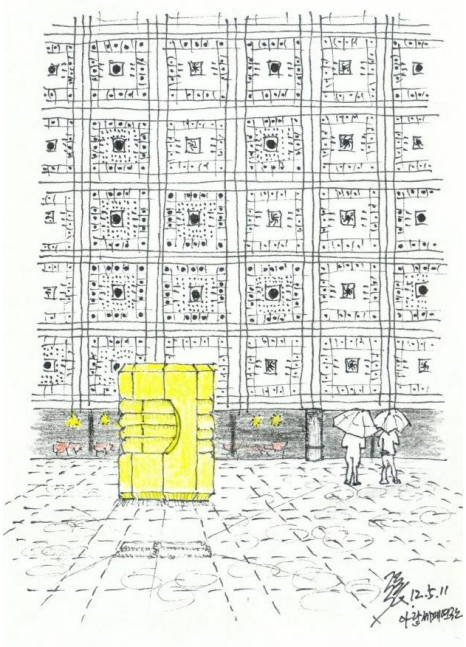

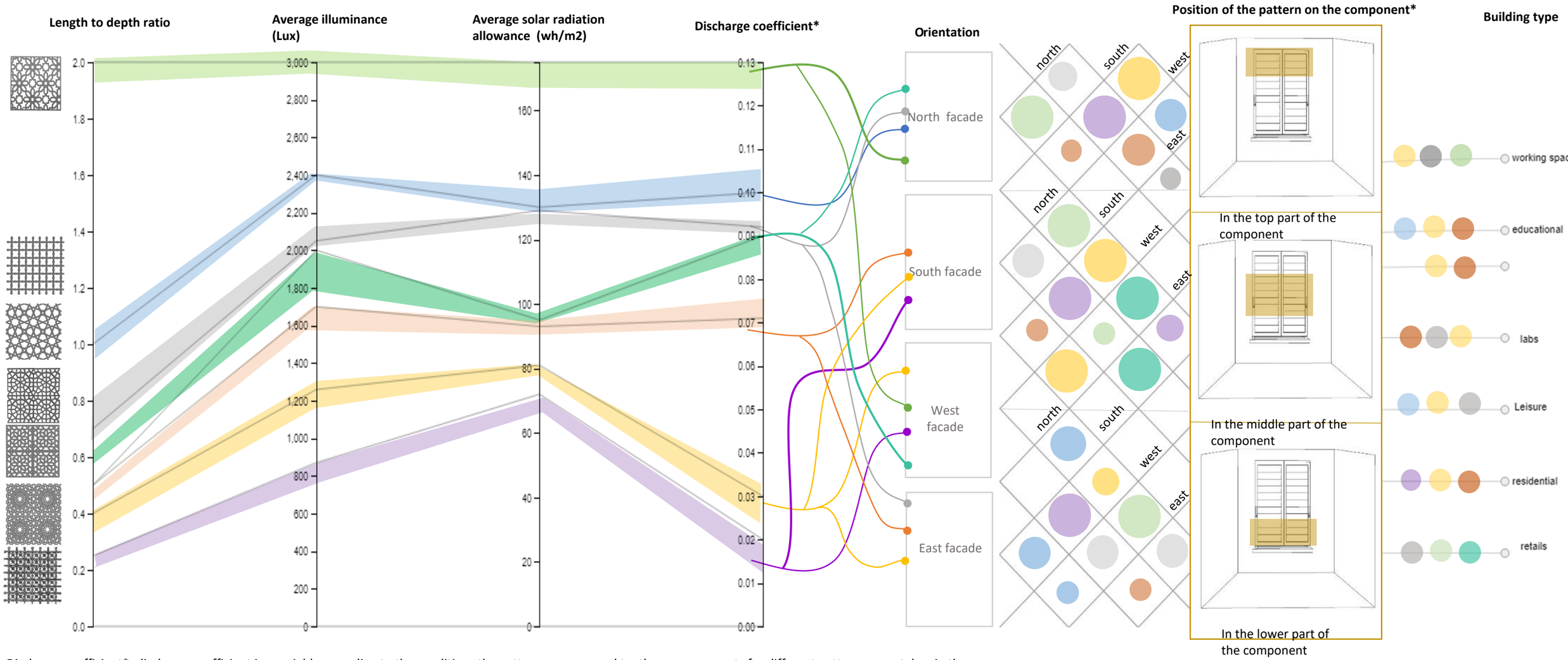

Discharge coefficient*: discharge coefficient is a variable according to the conditions the patterns are exposed to, the measurements for different patterns were taken in the same affecting conditions in order to provide a fair comparison between the patterns and state the difference in the performance according to the patterns

Position of the pattern on the component* : this matrix is comparing what is the preferred patterns for each part of the component in the four orientations ,the size of the circles represent the desirable level

critical Desirable acceptable

Fig 6.2.3.a summary table that gathers all the analysis and studies made
source (author)

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